CS1231S Guidebook

The plain-language companion to the Midterm Companion cheat sheet: every symbol, every word, every example on the sheet, and every tutorial question, broken down step by step.

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How to use this with the cheat sheet

- Every heading here carries a tag like **Sheet 1B §2**. That is the cheat sheet page and section it explains. Read the two side by side.
- Read G0 to G2 once, in order. They are the vocabulary for everything else.
- Then read one topic chapter, then its tutorial walkthrough, then try the matching past-paper questions on the sheet.
- Boxes: **blue-grey** = key idea or definition, **pale red** = trap that loses marks, **navy bar** = trick, **pale yellow** = check yourself.
- Where your uploads have no official answer (all of Tutorials 5 and 6, and a few discussion questions), the answer is worked out here and marked “worked out here, no official key”.

Built from Lectures 1–6, Tutorials 1–6, the proof-writing guideline, the in-tutorial quiz notes, and the AY23/24–AY25/26 midterms. Course conventions: ℕ = {0, 1, 2, …}; negation is written ~; S∘R means R first, then S; proofs are numbered lines with a reason on each.

G0 · Start here: the one idea that links everything

read this first · 10 minutes

Why this module feels like a foreign language

It is one. CS1231S is not testing whether you can *see* that something is true. It tests whether you can **say it in a fixed grammar** so that nobody could misread it. Think of it as a programming language for arguments:

programming	CS1231S
a Boolean value	a **statement** (true or false)
`&&`, `||`, `!`, `if`	$\wedge$, $\vee$, $\sim$, $\rightarrow$
a `for` loop over all items	$\forall$ ("for all")
a search that finds one item	$\exists$ ("there exists")
a function definition you must call exactly	a **definition** ("even", "$\subseteq$", "reflexive")
code that compiles and type-checks	a **proof** with a reason on every line
the compiler	the grader

Your intuition is fine. What you have not had is the grammar. This guidebook teaches the grammar, then shows it running on every example you have.

The one idea: everything is a statement

A **statement** is a sentence that is either true or false, never both. "7 is odd" is a statement. "x is odd" is not (yet): it depends on x. That second kind is a **predicate**; it becomes a statement once you plug in x or put "for all x" / "there is an x" in front.

Every topic in the module is statements in a different costume:

topic	its statements look like	under the costume
Logic (L2)	$p \wedge q \rightarrow r$	true/false combinations
Quantifiers (L3)	$\forall x\, \exists y\, (xy = 1)$	"for every… there is…"
Proofs (L1, L4)	"n even $\rightarrow$ n^2 even"	a $\forall$ statement you must justify
Sets (L5)	$A \subseteq B$	$\forall x\, (x \in A \rightarrow x \in B)$
Relations (L6)	"R is symmetric"	$\forall x\, \forall y\, (xRy \rightarrow yRx)$

So when a set or relation question confuses you, **translate it into logic**. Once it is logic, you already know what to do.

The chain of chapters

1. **Logic** tells you how "and", "or", "not", "if" behave.
2. **Quantifiers** add "for all" and "there exists".
3. **Proofs** are how you show a "for all" statement is true when you cannot check infinitely many cases.
4. **Sets** are defined with logic, so set proofs are logic proofs.
5. **Relations** are sets of pairs, and their properties are "for all" statements, so they are proved the same way.
6. **Equivalence relations** and **partial orders** are just relations with a particular bundle of properties.

How to read any definition

Definitions in this course are written "X is [word] **iff** [condition]". **Iff** means "if and only if": the word and the condition are interchangeable. Whenever you see the word, you may replace it by the condition, and vice versa. That swap is the most common step in every proof you will write.

```
n is even  ⇔  ∃k ∈ ℤ, n = 2k
read: n is even exactly when some integer k
has n = 2k

"14 is even" → swap in the condition
             → 14 = 2·7, k = 7
"n = 2(m+1)"   → swap back            → n is even
```

Unwrapping a word into its condition and wrapping a condition back into its word is 80% of this module. Proofs in G6 are literally: unwrap the hypothesis, do algebra, rewrap into the goal.

How to read a line of maths aloud

If you cannot say a line aloud, you cannot use it. Practise with these (G1 has every symbol):

```
∀x ∈ ℝ ∃y ∈ ℝ (x + y = 0)
for every real x there is a real y with x + y = 0

A ⊆ B ⇔ ∀x (x ∈ A → x ∈ B)
A is a subset of B iff every element of A is in B

x R y
x is related to y by R, i.e. the pair (x, y) is in R

d | n
d divides n (a true/false statement, not a number)
```

Trick: when lost, go down one level

Stuck on a relation question? Write the property as a $\forall$ statement. Stuck on a set identity? Write each side with $\in$ and turn $\cup$, $\cap$ into $\vee$, $\wedge$. Stuck on a proof? Write the definition of every word in the claim. Going one level more concrete is almost always the move.

What the exam wants from you

- **MCQs** (15 × 2 marks): mostly "which of these are true?". Almost every one is killed by one counterexample or confirmed by one standard shape. The speed-trick pages 0A and 0B are built for these; G3 to G10 explain every trick.
- **Written part** (20 marks): roster sets, a Hasse diagram, relation closures, and one short proof. The proof is marked on format as much as on idea: numbered lines, a reason on each, correct first and last lines. G6 teaches exactly that.

The mindset shift that matters most

In everyday life "it seems true in every case I tried" is convincing. Here it proves nothing about a $\forall$ statement. But **one** failing case is a complete disproof. So: examples can only **disprove** "for all" claims, and can only **prove** "there exists" claims. Keep that asymmetry in your head and half the MCQs become easy.

G1 · Every symbol, read aloud

covers every symbol used on the cheat sheet

Each row gives the symbol, how to say it, what it means, and a tiny example. The "say it" column matters: if you can say a line aloud in English, you understand it.

The "straight line" symbols

all pages

Several different symbols are built from a vertical line. They mean completely different things, so this is the first thing to get straight.

symbol	say it	meaning and example
$\nmid$	"does not divide"	A vertical line with a slash through it. $5 \nmid 12$: 12 is not a multiple of 5. Used in Lecture 1's exhaustion proof ("$5 \nmid 6$").
$\mid$ between numbers	"divides"	$3 \mid 12$: 12 is 3 times an integer. It is a **true/false statement**, not a division. $3 \mid 10$ is false; it is not "3.33".
$\vdash$	"yields", "proves"	A vertical line with a dash sticking out to the right (a *turnstile*). $p \to q, p \vdash q$: from the premises on the left you may conclude the right. Used in the rule-of-inference tables (Sheet 1 §5, 2 §7).
$\therefore$	"therefore"	Marks the conclusion of an argument. Lectures write rules as premises above a line, then $\therefore$ conclusion. Same job as $\vdash$.
$\lvert x \rvert$	"absolute value of x"	Around a **number**: its size without sign. $\lvert -3 \rvert = 3$.
$\lvert A \rvert$	"size of A"	Around a **set**: its number of elements (cardinality). $\lvert \{a, b\} \rvert = 2$.
$\{x \in \mathbb{Z} \mid x > 0\}$	"such that"	Inside set braces the bar means "such that"; a colon means the same. Read: "the set of integers x such that x > 0".
$\mid$ in $P(A \mid B)$	"given"	Probability only (later topic): chance of A given B.

Trick: decide by what is on each side

Numbers on both sides ⇒ divides. Wrapped around one thing ⇒ size or absolute value. Inside $\{\}$ after a variable ⇒ "such that". Formulas on both sides, with a dash ⇒ "yields" ($\vdash$).

Logic symbols

Sheet 0A, 1, 1B

symbol	say it	meaning and example
$\sim p$	"not p"	True exactly when p is false. Other books write $\neg p$; same thing. Binds tightest: $\sim p \wedge q$ means $(\sim p) \wedge q$.
$p \wedge q$	"p and q"	True only when both are true. Looks like an A without the bar: **A**nd.
$p \vee q$	"p or q"	True when at least one is true (inclusive or). Looks like a cup that catches either.
$p \to q$	"if p then q", "p implies q"	False only when p is true and q is false. p is the **hypothesis**, q the **conclusion**.
$p \leftrightarrow q$	"p if and only if q"	True when p and q have the same truth value.
$P \equiv Q$	"P is logically equivalent to Q"	A claim **about** two formulas: same truth value on every row. Not a connective inside a formula.
$\not\equiv$	"not equivalent"	Some row gives different values.
$\Rightarrow$, $\Longrightarrow$	"implies" (in your working)	Used in explanations between claims: "p = F ⇒ the implication is true". Informal cousin of $\to$.
$\Leftrightarrow$, $\Longleftrightarrow$	"iff" (in definitions)	"n is even $\Longleftrightarrow \exists$k, n = 2k": the two sides can be swapped.
true, false		The constant statements. Truth tables write T and F. In law proofs write **true**/**false**, never t/c.
■ ■	"end of proof"	Also written QED. Put it after your last line.

Quantifier symbols

Sheet 2

symbol	say it	meaning and example
$\forall x \in D$	"for all x in D"	An upside-down A: **A**ll. $\forall x \in \mathbb{Z} (x^2 \geq 0)$.
$\exists x \in D$	"there exists x in D"	A backwards E: **E**xists. At least one, possibly more. $\exists x \in \mathbb{Z} (x^2 = 4)$.
$\exists! x$	"there is exactly one x"	Existence plus uniqueness. $\exists! x \in \mathbb{R} (x + 3 = 5)$.
$P(x)$	"P of x"	A predicate: true or false once x is fixed. $Even(x)$.
s.t.	"such that"	Written after ∃: "∃k ∈ ℤ s.t. n = 2k".

Number sets and number notation

Sheet 3, 4 §1

symbol	say it	meaning
$\mathbb{N}$	naturals	{0, 1, 2, ...}. **Includes 0 in this course.**
$\mathbb{Z}$	integers	{..., −2, −1, 0, 1, 2, ...} (Z from German *Zahlen*).
$\mathbb{Q}$	rationals	fractions a/b with a, b ∈ $\mathbb{Z}$, b ≠ 0 (Q for quotient).
$\mathbb{R}$	reals	all points on the number line, including $\sqrt{2}$, π.
$\mathbb{Z}^+$, $\mathbb{Z}^-$	positive / negative integers	0 is in neither. $\mathbb{Z}_{\geq 0}$ = non-negative = {0, 1, 2, ...} = $\mathbb{N}$.
$\mathbb{Z}_{\geq 12}$		integers ≥ 12. Subscripts restrict the set.
$\lfloor x \rfloor$	floor of x	largest integer ≤ x. $\lfloor 2.7 \rfloor = 2$, $\lfloor -2.7 \rfloor = -3$.
$\lceil x \rceil$	ceiling of x	smallest integer ≥ x. $\lceil 2.1 \rceil = 3$, $\lceil 5 \rceil = 5$.
$a \equiv b \pmod{n}$	"a is congruent to b mod n"	$n \mid (a-b)$: same remainder on division by n. $17 \equiv 2 \pmod{5}$. **Different** from logical ≡.

Set symbols

Sheet 4, 4B, 0B §1–3

symbol	say it	meaning and example
$\{1, 2, 3\}$	"the set 1, 2, 3"	Roster form. Order and repeats do not matter: {1, 2} = {2, 1, 1}.
$x \in A$	"x is an element of A"	x is one of the items listed in A. $2 \in \{1, 2\}$.
$x \notin A$	"x is not in A"	$5 \notin \{1, 2\}$.
$A \subseteq B$	"A is a subset of B"	Every element of A is in B. $\{1\} \subseteq \{1, 2\}$. The line under it is like the line under ≤: equality allowed.
$A \subsetneq B$	"proper subset"	A ⊆ B and A ≠ B. The slash on the bottom line says "not equal".
$A \not\subseteq B$	"not a subset"	Some element of A is missing from B.
$\subset$		Ambiguous in textbooks; the lecture says avoid it. Use ⊆ or ⊊.
$\varnothing$	"the empty set"	The set with no elements, {}. A circle with a slash. Not the number 0.
U	"universal set"	Everything under discussion.
$A \cup B$	"A union B"	in A or in B (or both). ∪ looks like ∨: same idea.
$A \cap B$	"A intersect B"	in A and in B. ∩ looks like ∧: same idea.
$A \setminus B$	"A minus B"	in A but not in B. Lecture also writes A − B.
$\bar{A}$, A^c	"complement of A"	in U but not in A.
$A \oplus B$	"symmetric difference"	in exactly one of A, B (Tutorial 3).
(a, b)	"the ordered pair a, b"	Order matters: (1, 2) ≠ (2, 1). Careful: (a, b) is also an open interval of reals; context tells.
$A \times B$	"A cross B"	all pairs (a, b) with a ∈ A, b ∈ B.
$\mathcal{P}(A)$	"power set of A"	the set of all subsets of A. $\mathcal{P}(\{1\}) = \{\varnothing, \{1\}\}$.
$A_1, A_2, \ldots$	"A one, A two"	Subscripts just label different sets.

Relation symbols

Sheet 5, 5B, 6

symbol	say it	meaning
$x\,R\,y$	"x is R-related to y"	shorthand for $(x, y) \in R$.
R^{-1}	"R inverse"	every pair of R flipped: (x, y) becomes (y, x).
$S \circ R$	"S after R", "S composed with R"	**R first, then S**. Read right to left, like f(g(x)).
R^2, R^n	"R squared"	R∘R, R∘R∘…∘R: follow n arrows in a row.
R^r, R^s, R^t	"reflexive / symmetric / transitive closure"	smallest relation containing R with that property.
$x \sim y$	"x is equivalent to y"	a name for an equivalence relation. Same squiggle as "not", different job: between two things it is a relation, in front of one statement it is negation.
$[a]$, $[a]_\sim$	"the class of a"	the set of everything equivalent to a.
$A/\sim$	"A mod tilde", "the quotient"	the set of all classes.
$x \preccurlyeq y$	"x precedes y"	a general partial order; think "$\leq$ but maybe not for numbers".
$x \prec y$	"x strictly precedes y"	$x \preccurlyeq y$ and $x \neq y$.
$\preccurlyeq^*$	"a linearization of $\preccurlyeq$"	a total order that respects $\preccurlyeq$.

Function symbols

Sheet 8

symbol	say it	meaning
$f\colon A \to B$	"f from A to B"	f takes inputs from A (domain) and gives outputs in B (codomain). Here $\to$ is **not** "implies".
$x \mapsto 2x$	"x maps to 2x"	the rule of the function.
$f(X)$	"image of X"	all outputs from inputs in X.
$f^{-1}(Y)$	"preimage of Y"	all inputs whose output lands in Y. Exists even when f has no inverse.
$g \circ f$	"g after f"	$(g \circ f)(x) = g(f(x))$: f first.

Symbols with two jobs

- $\sim$: "not" in front of a statement; "is equivalent to" between two objects.
- $\equiv$: "logically equivalent" between formulas; "congruent mod n" between numbers.
- $\to$: "implies" between statements; "from … to …" in $f\colon A \to B$.
- $|$: divides / size / absolute value / such that (table at the top).
- (a, b): ordered pair, or open interval.
- $^{-1}$: inverse relation / inverse function / preimage.

G2 · The words, in plain English

Lecture 1 terminology · Lecture 2–4 vocabulary

Kinds of mathematical fact

Lecture 1 §1.3.2

word	what it is	course example
Definition	The exact meaning of a word. Not proved, just agreed. You must quote it exactly in proofs.	"n is even iff $\exists k \in \mathbb{Z}$, n = 2k".
Axiom (postulate)	A statement assumed true without proof. The starting blocks.	Euclid's postulates; "every integer is even or odd, not both" is used like one (Assumption 1).
Theorem	An important statement that has been proved.	Theorem 4.2.2: the sum of any two rationals is rational.
Lemma	A small proved result whose job is to help prove a theorem. A stepping stone.	Lemma 4.4.4: $-\lvert r\rvert \le r \le \lvert r\rvert$, used to prove the triangle inequality.
Corollary	A result that falls straight out of a theorem, with almost no extra work.	Corollary 4.2.3: the double of a rational is rational. Proof: $2r = r + r$, and the theorem says a sum of two rationals is rational. Done.
Proposition	A proved statement of medium importance.	Prop. 4.6.4: if n^2 is even then n is even.
Conjecture	Believed true, not yet proved.	Goldbach: every even integer > 2 is a sum of two primes.
Assumption	Something the course lets you use without proof.	Assumption 1 (even or odd, not both); Assumption 2 (every rational can be put in lowest terms).

Why the corollary idea matters to you

Lecture 1 proves "the product of two *consecutive* odd numbers is odd". If you had already proved "the product of *any* two odd numbers is odd", the consecutive case is a corollary: it is a special case of a proved general fact. In proofs you may always cite a proved theorem instead of redoing it: "(by Theorem 4.2.2)".

Words about statements

Sheet 1 §6, 2 §1

word	meaning
Statement (proposition)	A sentence that is true or false, not both. "Close the door" and "x + 3 = 5" are not statements.
Statement variable	A letter p, q, r standing for any statement.
Compound statement / statement form	Statements joined with ~, ∧, ∨, →, ↔.
Predicate	A sentence with variables, like "x > 3". It becomes a statement once x is given a value or quantified.
Domain	The set the variable ranges over ($\mathbb{Z}$, $\mathbb{R}$, the students in the room).
Truth set	All values in the domain that make the predicate true.
Tautology	A form that is true on every row, whatever the letters are. $p \vee \sim p$.
Contradiction	A form that is false on every row. $p \wedge \sim p$. (Also the name of a proof method, G6.)
Logically equivalent (≡)	Two forms with identical truth values on every row.
Universal statement	"for all …" (∀). All positive integers are > 0.
Existential statement	"there is at least one …" (∃). There is an even prime.
Conditional statement	"if … then …" (→).
Universal conditional	"for all x, if P(x) then Q(x)". "All dogs are mammals" is secretly this.
Universal existential	∀ then ∃: every real number has an additive inverse.
Existential universal	∃ then ∀: there is a positive integer ≤ every positive integer (it is 1).

The if-then family

Sheet 1 §3

Start from $p \to q$: "if it rains (p), the ground is wet (q)".

name	form	example	same as original?
Hypothesis / conclusion	p / q	"it rains" / "ground wet"	
Contrapositive	$\sim q \to \sim p$	if the ground is not wet, it did not rain	**yes**, always
Converse	$q \to p$	if the ground is wet, it rained	no (a sprinkler)
Inverse	$\sim p \to \sim q$	if it does not rain, the ground is not wet	no (a sprinkler)
Negation	$p \wedge \sim q$	it rains and the ground is not wet	opposite

Memory hook: **contra**positive = flip **and** negate (both moves, stays true). Converse = flip only. Inverse = negate only. Converse and inverse are each other's contrapositive, so they are equivalent to each other.

Words about arguments

Sheet 1 §5–6

word	meaning
Argument	A list of statements (premises) followed by a conclusion.
Premise	A statement you are given to start from.
Valid	The form guarantees: whenever all premises are true, the conclusion is true. Validity is about the **shape**, not whether the premises are actually true.
Sound	Valid **and** all premises actually true. So the conclusion is actually true.
Critical row	A truth-table row where all premises are true. Validity is checked only on these.
Rule of inference	A small valid argument form you may use as one step: modus ponens, modus tollens, etc. (G4).
Fallacy	An invalid pattern that looks valid: converse error, inverse error.
Syllogism	An argument with two premises and a conclusion. **Hypothetical** syllogism: both premises are if-thens ("hypotheticals"), and you chain them: $p \to q$, $q \to r$ $\therefore p \to r$.
Modus ponens	Latin "method of affirming": $p \to q$ and p, so q.
Modus tollens	Latin "method of denying": $p \to q$ and $\sim q$, so $\sim p$.

Words inside proofs

Sheet 0C, 3

word	meaning
Arbitrary / "particular but arbitrarily chosen"	You fix one element but use nothing about it except what every element has. That is what lets one element stand for all (G6).
Generic particular	The name of that method: prove it for a generic element, conclude it for all.
Witness	The concrete value that makes an ∃ true. In "n = 2k", k is the witness that n is even.
Counterexample	One value that makes a ∀ statement fail. One is enough.
Vacuously true	True because there is nothing to check: "if p then q" with p false, or "all elements of ∅ are purple".
Closure	"$\mathbb{Z}$ is closed under + and ×": adding or multiplying integers gives an integer. The reason you write whenever you claim a new expression is an integer.
Basic algebra	The reason for rearranging, expanding, factorising.
WLOG	"without loss of generality": you may assume a special case (say a < b) because the other case is the same with names swapped.
Suppose not	Opening line of a proof by contradiction: assume the negation of the claim.
iff	"if and only if": both directions.
Necessary / sufficient	"p is sufficient for q": p → q (having p is enough). "p is necessary for q": q → p (you cannot have q without p).

Words about sets and relations

Sheet 4, 5, 5B, 6

word	meaning
Element / member	Something inside a set.
Cardinality	Number of elements, |A|.
Disjoint	No common elements: A ∩ B = ∅.
Partition	Cutting a set into non-empty, non-overlapping pieces that cover it.
Relation from A to B	Any set of pairs (a, b) with a ∈ A, b ∈ B. Think: a set of arrows.
Reflexive / symmetric / transitive / antisymmetric	Properties of a relation (G8): every loop present / every arrow has a return arrow / every two-step path has a shortcut / no two-way arrows between different points.
Equivalence relation	Reflexive + symmetric + transitive. Means "has the same X as".
Partial order	Reflexive + antisymmetric + transitive. Means "comes before or equal to".
Poset	Partially ordered set: a set with a partial order.

G3 · Logic I: connectives, truth tables, the if-then family

Lecture 2 · explains Sheet 1 §1, §3, §4, §6 and the reflexes on 0A

1. Five connectives, five truth tables

Sheet 1 §1

A connective builds a bigger statement out of smaller ones. Its truth table lists every combination of T/F for the parts and says what the whole is.

p	q	~p	p ∧ q	p ∨ q	p → q	p ↔ q
T	T	F	T	T	T	T
T	F	F	F	T	**F**	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

- $\wedge$ is picky: needs both. $\vee$ is generous: needs one. $\vee$ is **inclusive**: T ∨ T is T.
- $\leftrightarrow$ is "same value": true on TT and FF.
- With n letters there are 2^n rows: 2 letters give 4 rows, 3 give 8, 4 give 16.

2. Why is "if F then …" true?

Sheet 1 §1

This is the row everyone finds weird. Read $p \rightarrow q$ as a **promise**: "If you score 90 (p), I will buy you a laptop (q)."

p	q	what happened	promise broken?
T	T	scored 90, got laptop	no → **T**
T	F	scored 90, no laptop	**yes → F**
F	T	didn't score 90, got laptop anyway	no → **T**
F	F	didn't score 90, no laptop	no → **T**

The only way to break an if-then is: **hypothesis true, conclusion false**. When the hypothesis is false, the promise was never tested, so it stands. That is called **vacuously true**. So "false → anything" is true, and "anything → true" is true.

> **Trick: the one-row rule**
>
> To decide if $A \rightarrow B$ can be false, ask only: can A be true while B is false? If not, it is always true (a tautology). This one question solves most logic MCQs (Sheet 0A reflex 4, 1B §2).

3. Order of operations

Sheet 0A reflex 1, 1 §1

Like × before + in arithmetic:

```
~        binds tightest   (like a minus sign)
∧  ∨     next, EQUAL rank  (so you must bracket)
→  ↔       loosest
```

- $\sim p \wedge q$ means $(\sim p) \wedge q$, not $\sim(p \wedge q)$.
- $a \rightarrow b \vee c$ means $a \rightarrow (b \vee c)$, because → is looser than ∨.
- $p \wedge q \vee r$ is **ambiguous** in this course. You must write $(p \wedge q) \vee r$ or $p \wedge (q \vee r)$. Writing it without brackets loses marks.

The **main connective** is the one applied last: the loosest one not inside brackets. It decides what kind of statement the whole thing is. In $(p \wedge q) \rightarrow (p \vee q) \vee junk$ the main connective is →, so the whole thing is an if-then, and the one-row rule applies.

4. Evaluating one row, carefully

Take $(p \vee \sim q) \rightarrow q$ at p = T, q = F. Work inside out, writing each value:

```
~q          = ~F = T
p ∨ ~q      = T ∨ T = T
(…) → q     = T → F = F      the one false row of →
```

So the formula is **not** a tautology: this row is a counterexample.

5. Tautology, contradiction, equivalence

Sheet 1 §6

- **Tautology**: the final column is all T. Example $p \vee \sim p$ (either it rains or it doesn't).
- **Contradiction**: final column all F. Example $p \wedge \sim p$.
- $P \equiv Q$: P and Q have identical final columns. Equivalently, $P \leftrightarrow Q$ is a tautology.

≡ versus ↔: ↔ is a connective that builds a new formula that may be T on some rows and F on others. ≡ is you making a claim that two formulas agree on every row. "$p \leftrightarrow q$" is a formula; "$p \rightarrow q \equiv \sim q \rightarrow \sim p$" is a fact.

> **Trick: disproving an equivalence needs ONE row**
>
> To show $P \not\equiv Q$, find one row where they differ. To show $P \equiv Q$, you need every row (truth table) or a chain of laws (G4 §4). Test rows first; only simplify if forced (Sheet 1B §1).

6. The if-then family, and why only the contrapositive survives

Sheet 1 §3

```
p q | p→q   ~q→~p | q→p   ~p→~q | p∧~q
T T |  T      T   |  T      T   |  F
T F |  F      F   |  T      T   |  T
F T |  T      T   |  F      F   |  F
F F |  T      T   |  T      T   |  F
      original = contrapositive; converse = inverse
```

Why the contrapositive is the same statement: "if it rains, the ground is wet" rules out exactly one situation: rain and dry ground. "If the ground is dry, it did not rain" rules out the same situation. Same forbidden row, same statement.

Why converse and inverse are not: "if the ground is wet then it rained" forbids a different situation (wet ground, no rain). A sprinkler makes that happen while the original is still true.

The **negation** of $p \rightarrow q$ is not an if-then. "It is not true that if it rains the ground is wet" means "it rained and the ground was dry": $p \wedge \sim q$.

> **Trap (Sheet 1 §3)**
>
> "Converse true but inverse false" can never happen: converse and inverse are contrapositives of each other, so they are always both true or both false. AY25/26 Q10 used this.

7. English into symbols

Sheet 1 §4

English	symbols	how to see it
if p then q; p implies q; q whenever p	$p \to q$	p is the trigger
p only if q	$p \to q$	p can happen **only** when q holds, so p guarantees q
p if q	$q \to p$	"if" introduces the **hypothesis**, wherever it sits
p is sufficient for q	$p \to q$	p alone is enough to get q
p is necessary for q	$q \to p$	no q without p, so q forces p
p iff q; necessary and sufficient	$p \leftrightarrow q$	both directions
p but q; although p, q	$p \wedge q$	"but" is just "and" with attitude
neither p nor q	$\sim p \wedge \sim q$	not p, and not q
p unless q	$\sim q \to p \equiv p \vee q$	if not q, then p

Sheet 1 §4, from Tutorial 1:

```
"I use the umbrella if it rains"
   "if" is in front of "it rains" → hypothesis
   rains → umbrella
"I use the umbrella only if it rains"
   umbrella can happen only when it rains
   umbrella → rains

"Either you are on time or you are fired"
   if you are not on time (~p) you are fired (q)
   ~p → q   ≡   p ∨ q          implication law
```

Trick for "only if"

Replace "only if" by **"→"** in the exact spot it sits: "p only if q" = p → q. Replace a lone "if" by moving what follows it to the front: "p if q" = q → p. Never reason about meaning under time pressure; use the substitution.

8. The four reflexes on Sheet 0A, explained

Sheet 0A top box

1. **Find the main connective** (§3 above): it tells you whether you are testing an if-then, an or, or an and.
2. **Rename big chunks.** If $p \vee q$ appears twice, call it A. $(p \vee q \vee \sim r) \wedge (s \vee \sim(p \vee q))$ becomes $(A \vee \sim r) \wedge (s \vee \sim A)$. The shape becomes recognisable (G4).
3. **"Is it true?"** When an MCQ asks whether a formula with letters "is true", it means true for every assignment: a tautology.
4. **One bad row kills it.** A single row making it false is a full disproof. To confirm, either match a known valid shape (G4 §2) or show that the bad row is impossible (G4 §6).

9. Definitions on Sheet 1 §6, unpacked

- **Statement**: "x + 3 = 5" is an open sentence, not a statement, because its truth depends on x. Questions and commands are not statements.
- **Valid argument**: whenever all premises are true, the conclusion is true. Equivalently, "(premise 1 ∧ premise 2 ∧ ...) → conclusion" is a tautology. So checking validity is the same as the one-row rule: can the premises all be true while the conclusion is false?
- **Critical row**: a row where every premise is true. Only these rows matter for validity.
- **Sound**: valid, and the premises really are true. "All cats fly; Tom is a cat; so Tom flies" is valid (good shape) but not sound (false premise).

Check yourself

1. Is "2 + 2 = 5 → the moon is cheese" true? *Yes: false hypothesis.*
2. Write "You pass only if you study" in symbols. *pass → study.*
3. Contrapositive of "if n^2 is even then n is even"? *if n is odd then n^2 is odd (using "not even" = odd, Assumption 1).*
4. Main connective of $\sim p \vee q \to r$? *→, so it reads $((\sim p) \vee q) \to r$.*

G4 · Logic II: arguments, laws, and every logic example

Lecture 2 · explains Sheet 0A §1–3, Sheet 1 §2 §5, all of Sheet 1B

1. Rules of inference: the legal moves

Sheet 1 §5

A rule of inference is a tiny argument that is always valid. Read "$p \to q,\ p \vdash q$" as: "if you already have $p \to q$ and p, you may write down q." In a proof, each line is justified by one of these.

rule	form	in everyday words
Modus ponens	$p \to q,\ p \vdash q$	If it rains the road is wet. It rains. So the road is wet.
Modus tollens	$p \to q,\ \sim q \vdash \sim p$	If it rains the road is wet. The road is dry. So it did not rain.
Hypothetical syllogism (transitivity)	$p \to q,\ q \to r \vdash p \to r$	If I study I pass. If I pass I graduate. So if I study I graduate. **Chain the ifs.**
Generalisation	$p \vdash p \vee q$	I am in Singapore, so I am in Singapore or on Mars. Adding an "or" can only make it easier to be true.
Specialisation	$p \wedge q \vdash p$	It is hot and sunny, so it is hot.
Elimination	$p \vee q,\ \sim p \vdash q$	It is in my bag or my pocket. Not my bag. So my pocket.
Division into cases	$p \vee q,\ p \to r,\ q \to r \vdash r$	I take bus or MRT. Either way I'm late. So I'm late.
Conjunction	$p,\ q \vdash p \wedge q$	Two facts may be joined with "and".
Contradiction rule	$\sim p \to \mathit{false} \vdash p$	If assuming "not p" leads to nonsense, p must be true. **This is why proof by contradiction works.**

Tutorial 1 calls hypothetical syllogism "transitivity". Both names are accepted.

2. The valid shapes on Sheet 0A §2

Sheet 0A §2

Each rule above, written as one formula "(premises joined by $\wedge$) $\to$ conclusion", is a tautology. That is Sheet 0A's table. Why each one can never be false (use the one-row rule: try to make the left side true and the right side false):

shape	why it can't fail
$p \wedge q \to p$	left true forces p true.
$p \to p \vee q$	p true makes p $\vee$ q true.
$p \wedge (p \to q) \to q$	q false and p true makes $p \to q$ false, so the left fails.
$\sim q \wedge (p \to q) \to \sim p$	~p false means p true; with ~q, $p \to q$ is T $\to$ F, false.
$(p \to q) \wedge (q \to r) \to (p \to r)$	right false needs p = T, r = F. Then q = T breaks $q \to r$, q = F breaks $p \to q$.
$(p \vee q) \wedge \sim p \to q$	q false and p false makes p $\vee$ q false.
$(p \vee q) \wedge (p \to r) \wedge (q \to r) \to r$	r false forces p false and q false, then p $\vee$ q is false.
Resolution $(p \vee q) \wedge (\sim p \vee r) \to (q \vee r)$	right false: q = r = F. Then first part needs p = T, second needs p = F. Clash.
$(\sim p \to \mathit{false}) \to p$	p false makes $\sim p \to \mathit{false}$ into T $\to$ F, false.
$(p \to q) \vee (q \to p)$	the first fails only at p = T, q = F, and there the second is F $\to$ T, true.
$p \to (q \to p)$	p true makes $q \to p$ true.

Resolution in words: one piece contains A, another contains ~A. One of A, ~A is false, so the "leftovers" of that piece must carry it. Hence leftovers joined by $\vee$.

3. Fallacies: shapes that look valid but are not

Sheet 0A "Look valid", 1 §5

fallacy	shape	killer row
Converse error (affirming the consequent)	$(p \to q) \wedge q \to p$	p = F, q = T. "If it rains the road is wet; the road is wet; so it rained." Sprinkler.
Inverse error (denying the antecedent)	$(p \to q) \wedge \sim p \to \sim q$	p = F, q = T. "It didn't rain, so the road is dry." Sprinkler.
Converse as a claim	$(p \to q) \to (q \to p)$	p = F, q = T.
"Or means only one"	$(p \vee q) \wedge p \to \sim q$	p = q = T. $\vee$ is inclusive.

Tutorial 1 Q7b: "If at least one of two numbers is divisible by 6, their product is. Neither is divisible by 6. So the product is not." That is the inverse error ($p \to q,\ \sim p \therefore \sim q$). The numbers 2 and 3 show it: neither is divisible by 6, yet $2 \times 3 = 6$ is. An invalid argument can still land on a true conclusion by luck for some numbers; it is invalid because the shape does not guarantee it.

4. The laws (Theorem 2.1.1) with the idea behind each

Sheet 1 §2

Laws let you replace a formula by an equivalent one. Each has an $\wedge$ version and an $\vee$ version: swap $\wedge \leftrightarrow \vee$ and true$\leftrightarrow$false to get the other (this is called **duality**).

law	form	idea
Commutative	$p \wedge q \equiv q \wedge p$	order doesn't matter
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	grouping doesn't matter (same connective only)
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	like $a(b+c) = ab + ac$, and it works both ways round
Identity	$p \wedge \mathit{true} \equiv p$, $p \vee \mathit{false} \equiv p$	"and true" / "or false" changes nothing
Negation	$p \vee \sim p \equiv \mathit{true}$, $p \wedge \sim p \equiv \mathit{false}$	something and its opposite
Double negative	$\sim(\sim p) \equiv p$	not not
Idempotent	$p \wedge p \equiv p$, $p \vee p \equiv p$	saying it twice
Universal bound	$p \vee \mathit{true} \equiv \mathit{true}$, $p \wedge \mathit{false} \equiv \mathit{false}$	a dominating constant wins
De Morgan's	$\sim(p \wedge q) \equiv \sim p \vee \sim q$ $\sim(p \vee q) \equiv \sim p \wedge \sim q$	push ~ inside, flip $\wedge \leftrightarrow \vee$
Absorption	$p \vee (p \wedge q) \equiv p$, $p \wedge (p \vee q) \equiv p$	the bigger piece is swallowed by p
Implication law	$p \to q \equiv \sim p \vee q$	turns every $\to$ into $\vee$ so the other laws apply
Variant absorption	$p \wedge (\sim p \vee q) \equiv p \wedge q$	once p is true, "~p or q" reduces to q

Law-proof house rules (proof-writing guideline)

- One law per line. The same law several times in one line is OK; two different laws are not.
- Name it exactly: "by De Morgan's law", "by absorption law".
- Write **true**/**false**, not t/c. Write ≡ between lines, never =.
- Don't skip the double-negative step after De Morgan.
- The classic wrong step: $(q \vee r) \wedge (q \vee \sim r) \equiv q \vee (r \wedge \sim r)$ is right; writing $q \wedge (r \vee \sim r)$ is wrong. The shared piece q is the one that comes out, and the connective between the pairs ($\wedge$) goes inside.

5. "Which is equivalent to ...?" rewrite rules

Sheet 0A §3

Each row can be checked with the implication law. The two you must know by heart:

```
(p ∨ q) → r   ≡ ~(p ∨ q) ∨ r
              ≡ (~p ∧ ~q) ∨ r          De Morgan
              ≡ (~p ∨ r) ∧ (~q ∨ r)    distributive
              ≡ (p → r) ∧ (q → r)
"If p or q then r" = "p gives r AND q gives r".

(p ∧ q) → r   ≡ ~p ∨ ~q ∨ r
              ≡ (~p ∨ r) ∨ (~q ∨ r)    idempotent, comm.
              ≡ (p → r) ∨ (q → r)
"If p and q then r" is an OR of the two. Trap: not ∧.
```

Also $p \to (q \vee r) \equiv (p \wedge \sim q) \to r$: both are $\sim p \vee q \vee r$. You may move a disjunct from the conclusion to the hypothesis by negating it.

Fast check (Sheet 0A §3): find the rows where the original is false (for $A \to B$ that is A = T, B = F). Any option true on such a row is wrong.

6. Quiz 1 "which are true?", fully explained

Sheet 0A §1

```
1. false → true
```

No letters, so just evaluate: F → T is true (G3 §2).

```
2. (p ∧ q) → (p ∨ q) ∨ (q ∧ (r ∨ (s ∧ ~t)))
```

Main connective: → (loosest). So it is $(p \wedge q) \to ((p \vee q) \vee \mathit{junk})$. Can the left be true and the right false? Left true means p = q = T. Then p ∨ q = T, so the right side (an "or" containing p ∨ q) is T. Never false: **true**. That is the **junk rule**: if the hypothesis forces one piece of an "or" in the conclusion, the rest of the "or" is irrelevant.

```
3. (p ∨ q ∨ ~r) ∧ (s ∨ ~(p ∨ q)) → (s ∨ ~r)
   rename A = p ∨ q:
   (A ∨ ~r) ∧ (~A ∨ s) → (~r ∨ s)
```

That is exactly resolution (A in one piece, ~A in the other, conclusion = leftovers ~r and s joined by ∨): **true**. Or break it: conclusion false needs s = F, r = T. Then the first piece needs A = T and the second needs ~A = T. Clash, so it can't be false.

All three are true, so the answer is "None of the options", since none offered "all three". This is the same question as AY24/25 Q3.

7. AY25/26 Q3: test rows first

Sheet 1B §1

```
(a ∧ ~(b ∧ c)) ∨ (~a ∧ (b ∨ (~b ∧ c)))
b ∨ (~b ∧ c) ≡ b ∨ c          variant absorption
≡ (a ∧ ~(b ∧ c)) ∨ (~a ∧ (b ∨ c))
```

Read it in words. If a is true, the second half dies and we need $\sim(b \wedge c)$: not both b and c true. If a is false we need $b \vee c$: not both b and c false. Together: **true exactly when a, b, c are not all equal**.

- Option C, $\sim((a \leftrightarrow b) \wedge (a \leftrightarrow c))$, says "not (a = b and a = c)": not all equal. Equivalent.
- Options A and B are "cyclic" forms like $(\sim a \wedge b) \vee (\sim b \wedge c) \vee (\sim c \wedge a)$: some variable is F while the next is T, which happens exactly when they are not all equal. Test all-T and all-F: both give F. Equivalent.
- Option D as extracted, $(\sim a \vee \sim b \vee \sim c) \vee (a \vee b \vee c)$, is a tautology (true on all-T), so it is not equivalent. If the printed paper has ∧ in the middle, D says "not all T and not all F", which is also equivalent, and E ("all equivalent") is the answer. Check your printed copy.

> **Trick**
>
> Before any algebra, evaluate the original on all-T and all-F. Most wrong options fail one of those two rows.

8. "Is it a tautology?": try to break it

Sheet 1B §2

```
AY23/24 Q3(iii)
((p ↔ q) ∧ (q → r) ∧ (r ∨ ~s) ∧ (~s → q)) → s
```

1. To make it false, the conclusion must be false: s = F.
2. Now try to make every conjunct of the hypothesis true. $\sim s \to q$: ~s is T, so q must be T.
3. $p \leftrightarrow q$ with q = T: p = T. $q \to r$ with q = T: r = T. $r \vee \sim s$: T.

4. Every hypothesis piece is true and s is false: we found a false row (p = q = r = T, s = F). **Not a tautology.**

```
AY24/25 Q3(iii)
(p ∨ q ∨ ~r) ∧ (s ∨ ~(p ∨ q)) → (s ∨ ~r)
conclusion F    → s = F, r = T
1st piece T     → p ∨ q must be T   (~r is F)
2nd piece T     → ~(p ∨ q) must be T (s is F)
clash: p ∨ q can't be both   → tautology
```

Trick

Breaking attempts are fast because the conclusion usually pins down 1–2 letters and each hypothesis piece then forces another. Either you finish with a consistent assignment (not a tautology, and you have the counterexample) or you hit a clash (tautology).

9. "Which conclusions make it valid?": critical rows

Sheet 1B §3

```
AY24/25 Q5. Premises:
(1) p → (q ∧ r)  (2) ~p → ~r  (3) ~p ∧ (r → ~p)
```

1. Find every row where all premises are true (the critical rows). Start with the strongest premise: (3) is an ∧, so both parts are true, so ~p is true: **p = F**.
2. (2): ~p is T, so ~r must be T: **r = F**.
3. (1): p = F, so it is vacuously true whatever q is. **q is free**.
4. Critical rows: (p, q, r) = (F, T, F) and (F, F, F).
5. A conclusion makes the argument valid iff it is true on **every** critical row.

```
(i)   p → q          F → …  = T, T       valid
(ii)  ~p ∨ ~r        T       = T, T      valid
(iii) q → (p → r)    p → r is F → F = T  valid
(iv)  ~p ∧ ~q ∧ r    r = F   = F         invalid
answer: only (i), (ii), (iii)   (C)
```

10. Simplifying with laws, line by line

Sheet 1B §4

```
Tutorial 1 Q3b:  (p ∨ ~q) → q
≡ ~(p ∨ ~q) ∨ q          by implication law
≡ (~p ∧ ~(~q)) ∨ q       by De Morgan's law
≡ (~p ∧ q) ∨ q           by double negative law
≡ q ∨ (~p ∧ q)           by commutative law
≡ q ∨ (q ∧ ~p)           by commutative law
≡ q                      by absorption law
```

Strategy: (1) remove every → with the implication law, (2) push every ~ inward with De Morgan and double negative, (3) look for absorption, idempotent, negation + identity to collapse. Commutative/associative lines are there only to put things in the exact shape a law needs; the graders want them written.

```
AY24/25 Q4 (official):
((X) ∨ p) ∧ (p → (p → q)) ∧ p      X = a long mess
≡ p ∧ (p → (p → q))   absorption (p ∧ (p ∨ X) ≡ p,
                      after comm./assoc. moves)
≡ p ∧ (~p ∨ (~p ∨ q))    implication law ×2
≡ p ∧ ((~p ∨ ~p) ∨ q)    associative law
≡ p ∧ (~p ∨ q)           idempotent law
≡ p ∧ q                  variant absorption   (C)
```

Trick: kill the monster first

When a huge sub-formula X sits inside $(X \vee p)$ and the whole thing is also ∧ p, absorption deletes X without ever reading it. Always look for a lone letter that appears both outside and inside an "or".

11. Knights and knaves: suppose, and look for a clash

Sheet 1B §5

Knights always tell the truth, knaves always lie, everyone is exactly one. So: if X is a knight, everything X says is true; if X is a knave, everything X says is false.

```
AY23/24 Q4
Aiken: "Dueet tells the truth."
Dueet: "Noasu never tells the truth."
Noasu: "Aiken is not my type."  "Aiken lies."
```

1. Pick the person who says the most: Noasu. **Suppose Noasu is a knave.** Then both Noasu claims are false.
2. "Aiken lies" false ⇒ Aiken is a knight.
3. "Aiken is not my type" false ⇒ Aiken *is* Noasu's type ⇒ Aiken is a knave.
4. Aiken can't be both: contradiction. So Noasu is a **knight**.
5. Noasu's claims are true: Aiken lies ⇒ Aiken is a **knave** (and indeed not Noasu's type).
6. Aiken's claim "Dueet tells the truth" is false ⇒ Dueet is a **knave**.
7. Check: Dueet (knave) says Noasu never tells the truth. That is false, as it should be. ✓

Answer B (Aiken is a knave).

Mark-losing step

In a written proof, "A is a knave" is not literally the negation of "A is a knight". Write the reason: "every native is a knight or a knave but not both", then conclude.

12. The glasses story: rules of inference in a chain

Sheet 1B §6

```
RK: reading in kitchen   GK: glasses on
kitchen table
SB: saw glasses at breakfast
RL: reading in living room  GC: glasses on coffee table
premises: RK → GK, GK → SB, ~SB, RL ∨ RK, RL → GC

1. RK → SB  from RK → GK, GK → SB
                                hypothetical syllogism
2. ~RK          from 1 and ~SB          modus tollens
3. RL           from RL ∨ RK and 2      elimination
4. GC           from RL → GC and 3       modus ponens
So the glasses are on the coffee table.
```

This is the shape of every "deduce X" question: chain the if-thens, deny with tollens, remove an option with elimination, then fire with ponens.

13. The mistakes box and self-test, explained

Sheet 1B bottom

- $(p \to q) \to r \not\equiv p \to (q \to r)$: at p = q = r = F the left is (F → F) → F = T → F = F, the right is F → … = T. So → is not associative: brackets matter.
- $\sim(p \to q) \equiv p \vee (p \to \sim q)$? At p = q = T: left ~T = F; right T ∨ … = T. Differ: no.
- $(p \to (q \vee r)) \leftrightarrow ((p \wedge \sim q) \to r)$: both sides are $\sim p \vee q \vee r$, so ↔ is always T: tautology.
- Sound ⇒ valid: yes (sound means valid plus true premises). Valid ⇒ sound: no.

G5 · Quantifiers: ∀ and ∃, and every example

Lecture 3 · Tutorial 2 · explains Sheet 2 and Sheet 0A §4

1. From predicate to statement

Sheet 2 §1

"x > 3" is a **predicate**: no truth value until x is known. Two ways to turn it into a statement:

- Plug in a value: "5 > 3" (true).
- Quantify: "$\forall x \in \mathbb{Z}\,(x > 3)$" (false: x = 0 fails) or "$\exists x \in \mathbb{Z}\,(x > 3)$" (true: x = 4).

statement	true when	false when
$\forall x \in D\ P(x)$	P(x) holds for every x in D	one x fails (a counterexample)
$\exists x \in D\ P(x)$	at least one x works (a witness)	every x fails
$\exists! x \in D\ P(x)$	exactly one x works	none works, or two do

On a finite domain {a, b, c}: $\forall x\,P(x)$ is just $P(a) \wedge P(b) \wedge P(c)$, and $\exists x\,P(x)$ is $P(a) \vee P(b) \vee P(c)$. That is why ∀ behaves like ∧ and ∃ like ∨ everywhere, including in negation.

Course formal style. $\forall x \in D\ P(x)$ is shorthand for $\forall x\,(x \in D \rightarrow P(x))$, and $\exists x \in D\ P(x)$ for $\exists x\,(x \in D \wedge P(x))$. Notice → goes with ∀ and ∧ with ∃. That pattern is the key to translation.

Bound and free. In $\forall x\,(P(x, y))$, x is **bound** (governed by ∀) and y is **free** (not quantified). Without brackets, a quantifier covers only the smallest formula right after it: $\forall x\,P(x) \wedge Q(x)$ means $(\forall x\,P(x)) \wedge Q(x)$, and the second x is free. Always bracket.

2. Translating English: ∀ with →, ∃ with ∧

Sheet 2 §2

English	logic
All birds fly	$\forall x\,(Bird(x) \rightarrow Fly(x))$
Some bird flies	$\exists x\,(Bird(x) \wedge Fly(x))$
No bird flies	$\forall x\,(Bird(x) \rightarrow \sim Fly(x))$
Not all birds fly	$\exists x\,(Bird(x) \wedge \sim Fly(x))$

Why ∀ needs →. Over a domain of all animals, $\forall x\,(Bird(x) \wedge Fly(x))$ says every animal is a flying bird. False because of cows, and not what you meant. With →, a cow gives F → ..., which is vacuously true, so cows are ignored. The → acts as a filter: "for everything, *if* it is a bird, then...".

Why ∃ needs ∧. $\exists x\,(Bird(x) \rightarrow Fly(x))$ is true as soon as there is any non-bird (a cow makes Bird false, so the → is true). It would be true even in a world with no birds at all. With ∧ you demand one thing that is both a bird and flies.

AY24/25 Q2: "All birds cannot fly"
means: every bird is a non-flier
∀x (Bird(x) → ~Fly(x)) *answer C*
"Some T did E and some T didn't" (Tutorial 2)
∃v (T(v) ∧ E(v)) ∧ ∃u (T(u) ∧ ~E(u))
two separate ∃, because they are different people

Beginner errors Lecture 3 calls out

- $Fly(Bird(x))$: Bird(x) is already T or F, so this asks "does *true* fly?". Predicates take objects, not statements. Use connectives.
- A comma is not a connective. Write ∧.
- One ∃ over "some did and some didn't" forces the same person to both do and not do E: always false.

3. Negation: flip the quantifier, negate the inside

Sheet 2 §3, 0A "Which is the negation?"

"Not everyone passed" = "someone failed". "Nobody passed" = "everyone failed". So:

```
~∀x P(x)   ≡  ∃x ~P(x)
~∃x P(x)   ≡  ∀x ~P(x)
~∀x ∃y P(x,y) ≡ ∃x ~∃y P(x,y) ≡ ∃x ∀y ~P(x,y)
```

Walk the ~ from left to right. Every quantifier it passes flips (∀↔∃); the domain stays; when it reaches the inside, negate with the logic laws:

- $\sim(P \rightarrow Q) \equiv P \wedge \sim Q$ (the one false row of →).
- De Morgan for ∧ and ∨.
- $\sim(x < y) \equiv x \geq y$. Careful: not $x > y$.

Lecture 3 example: every circle has a square of the same colour. Negate:
~∀x (Circle(x) → ∃y (Square(y) ∧ Same(x,y)))
≡ ∃x ~(Circle(x) → ∃y (…)) *flip ∀*
≡ ∃x (Circle(x) ∧ ~∃y (…)) *~(P→Q) ≡ P∧~Q*
≡ ∃x (Circle(x) ∧ ∀y ~(Square(y) ∧ Same(x,y)))
≡ ∃x (Circle(x) ∧ ∀y (~Square(y) ∨ ~Same(x,y)))
"some circle has no same-coloured square"

Trick for MCQ negations

An option that still contains → after negating an if-then is wrong: the negation of "if P then Q" is "P and not Q". An option that keeps the original quantifiers (∀ stays ∀) is wrong.

4. Order of quantifiers: the game

Sheet 2 §4, 0A §4

Treat a nested statement as a game. ∀ is your **opponent's** move (they pick the nastiest value). ∃ is **your** move (you pick a value that works). Moves happen left to right, and each player sees earlier moves.

- $\forall x\,\exists y\,P$: opponent picks x, then you pick y knowing x. **y may depend on x.**
- $\exists y\,\forall x\,P$: you commit to one y first, then the opponent picks x to beat it. **One y must work for every x.** Much stronger.

"Everyone loves someone" ($\forall x\,\exists y\,L(x,y)$): each person may love a different someone. "Someone is loved by everyone" ($\exists y\,\forall x\,L(x,y)$): one celebrity everyone loves. The second implies the first, not the other way round: $\exists x \forall y\,P \Rightarrow \forall y \exists x\,P$ always holds.

Same-type quantifiers commute: $\forall x \forall y \equiv \forall y \forall x$, $\exists x \exists y \equiv \exists y \exists x$. Only mixing ∀ and ∃ makes order matter.

5. Quiz 1 / AY23/24 Q5 (domain ℝ), fully played

Sheet 0A §4, 2 §4

(i) ∀x ∃y (xy = 1)

Opponent moves first and picks the nastiest x. Try the usual suspects: x = 0. Now you need y with 0 · y = 1. Impossible. So **false**. (For any other x you would win with y = 1/x, but one bad x kills a ∀.)

> (ii) ∃x ∀y ($x^2 \geq y^2$)

You must commit to one x first. Whatever you pick, the opponent replies with y = |x| + 1, and then $y^2 > x^2$. So no x works: **false**. The trick to kill an ∃∀: let the ∀ player's value **depend on** your x.

> (iii) ∃x ∀y ∃z ($xy^2 = yz$)

You pick x first. Choose x = 0: the left side becomes 0 for every y. Opponent picks any y. You then pick z = 0: right side y · 0 = 0. Equal. You always win: **true**.

Answer: only (iii) (B in AY23/24).

> **Trick: the suspects list**
>
> For a ∀, first try 0, 1, −1, a negative, and the edge of the domain. For an ∃, try 0 and 1 as your pick, because they make products vanish or stay the same. Sheet 0A §4's table is this list, organised by prefix.

6. The domain changes the answer

Sheet 0A §4 box

- $\exists x\, \forall y\, (x \leq y)$: "there is a smallest element". On ℕ true (x = 0 is ≤ everything). On ℤ and ℝ false (for any x, y = x − 1 beats it).
- $\forall x\, \exists y\, (y > x)$: "there is always something bigger". True on ℕ, ℤ, ℝ (y = x + 1).
- $\exists y\, \forall x\, (y > x)$: "something is bigger than everything, including itself". False everywhere, because x = y gives y > y.
- $\forall x \neq 0\, \exists y\, (xy = 1)$: "every non-zero number has a reciprocal". On ℝ, y = 1/x. On ℤ, x = 2 needs y = ½, not an integer: false.
- Empty domain: every ∀ is true (nothing to fail), every ∃ is false (nothing to be the witness).

7. Empty domain and vacuous truth: AY25/26 Q2

Sheet 2 §5

D = all objects in LT11. Statement: $\forall x \in D\, (Red(x) \wedge Flying(x))$. Which option is false?

- A: if true, LT11 only contains red objects. Yes: every object is red (and flying).
- B: if true, only flying objects. Yes, same reason.
- C: true if LT11 is empty. Yes: ∀ over an empty domain is vacuously true.
- D: if false, LT11 has at least one object. Yes: a false ∀ needs a counterexample, which is an object.
- So A–D are all correct, and E ("at least one of A–D is incorrect") is the false one. **Answer E.**

> **Trap**
>
> A true ∀ statement does **not** tell you the domain has anything in it. Only a false ∀, or a true ∃, guarantees an element.

8. Find the witness (write-in)

Sheet 2 §6

> AY23/24 Q16c: ∀x ∈ ℤ ∃y ∈ ℝ\{0} (xy < 1)
> *For each x, give a y that works.*

The ∃ comes after the ∀, so your y may be a formula in x. Need: y ≠ 0 and xy < 1.

- Try $y = 1/(|x| + 1)$. Never 0 (numerator 1). Then $xy = x/(|x| + 1)$. Since $x \leq |x| < |x| + 1$, the fraction is less than 1. Works for every x, including 0 and negatives.
- Also accepted: y = −x when x ≠ 0 (then $xy = -x^2 < 0 < 1$), and y = 1 when x = 0.

Always test your formula at x = 0 and at a negative x before writing it.

9. Rules of inference with quantifiers

Sheet 2 §7

rule	what it lets you do
Universal instantiation	From "all humans are mortal", conclude "Socrates is mortal". Use a ∀ fact on a specific object.
Universal generalisation	If you proved P(a) for an **arbitrary** a (you used nothing special about it), conclude ∀x P(x). This is the engine of every ∀ proof (G6).
Existential instantiation	From ∃x P(x), give the thing a name: "let a be such an x". (This is why "∃k ∈ ℤ, n = 2k" lets you write "n = 2k".)
Existential generalisation	From P(17), conclude ∃x P(x). A proof by construction.
Universal modus ponens	$\forall x\, (P(x) \to Q(x))$ and P(a), so Q(a).
Universal modus tollens	$\forall x\, (P(x) \to Q(x))$ and ~Q(a), so ~P(a).
Universal transitivity	$\forall x\, (P \to Q)$ and $\forall x\, (Q \to R)$, so $\forall x\, (P \to R)$.

"Reorder the premises" questions (Tutorial 2): write every premise as "for all x, if … then …". Some will point the wrong way; replace those by their contrapositive. Then line them up so each conclusion is the next hypothesis, and chain them with universal transitivity.

10. Variants of a universal conditional

Sheet 2 last box

$\forall x\, (P(x) \to Q(x))$ has the same family as G3 §6, applied inside the ∀. Tutorial 2: "every rational is an integer", $\forall x\, (x \in \mathbb{Q} \to x \in \mathbb{Z})$, is false (x = ½). Its converse "every integer is rational" is true. Its contrapositive "every non-integer is irrational" is false, like the original.

> **Check yourself**
>
> 1. Negate "every student has a friend who likes maths". *Some student has no friend who likes maths: $\exists s\, \forall f\, (\sim Friend(s,f) \vee \sim Likes(f))$.*
> 2. True on ℕ? $\forall x\, \exists y\, (x = y + 1)$. *False: x = 0 needs y = −1 ∉ ℕ.*
> 3. True on ℤ? $\exists x\, \forall y\, (x + y = y)$. *True: x = 0.*

G6 · Proofs from zero

Lectures 1, 4 · proof-writing guideline · in-tutorial quiz · explains Sheet 0C and Sheet 3

1. The question that blocks everyone

"To show S ⊆ P you take **one** element of S and show it is in P. How does one element prove it for all of them?"

Because it is not one *particular* element you picked. It is a **blank**. You write the proof without knowing which element it is, and you only use facts that every element of S has. So the same lines work, word for word, for every element. The proof is a template: fill in any element and every line stays true.

See it run. Claim: every multiple of 4 is even, i.e. $S \subseteq P$ with S = multiples of 4, P = even integers.

```
1. Let x ∈ S.                          x is a blank
2. x = 4k for some k ∈ ℤ               (def. of S)
3. x = 2(2k)                           (basic algebra)
4. 2k ∈ ℤ                              (closure under ×)
5. So x is even, x ∈ P                 (def. of even)
```

Now "run" it on two actual elements:

```
x = 12:  12 = 4·3,  12 = 2·6,  6 ∈ ℤ,  12 even ✓
x = −8:  −8 = 4·(−2), −8 = 2·(−4), −4 ∈ ℤ, even ✓
```

Nothing in the proof depended on the element being 12 or −8. That is exactly what "let x be arbitrary" promises. The proof is a **function**: give it any element of S, it returns a certificate that the element is in P.

Why "for all" is proved this way (Lecture 4: generalizing from the generic particular)

You cannot check infinitely many cases one by one. So instead you prove the case of a **generic** element: one about which you assume nothing except membership in the set. Whatever you derive about it is true for all members. The rule of inference behind it is **universal generalisation** (G5 §9).

What breaks the template

Writing "let x = 12" is not a blank; you chose it. A proof that uses special facts about 12 says nothing about 13. That is why "proof by example" never proves a ∀ statement (it can only prove ∃, or disprove ∀).

2. Anatomy of a proof

Sheet 0C §2, 3 house style

Every proof has the same skeleton. Write the first and last lines **before thinking**, then fill the middle.

1. **First line**: introduce your blank and what you are allowed to assume. "Let x, y be colorful integers."
2. **Unwrap** each assumption with its definition, using a fresh letter for each ∃: "∃a, b ∈ ℤ with x = 3a, y = 3b".
3. **Unwrap the goal** too, so you know the target shape: "need c ∈ ℤ with x + 2y = 3c".
4. **Algebra** from the unwrapped assumptions toward the target shape.
5. **Name** the new witness and justify that it is an integer (closure).
6. **Rewrap** into the word, citing the definition. **Last line** restates the claim.

House style (the graders enforce it): number every line (1, 1.1, 1.2, 2 ...), with sub-steps indented; put the reason in brackets on every line: (by definition of ...), (by basic algebra), (by closure of integers under + and ×), (by the ... law), (by line 1.2). Finish with a line restating the claim, then ■.

3. The colorful example, line by line

Sheet 0C §2

Definition (Lecture 1 only): n is **colorful** iff ∃k ∈ ℤ, n = 3k. So "colorful" is just "multiple of 3" in disguise; the made-up word forces you to use the definition instead of intuition.

```
Claim: if x, y are colorful then x + 2y is colorful.
1. Let x, y be colorful integers.
 1.1 ∃a, b ∈ ℤ s.t. x = 3a, y = 3b  (def. of colorful)
 1.2 x + 2y = 3a + 6b = 3(a + 2b)  (basic algebra)
2. Let c = a + 2b. Then c ∈ ℤ
                                    (closure under +, ×)
3. So ∃c ∈ ℤ s.t. x + 2y = 3c.
4. So x + 2y is colorful          (def. of colorful) ■
```

- Line 1 is the blank: x, y are any colorful integers.
- Line 1.1 unwraps. We used **two letters** a and b, because x and y are different multiples of 3. Writing x = 3k and y = 3k would secretly assume x = y.
- Line 1.2 is aimed at the goal shape "3 × (integer)": factor out the 3.
- Line 2 is the step people skip: a + 2b is an integer because integers added and multiplied stay integers. Without it, "3(a + 2b)" isn't known to fit the definition.
- Line 4 rewraps with the definition, in the words of the claim.

4. Read the claim, pick the method

Sheet 0C §1, 3 §3

the claim looks like	method	why this method
∀x, if P(x) then Q(x), and P unpacks into a formula	direct	P gives an equation to work with
P gives nothing, but "not Q" gives a formula	contraposition	"n^2 even → n even": "n odd" gives n = 2k + 1, "n^2 even" gives nothing about n
"irrational", "infinitely many", "no largest", "there is no"	contradiction	these are absences; assuming the opposite gives you something concrete (a fraction, a finite list, a largest one)
∃x such that ...	construction	just exhibit one
a small finite set of cases	exhaustion	check each
"always true?" and it isn't	counterexample	one failing case disproves ∀
P iff Q	two directions	↔ is (→) ∧ (←)
involves |x|, odd/even, remainders	cases	each case gives a formula
∃! x	existence + uniqueness	two separate facts
A ⊆ B, A = B	element chase	the definition of ⊆ is a ∀ statement
R is an equivalence / partial order	one property at a time	each property is its own ∀ statement

Lecture advice: if a simple direct proof exists, use it.

5. Direct proof

Sheet 3 §4

```
Claim: the sum of any two even integers is even.
1. Let m, n be arbitrarily chosen even integers.
2. m = 2r, n = 2s for some r, s ∈ ℤ   (def. of even)
3. m + n = 2r + 2s = 2(r + s)         (basic algebra)
4. r + s ∈ ℤ                          (closure under +)
5. So m + n is even                   (def. of even)
6. Therefore the sum of any two evens is even. ■
```

Same skeleton as colorful. Lecture 1's product of consecutive odds is the same again: a = 2k + 1, b = a + 2 = 2k + 3 (the WLOG line says we may call the smaller one a), ab = $4k^2 + 8k + 3 = 2(2k^2 + 4k + 1) + 1$, and $2k^2 + 4k + 1 \in \mathbb{Z}$ by closure, so ab is odd.

6. Construction, counterexample, exhaustion

Lecture 1 examples

```
∃x ∈ ℤ s.t. x > 2 and x² − 5x + 6 > 0
1. Let x = 17.
2. 17 ∈ ℤ and 17 > 2.
3. 17² − 5·17 + 6 = 210 > 0. ■
```

For ∃ you only show the chosen value works. You never explain how you found 17.

```
Disprove: product of two irrationals is irrational.
1. Let the numbers be √2 and √8 (both irrational).
 1.1 √2 × √8 = √16 = 4, which is rational. ■
```

A ∀ claim dies with one counterexample. Say what it is and check both parts: the hypothesis holds (both irrational), the conclusion fails (product rational).

```
6a is not divisible by 5, for integers 1 ≤ a ≤ 4
1.1 a = 1: 6a = 6,  5 ∤ 6
1.2 a = 2: 6a = 12, 5 ∤ 12
1.3 a = 3: 6a = 18, 5 ∤ 18
1.4 a = 4: 6a = 24, 5 ∤ 24 ■
```

Exhaustion is allowed only because the domain is finite. $5 \nmid 6$ reads "5 does not divide 6".

7. Proof by cases: the √2^√2 trick

Lecture 1 Example #7

Claim: there exist irrational p, q with p^q rational. You don't know whether $\sqrt{2}^{\sqrt{2}}$ is rational, and you don't need to.

```
1. √2 is irrational                  (Theorem 4.7.1)
2. √2^√2 is rational or irrational   (every real is one)
3. Case 1: √2^√2 is rational.
   Take p = q = √2. Done.
4. Case 2: √2^√2 is irrational.
   Take p = √2^√2, q = √2 (both irrational).
   p^q = √2^(√2·√2) = √2² = 2, rational. Done.
5. Either way such p, q exist. ■
```

Division into cases works because the cases cover every possibility (step 2). This is the rule $p \vee q,\ p \to r,\ q \to r \vdash r$ from G4.

8. Contraposition

Sheet 0C §5, 3 §5

Because $P \to Q \equiv \neg Q \to \neg P$ (G3 §6), proving the contrapositive proves the original. Use it when "not Q" is easier to unwrap than P.

```
Claim: ∀n ∈ ℤ, n² even → n even.
1. Contrapositive: ∀n ∈ ℤ, n odd → n² odd.
   (not even = odd, by Assumption 1)
2. Let n be any odd integer.
 2.1 n = 2k + 1, k ∈ ℤ              (def. of odd)
 2.2 n² = 4k² + 4k + 1 = 2(2k² + 2k) + 1   (algebra)
 2.3 2k² + 2k ∈ ℤ                   (closure)
 2.4 So n² is odd                   (def. of odd)
3. Hence n² even → n even (contrapositive). ■
```

Why not direct? Start "$n^2 = 2k$". To get n you would need $\sqrt{(2k)}$, which is not a formula you can use. Starting from "n odd" gives a formula straight away.

9. Contradiction: assume the negation, find a clash

Sheet 0C §3, 3 §6

To prove S: assume ~S, derive something false, conclude S. This is the **contradiction rule** $(\neg S \to \mathit{false}) \vdash S$ (G4 §1). The whole proof depends on writing ~S correctly, so write it in logic first.

claim S	"suppose not" means
$\forall x\,(P(x) \to Q(x))$	$\exists x\,(P(x) \wedge \neg Q(x))$: some x has P but not Q
$\sqrt{2}$ is irrational	$\sqrt{2}$ = m/n in lowest terms
there are infinitely many primes	the primes are exactly $p_1, \ldots, p_n$
there is no largest even integer	M is the largest even integer (then M + 2 is even and larger: clash)
P ∨ Q	~P ∧ ~Q
at most one x works	two different $x_1 \neq x_2$ both work

In-tutorial quiz (official):

```
S: ∀x, y ∈ ℤ (Even(x) ∧ Even(y) → Even(x + y))
~S: ∃x, y ∈ ℤ (Even(x) ∧ Even(y) ∧ ~Even(x + y))
1. Suppose not: there exist even integers x, y
   with x + y not even.
 1.1 x = 2k, y = 2m for some k, m ∈ ℤ  (def. of even)
 1.2 x + y = 2k + 2m = 2(k + m)       (basic algebra)
 1.3 k + m ∈ ℤ (closure under +), so x + y is even
                                      (def. of even)
 1.4 This contradicts line 1 (x + y not even).
2. Since ~S is false, S is true. ■
```

The three wrong negations students wrote (quiz)

- (I) $\forall x, y\,(\mathit{Even}(x) \wedge \mathit{Even}(y) \wedge \neg \mathit{Even}(x+y))$. Most common. ∀ must flip to ∃: "not all pairs work" means "some pair fails", not "every pair fails".
- (II) $\forall x, y\,(\mathit{Even}(x) \wedge \mathit{Even}(y) \to \neg \mathit{Even}(x+y))$. Still ∀, and still an if-then. The negation of an if-then is an "and".
- (III) $\forall x, y\,(\neg \mathit{Even}(x+y) \to \neg(\mathit{Even}(x) \wedge \mathit{Even}(y)))$. That is the contrapositive: the **same** statement, not its negation.
- Also wrong: "Let x, y be arbitrarily chosen even integers" inside a contradiction proof. "Arbitrary" means ∀; your assumption is ∃ ("there exist").
- $\mathit{Even}(x \wedge y)$ is meaningless: ∧ joins statements, not numbers.

```
√2 is irrational (Lecture 1 Example #6)
1. Suppose not: √2 is rational.
 1.1 √2 = a/b, a, b ∈ ℤ, b ≠ 0   (def. of rational)
 1.2 Reduce to lowest terms m/n     (Assumption 2)
 1.3 m² = 2n²                       (basic algebra)
 1.4 m² is even (def. of even; n² ∈ ℤ by closure)
 1.5 m is even                      (Prop. 4.6.4)
 1.6 m = 2k: 4k² = 2n², so n² = 2k²
 1.7 n² even, so n even           (def.; Prop. 4.6.4)
 1.8 m, n both even contradicts lowest terms (1.2)
2. So √2 is irrational. ■
```

Why contradiction here: "irrational" means "not a fraction", which gives nothing to write down. Assuming it *is* a fraction gives m and n to work with.

10. Iff and exactly-one

Sheet 3 §7

- **P iff Q**: two proofs. (⇒) assume P, reach Q. (⇐) assume Q, reach P. Proving only one direction gets half marks at best.
- **Three equivalent statements** (i), (ii), (iii): prove a cycle (i) ⇒ (ii) ⇒ (iii) ⇒ (i).
- **∃! x P(x)**: existence (show one), then uniqueness (suppose y and y′ both work, show y = y′).

```
Every integer has a unique additive inverse
Existence: 1. Let x ∈ ℤ. 2. Let y = −x (∈ ℤ, closure).
           3. x + y = 0.
Uniqueness: 1. Let x ∈ ℤ, and y, y′ ∈ ℤ with
               x + y = 0 and x + y′ = 0.
            2. x + y = x + y′, so y = y′ (subtract x). ■
```

11. Spot the flaw

Sheet 0C §4

```
"Proof" that 2 = 1. Let a = b (nonzero).
a² = ab
a² − b² = ab − b²
(a − b)(a + b) = (a − b)b
a + b = b          ← divided both sides by a − b
2b = b, so 2 = 1
```

Since a = b, a − b = 0. Dividing by zero is not allowed, so the marked line is invalid. General rule: before cancelling anything, make sure it cannot be 0.

flaw	what it looks like
one letter for two ∃	"a ∣ b and a ∣ c, so b = ak and c = ak". That quietly forces b = c. Use k and l.
cancelling 0	the 2 = 1 proof; "all integers are equal" cancels d = a − b.
non sequitur	ends at "a + b = a + b": true for all a, b, so proves nothing.
proves the converse	claim P → Q, but the proof assumes Q and reaches P.
proof by example	"let x = 2 ..." for a ∀ claim.
circular	uses the claim (or an equivalent) as a step. E.g. to prove n² odd, starting "n² = 2c + 1".
wrong negation	contradiction assumes ∀x ~P instead of ∃x ~P.
undefined letter	k appears without "for some k ∈ ℤ".

12. Definitions and citable results on Sheet 3

Sheet 3 §1–2

- **d ∣ n** iff ∃k ∈ ℤ, n = dk. No "d ≠ 0" condition: 0 ∣ 0 is true (0 = 0·k), 0 ∣ 5 is false, and every d divides 0 (0 = d·0).
- **Prime**: n > 1 and whenever n = rs with r, s positive, one of r, s is n. So 1 is not prime (fails n > 1), 2 is.
- **Composite**: n > 1 and n = rs with 1 < r, s < n. Every integer > 1 is exactly one of prime/composite.
- **Rational**: r = a/b with a, b ∈ ℤ, b ≠ 0. Every integer n is rational (n = n/1).
- **Floor** ⌊x⌋: the unique integer n with n ≤ x < n + 1. **Ceiling** ⌈x⌉: unique n with n − 1 < x ≤ n.
- **Quotient-remainder**: for n ∈ ℤ, d ∈ ℤ⁺ there are unique q, r with n = dq + r and 0 ≤ r < d. The remainder is never negative: −27 = 5·(−6) + 3, not 5·(−5) − 2.
- **Transitivity of divides**: a ∣ b and b ∣ c ⇒ a ∣ c. Proof: b = ak, c = bl, so c = a(kl).
- **Closure**: ℤ is closed under +, −, × but not ÷ (1 ÷ 2 ∉ ℤ). ℚ is closed under +, −, ×, and ÷ by non-zero.

Marks lost on correct ideas

- A line without a reason in brackets.
- No closing line restating the claim.
- Reusing k for two different integers.
- Skipping the closure line before claiming "so it is even/odd/rational".
- Calling a contraposition proof a contradiction proof, or vice versa. Say in line 1 which one you are doing.

G7 · Sets: ∈ versus ⊆, power sets, products, proofs

Lecture 5 · Tutorial 3 · explains Sheet 4, Sheet 4B, Sheet 0B §1–3

1. A set is a bag; ∈ and ⊆ are different questions

Sheet 4 §1, §5

Picture a set as a bag with things inside. $\{1, 2, 3\}$ is a bag holding 1, 2 and 3. Order and repeats don't matter: $\{3, 1, 1, 2\}$ is the same bag.

- $x \in A$ asks: **is x one of the things in the bag?** Look at the top-level items only.
- $X \subseteq A$ asks: **is every thing in bag X also in bag A?** X must itself be a bag, and you compare contents.

A bag can hold other bags. $\{\{1\}, 2\}$ holds two things: the bag $\{1\}$ and the number 2. It does **not** hold 1 directly; 1 is inside an inner bag.

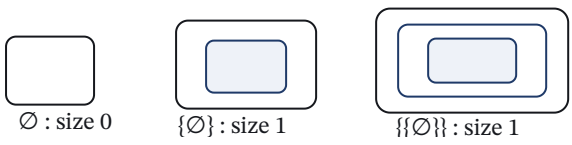
claim		why
$\varnothing \in \{1,2,3\}$	F	the items are 1, 2, 3; none of them is ∅
$\varnothing \subseteq \{1,2,3\}$	T	∅ has no items, so "every item of ∅ is in A" is vacuously true. ∅ ⊆ every set.
$\{2\} \in \{1,2,3\}$	F	the bag {2} is not one of the items (2 is)
$\{2\} \subseteq \{1,2,3\}$	T	its only item, 2, is in A
$2 \subseteq \{1,2,3\}$	F	2 is a number, not a set; ⊆ compares sets
$\{2\} \in \{\{1\}, \{2\},\{3\}\}$	T	now the items are bags, and {2} is one of them
$\{2\} \subseteq \{\{1\}, \{2\},\{3\}\}$	F	the item 2 is not among the items {1}, {2}, {3}
$a \in \{\{a\}, b\}$	F	items are {a} and b; a is only inside the inner bag

> **Trick: strip one pair of braces**
> To test $X \in A$: list A's items (strip A's outer braces) and look for X exactly. To test $X \subseteq A$: strip X's outer braces too, and check each of X's items is on A's list.

2. Braces count: ∅, {∅}, {{∅}}

Sheet 4B top

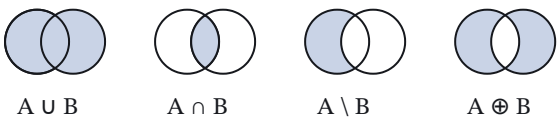
∅ is an empty bag. $\{\varnothing\}$ is a bag containing one empty bag, so it has 1 element. $\{\{\varnothing\}\}$ is a bag containing a bag containing an empty bag: also 1 element. All three are different.



Each pair of braces is a bag. $|\varnothing| = 0$, $|\{\varnothing\}| = 1$.

3. Operations, and their logic twins

Sheet 4 §2, 0B top box



Shaded = the result. Left circle A, right circle B.

Every set operation is defined by a logic connective on "x ∈ A". Write p for "x ∈ A", q for "x ∈ B":

set	x is in it iff	logic twin
$A \cup B$	x ∈ A or x ∈ B	$p \vee q$
$A \cap B$	x ∈ A and x ∈ B	$p \wedge q$
$A \setminus B$	x ∈ A and x ∉ B	$p \wedge \sim q$
$\bar{A}$	x ∉ A (x ∈ U)	$\sim p$
$A \oplus B$	in exactly one	$(p \wedge \sim q) \vee (q \wedge \sim p)$
$A \subseteq B$	(a statement)	$\forall x\,(p \to q)$
$A = B$	(a statement)	$A \subseteq B \wedge B \subseteq A$

That is why the set identities (Theorem 6.2.2) look exactly like the logic laws: ∪ ↔ ∨, ∩ ↔ ∧, complement ↔ ~, U ↔ true, ∅ ↔ false. De Morgan: $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$ is $\sim(p \vee q) \equiv \sim p \wedge \sim q$. A set identity holds iff its logic twin is an equivalence, and a failing truth row becomes a counterexample.

4. Ordered pairs and A × B

Sheet 4 §2, 0B §3

- (a, b) is an ordered pair: order matters, and $(a, b) = (c, d)$ iff a = c and b = d. Unlike {a, b} = {b, a}.
- $A \times B$ = every pair with first coordinate from A, second from B. $\{1, 2\} \times \{x\} = \{(1, x), (2, x)\}$. Size: $|A \times B| = |A| \cdot |B|$.
- $\varnothing \times X = \varnothing$: no first coordinate to pick.
- $A \times B \neq B \times A$ in general: (1, x) ≠ (x, 1). Equal only if A = B or one of them is empty.

5. Power set 𝒫(A)

Sheet 4 §2

$\mathcal{P}(A)$ is the bag of all sub-bags of A. So $X \in \mathcal{P}(A)$ iff $X \subseteq A$. That one line converts every power-set question into a subset question.

```
𝒫({a, b}) = {∅, {a}, {b}, {a, b}}      size 2² = 4
𝒫(∅)      = {∅}                         size 2⁰ = 1
𝒫({∅})    = {∅, {∅}}                    size 2¹ = 2
```

Why $2^{|A|}$: for each element you decide "in or out" independently. Always: $\varnothing \in \mathcal{P}(A)$ and $A \in \mathcal{P}(A)$.

6. Nested sets: innermost first

Sheet 4B §1

Rule: evaluate the innermost $\mathcal{P}$ or operation, write it down, check its size, then move outward. Never do two layers in your head.

```
AY24/25 Q16, A = {∅, x}
𝒫(∅) = {∅}          𝒫(𝒫(∅)) = 𝒫({∅}) = {∅, {∅}}
𝒫(A) = {∅, {∅}, {x}, {∅,x}}          (2² = 4 ✓)

(a) 𝒫(A) \ 𝒫(𝒫(∅))
    remove ∅ and {∅} from 𝒫(A)
    = {{x}, {∅,x}}

(b) 𝒫(A) \ 𝒫(∅) = 𝒫(A) \ {∅}
    = {{∅}, {x}, {∅,x}}
    … \ {A}: remove the item A = {∅,x}
    = {{∅}, {x}}                        2 items
    𝒫 of that: all subsets of a 2-item set
    = {∅, {{∅}}, {{x}}, {{∅},{x}}}   (2² = 4 ✓)

(c) 𝒫(∅) × 𝒫(A) = {∅} × 𝒫(A)
    = {(∅,∅), (∅,{∅}), (∅,{x}), (∅,{∅,x})}
    𝒫(A) × 𝒫(∅)
    = {(∅,∅), ({∅},∅), ({x},∅), ({∅,x},∅)}
    difference: drop the one common pair (∅,∅)
    = {(∅,{∅}), (∅,{x}), (∅,{∅,x})}
```

In (b), "\ $\mathcal{P}(\emptyset)$" removes the item ∅ (since $\mathcal{P}(\emptyset) = \{\emptyset\}$); it does not remove nothing. In (b) again, the $\mathcal{P}$ of a 2-item set makes each item into a singleton bag: {{∅}} and {{x}} have double braces because the items were already {∅} and {x}.

```
AY23/24 Q17: A₁ = {∅}, A₂ = {a}, A₃ = {a, {a}}
(A₁ ∩ A₂) \ A₃  = ∅ \ A₃ = ∅       no common item
(A₁ ∪ A₂) \ A₃  = {∅, a} \ {a, {a}} = {∅}
(A₁×A₂) ∪ A₃    = {(∅,a)} ∪ {a, {a}}
                = {(∅,a), a, {a}}   mixing is fine in ∪
(A₁ ∪ A₂) × A₃  = {∅, a} × {a, {a}}
   = {(∅,a), (∅,{a}), (a,a), (a,{a})}   (2·2 = 4 ✓)

AY23/24 Q6: which equal ∅ ∪ 𝒫(∅) = ∅ ∪ {∅} = {∅}?
∅ × {∅}              = ∅          no
𝒫(∅ ∪ 𝒫(∅)) = 𝒫({∅}) = {∅, {∅}}  no
𝒫(∅) \ ∅             = {∅}        yes
(∅ ∩ {∅}) ∪ (∅ ∪ {∅}) = ∅ ∪ {∅} = {∅}  yes   (D)
```

7. Counting with partitions and power sets

Sheet 4B §2

A **partition** of A cuts A into pieces that are non-empty, don't overlap, and together make A. So the sizes add up: if {B, C, D} partitions A then |A| = |B| + |C| + |D|.

```
AY25/26 Q8: |A| = 2|B|, |B| = |C| + 2,
            {B,C,D} partitions A
|A| = |B| + |C| + |D|
2|B| = |B| + (|B| − 2) + |D|
0 = −2 + |D|   ⇒   |D| = 2                  (C)
```

```
AY25/26 Q7: any two of A, B, C share exactly
            one element.
D = 𝒫(A) ∩ 𝒫(B) ∩ 𝒫(C)
```

X is in all three power sets iff X ⊆ A, X ⊆ B and X ⊆ C, iff X ⊆ A ∩ B ∩ C. So D = $\mathcal{P}$(A ∩ B ∩ C). Each pair shares one element, so the triple intersection has either that one element (if it is the same one for all pairs) or none. |A ∩ B ∩ C| ∈ {0, 1}, so |D| ∈ {2^0, 2^1} = {1, 2}. Answer B.

8. "True for any sets?" The ∈/⊆ chains

Sheet 0B §1, Quiz 1

Only two chains are safe:

- $A \in B$ and $B \subseteq C \Rightarrow A \in C$. A is an item of B, and every item of B is an item of C.
- $A \subseteq B$ and $B \subseteq C \Rightarrow A \subseteq C$. Containment passes along.

Everything else can fail. Build killers with the kit "C = {B}, then A = B or A = ∅":

```
Quiz 1 (= AY23/24 Q8):
(i)  A ∈ B ∧ B ⊆ C → A ∈ C    safe chain      true
(ii) A ∈ B ∧ B ⊆ C → A ⊆ C
     A = {1}, B = C = {{1}}:
     A ∈ B ✓, B ⊆ C ✓, but 1 ∉ C, so A ⊄ C    false
(iii) A ⊆ B ∧ B ∈ C → A ∈ C
     A = ∅, B = {1}, C = {B} = {{1}}:
     ∅ ⊆ B ✓, B ∈ C ✓, ∅ ∉ C (C's item is {1}) false
(iv) A ⊆ B ∧ B ∈ C → A ⊆ C
     A = B = {1}, C = {{1}}:
     A ⊆ B ✓, B ∈ C ✓, 1 ∉ C so A ⊄ C       false
Answer: only (i).
```

"For any sets A and B" still means C is anything you like when C appears. Also $A \in B \wedge B \in C \rightarrow A \in C$ fails: A = ∅, B = {∅}, C = {{∅}}. Membership does not pass through two layers of bags.

9. "Is this identity true for all sets?"

Sheet 0B §2–3

Translate to logic and test rows. Three sets make 8 regions, like 8 truth rows.

```
A \ (B ∩ C) = (A \ B) ∩ (A \ C) ?
left:  p ∧ ~(q ∧ r)    right: p ∧ ~q ∧ ~r
row p = T, q = T, r = F:  left T, right F   differs
turn that row into sets: 1 ∈ A, 1 ∈ B, 1 ∉ C
A = B = {1}, C = ∅:  left {1}, right ∅   → false
(the true version has ∪ on the right: De Morgan)
```

claim	verdict and reason
$\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$	true: X ⊆ A ∩ B iff X ⊆ A and X ⊆ B
$\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$	false: A = {1}, B = {2}. {1, 2} is a subset of A ∪ B but of neither alone. Only ⊇ holds.
$A \times (B \cup C) = (A \times B) \cup (A \times C)$	true (also with ∩ and \)
$(A \times B) \cap (C \times D) = (A \cap C) \times (B \cap D)$	true: both coordinates must be in both
$(A \times B) \cup (C \times D) = (A \cup C) \times (B \cup D)$	false: A = B = {1}, C = D = {2}. (1, 2) is in the right, not the left. Only ⊆.
$A \cup B = A \cup C \rightarrow B = C$	false: A = B = {1}, C = ∅. You cannot "cancel" A.
$A \setminus B = \emptyset \leftrightarrow A \subseteq B$	true: nothing of A is outside B

10. The counterexample kit

Sheet 4B §3

try	kills	why
B = ∅	$A \times B \subseteq B \times C \rightarrow A \subseteq C$	A × ∅ = ∅ ⊆ anything, whatever A is
B = {A}	$A \in B \wedge B \subseteq C \rightarrow A \subseteq C$	makes A ∈ B automatic
A = {x}, B = {x, y}, C = {{x, y}}	$A \subseteq B \wedge B \in C \rightarrow A \in C$ (or ⊆ C)	C's only item is B
S = {a, {a}}	$S \cap \mathcal{P}(S) = \emptyset$	{a} ∈ S and {a} ⊆ S, so {a} is in both
A = B	shows A ∪ B = A ∩ B is possible	
A ∩ B = ∅	shows A ⊕ B = A ∪ B is possible	

Impossible for non-empty A, B: $A \setminus B = A \cap B$ (an x ∈ A would have to be both outside and inside B) and $A \setminus B = A \cup B$ (an element of B would have to be outside B). Always true, seen

as options: $\mathcal{P}(A) \subseteq \mathcal{P}(B) \to A \subseteq B$ (A ∈ 𝒫(A) ⊆ 𝒫(B)); $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$; $|\mathcal{P}(S) \setminus \mathcal{P}(\varnothing)| = 2^{|S|} - 1$ (remove just ∅).

11. Partition checks

Sheet 4 §2, 4B §4

- Three conditions: every piece non-empty; pieces pairwise disjoint; union is the whole set.
- $\{\{2,3\},\{4\},\{3,2\},\{1\}\}$: {2,3} and {3,2} are the same set, and repeats collapse, so this is $\{\{2,3\},\{4\},\{1\}\}$, a partition of {1,2,3,4}.
- $\{\mathbb{Z}\}$ is a partition of $\mathbb{Z}$ (one piece, the whole thing). $\mathbb{Z}$ itself is not a partition: its items are numbers, not pieces (sets).
- T_0, T_1, T_2 = integers with remainder 0, 1, 2 on division by 3 partition $\mathbb{Z}$, by quotient-remainder: every integer has exactly one remainder.

12. Proving A = B with elements

Sheet 4B §5, 0C §1

$A = B$ means $A \subseteq B$ and $B \subseteq A$. Each ⊆ is a ∀ statement, so each gets a generic-element proof (G6 §1): take an arbitrary element of one side, unwrap its definition, rewrap into the other side.

```
Tut 3 Q4: A = {2n+1 : n ∈ ℤ},  B = {2n−5 : n ∈ ℤ}
1. (⊆) Let a ∈ A. Then a = 2n + 1 for some n ∈ ℤ.
                                        (def. of A)
 1.1 a = 2(n + 3) − 5                  (basic algebra)
 1.2 n + 3 ∈ ℤ                      (closure under +)
 1.3 So a ∈ B                           (def. of B)
2. (⊇) Let b ∈ B. Then b = 2m − 5 for some m ∈ ℤ.
 2.1 b = 2(m − 3) + 1, m − 3 ∈ ℤ, so b ∈ A.
3. A = B                    (def. of set equality) ■
```

Both sets are the odd integers, but "they are both the odd numbers" is not a proof. The algebra trick: rewrite a in the **shape** B uses ("2 × something − 5"). Solve 2n + 1 = 2m − 5 for m: m = n + 3.

Mark-losing step

Going from "z ∈ A" straight to "z ∈ A ∪ B". Insert the logic: "z ∈ A ∨ z ∈ B (by generalisation)", then "so z ∈ A ∪ B (by definition of ∪)". Each definition is one line.

13. Proving an identity with laws

Sheet 4B §6

```
(A ∩ B) ∪ (A \ B)
 = (A ∩ B) ∪ (A ∩ B̄)   by set difference law
 = A ∩ (B ∪ B̄)         by distributive law
 = A ∩ U               by complement law
 = A                   by identity law
```

Same rules as logic law proofs: one law per line, named. In words: "in A and B, or in A and not B" is just "in A".

14. Builder and replacement notation

Sheet 4 §1, 4B §7

- **Set builder** $\{x \in U : P(x)\}$: filter U, keep what satisfies P.
- **Replacement** $\{t(x) : x \in A\}$: run each x through a formula, collect outputs.

```
AY25/26 Q17b (many answers accepted)
{0, 1, 3, 7, 15, 31, …}: each is one less
  than 1, 2, 4, 8, 16, 32
  = {2ⁿ − 1 : n ∈ ℕ}          n = 0 gives 0 (ℕ has 0)
{3, 9, 15, 21, 27, …}: multiples of 3 that are odd
  = {n ∈ ℕ : 3 | n ∧ n is odd}
```

Check yourself

1. $|\mathcal{P}(\mathcal{P}(\varnothing))|$? *$\mathcal{P}(\varnothing) = \{\varnothing\}$ has size 1, so $2^1 = 2$.*
2. Is {∅} ⊆ {∅, {∅}}? Is {∅} ∈ {∅, {∅}}? *Both yes: its item ∅ is listed, and {∅} itself is listed.*
3. Is (1, 2) ∈ {1, 2} × {1}? *No: second coordinate must be 1.*

G8 · Relations: arrows, five properties, closures, composition

Lecture 6.1–6.2 · Tutorials 4, 5 · explains Sheet 5 and Sheet 0B §4

1. A relation is a set of arrows

Sheet 5 §1

A **relation from A to B** is any subset $R \subseteq A \times B$: a set of ordered pairs. Draw each pair (x, y) as an arrow x → y. A **relation on A** has both ends in A. Writing $x\,R\,y$ just means $(x, y) \in R$.

```
A = {1, 2, 3},  "less than" on A:
R = {(1,2), (1,3), (2,3)}
1 R 2 is true;  2 R 1 is false;  (3,3) ∉ R
```

That is all a relation is. No formula is required: any set of pairs counts, including ∅ (no arrows) and A × A (every arrow). When a formula is given ("x R y iff x − y is even"), it is just a rule for which arrows exist.

Inverse: R^{-1} reverses every arrow: $(y, x) \in R^{-1}$ iff $(x, y) \in R$. The inverse of "less than" is "greater than".

An **n-ary relation** is a set of n-tuples: a database table where each row is a tuple.

2. The five properties in pictures and words

Sheet 5 §1, 0B §4

property	definition	in the arrow picture	fails as soon as
reflexive	$\forall x\,(xRx)$	a loop on **every** dot	one dot has no loop
symmetric	$\forall x,y\,(xRy \to yRx)$	every arrow has a return arrow	one arrow without its return
transitive	$\forall x,y,z\,(xRy \wedge yRz \to xRz)$	every two-step path has a direct shortcut	a → b → c with no a → c
antisymmetric	$\forall x,y\,(xRy \wedge yRx \to x = y)$	no two-way arrows between **different** dots (loops fine)	a ⇄ b with a ≠ b
asymmetric	$\forall x,y\,(xRy \to \sim yRx)$	no two-way arrows and no loops	any loop or two-way pair

Everyday examples on people: "is the same age as" is reflexive, symmetric, transitive. "Is at most as tall as" is reflexive and transitive but not antisymmetric: two different people of equal height are related both ways. "Is a parent of" is none of reflexive, symmetric, transitive (your parent's parent is not your parent), but it is asymmetric.

Traps

- Antisymmetric is **not** "not symmetric". A relation can be both (only loops, e.g. identity) or neither ({(1,2),(2,1), (1,3)}).
- Transitive with a → b → a needs the loop (a, a). {(1,2), (2,1)} is **not** transitive: 1 → 2 → 1 requires (1,1).
- Transitive checks every x, y, z, including x = z and repeats.
- Say "R is reflexive", never "x is reflexive". The property belongs to the relation.

Vacuous truth shows up here. Symmetric, transitive, antisymmetric, asymmetric are all "if … then …" statements. If no arrow matches the "if", the property holds automatically. Reflexive has no "if": it demands loops, so it can fail even for tiny relations.

3. Instant answers

Sheet 0B §4 box

- **Empty relation** on non-empty A: not reflexive (loops missing); symmetric, transitive, antisymmetric, asymmetric all vacuously true (no arrows to test).
- **Identity** {(x, x) : x ∈ A}: only loops. Reflexive, symmetric (a loop is its own return), transitive (x → x → x has shortcut x → x), antisymmetric. So it is an equivalence relation and a partial order.
- **A × A** (all arrows): reflexive, symmetric, transitive. Antisymmetric only if A has at most 1 element.
- **Symmetric and antisymmetric** together: every arrow must have a return, and no two-way arrows between different dots, so the only arrows are loops.
- R is symmetric iff $R = R^{-1}$ (reversing changes nothing). R is transitive iff $R \circ R \subseteq R$ (every two-step trip is already a one-step arrow).
- Asymmetric ⇒ antisymmetric (the "if" of antisymmetric never fires). Asymmetric = antisymmetric + no loops.

4. Checking a formula relation on $\mathbb{Z}$

Sheet 5 §4, 0B §4

For "x R y iff [formula]", go property by property, and write the one failing pair or triple for each "no".

1. **Reflexive**: put y = x. Is the formula true for every x? Try 0 and a negative.
2. **Symmetric**: swap x and y. Does the formula survive the swap?
3. **Transitive**: pick three spaced numbers. For $|x - y| \leq 2$: −2 R 0 and 0 R 2, but $|-2 - 2| = 4$: not transitive.
4. **Antisymmetric**: find x ≠ y with both x R y and y R x.

```
AY25/26 Q17c (population):
x R₁ y iff x, y share biological father AND mother
x R₂ y iff x, y share a biological parent
R₁: reflexive (you share both with yourself),
    symmetric, transitive (same two parents pass
    along): equivalence relation.  Yes
R₂: reflexive, symmetric, NOT transitive:
    half-siblings. x and y share a mother, y and z
    share a father, x and z share nothing.     No
```

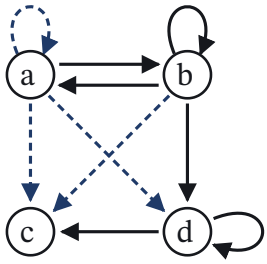
5. Closures: add the fewest arrows

Sheet 5 §2

The **closure** of R under a property is the smallest relation that contains R and has the property. You only ever **add** arrows.

- **Reflexive closure** R^r: add every missing loop. $R \cup \{(x, x) : x \in A\}$.
- **Symmetric closure** R^s: add every missing return arrow. $R \cup R^{-1}$.
- **Transitive closure** R^t: add shortcuts for every two-step path, then check again (new arrows can create new two-step paths), until nothing changes. Equivalently: x → y is in R^t iff there is a path of any length from x to y.

```
AY24/25 Q17a: R = {(a,b), (a,c)} on {a,b,c}
|Rʳ| = 2 + 3 loops              = 5
|Rˢ| = 2 + (b,a), (c,a)         = 4
|Rᵗ| = 2   no two-step path (b, c have no
           outgoing arrows), already transitive
```



AY24/25 Q17b: solid = S, dashed = arrows the transitive closure adds (the set T).

```
AY24/25 Q17b: S = {(a,b),(b,a),(b,d),(d,c),(b,b),(d,d)}
S ∪ T is the transitive closure, S ∩ T = ∅: T = added
a → b → a   add (a,a)
a → b → d   add (a,d)
b → d → c   add (b,c)
a → d → c   add (a,c)    uses the new (a,d)
re-check: nothing new appears
T = {(a,a), (a,d), (a,c), (b,c)}
```

Trick: transitive closure = "can I get there?"

For each dot, list every dot reachable by following arrows (any number of steps). Each reachable dot is an arrow in R^t. A cycle a → b → a means a can reach itself, so the loop (a, a) is added. A cycle 1 → 2 → 3 → 1 puts both (1, 3) and (3, 1) in R^t, so R^t is not antisymmetric (AY23/24 Q14).

6. Composition: a two-leg journey

Sheet 5 §1, §3

$S \circ R$ = "do an R-arrow, then an S-arrow". Read it right to left, like $f(g(x))$: the one written nearest x goes first.

```
S∘R = {(x, z) : ∃y (x R y ∧ y S z)}
x goes to y by R, then y goes to z by S
```

Recipe: for each pair (x, y) of the relation on the **right**, find every pair (y, z) of the relation on the **left** that starts where the first one ended. Output (x, z).

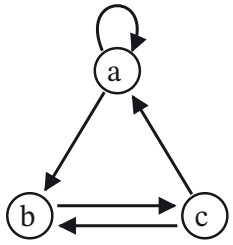
```
AY25/26 Q17a:
S = {(1,3), (3,6), (6,9), (9,12)},  R = {(6,6), (9,9)}
R∘S  (S first, then R)
  (1,3) then R from 3?   none
  (3,6) then (6,6)       → (3,6)
  (6,9) then (9,9)       → (6,9)
  (9,12) then R from 12? none
  R∘S = {(3,6), (6,9)}
S∘R  (R first, then S)
  (6,6) then (6,9)       → (6,9)
  (9,9) then (9,12)      → (9,12)
  S∘R = {(6,9), (9,12)}
(i)  (R∘S) \ (S∘R) = {(3,6)}
(ii) (R⁻¹∘S⁻¹)∘(S∘R)
     R⁻¹∘S⁻¹ = (S∘R)⁻¹ = {(9,6), (12,9)}
     now do S∘R first, then its inverse:
     (6,9) then (9,6)    → (6,6)
     (9,12) then (12,9)  → (9,9)
     = {(6,6), (9,9)}
```

Why $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$: to undo "R then S" you undo S first, then R. Socks then shoes; to reverse, shoes off then socks off.

```
AY24/25 Q17e on ℤ⁺:
  x R₁ y ⇔ y = 2x,   x R₂ y ⇔ x = y²
x (R₁∘R₂) y: R₂ first, then R₁
  x R₂ z   means x = z²
  z R₁ y   means y = 2z
  so z = y/2 and x = (y/2)² = y²/4
R₁∘R₂ = {(x, y) ∈ ℤ⁺×ℤ⁺ : x = y²/4}
      = {(z², 2z) : z ∈ ℤ⁺}
```

7. Powers: follow n arrows

Sheet 5 §3



AY24/25 Q11: R = {(a,a),(a,b),(b,c),(c,a),(c,b)}

```
where can each dot get in exactly 2 steps?
a: a→a→{a,b}, a→b→c        {a,b,c}
b: b→c→{a,b}               {a,b}
c: c→a→{a,b}, c→b→c        {a,b,c}
exactly 3 steps (one more step from those):
a, b, c each reach {a,b,c}
R∘R∘R = A × A                              (B)
```

R^n is "exactly n arrows in a row". Keep a list of reachable dots per start dot and extend it one step at a time.

8. Which properties survive an operation?

Sheet 5 §5

R, S both have it	reflexive	symmetric	transitive
R ∪ S	yes	yes	no
R ∩ S	yes	yes	yes
R^{-1}	yes	yes	yes
R∘S	yes	no	no

- Union breaks transitivity: R = {(1,2)}, S = {(2,3)}, each transitive, but R ∪ S has 1 → 2 → 3 without (1,3).
- Composition breaks symmetry: R = {(a,b),(b,a)}, S = {(b,c), (c,b)}. S∘R contains (a, c) (a → b by R, b → c by S) but not (c, a) (c has no R-arrow).
- Intersection keeps everything: an arrow in both relations has its return / shortcut in both.
- |R∘R| has no fixed link to |R| (AY24/25 Q12): it can be smaller, equal or bigger.

9. Counting relations on an n-element set

Sheet 5 §6

There are n^2 possible arrows (n loops, n(n−1) non-loops forming n(n−1)/2 pairs {a, b}). Count the free choices:

kind	count	why
all	$2^{(n^2)}$	each arrow in or out
reflexive	$2^{(n^2-n)}$	loops forced in, others free
symmetric	$2^{(n(n+1)/2)}$	each loop free; each pair {a,b}: both arrows or neither
reflexive + symmetric	$2^{(n(n-1)/2)}$	loops forced; pairs: both or neither
antisymmetric	$2^n \cdot 3^{(n(n-1)/2)}$	loops free; each pair: none, a→b, or b→a (3 ways)
asymmetric	$3^{(n(n-1)/2)}$	no loops; each pair 3 ways
equivalence	1, 2, 5, 15, 52	= number of partitions (G9)

Relations from A to B: $2^{|A||B|}$, and the set of all of them is $\mathcal{P}(A \times B)$ (each relation is a subset of A $\times$ B).

G9 · Equivalence relations: classes, quotients, partitions

Lecture 6.3 · Tutorial 4 · partitions note · explains Sheet 5B and Sheet 0B §5

1. What an equivalence relation really is

Sheet 5B §1

An **equivalence relation** is reflexive, symmetric and transitive. In plain words, it always means **"has the same X as"** for some feature X:

- "has the same birthday as" (feature: birthday)
- "has the same remainder mod 3 as" (feature: remainder)
- $x \sim y$ iff $f(x) = f(y)$ (feature: the output of f). This is **always** an equivalence relation, because "=" is reflexive, symmetric and transitive.

Why the three properties fit "same X": you have the same X as yourself (reflexive); if you match me, I match you (symmetric); if you match me and I match them, you match them (transitive).

The symbol ~ (tilde) is a common name for an equivalence relation. Between two objects it is a relation, not "not".

2. Classes are buckets

Sheet 5B §1

Sort everything into buckets by the feature. Each bucket is an **equivalence class**:

```
[a] = {x ∈ A : a ~ x}
the class of a: everything equivalent to a
x ∈ [a]   ⇔   a ~ x
```

The **quotient** $A/\sim$ is the set of all buckets: $\{[a] : a \in A\}$. It is a **set of sets**.

```
"same remainder mod 3" on ℤ
[0] = {…, −3, 0, 3, 6, …}
[1] = {…, −2, 1, 4, 7, …}
[2] = {…, −1, 2, 5, 8, …}
[3] = [0], [4] = [1] …    same bucket, different names
ℤ/~ = {[0], [1], [2]}      3 buckets
```

- **Lemma**: $x \sim y \Leftrightarrow [x] = [y]$. Two things are equivalent exactly when they sit in the same bucket.
- Two classes are either **equal** or **disjoint**, never partly overlapping. (If they shared one element, by transitivity everything in one matches everything in the other.)
- **Theorem**: the classes form a partition of A. Conversely, any partition gives an equivalence relation: "in the same piece as". Equivalence relations and partitions are the same thing seen two ways.

The class-versus-quotient trap (AY25/26 Q15)

$[x] \subseteq A$ (a class is a set of elements of A) and $[x] \in A/\sim$ (a class is one item of the quotient). Writing $[x] \subseteq A/\sim$ is wrong: the classes are items of the quotient, not collections of its items.

3. Congruence mod n

Sheet 5B §1

$a \equiv b \pmod n$ iff $n \mid (a - b)$: a and b differ by a multiple of n, i.e. same remainder on division by n. It has exactly n classes: [0], [1], ..., [n − 1].

Is $-4 \equiv 5 \pmod 7$? $-4 - 5 = -9$, and $7 \nmid -9$. No. (Remainders: $-4 = 7\cdot(-1) + 3$, so remainder 3; 5 has remainder 5.)

4. Proving ~ is an equivalence relation

Sheet 5B §2

Three separate mini-proofs, each a generic-element proof (G6 §1). The transitive one is where marks are.

```
1. (Reflexive)  Let x ∈ A. … so x ~ x.
2. (Symmetric)  Let x, y ∈ A with x ~ y. … so y ~ x.
3. (Transitive) Let x, y, z ∈ A with x ~ y, y ~ z.
   3.1 unfold both with the definition of ~
       (fresh letters for each witness)
   3.2 combine them (usually add)
   3.3 the new witness is in ℤ (closure)
   3.4 so x ~ z (def. of ~)
4. Therefore ~ is an equivalence relation.
```

```
AY23/24 Q19a (official): on ℚ,
a S b ⇔ ∃k ∈ ℤ (a − b = k/2).  Prove S transitive.
1. Let a, b, c ∈ ℚ with aSb and bSc.
2. a − b = m/2 and b − c = n/2 for some m, n ∈ ℤ
                                        (def. of S)
3. a − c = (a − b) + (b − c) = (m + n)/2
                                        (basic algebra)
4. m + n ∈ ℤ                          (closure under +)
5. Therefore aSc                        (def. of S) ■
```

Why this works: "a S b" says a and b differ by a multiple of ½. Adding two such differences gives a multiple of ½. The trick in every transitive proof: $a - c = (a - b) + (b - c)$. Different letters m and n, because the two witnesses need not be equal.

5. Writing the classes and the quotient

Sheet 5B §3

1. Unfold: $[a] = \{x : a \sim x\}$, then substitute the rule.
2. Pick one representative per class, no repeats.
3. Write $A/\sim = \{[r] : r$ in that representative set$\}$.

```
AY23/24 Q19b,c: a S b ⇔ a − b is a multiple of ½, on ℚ
[0]   = {…, −½, 0, ½, 1, …} = {k/2 : k ∈ ℤ}
[0.1] = {…, −0.4, 0.1, 0.6, …}
two distinct classes: [0], [0.1]
each class has exactly one member x with 0 ≤ x < ½
ℚ/S = { [x] : x ∈ ℚ ∧ 0 ≤ x < ½ }
```

Like choosing remainders 0, 1, 2 for mod 3: each bucket is named by its member in the "first window" [0, ½).

```
AY25/26 Q17d: on ℤ⁺, x ~ y ⇔ ∃d ∈ ℤ⁺ (d | x ∧ d | y)
d = 1 divides every positive integer
so every x ~ every y: one single class
ℤ⁺/~ = {ℤ⁺} = {[1]}
```

Trick: always test the witness 1 (or 0) first. If it always works, the relation is everything, and there is one class.

```
Tut 4 Q8: on ℤ⁺ × ℤ⁺, (a,b) ~ (c,d) ⇔ ab = cd
feature: the product
[(1,1)] = pairs with product 1 = {(1,1)}
[(4,3)] = pairs with product 12
        = {(1,12),(2,6),(3,4),(4,3),(6,2),(12,1)}

Tut 4 Q10: on ℤ \ {0}, a ~ b ⇔ ab > 0
feature: the sign
two classes: positives [1], negatives [−1]
```

6. From a partition to the relation

Sheet 5B §4

Given the pieces, the relation is "every pair inside the same piece", including loops.

```
Partition {{0,3,4}, {1}, {2}} of {0,1,2,3,4}
piece {0,3,4}: 3 × 3 = 9 pairs
  (0,0),(0,3),(0,4),(3,0),(3,3),(3,4),(4,0),(4,3),(4,4)
piece {1}: (1,1)     piece {2}: (2,2)
R has 11 pairs
```

Because equivalence relations and partitions match one to one, the number of equivalence relations on an n-element set equals the number of partitions: 1, 2, 5, 15, 52 for n = 1 to 5 (the Bell numbers). For {a, b, c}, the 5 partitions are {{a},{b},{c}}, {{a},{b,c}}, {{b},{a,c}}, {{c},{a,b}}, {{a,b,c}}. AY25/26 Q4 asked exactly this: answer 5.

7. Consequences used in MCQs

Sheet 5B box

- For an equivalence relation, $R^{-1} = R$ (symmetric) and $R \circ R = R$ (transitive gives $\subseteq$; reflexive gives $\supseteq$, since x → x → y). So every power $R^n = R$, and $R \circ R^{-1} = R$. AY23/24 Q12: all four statements true.
- |A| is the sum of the class sizes (the classes partition A).
- R = {(1,1),(1,2),(2,1),(2,2),(3,3)} is an equivalence relation (classes {1,2} and {3}), so $R^7 = R$.
- "|A ∩ B| = 1" on $\mathcal{P}(\mathbb{Z})$: symmetric yes; not reflexive (∅ ∩ ∅ = ∅ has size 0); not transitive ({1} and {1,2} share 1, {1,2} and {2} share 2, {1} and {2} share nothing). Not an equivalence relation (AY25/26 Q11).

Check yourself

1. How many classes does "x − y is even" have on $\mathbb{Z}$? *2: evens and odds.*
2. On {1,2,3,4}, how many equivalence relations? *15 (Bell number).*
3. Is "x ~ y iff |x| = |y|" on $\mathbb{Z}$ an equivalence relation? Classes? *Yes (same feature |x|). Classes {0}, {1, −1}, {2, −2}, ...*

G10 · Partial orders: Hasse diagrams, chains, linearizations

Lecture 6.4 · Tutorial 5 · explains Sheet 6 and Sheet 0B §6

1. What a partial order is

Sheet 6 §1

A **partial order** $\preccurlyeq$ is reflexive, antisymmetric and transitive. It means "comes before or is equal to", generalising $\leq$. The pair $(A, \preccurlyeq)$ is a **poset**.

- Reflexive: everything is $\preccurlyeq$ itself (like $x \leq x$).
- Antisymmetric: if $a \preccurlyeq b$ and $b \preccurlyeq a$ then $a = b$. No two different things are each "before" the other.
- Transitive: before-before means before.

"**Partial**" because some pairs may be **incomparable**: neither $a \preccurlyeq b$ nor $b \preccurlyeq a$. Under divisibility, 4 and 6 are incomparable ($4 \nmid 6$, $6 \nmid 4$).

poset	notes
$(\mathbb{R}, \leq)$	every pair comparable: a **total** order
$(\mathcal{P}(A), \subseteq)$	{1} and {2} incomparable
$(\mathbb{Z}^+, \mid)$	divisibility; 4, 6 incomparable
$(\mathbb{Z}, \mid)$	**not** a partial order: $1 \mid -1$ and $-1 \mid 1$ but $1 \neq -1$ (antisymmetry fails)

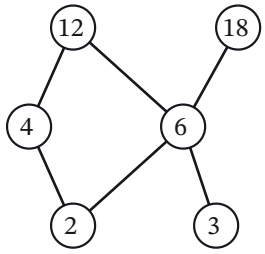
2. Hasse diagrams: the order with the clutter removed

Sheet 6 §1 recipe

The arrow diagram of a partial order is crowded: loops everywhere, and a shortcut for every path. A Hasse diagram keeps only what you cannot work out:

1. Draw every arrow pointing up (smaller things lower).
2. Remove the loops (reflexivity says they are always there).
3. Remove every arrow implied by transitivity: keep only **covers**, where b sits directly above a with nothing in between.
4. Remove the arrowheads (up is the direction).

Read it back: $a \preccurlyeq b$ iff you can climb from a to b along lines going up.



Divisibility on {2,3,4,6,12,18} (Sheet 0B §6). Covers: 2–4, 2–6, 3–6, 4–12, 6–12, 6–18.

> **Trick: levels for divisibility**
>
> Count prime factors with repeats: 2, 3 have one; 4 = 2·2, 6 = 2·3 have two; 12 = 2·2·3, 18 = 2·3·3 have three. Put each number on that level. A cover always goes up exactly one level (you multiply by one prime). Then draw a line wherever the lower one divides the upper one.

3. Special elements

Sheet 6 §1 table

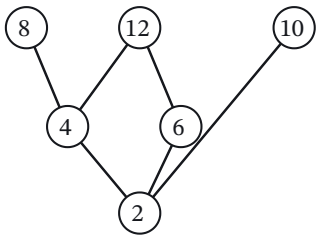
c is …	definition	in the diagram	how many
minimal	nothing is strictly below c	no line going down from c	could be several
maximal	nothing is strictly above c	no line going up from c	could be several
smallest (least)	$c \preccurlyeq$ everything	every element can be reached climbing up from c	at most one
largest (greatest)	everything $\preccurlyeq c$	c can be reached climbing up from every element	at most one

Minimal is "nobody beats me from below". Smallest is "I am below everybody". Two minimal elements means there is no smallest: neither is below the other. In the diagram above: minimal 2 and 3, so smallest is **None**; maximal 12 and 18, so largest is **None**.

In a finite poset, if there is exactly one maximal element, it is the largest (every element climbs to some maximal element, and there is only one). This fails for infinite posets.

4. Worked: read everything off one diagram

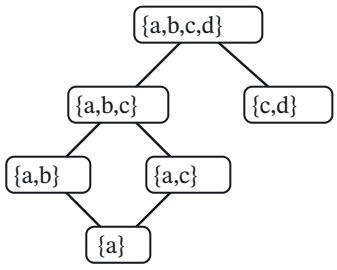
Sheet 6 §2



AY24/25 Q17c,d: divisibility on {2,4,6,8,10,12}

```
covers: 2–4, 2–6, 2–10, 4–8, 4–12, 6–12
minimal: 2          maximal: 8, 10, 12
smallest: 2         largest: None
maximal chains: {2,4,8}, {2,4,12}, {2,6,12}  length 2
                {2,10}                       length 1
```

10 = 2·5 has only 2 below it (5 is not in the set), so it can sit at any height above 2; it is maximal because nothing in the set is a multiple of 10. The question asked for two maximal chains of **different** lengths: {2,4,8} and {2,10}.



AY25/26 Q17e: $\subseteq$ on six sets

```
smallest?  No. {c,d} is not above {a} ({a} ⊄ {c,d})
maximal?   Yes, only {a,b,c,d} (it is also largest)
minimal:   {a} and {c,d}
```

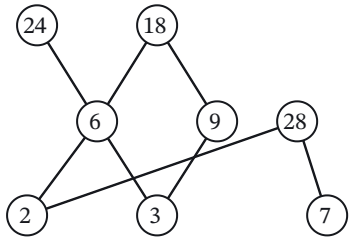
```
AY23/24 Q18: A = {11,…,16}, F_x = positive divisors
x ≼ z ⇔ F_x = F_z ∨ |F_x| < |F_z|
divisor counts: 11:2 13:2 14:4 15:4 16:5 12:6
levels: {11,13} < {14,15} < 16 < 12
minimal: 11, 13   smallest: None
maximal: 12       largest: 12
```

Unpack the definition: x comes before z if they have the same divisor set (only when x = z) or x has fewer divisors. 11 and 13 both have 2 divisors but different divisor sets, so they are incomparable. Same for 14 and 15.

5. Chains and antichains

Sheet 6 §1, §3

- **Chain**: a subset where every pair is comparable: a vertical path in the diagram. **Length = number of elements − 1** (count the steps, not the dots).
- **Maximal chain**: a chain you cannot add any element to. Not the same as "longest".
- **Antichain**: a subset where no two different elements are comparable: no two are joined by an upward path. A single element is an antichain (nothing to compare, vacuous). A whole level is usually one.



AY24/25 Q13–15: divisibility on {2,3,6,7,9,18,24,28}

```
covers: 2–6, 3–6, 3–9, 6–18, 9–18, 6–24, 2–28, 7–28
longest chains (length 2): {3,9,18} {3,6,18}
  {3,6,24} {2,6,18} {2,6,24}            → 5   (B)
antichains?
  {2,3,28}: 2 | 28, comparable → not an antichain
  {7,9,24}: no divisibility at all → antichain
  {6}: one element → antichain     → (ii),(iii) (D)
```

6. Linearizations

Sheet 6 §4, 0B §6

A **linearization** $\preccurlyeq^*$ is a total order (a single list) that respects $\preccurlyeq$: if $a \preccurlyeq b$ then a comes before b in the list. Think: an order to take modules that respects every prerequisite.

- **Check a list**: for every cover a–b (a below b), a must appear before b. One violation kills it.
- **Build one** (Kahn's algorithm): repeatedly write down any minimal element of what is left, then remove it.

```
AY24/25 Q15, same poset:
(i)   2 3 6 7 9 18 24 28   every cover respected ✓
(ii)  7 3 2 28 9 6 18 24   ✓
(iii) 7 2 3 6 18 9 24 28   18 before 9, but 9 | 18 ✗
→ only (i) and (ii)   (B)

AY23/24 Q15 (levels from §4 above):
{11,13} first in either order, then {14,15} in
either order, then 16, then 12.
(ii)  13 11 15 14 16 12  ✓
(iii) 11 13 14 15 16 12  ✓
(i)  has 12 second, (iv) has 12 before 16  ✗
```

7. Comparable versus compatible (Tutorial 5 definition)

Sheet 6 §5

- **Comparable**: $a \preccurlyeq b$ or $b \preccurlyeq a$.
- **Compatible**: some c has $a \preccurlyeq c$ and $b \preccurlyeq c$ (they have a common upper bound).

Comparable ⇒ compatible (take c to be the larger of the two). Not the other way: in divisibility on {2, 4, 6, 12}, 4 and 6 are compatible (12 is above both) but not comparable.

```
AY24/25 Q17f,g on ℤ × ℤ:
(a,b) W (c,d) ⇔ a ≤ c ∧ b ≥ d
"all pairs comparable": FALSE
  (1,3), (2,4): 1 ≤ 2 but 3 ≰ 4;  2 ≰ 1.  Neither.
"all pairs compatible": TRUE
1. Let x = (x₁,x₂), y = (y₁,y₂) ∈ ℤ×ℤ be arbitrary.
2. Let c = (max(x₁,y₁), min(x₂,y₂)) ∈ ℤ×ℤ.
3. x₁, y₁ ≤ c₁ (def. of max); x₂, y₂ ≥ c₂ (def. of min)
4. So x W c and y W c (def. of W).
5. So every two elements are compatible. ■
```

How to find c: W wants the first coordinate to go up and the second to go down. A common upper bound takes the bigger first coordinate and the smaller second coordinate.

8. Other words, and the true/false facts

Sheet 6 box

- **Total order**: every pair comparable; the Hasse diagram is one vertical line.
- **Well-ordered**: a total order where every non-empty subset has a smallest element. $(\mathbb{N}, \leq)$ is; $(\mathbb{Z}, \leq)$ is not ($\mathbb{Z}$ itself has no smallest).
- A one-element poset: its element is both smallest and largest.
- ({a, b}, identity): a and b are each maximal and minimal, but there is neither smallest nor largest.
- Two different maximal elements are never comparable (if one were below the other, it would not be maximal).
- $(\mathbb{Z}, \leq)$ has no maximal or minimal element.
- A finite non-empty poset always has at least one maximal and one minimal element (climb until you can't).
- $\{0,1\}^2$ with $(a, b) \preccurlyeq (c, d)$ iff $a \leq c$ and $b \leq d$ is a diamond: (0,1) and (1,0) are incomparable, so it is not total and not well-ordered.

Mistakes that lose the mark

- Two minimal elements, yet writing "smallest = …".
- Drawing loops or transitive shortcut lines in a Hasse diagram.
- Chain length = number of elements (it is one less).
- Leaving "largest" blank. Write **None**; a blank is marked as no answer.

G11 · Functions: the words and the two proof templates

Tutorial 6 · Epp ch. 7 · explains Sheet 8

1. A function is a relation with one rule

Sheet 8 §1

$f: A \to B$ is a relation from A to B in which **every input has exactly one output**: $\forall x \in A\ \exists! y \in B\ (x, y) \in f$. In the arrow picture: every dot in A sends out exactly one arrow.

- **Domain** A: the allowed inputs. **Codomain** B: where outputs are allowed to land.
- **Range**: the outputs that actually occur, $\{f(x) : x \in A\}$. Always $\subseteq$ B, may be smaller.
- **Image** of $X \subseteq$ A: $f(X) = \{f(x) : x \in X\}$, the outputs of the inputs in X.
- **Preimage** of $Y \subseteq$ B: $f^{-1}(Y) = \{x \in A : f(x) \in Y\}$, the inputs that land in Y. Defined for every f, even when f has no inverse.

```
f: ℤ → ℤ, f(x) = x²
range = {0, 1, 4, 9, …}          (not all of ℤ)
f({−2, 2}) = {4}
f⁻¹({4}) = {−2, 2}    f⁻¹({3}) = ∅
```

Well-defined means the rule really gives exactly one output in B for each input. It breaks when cases overlap and disagree ("f(z) = 0 if 2 | z, 1 if 3 | z": z = 6 gets two outputs) or an output falls outside the codomain ("f(x) = 2x from $\mathbb{R}$ to $\mathbb{Z}$": f(0.3) = 0.6 $\notin \mathbb{Z}$).

Equal functions: same domain, same codomain, same output for every input. $\lfloor x \rfloor + 1$ and $\lceil x \rceil$ on $\mathbb{R}$ agree on non-integers but differ at integers (x = 2: 3 versus 2), so they are not equal (Tutorial 6 D3).

2. Injective, surjective, bijective

Sheet 8 §1

word	definition	arrow picture	everyday
injective (one-to-one)	$f(x_1) = f(x_2) \to x_1 = x_2$	no dot in B is hit twice	no two students share a matric number
surjective (onto)	$\forall y \in B\ \exists x \in A$ $f(x) = y$	every dot in B is hit	every seat is taken
bijective	both	a perfect pairing	every seat has exactly one student

Injective is stated backwards on purpose: "same output forces same input". Its contrapositive, $x_1 \neq x_2 \to f(x_1) \neq f(x_2)$, is the everyday version (different inputs give different outputs).

The **inverse** $f^{-1}: B \to A$ exists exactly when f is bijective: $f^{-1}(y) = x \Leftrightarrow f(x) = y$. **Composition**: $(g \circ f)(x) = g(f(x))$, f first, same right-to-left rule as relations.

3. Template: prove injective

Sheet 8 §2

```
f: ℝ → ℝ, f(x) = 2x + 3.  Claim: f is injective.
1. Let x₁, x₂ ∈ ℝ with f(x₁) = f(x₂).
2. 2x₁ + 3 = 2x₂ + 3                  (def. of f)
3. x₁ = x₂                            (basic algebra)
4. So f is injective.            (def. of injective) ■
```

First line: two blank inputs with equal outputs (that is the "if" of the definition). Goal: they were the same input. Disprove with one pair: $f(x) = x^2$ on $\mathbb{Z}$ is not injective because $f(-1) = f(1)$ but $-1 \neq 1$.

4. Template: prove surjective

Sheet 8 §2

```
f: ℝ → ℝ, f(x) = 2x + 3.  Claim: f is surjective.
1. Let y ∈ ℝ be arbitrary.      start in the codomain
2. Let x = (y − 3)/2.        solved f(x) = y on scrap
3. x ∈ ℝ (ℝ closed under −, ÷ by non-zero)
4. f(x) = 2·(y − 3)/2 + 3 = y       (basic algebra)
5. So f is surjective.          (def. of surjective) ■
```

This is a ∀∃ statement: for every y, build an x. So the blank is y, and x is a formula in y. On scrap paper, solve $f(x) = y$ for x; in the proof, just state x and check it.

Tutorial 6 D4 mistake

Starting from "let x ∈ A" and writing "then y = f(x), so f is surjective" proves nothing: every function has outputs. Surjectivity is about every **y** being reached, so the proof must start from an arbitrary y in the codomain. Also check your x is really in the domain: $f(x) = 2x$ from $\mathbb{Z}$ to $\mathbb{Z}$ is not surjective, because y = 3 needs x = 1.5 $\notin \mathbb{Z}$.

5. Facts worth quoting

Sheet 8 box

- f, g injective ⇒ g∘f injective. Same for surjective, hence bijective.
- g∘f injective ⇒ f injective (if f merged two inputs, g∘f would too). g∘f surjective ⇒ g surjective (the last step must reach everything).
- $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$ for bijections: undo in reverse order (socks and shoes, G8 §6).
- Left inverse exists ⇔ injective (A ≠ ∅). Right inverse exists ⇔ surjective.
- $X \subseteq f^{-1}(f(X))$ always, equality when f is injective. $f(f^{-1}(Y)) \subseteq Y$ always, equality when f is surjective. Example: f(x) = x², X = {2}: f(X) = {4}, $f^{-1}(\{4\}) = \{-2, 2\} \supsetneq X$.

G12 · The past-MCQ answer bank, decoded

explains every row of Sheet 7 · "official" = from the answer-sheet explanation

Each row of Sheet 7 in full sentences: what the question asks, the key idea, and why the answer is what it is. Rows fully worked elsewhere point there instead of repeating.

AY25/26

Sheet 7

Q	answer	why
1	D	Modern set theory: Cantor, Russell, Zermelo, Fraenkel. Not Gauss. Reading-only trivia.
2	E	∀ over an empty room. G5 §7.
3	check paper	Original = "a, b, c not all equal". G4 §7.
4	C (5)	Equivalence relations on {1,2,3} = partitions of {1,2,3} = 5. G9 §6.
5	C	Readers-and-titles diagram (appendix, not in the extracted text). Official: (i) every female reader reads some Mystery title; (iii) Mrs Duett reads every Mystery title, and every male reader reads at least one Mystery title, so they share one. (ii) fails: no male reader reads all titles of any genre. Method: turn each sentence into ∃/∀ and check it against the picture.
6	C or D	n a perfect square. (i) "n is prime": 4 is a square and not prime. (iii) "$p \mid n \to p^2 \mid n$": in a square every prime appears an even number of times, so yes. (ii) "odd number of positive factors" is true for positive squares (factors pair up d with n/d except $\sqrt{n}$), but 0 is a perfect square in ℕ with infinitely many factors, so both C and D were accepted.
7	B	D = $\mathcal{P}(A \cap B \cap C)$, size 1 or 2. G7 §7.
8	C (2)	Partition sizes add up. G7 §7.
9	A	A = {1,2,3}, B = {2,5}. $S_1 = \mathcal{P}(A) \times \mathcal{P}(B)$ (pairs of subsets), $S_2 = \mathcal{P}(A \times B)$, S_3 = all relations from A to B, S_4 = symmetric ones. A relation from A to B is a subset of A × B, so $S_3 = S_2$; $S_4 \subseteq S_3$ trivially. So $S_4 \subseteq S_3 \subseteq S_2$ (A). S_1's elements are pairs, S_2's are sets of pairs: the two never mix, which kills C and E.
10	B or D	→ as a relation on {true, false}: pRq iff p → q, so R = {(T,T), (F,T), (F,F)}. Reversing gives {(T,T), (T,F), (F,F)}: q → p, the converse. Converse ≡ inverse, so both accepted.
11	B	"$\lvert A \cap B \rvert = 1$" on $\mathcal{P}(\mathbb{Z})$: symmetric, not reflexive (∅), not transitive. G9 §7.
12	B	Official key counts relations on {0,1}: equivalence relations are {(0,0),(1,1)} and all four pairs, so $\lvert F \rvert = 2$; partial orders are the identity plus one of (0,1) or (1,0), so $\lvert G \rvert = 3$. $\lvert F \rvert + 1 = \lvert G \rvert$.
13	D	Sound means valid and true premises. I (sound → valid) true. II (valid → sound) false. III (sound but not valid) impossible, false. IV (valid but not sound) true: a valid argument with a false premise.
14	B (1)	Reflexive relations on {1,2,3,4} where each x relates to at most one thing: x R x is forced, so x can relate to nothing else. Only the identity: 1.
15	E	I: some class has all its members in A: true (every class ⊆ A, A non-empty). II: [x] ⊆ A/~ is false: classes are items of the quotient. III: $\lvert A \rvert$ = sum of class sizes: true. G9 §2.

AY24/25

Sheet 7

Q	answer	why
2	C	"All birds cannot fly" = ∀x (Bird(x) → ~Fly(x)). G5 §2.
3	E	All three true. G4 §6.
4	C (p ∧ q)	Absorption kills the monster. G4 §10.
5	C	Critical rows. G4 §9.
6	A	(i) $\mathcal{P}(A) \subseteq \mathcal{P}(B) \to A \subseteq B$: true ($A \in \mathcal{P}(A) \subseteq \mathcal{P}(B)$, so $A \subseteq B$). (ii) $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$: true (De Morgan). (iii) $A \times B \subseteq B \times C \to A \subseteq C$ fails when B = ∅: both products are ∅ whatever A and C are.
7	C	(i) {{2,3},{4},{3,2},{1}} = {{2,3},{4},{1}}: a partition. (ii) {ℤ}: yes. (iii) ℤ: no. G7 §11.
8	D	(i) $\lvert \mathcal{P}(S) \setminus \mathcal{P}(\varnothing) \rvert = 2^{\lvert S \rvert} - 1$: removes only ∅. (ii) $\lvert S \cap \mathcal{P}(S) \rvert = 0$ fails for S = {a, {a}}. (iii) $\lvert S \rvert \cdot \lvert \mathcal{P}(S) \rvert = \lvert S \times \mathcal{P}(S) \rvert$: product rule. So (i), (iii).
9	C	Divisibility on the first 10 primes: a prime divides another prime only if they are equal, so the relation is just the identity. The identity is an equivalence relation (10 singleton classes) and a partial order (all 10 elements minimal). Both true; E is the trap (a relation can be both).
10	D	pRq iff p → q is true: R = {(T,T), (F,F), (F,T)}. Reflexive (both loops), not symmetric ((F,T) without (T,F)), antisymmetric, transitive.
11	B	$R \circ R \circ R = A \times A$. G8 §7.
12	E	A = {a, b}. $\lvert R \circ R \rvert = \lvert R \rvert$ fails for R = {(a,a),(a,b),(b,a)} ($R \circ R$ has 4). $\lvert R \circ R \rvert = \lvert R \rvert^2$ fails for R = {(a,a),(a,b)} ($R \circ R$ has 2, not 4). $\lvert R \circ R \rvert \neq \lvert R \rvert^2$ fails for R = {(a,a)} ($1 = 1^2$). No fixed link.
13	B (5)	Five longest chains. G10 §5.
14	D	Antichains (ii), (iii). G10 §5.
15	B	Linearizations (i), (ii). G10 §6.

AY23/24

Sheet 7

Q	answer	why
2	E	Given $\forall x \in S$ (SoC(x) $\wedge$ CS1231(x)) is true. (i) all computing students took CS1231S: true. (ii) every student is computing: true. (iii) S non-empty: not guaranteed (vacuous truth). (iv) some non-computing student took it: false, there are none. Only (i), (ii), which is not an option, so E.
3	A	(i) $\sim(p \to q)$ vs $p \vee (p \to \sim q)$: differ at p = q = T. (ii) tautology. (iii) not a tautology (p = q = r = T, s = F). Only (ii). G4 §8, §13.
4	B	Aiken and Dueet knaves, Noasu knight. G4 §11.
5	B	Only (iii). G5 §5.
6	D	(iii), (iv) equal $\{\varnothing\}$. G7 §6.
7	B	Which cannot hold for non-empty A, B? (i) $A \setminus B = A \cap B$ and (ii) $A \setminus B = A \cup B$ are impossible (G7 §10). (iii) $A \cup B = A \cap B$ works with A = B. (iv) $A \oplus B = A \cup B$ works when $A \cap B = \varnothing$. So only (i), (ii).
8	A	$\in/\subseteq$ chains, only (i). G7 §8.
9	A	R reflexive, S symmetric. (i) $R \cup S$ reflexive: the loops of R survive. Official counterexample for the rest on {a,b,c}: R = {(a,a),(b,b),(c,c),(a,c)}, S = {(a,b),(b,a)}. $R \circ S$ = {(a,b),(b,a),(b,c)} not symmetric; $R \cup S$ has (a,c) without (c,a); $(R \circ S)^{-1}$ = {(b,a),(a,b),(c,b)} has $c \to b \to a$ without (c,a), not transitive.
10	A	(i) asymmetric $\Rightarrow$ antisymmetric: true. (ii) "not reflexive $\Rightarrow$ not asymmetric": R = {(a,b)} on {a,b} is not reflexive yet asymmetric. (iii) "not reflexive $\Rightarrow$ asymmetric": R = {(a,a)} on {a,b} is neither. (iv) "= is both an equivalence and a partial order" refutes it.
11	B (5)	Graph not in the extracted text. Official: $R \circ R$ = {(1,2),(1,4),(3,3),(4,2),(4,4)}, and $R^4 = (R \circ R) \circ (R \circ R)$ turns out to be the same 5 pairs. Tip: compute R^2, then square it.
12	D	All four true for equivalence relations. G9 §7.
13	C	(i) false (one-element poset). (ii) true ({a, b} with identity). (iii) false (distinct maximals are never comparable). (iv) true ($(\mathbb{Z}, \leq)$). G10 §8.
14	B	A = {1,2,3}. P misses (3,3), not reflexive. Q is reflexive, antisymmetric, transitive: partial order. R has $1 \to 2 \to 3 \to 1$; its transitive closure contains (1,3) and (3,1), not antisymmetric. Only Q.
15	B	Levels $\{11,13\} < \{14,15\} < 16 < 12$. G10 §6.

Patterns across three years

- Every paper has: a "which are true" logic MCQ (junk rule / resolution), a quantifier-order MCQ over $\mathbb{R}$, an $\in/\subseteq$ or $\mathcal{P}$ question, a relation-property question with a small counterexample, and 2–3 poset questions.
- Most "always true?" options die on the smallest possible example: $\varnothing$, a one- or two-element set, the identity relation, or R = {(a, a)}.
- When two answers are both defensible (0 as a perfect square, converse $\equiv$ inverse), the graders accept both. Pick the one that follows the definitions strictly.

T1 · Tutorial 1 walkthrough

Propositional logic and proofs · answers from the official key (D1 to D4 worked out here)

This tutorial trains four moves you will use all semester: turning English into symbols, rewriting a statement one law per line, judging an argument by its critical rows, and writing a numbered proof with a reason on every line. Three tricks carry most of it. An if-then is false in exactly one situation (hypothesis true, conclusion false). "Only if" points forward: p only if q is $p \to q$. And $\wedge$ next to $\vee$ always needs brackets.

D1 · English into symbols

Sheet 1 §4

Question. p = "I went to Universal Studios Singapore (USS)", q = "I rode the Battlestar Galactica (BG)". Write in symbols. (a) I did not go to USS. (b) I went to USS but I did not ride BG. (c) If I rode BG, then I did not go to USS. (d) If I did not go to USS, then I did not ride BG. (e) I went to USS or I rode BG. (f) I went to USS or I rode BG, but not both. (g) I did not go to USS or I went to USS and I rode BG.

Answer (worked out here, no official key)

part	symbols	part	symbols
(a)	$\sim p$	(e)	$p \vee q$
(b)	$p \wedge \sim q$	(f)	$(p \vee q) \wedge \sim(p \wedge q)$
(c)	$q \to \sim p$	(g)	$\sim p \vee (p \wedge q)$
(d)	$\sim p \to \sim q$		

Break it down

1. Words first. A **statement** is a sentence that is either true or false, never both. A **statement variable** such as p is a letter standing for one statement. A **compound statement** joins statements with **connectives**: ~ (not), $\wedge$ (and), $\vee$ (or), $\to$ (if-then), $\leftrightarrow$ (if and only if).
2. (a) "Did not go" flips the truth of "went". Flipping true and false is **negation**, written ~. So $\sim p$.
3. (b) Logic has no "but". In English "but" adds surprise (you went, yet skipped the ride). The logical content is only "both of these happened", which is $\wedge$, called **conjunction**. "Did not ride" is $\sim q$. So $p \wedge \sim q$.
4. (c) "If A, then B" is a **conditional**, written $A \to B$. The part after "if" is the **hypothesis**; the part after "then" is the **conclusion**. Here A = "I rode BG" = q and B = "I did not go to USS" = $\sim p$. So $q \to \sim p$. The arrow starts at the "if" part, whatever order the letters were named in.
5. (d) Same recipe: hypothesis $\sim p$, conclusion $\sim q$, so $\sim p \to \sim q$. No brackets are needed: ~ is always done first and only grabs the letter right after it.
6. (e) $\vee$, called **disjunction**, is **inclusive**: $p \vee q$ is true when at least one part is true, including when both are. Plain "or" always means this in the course.
7. (f) "But not both" throws out the both-true case. Say it as two pieces: "at least one" is $p \vee q$, and "not both" is $\sim(p \wedge q)$. Both pieces must hold, so join them with $\wedge$. This is called **exclusive or**.
8. (g) English has no brackets, so you choose the grouping from the meaning. The sentence offers two options: "I did not go", or "I went and I rode". The "and" glues the last two clauses into one option, giving $\sim p \vee (p \wedge q)$. The other grouping, $(\sim p \vee p) \wedge q$, starts with something that is always true, which nobody would bother to say. Whatever you decide, the brackets must show it.

Bonus: (g) collapses to a plain if-then. Every line uses exactly one law.

```
   ~p ∨ (p ∧ q)
≡  (~p ∨ p) ∧ (~p ∨ q)   (by distributive law)
≡  (p ∨ ~p) ∧ (~p ∨ q)   (by commutative law)
≡  true ∧ (~p ∨ q)       (by negation law)
≡  (~p ∨ q) ∧ true       (by commutative law)
≡  ~p ∨ q                (by identity law)
≡  p → q                 (by implication law)
```

So (g) just says "if I went to USS, then I rode BG".

Trick

Translate with a fixed dictionary: "not" is ~; "and", "but", "although" are $\wedge$; "or" is $\vee$ (inclusive); "if A then B" is $A \to B$ with the arrow leaving the "if" part. Then add brackets that match the meaning. Every place where $\wedge$ and $\vee$ meet gets a bracket.

D2 · A tautology by truth table

Sheet 1B §2

Question. Draw a truth table to show that $((a \to x) \wedge (b \to y)) \to ((a \wedge b) \to (x \wedge y))$ is a tautology.

Answer (worked out here, no official key)

Write $H = (a \to x) \wedge (b \to y)$ and $C = (a \wedge b) \to (x \wedge y)$, so the statement is $H \to C$.

#	a	b	x	y	a→x	b→y	H	a∧b	x∧y	C	H→C
1	T	T	T	T	T	T	T	T	T	T	T
2	T	T	T	F	T	F	F	T	F	F	T
3	T	T	F	T	F	T	F	T	F	F	T
4	T	T	F	F	F	F	F	T	F	F	T
5	T	F	T	T	T	T	T	F	T	T	T
6	T	F	T	F	T	T	T	F	F	T	T
7	T	F	F	T	F	T	F	F	F	T	T
8	T	F	F	F	F	T	F	F	F	T	T
9	F	T	T	T	T	T	T	F	T	T	T
10	F	T	T	F	T	F	F	F	F	T	T
11	F	T	F	T	T	T	T	F	F	T	T
12	F	T	F	F	T	F	F	F	F	T	T
13	F	F	T	T	T	T	T	F	T	T	T
14	F	F	T	F	T	T	T	F	F	T	T
15	F	F	F	T	T	T	T	F	F	T	T
16	F	F	F	F	T	T	T	F	F	T	T

The last column is T in all 16 rows, so the statement is a tautology.

Break it down

1. A **truth table** lists every possible combination of T/F for the variables, one per row, and works out the statement in each row. Each variable doubles the number of combinations, so 4 variables give $2^4 = 16$ rows.
2. A **tautology** is a statement that is true in every row, whatever the variables are. Its opposite, a **contradiction**, is false in every row. So the job is to show the final column is all T.
3. Fill the variable columns in a fixed pattern so no row is missed: the first variable changes slowest (8 T then 8 F),

the last changes fastest (T, F, T, F ...).

4. Work from the inside out: the small pieces $a \to x$, $b \to y$, $a \wedge b$, $x \wedge y$ first, then H and C, then $H \to C$.
5. The only rule you need for any arrow: $P \to Q$ is F only when P is T and Q is F. Every other row is T. When P is F, the arrow is **vacuously true**: true because the "if" part never happened, so no promise was broken.
6. Reading the table: $H \to C$ could only be F in a row with H = T and C = F. C is F only in rows 2, 3, 4 (where a and b are both T but $x \wedge y$ is F), and H is F in each of those rows. The deadly combination never happens.

> **Trick: try to break it**
>
> This is faster than 16 rows and is how you answer "is it a tautology?" MCQs. Try to make it false: you need H = T and C = F. C = F forces $a = b$ = T and $x \wedge y$ = F. Now H = T needs $a \to x$ = T, and with a = T that forces x = T; the same way y = T. Then $x \wedge y$ = T, clashing with $x \wedge y$ = F. No row can make it false, so it is a tautology. Had the attempt succeeded, that row would be your counterexample.

D3(a) · $(p \wedge q) \to p$ by laws

Sheet 1B §4

Question. Using the laws of Theorem 2.1.1 (Epp) and the implication law, prove that $(p \wedge q) \to p$ is a tautology.

Answer (worked out here, no official key)

```
   (p ∧ q) → p
≡  ~(p ∧ q) ∨ p     (by implication law)
≡  (~p ∨ ~q) ∨ p    (by De Morgan's law)
≡  (~q ∨ ~p) ∨ p    (by commutative law)
≡  ~q ∨ (~p ∨ p)    (by associative law)
≡  ~q ∨ (p ∨ ~p)    (by commutative law)
≡  ~q ∨ true        (by negation law)
≡  true             (by universal bound law)
```

The statement is equivalent to **true**, so it is a tautology.

Break it down

1. **Logically equivalent**, written $P \equiv Q$: P and Q have the same truth value in every row. Each law in Theorem 2.1.1 is a ready-made equivalence, so swapping one side for the other anywhere inside an expression never changes its truth value.
2. So "prove it is a tautology" becomes "rewrite it one law at a time until it is **true**". Something equivalent to **true** is true in every row, which is exactly what tautology means.
3. **Implication law**: $p \to q \equiv \sim p \vee q$. Use it first, because every other law talks only about ~, $\wedge$, $\vee$. Arrows must go before those laws can touch anything.
4. **De Morgan's law**: $\sim(p \wedge q) \equiv \sim p \vee \sim q$ and $\sim(p \vee q) \equiv \sim p \wedge \sim q$. A ~ entering a bracket negates each part and swaps $\wedge$ with $\vee$.
5. Now hunt for a letter next to its own negation, because the **negation law** $p \vee \sim p \equiv$ ***true*** wipes out the pair. The **commutative law** swaps the two sides of one connective. The **associative law** moves brackets when the same connective repeats: $(p \vee q) \vee r \equiv p \vee (q \vee r)$. One law per line.
6. Why the second commutative step? The negation law is printed as $p \vee \sim p$, plain letter first. The graders want each line to match its law exactly as printed, so $\sim p \vee p$ must be flipped first. Q3(a) below is about losing marks for skipping exactly this kind of step.
7. **Universal bound law**: $p \vee$ ***true*** $\equiv$ ***true***. Anything "or" true is true. Here the law's p is $\sim q$.
8. In words: "if p and q both hold, then p holds". Of course that is always true. As an argument it is called **specialization** (see Q7(a)).

> **Trick**
>
> The standard route for "prove it is a tautology": arrows out (implication law), push every ~ inward (De Morgan, double negative), then regroup until some $X \vee \sim X$ appears. That is **true**, and **true** swallows anything it is $\vee$-ed with.

D3(b) · $((p \vee q) \wedge \sim p) \to q$ by laws

Sheet 1B §4

Question. Using the laws, prove that $((p \vee q) \wedge \sim p) \to q$ is a tautology.

Answer (worked out here, no official key)

```
   ((p ∨ q) ∧ ~p) → q
≡  ~((p ∨ q) ∧ ~p) ∨ q      (by implication law)
≡  (~(p ∨ q) ∨ ~(~p)) ∨ q   (by De Morgan's law)
≡  (~(p ∨ q) ∨ p) ∨ q       (by double negative law)
≡  ~(p ∨ q) ∨ (p ∨ q)       (by associative law)
≡  (p ∨ q) ∨ ~(p ∨ q)       (by commutative law)
≡  true                     (by negation law)
```

Break it down

1. Implication law, then De Morgan's law pushes the ~ into $((p \vee q) \wedge \sim p)$: the $\wedge$ becomes $\vee$ and each part is negated, giving $\sim(p \vee q) \vee \sim(\sim p)$.
2. **Double negative law**: $\sim(\sim p) \equiv p$. Two flips cancel. It is a separate step with its own name; never fold it silently into De Morgan.
3. The key idea: the letters in a law can stand for whole blocks, not just single letters. Call the block $(p \vee q)$ by one name, X. After the associative step the line is $\sim X \vee X$. Commute to $X \vee \sim X$, and the negation law makes it **true**.
4. You could instead expand $\sim(p \vee q)$ with De Morgan, but then you need distributive steps to rebuild. Spotting $X \vee \sim X$ with a block X is shorter and just as valid.
5. In words: "p or q; not p; so q". As an argument this is **elimination**: two options, one ruled out, the other must hold. The tutorial points out this is the engine of knights and knaves (Q9): everyone is a knight or a knave, so ruling one out gives the other.

> **Trick**
>
> Before expanding a bracket, check whether the same bracket appears elsewhere with a ~ in front. If so, treat it as a single letter and aim for $X \vee \sim X$.

D3(c) · $((p \to q) \wedge p) \to q$ by laws

Sheet 1B §4

Question. Using the laws, prove that $((p \to q) \wedge p) \to q$ is a tautology.

Answer (worked out here, no official key)

```
   ((p → q) ∧ p) → q
≡ ((~p ∨ q) ∧ p) → q     (by implication law)
≡ ~((~p ∨ q) ∧ p) ∨ q    (by implication law)
≡ (~(~p ∨ q) ∨ ~p) ∨ q   (by De Morgan's law)
≡ ~(~p ∨ q) ∨ (~p ∨ q)   (by associative law)
≡ (~p ∨ q) ∨ ~(~p ∨ q)   (by commutative law)
≡ true                   (by negation law)
```

Break it down

1. There are two arrows, so the implication law is used twice, inner one first. You may do both in one line and write "(by implication law, twice)": the course lets you repeat one law in a line, but never mix two different laws in one line.
2. De Morgan on $\sim((\sim p \vee q) \wedge p)$: treat $(\sim p \vee q)$ as a block Y. Then $\sim(Y \wedge p) \equiv \sim Y \vee \sim p$.
3. The associative step regroups $(\sim Y \vee \sim p) \vee q$ as $\sim Y \vee (\sim p \vee q)$, and $(\sim p \vee q)$ is Y again. So the line is $\sim Y \vee Y$: commute, then negation law. Same block trick as D3(b).
4. In words: "if p then q; p; so q". This is **modus ponens**. The tutorial's Additional Notes give the general link: an argument with premises $p_1, \ldots, p_k$ and conclusion q is valid exactly when $(p_1 \wedge \ldots \wedge p_k) \rightarrow q$ is a tautology. So this proof shows modus ponens is a valid argument.

Trick

Any argument can be checked as one big statement: AND the premises, arrow to the conclusion, and test for a tautology. Laws, truth table, or "try to break it" all work.

D3(d) · $(\sim p \rightarrow (q \wedge \sim q)) \rightarrow p$ by laws

Sheet 1B §4

Question. Using the laws, prove that $(\sim p \rightarrow (q \wedge \sim q)) \rightarrow p$ is a tautology.

Answer (worked out here, no official key)

```
   (~p → (q ∧ ~q)) → p
≡ (~p → false) → p     (by negation law)
≡ (~(~p) ∨ false) → p  (by implication law)
≡ (p ∨ false) → p      (by double negative law)
≡ p → p                (by identity law)
≡ ~p ∨ p               (by implication law)
≡ p ∨ ~p               (by commutative law)
≡ true                 (by negation law)
```

Break it down

1. Look for anything obviously constant first. $q \wedge \sim q$ (a thing and its opposite joined by "and") is always false: the negation law in its $\wedge$ form, $p \wedge \sim p \equiv \mathit{false}$. Replacing it at once shrinks everything.
2. Inner arrow by the implication law gives $\sim(\sim p) \vee \mathit{false}$. Double negative gives $p \vee \mathit{false}$. **Identity law**: $p \vee \mathit{false} \equiv p$ ("or false" adds nothing), and likewise $p \wedge \mathit{true} \equiv p$.
3. What is left, $p \rightarrow p$, is true by the implication law, a commutative swap and the negation law.
4. In words: "if not-p leads to something impossible, then p". $q \wedge \sim q$ is a **contradiction** (false in every row). This is the **contradiction rule**, the logic behind proof by contradiction, which Q9(a) uses.

Trick

Collapse constants early: $X \wedge \sim X$ becomes **false**, $X \vee \sim X$ becomes **true**, then identity and universal bound laws finish them off.

D4 · Where is Mala's treasure?

Sheet 1B §6

Question. (a) If this house is next to a lake, then the treasure is not in the kitchen. (b) If the tree in the front yard is an elm, then the treasure is in the kitchen. (c) If the tree in the back yard is an oak, then the treasure is in the garage. (d) The tree in the front yard is an elm or the treasure is buried under the flagpole. (e) The house is next to a lake. Where is the treasure?

Answer (worked out here, no official key)

Let L: house is next to a lake; K: treasure in the kitchen; E: front-yard tree is an elm; O: back-yard tree is an oak; G: treasure in the garage; F: treasure under the flagpole. The premises are (a) $L \rightarrow \sim K$, (b) $E \rightarrow K$, (c) $O \rightarrow G$, (d) $E \vee F$, (e) L.

```
1. L → ~K   (premise a)
   L        (premise e)
   ∴ ~K     (by modus ponens)
2. E → K    (premise b)
   ~K       (by line 1)
   ∴ ~E     (by modus tollens)
3. E ∨ F    (premise d)
   ~E       (by line 2)
   ∴ F      (by elimination)
```

The treasure is buried under the flagpole. Premise (c) is never used.

Break it down

1. An **argument** is a list of statements: the given ones are **premises**, the last one is the **conclusion** (marked ∴, "therefore"). An argument is **valid** if the conclusion is true whenever all premises are true. A **rule of inference** is a small valid argument pattern you may use as one step.
2. **Modus ponens**: from $p \rightarrow q$ and p, get q. You have the "if" part, so you get the "then" part.
3. **Modus tollens**: from $p \rightarrow q$ and $\sim q$, get $\sim p$. The "then" part failed, so the "if" part cannot have happened; if it had, the "then" part would have happened too.
4. **Elimination**: from $p \vee q$ and $\sim p$, get q.
5. Where to start: find a premise that is a bare fact, with no "if" and no "or". Only (e) is. Feed it into the if-then that begins with it, (a). Each new fact unlocks the next premise.
6. Why not "kitchen" from (b)? Because (b) needs "the front tree is an elm", which we never got; line 2 actually proves it false. Premise (c) is a distractor. Unused premises are normal in these puzzles.

Trick

Name every atom with a letter, start from the bare fact, then chain: modus ponens forwards, modus tollens backwards, elimination on the "or". Cite the premise or line used at every step.

Q1(a) · "if" versus "only if"

Sheet 1 §4

Question. With p = "I use the umbrella" and q = "it rains", write using $\rightarrow$: "I use the umbrella if it rains" and "I use the umbrella only if it rains".

Answer

"I use the umbrella if it rains" means "if it rains, then I use the umbrella": $q \to p$.

"I use the umbrella only if it rains" means "if I use the umbrella, then it rains": $p \to q$. It also means "if it doesn't rain, then I don't use the umbrella", $\sim q \to \sim p$, which is equivalent to $p \to q$ by contraposition.

Break it down

1. "A if B": the word "if" marks the hypothesis. Whatever comes right after "if" goes on the left of the arrow, wherever it sits in the sentence. After "if" is "it rains", so $q \to p$. English simply put the conclusion first.
2. "A only if B": B is a must for A. A cannot happen without B: no rain, no umbrella. That is $\sim q \to \sim p$.
3. The **contrapositive** of $A \to B$ is $\sim B \to \sim A$ (swap the two sides and negate both). It always has the same truth table as the original. So $\sim q \to \sim p \equiv p \to q$, and "A only if B" is written $A \to B$: the arrow runs from A to B, in reading order.
4. Test it on a real situation. "I use the umbrella only if it rains." You see me with an umbrella: you may conclude it is raining ($p \to q$). You see rain: can you conclude I have an umbrella? No; "only if" never promised that.

Trick

"if" introduces the **tail** of the arrow; "only if" introduces the **head**. So "A if B" is $B \to A$ and "A only if B" is $A \to B$. Adding the word "only" reverses the arrow.

Trap

"Only if" is not a stronger "if". The two point in opposite directions.

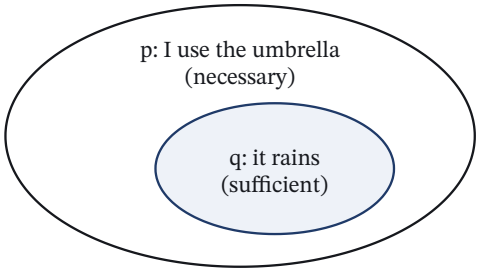
Q1(b) · Necessary and sufficient

Sheet 1 §4

Question. For "I use the umbrella if it rains": is "I use the umbrella" a necessary or a sufficient condition for "it rains"? Is "it rains" a necessary or a sufficient condition for "I use the umbrella"?

Answer

The sentence is $q \to p$. "I use the umbrella" is a **necessary** condition for "it rains". "It rains" is a **sufficient** condition for "I use the umbrella". (The other two options, umbrella sufficient for rain and rain necessary for umbrella, would both be $p \to q$, which this sentence does not say.)



$q \to p$ as a picture: every rainy day lies inside the umbrella days. Being inside the small region guarantees being in the big one.

Break it down

1. **Sufficient condition**: "r is sufficient for s" means r on its own guarantees s. That is $r \to s$.
2. **Necessary condition**: "r is necessary for s" means s cannot happen without r: $\sim r \to \sim s$, which is equivalent to $s \to r$.
3. Map these onto $q \to p$. The tail q (rain) guarantees the head p (umbrella), so rain is sufficient for the umbrella. By the contrapositive $\sim p \to \sim q$, no umbrella means no rain, so the umbrella is necessary for rain.
4. "The umbrella is necessary for rain" sounds silly because umbrellas do not cause rain. Logic ignores cause. It only records which combinations can occur, and under this sentence "rain without umbrella" never occurs. That is all "necessary" means here.
5. Why the other two are "no": from $q \to p$ you cannot get $p \to q$ (that is the **converse**, see Q5). I might carry the umbrella on a sunny day for shade.

Trick

In any arrow $A \to B$: the tail is **sufficient**, the head is **necessary**. "A is sufficient for B" and "B is necessary for A" are the same arrow $A \to B$.

Q1(c)(d) · "if and only if"

Sheet 1 §3

Question. (c) Write "I use the umbrella if and only if it rains" using p, q, $\to$ and $\wedge$ or $\vee$. Is there a shorter way with another connective? (d) What kind of condition is "I use the umbrella" for "it rains" in this sentence?

Answer

(c) It is the conjunction of "I use the umbrella if it rains" and "I use the umbrella only if it rains": $(q \to p) \wedge (p \to q)$, which shortens to $p \leftrightarrow q$.

(d) "I use the umbrella" is a **necessary and sufficient** condition for "it rains".

Break it down

1. "If and only if" is literally "if" plus "only if". Both sentences from (a) hold at once, so join them with $\wedge$. The order of the two halves does not matter (commutative law).
2. The **biconditional** $p \leftrightarrow q$ is true exactly when p and q have the same truth value (both T or both F). By definition $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$. The official answer notes that many students still do not know this after weeks of logic, so learn it now.
3. (d) From (b), the "if" half $q \to p$ makes the umbrella necessary for rain. The "only if" half $p \to q$ has the umbrella as its tail, so the umbrella is also sufficient for rain. Both together: necessary and sufficient. It works the other way round too: rain is necessary and sufficient for the umbrella.

Trick

"iff", "if and only if", "necessary and sufficient" and "exactly when" all mean $\leftrightarrow$. To prove or use a $\leftrightarrow$, split it into two arrows and handle each one separately (Q11 does exactly this).

Q2 · Brackets after De Morgan

Sheet 1 §1

Question. Explain and correct these mistakes. (a) $a \wedge \sim(b \wedge c) \equiv a \wedge \sim b \vee \sim c$, by De Morgan's law. (b) $\sim(x \vee y) \vee z \equiv \sim x \wedge \sim y \vee z$, by De Morgan's law.

Answer

(a) $a \wedge \sim b \vee \sim c$ is ambiguous. Corrected: $a \wedge \sim(b \wedge c) \equiv a \wedge (\sim b \vee \sim c)$.

(b) $\sim x \wedge \sim y \vee z$ is ambiguous. Corrected: $\sim(x \vee y) \vee z \equiv (\sim x \wedge \sim y) \vee z$.

Break it down

1. **Order of operations** in this course: ~ first; then $\wedge$ and $\vee$, which have **equal** rank; then $\rightarrow$ and $\leftrightarrow$ last. Equal rank means no rule says which of $\wedge$, $\vee$ goes first, so $a \wedge b \vee c$ has two possible readings.
2. The two readings of (a) really differ. Take a = F and c = F. Then $(a \wedge \sim b) \vee \sim c$ = F $\vee$ T = T, but $a \wedge (\sim b \vee \sim c)$ = F $\wedge$ (...) = F. An expression with two possible values has no single meaning, so it earns no marks.
3. The fix: De Morgan replaces the block $\sim(b \wedge c)$ by $\sim b \vee \sim c$. The block was a single unit inside the bigger expression, so its replacement must stay a single unit. Keep a bracket around it.
4. (b) is the same story: $\sim(x \vee y)$ becomes $\sim x \wedge \sim y$, then bracket it. The readings differ at x = T, z = T: $(\sim x \wedge \sim y) \vee z$ = T but $\sim x \wedge (\sim y \vee z)$ = F.
5. Other modules (CS2100) let $\wedge$ bind tighter than $\vee$. This course does not, so never rely on that habit here.

Trick

When a law rewrites a piece, put the new piece in brackets. Then scan the line: wherever $\wedge$ and $\vee$ sit side by side at the same bracket level, a bracket is missing.

Q3(a) · Aiken's skipped steps

Sheet 1B §4

Question. Simplify using Theorem 2.1.1 and the implication law, final answer using only ~, $\wedge$, $\vee$, with a justification for every step. (a) $\sim a \wedge (\sim a \rightarrow (b \wedge a))$. Aiken wrote: $\equiv \sim a \wedge (a \vee (b \wedge a))$ by the implication law; $\equiv \sim a \wedge a$ by the absorption law; $\equiv$ **false** by the negation law. What did he omit? (Note: use $\equiv$, not $=$, between statements.)

Answer

He skipped three steps: a double negative law step, and two commutative law steps.

```
   ~a ∧ (~a → (b ∧ a))
≡ ~a ∧ (~(~a) ∨ (b ∧ a))   (by implication law)
≡ ~a ∧ (a ∨ (b ∧ a))       (by double negative law)
≡ ~a ∧ (a ∨ (a ∧ b))       (by commutative law)
≡ ~a ∧ a                   (by absorption law)
≡ a ∧ ~a                   (by commutative law)
≡ false                    (by negation law)
```

Break it down

1. The implication law turns $\sim a \rightarrow (b \wedge a)$ into $\sim(\sim a) \vee (b \wedge a)$, not $a \vee (b \wedge a)$. Getting from $\sim(\sim a)$ to a is the double negative law, a separate line.
2. The **absorption law** is $p \vee (p \wedge q) \equiv p$ (and $p \wedge (p \vee q) \equiv p$). In words: "$p$, or ($p$ and something)" is just p, since whenever $p \wedge q$ is true, p is already true. The repeated letter comes first inside the bracket. Aiken had $a \vee (b \wedge a)$, so a commutative step must first make it $a \vee (a \wedge b)$.
3. The negation law is printed $p \wedge \sim p \equiv$ ***false***, plain letter first. Aiken had $\sim a \wedge a$, so another commutative step is needed.
4. Why so picky? The graders check each line as "exactly one named law applied to the previous line". A law is a pattern, and you may only use it when the expression matches the pattern as printed. That is why these steps are required: the course says so to make sure you do not reach the answer by coincidence.
5. Also: $\equiv$ means "same truth value in every row". $=$ is for numbers and sets, not for statements.

Trick

Before citing a law, hold the line next to the law's exact printed shape. Letters in a different order: insert a commutative step. A $\sim(\sim X)$ anywhere: insert a double negative step.

Q3(b) · Simplify $(p \vee \sim q) \rightarrow q$

Sheet 1B §4

Question. Simplify $(p \vee \sim q) \rightarrow q$ using the laws, with only ~, $\wedge$, $\vee$ in the final answer and a justification for every step.

Answer

```
   (p ∨ ~q) → q
≡ ~(p ∨ ~q) ∨ q        (by implication law)
≡ (~p ∧ ~(~q)) ∨ q     (by De Morgan's law)
≡ (~p ∧ q) ∨ q         (by double negative law)
≡ q ∨ (~p ∧ q)         (by commutative law)
≡ q ∨ (q ∧ ~p)         (by commutative law)
≡ q                    (by absorption law)
```

Break it down

1. Implication law first: the final answer may not contain $\rightarrow$.
2. De Morgan on $\sim(p \vee \sim q)$: negate each part and swap $\vee$ to $\wedge$, giving $\sim p \wedge \sim(\sim q)$. Keep it in brackets, since it sits next to $\vee q$ (Q2).
3. Double negative on its own line. The official answer specifically asks: did you jump straight from line 1 to line 3 citing only De Morgan? That loses marks.
4. Now $(\sim p \wedge q) \vee q$. The target is absorption $p \vee (p \wedge q) \equiv p$ with the law's p being q. That needs q outside on the left and q first inside, so two commutative steps (the official check: did you skip them?).
5. Sanity check by meaning: if q is T, the arrow's conclusion is T, so the whole thing is T. If q is F, the hypothesis $p \vee \sim q$ is T and the conclusion is F, so it is F. It always equals q.

Trick

Spot absorption: a letter X on its own, joined to a bracket that also contains X with the other connective. $X \vee (X \wedge \ldots)$ and $X \wedge (X \vee \ldots)$ both collapse to X.

Q3(c) · Simplify $\sim(p \vee \sim q) \vee (\sim p \wedge \sim q)$

Sheet 1B §4

Question. Simplify $\sim(p \vee \sim q) \vee (\sim p \wedge \sim q)$ using the laws, with a justification for every step.

Answer

```
   ~(p ∨ ~q) ∨ (~p ∧ ~q)
≡  (~p ∧ ~(~q)) ∨ (~p ∧ ~q)
                          (by De Morgan's law)
≡  (~p ∧ q) ∨ (~p ∧ ~q)   (by double negative law)
≡  ~p ∧ (q ∨ ~q)          (by distributive law)
≡  ~p ∧ true              (by negation law)
≡  ~p                     (by identity law)
```

Break it down

1. De Morgan on the first bracket, then double negative, exactly as in Q3(b).
2. Now $(\sim p \wedge q) \vee (\sim p \wedge \sim q)$: both brackets share $\sim p$. The **distributive law** $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ read from right to left pulls the common piece out, like factoring $xy + xz = x(y + z)$ in algebra. Laws are equivalences, so you may use them in either direction.
3. $q \vee \sim q \equiv \textbf{\textit{true}}$ is the negation law; $\sim p \wedge \textbf{\textit{true}} \equiv \sim p$ is the identity law.

> **Trick**
>
> Two brackets sharing a piece, joined by the other connective: factor the shared piece out with the distributive law, then look inside for $X \vee \sim X$ or $X \wedge \sim X$.

Q3(d) · Simplify $(p \to q) \to r$

Sheet 1B §4

Question. Simplify $(p \to q) \to r$ using the laws, with only $\sim$, $\wedge$, $\vee$ in the final answer.

Answer

```
   (p → q) → r
≡  (~p ∨ q) → r      (by implication law)
≡  ~(~p ∨ q) ∨ r     (by implication law)
≡  (~(~p) ∧ ~q) ∨ r  (by De Morgan's law)
≡  (p ∧ ~q) ∨ r      (by double negative law)
```

Break it down

1. Two arrows: remove the inner one, then the outer one. The outer one has the whole block $(\sim p \vee q)$ as its hypothesis, so it becomes $\sim(\sim p \vee q) \vee r$.
2. De Morgan then double negative turns $\sim(\sim p \vee q)$ into $p \wedge \sim q$. Note what that says: the negation of $p \to q$ is $p \wedge \sim q$.
3. Meaning check: $(p \to q) \to r$ is F only when $p \to q$ is T and r is F. $(p \wedge \sim q) \vee r$ is F only when r is F and $p \wedge \sim q$ is F, that is, $p \to q$ is T. Same.

> **Trick**
>
> Memorise $\sim(p \to q) \equiv p \wedge \sim q$: an if-then fails only when the hypothesis holds and the conclusion does not. It also gives the shortcut $X \to r \equiv \sim X \vee r$ for any block X, however big.

Q4 · Is $\to$ associative?

Sheet 1B §1

Question. Prove, or disprove, that $(p \to q) \to r$ is logically equivalent to $p \to (q \to r)$.

Answer

They are **not** logically equivalent. Counterexample: let p, q, r all be false. Then $(p \to q) \to r$ is false (T → F), but $p \to (q \to r)$ is true (its hypothesis p is false).

```
p q r | p→q (p→q)→r | q→r p→(q→r)
T T T |   T     T   |   T     T
T T F |   T     F   |   F     F
T F T |   F     T   |   T     T
T F F |   F     T   |   T     T
F T T |   T     T   |   T     T
F T F |   T     F   |   F     T   differ
F F T |   T     T   |   T     T
F F F |   T     F   |   T     T   differ
```

Rows 6 and 8 are both counterexamples. The official key adds: filling the whole table is tedious and should be avoided.

Break it down

1. "Prove or disprove": decide which first. To prove $\equiv$ you need all rows to agree (laws or a full table). To disprove, one row where they differ, a **counterexample**, is enough, because $\equiv$ demands agreement in every row.
2. Hunt smartly. Each side is an if-then, and an if-then is F in only one pattern. $p \to (q \to r)$ is F only when p = T and $q \to r$ = F, so only at p = T, q = T, r = F.
3. $(p \to q) \to r$ is F when $p \to q$ = T and r = F. $p \to q$ is T for (p, q) = TT, FT, FF. So the left side is F in three rows, while the right side is F in only one. The extra two, (F, T, F) and (F, F, F), are the counterexamples.
4. Lesson: moving the brackets on $\to$ changes the meaning. $\wedge$ and $\vee$ have an associative law; $\to$ does not. So never write $p \to q \to r$.
5. A fact worth keeping: $p \to (q \to r) \equiv (p \wedge q) \to r$. Both are F only at $p = q$ = T, r = F.

> **Trick**
>
> To compare two if-then statements, list the rows where each is false (usually very few). If those lists differ, any row in one list but not the other is a counterexample.

Q5 · The four forms

Sheet 1 §3

Question. Given "If $12x - 7 = 29$, then $x = 3$", write its negation, contrapositive, converse and inverse. Is the statement true? Is its converse true? (Prove, or give a counterexample.) In general, can the converse of a conditional be true while its inverse is false?

Answer (the four forms)

form	statement
negation	$12x - 7 = 29$ and $x \neq 3$
contrapositive	If $x \neq 3$, then $12x - 7 \neq 29$
converse	If $x = 3$, then $12x - 7 = 29$
inverse	If $12x - 7 \neq 29$, then $x \neq 3$

Break it down

1. Name the parts. Hypothesis P: "$12x - 7 = 29$". Conclusion Q: "$x = 3$". The statement is $P \to Q$. Negating "=" gives "$\neq$".
2. **Negation**: $\sim(P \to Q) \equiv P \wedge \sim Q$ (Q3(d)). It is an "and", not an if-then. It describes the one way an if-then can fail: the

hypothesis happened and the conclusion did not.

3. **Contrapositive**: $\sim Q \to \sim P$, swap and negate both. Always equivalent to the original.
4. **Converse**: $Q \to P$, swap only. **Inverse**: $\sim P \to \sim Q$, negate only. Neither is equivalent to the original in general.

Trick

Swap = converse. Negate = inverse. Swap and negate = contrapositive, the only one that always matches the original.

Trap

"If $12x - 7 = 29$, then $x \neq 3$" is **not** the negation. The negation must be true exactly when the original is false, and only "P and not Q" does that.

Q5 (cont.) · Is it true? Is the converse?

Sheet 3 §4

Question. Is "If $12x - 7 = 29$, then $x = 3$" true? Is its converse "If $x = 3$, then $12x - 7 = 29$" true? Prove, or give a counterexample.

Answer

Both are true.

```
Proof of "if 12x − 7 = 29, then x = 3":
1. 12x − 7 = 29              (given)
2. Then 12x = 29 + 7 = 36    (by basic algebra)
3. Then x = 3                (by basic algebra)

Proof of "if x = 3, then 12x − 7 = 29":
1. Substitute x = 3 into 12x − 7.
2. 12(3) − 7 = 36 − 7 = 29   (by basic algebra)
```

Break it down

1. To prove "if P then Q", assume P and derive Q. You never need to handle the case where P is false: there the if-then is vacuously true.
2. The sentence has a hidden "for every real number x". For $x \neq 3$, the hypothesis $12x - 7 = 29$ is false, so the statement is vacuously true for those x. For the one x that makes the hypothesis true, the algebra shows it is 3.
3. A counterexample would be an x with $12x - 7 = 29$ but $x \neq 3$. None exists, because the equation has exactly one solution.
4. Both directions hold, so in fact $12x - 7 = 29 \leftrightarrow x = 3$.

Trick

To prove an if-then, assume the hypothesis and derive the conclusion. To disprove it, give one case with the hypothesis true and the conclusion false. The converse never comes for free: redo the work in the other direction.

Q5 (cont.) · Converse true, inverse false?

Sheet 1 §3

Question. In general, is it possible for the converse of a conditional statement to be true while the inverse of the same statement is false? Why?

Answer

No. The converse is logically equivalent to the inverse, so they always have the same truth value.

```
   q → p        (converse of p → q)
≡  ~q ∨ p       (by implication law)
≡  p ∨ ~q       (by commutative law)
≡  ~(~p) ∨ ~q   (by double negative law)
≡  ~p → ~q      (by implication law)
```

Break it down

1. Faster reason: take the converse $q \to p$ and form its contrapositive (swap and negate): $\sim p \to \sim q$. That is the inverse. A statement and its contrapositive are always equivalent.
2. The last two lines use laws from right to left: the double negative law turns p into $\sim(\sim p)$, and the implication law turns $\sim(\sim p) \vee \sim q$ back into an arrow. That is allowed, because $\equiv$ works both ways.
3. So the four relatives form two equivalent pairs: {original, contrapositive} and {converse, inverse}. In Q5 the original and converse are both true, so all four are true.

Trick

MCQs asking "which two must have the same truth value?" are testing the two pairs: original with contrapositive, converse with inverse.

Q6 · Why → is defined the way it is

Sheet 1B §2

Question. $p \to q$ is T, F, T, T on rows TT, TF, FT, FF. This makes the transitive rule $((p \to q) \wedge (q \to r)) \to (p \to r)$ a tautology. Show that the rule no longer holds if the conditional is redefined as: $\to_1$ (T, F, F, F), $\to_2$ (T, F, T, F), or $\to_3$ (T, F, F, T), on the same rows.

Answer

Replace every arrow in the rule, the main one included, by the new arrow. One row where the rule comes out F shows it is not a tautology.

```
p q | p→q | p→₁q | p→₂q | p→₃q
T T |  T  |  T   |  T   |  T
T F |  F  |  F   |  F   |  F
F T |  T  |  F   |  T   |  F
F F |  T  |  F   |  F   |  T
```

```
Alternative 1, row p = T, q = T, r = F:
  p →₁ q = T →₁ T = T
  q →₁ r = T →₁ F = F
  p →₁ r = T →₁ F = F
  (p →₁ q) ∧ (q →₁ r) = T ∧ F = F
  whole rule = F →₁ F = F
```

```
Alternative 2, row p = T, q = T, r = F:
  p →₂ q = T →₂ T = T
  q →₂ r = T →₂ F = F
  p →₂ r = T →₂ F = F
  (p →₂ q) ∧ (q →₂ r) = T ∧ F = F
  whole rule = F →₂ F = F
```

```
Alternative 3, row p = F, q = T, r = F:
  p →₃ q = F →₃ T = F
  q →₃ r = T →₃ F = F
  p →₃ r = F →₃ F = T
  (p →₃ q) ∧ (q →₃ r) = F ∧ F = F
  whole rule = F →₃ T = F
```

In each case $((p \to_i q) \wedge (q \to_i r)) \to_i (p \to_i r)$ is not a tautology, so the transitive rule fails.

Break it down

1. The **transitive rule** (the lecture's summary table calls it **hypothetical syllogism**): from $p \to q$ and $q \to r$, conclude $p \to r$. A rule of inference must work in every situation, which as a single statement means it must be a tautology. So "no longer holds" means "find one row where it is F".
2. Recognise the three impostors. $\to_1$ is T only on TT, so it is really $\wedge$. $\to_2$ is T exactly when the right side is T, so it just copies q. $\to_3$ is T when both sides match, so it is really $\leftrightarrow$.
3. Choosing the row. Every version gives T $\to_i$ F = F. For $\to_1$ and $\to_2$, F $\to_i$ F is also F, so aim for "premise part F and conclusion $p \to_i r$ F". Row T, T, F does it: both $q \to_i r$ and $p \to_i r$ land on the TF row.
4. For $\to_3$, F $\to_3$ F is T, so that aim fails. Instead use its FT row, which is F: make the premise part F and $p \to_3 r$ T. $p \to_3 r$ is T when $p = r$; take $p = r =$ F and q = T, which makes both premises F.
5. Takeaway: the "strange" rows F → T = T and F → F = T are exactly what make chains of if-thens work.

Trap

If you keep the main arrow as the ordinary →, all three versions are tautologies (for example, with $\to_3$ it reads "$p \leftrightarrow q$ and $q \leftrightarrow r$, so $p \leftrightarrow r$", which is always true). You must redefine the conditional everywhere, main arrow included, as the official answer does.

Trick

To show something is not a tautology you need only one row that makes it F. Pick the row by working backwards from the rows where the main connective is F.

Q7(a) · Sandra knows C++

Sheet 1 §5

Question. Write the logical form. If valid, name the rule of inference; if not, say whether it is the converse or inverse error. Sandra knows Java and Sandra knows C++. ∴ Sandra knows C++.

Answer

Let p: Sandra knows Java; q: Sandra knows C++.

```
p ∧ q    (premise)
∴ q      (valid by specialization)
```

Break it down

1. **Specialization**: from $p \wedge q$, conclude q (or p). If both hold, each one holds. It lets you drop information you do not need.
2. Why valid: in every row where $p \wedge q$ is T, q is T. There is no row with the premise T and the conclusion F.
3. Its partner is **generalization**: from p, conclude $p \vee q$. Having one option is enough for an "or".

Trick

An "and" in a premise: use specialization to pull out a piece. An "or" in a conclusion built from a single fact: use generalization.

Q7(b) · Divisible by 6

Sheet 1 §5

Question. If at least one of these two numbers is divisible by 6, then the product of these two numbers is divisible by 6. Neither of these two numbers is divisible by 6. ∴ The product of these two numbers is not divisible by 6.

Answer

Let p: the first number is divisible by 6; q: the second number is divisible by 6; r: the product is divisible by 6.

```
(p ∨ q) → r   (premise)
~p ∧ ~q       (premise)
∴ ~r          (invalid: inverse error)
```

Explicit counterexample: 2 and 3. Neither is divisible by 6, yet the product 6 is.

Break it down

1. "At least one of the two" is $p \vee q$. "Neither … nor" is $\sim p \wedge \sim q$.
2. By De Morgan, $\sim p \wedge \sim q \equiv \sim(p \vee q)$: the negation of the hypothesis. So the shape is "$A \to r$; not A; therefore not r".
3. That shape is the **inverse error**: it acts as if $A \to r$ also promised $\sim A \to \sim r$ (the inverse). An if-then only says what happens when A holds; when A fails, it says nothing.
4. The official note: this deduction is invalid even for pairs where the conclusion happens to be true (say 1 and 5). **Invalid** means the form allows true premises with a false conclusion; it is about the pattern, not about these particular numbers.

Trick

If the second premise denies the "if" part and the conclusion denies the "then" part, it is the inverse error. Rewrite "neither", "not both" and so on with De Morgan first so the pattern becomes visible.

Q7(c) · Rationals versus irrationals

Sheet 1 §5

Question. If there are as many rational numbers as there are irrational numbers, then the set of all irrational numbers is infinite. The set of all irrational numbers is infinite. ∴ There are as many rational numbers as there are irrational numbers.

Answer

Let p: there are as many rationals as irrationals; q: the set of all irrational numbers is infinite.

```
p → q    (premise)
q        (premise)
∴ p      (invalid: converse error)
```

In fact p is false while both premises are true.

Break it down

1. The **converse error** (also called affirming the consequent): from $p \to q$ and q, conclude p. It treats $p \to q$ as if it were its converse $q \to p$.
2. The critical row that breaks it: p = F, q = T. Then $p \to q$ is vacuously T, q is T, but the conclusion p is F.
3. Real-world version: there are more irrationals than rationals (you meet this later in the course, under cardinality), so p is false. q is true. The first premise is true because its hypothesis is false. True premises, false conclusion.

Trick

If the second premise confirms the "then" part and the conclusion claims the "if" part, it is the converse error. Modus ponens goes forwards; the converse error tries to walk the arrow backwards.

Q7(d) · Bonus or motorcycle

Sheet 1 §5

Question. If I get a Christmas bonus, I'll buy a stereo. If I sell my motorcycle, I'll buy a stereo. ∴ If I get a Christmas bonus or I sell my motorcycle, I'll buy a stereo.

Answer

Let p: I get a Christmas bonus; q: I sell my motorcycle; r: I'll buy a stereo. Valid:

```
p → r                 (premise)
q → r                 (premise)
(p → r) ∧ (q → r)     (conjunction of premises)
(~p ∨ r) ∧ (~q ∨ r)   (implication law)
(r ∨ ~p) ∧ (r ∨ ~q)   (commutative law)
r ∨ (~p ∧ ~q)         (distributive law)
(~p ∧ ~q) ∨ r         (commutative law)
~(p ∨ q) ∨ r          (De Morgan's law)
∴ (p ∨ q) → r         (implication law)
```

Break it down

1. First combine the two premises into one with $\wedge$ (both are given, so both hold: **conjunction introduction**).
2. Each later line is logically equivalent to the one before it, so each is true whenever the premises are. That is the rule "**logical equivalence**": from P, conclude any P' with $P \equiv P'$.
3. The distributive step is read right to left: $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ with the law's p being r pulls the shared r out.
4. De Morgan is also read right to left: $\sim p \wedge \sim q$ becomes $\sim(p \vee q)$. The final implication law reads $\sim X \vee r$ as $X \to r$.
5. Intuition: whichever of the two things happens, a stereo follows. That is the idea behind **proof by division into cases**.

Trick

Memorise $(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$: two arrows into the same conclusion are one arrow out of the "or".

Q8(a) · Critical rows without a table

Sheet 1B §3

Question. Premises $p \vee (q \wedge r)$ and $\sim p$; conclusion $q \wedge r$. Without drawing the truth table, find p, q, r in the critical row(s). Is the argument valid?

Answer

Critical rows are where all premises are T. $\sim p$ = T forces p = F. Then $p \vee (q \wedge r)$ = T forces $q \wedge r$ = T, so $q = r =$ T. There is one critical row, $(p, q, r) = (F, T, T)$, and the conclusion $q \wedge r$ is T there. The argument is valid.

Break it down

1. A **critical row** is a row of the truth table where every premise is T. An argument is valid exactly when the conclusion is T in every critical row. The other rows do not matter: an argument only promises something when all its premises hold.
2. So skip the table and solve the premises like equations. Start with the premise that pins a variable down immediately ($\sim p$ fixes p), then push that value into the others.
3. $F \vee (q \wedge r)$ is T only when $q \wedge r$ is T, and an "and" is T only when both parts are.
4. This argument is elimination with $q \wedge r$ playing the second option.

Trick

Order of attack: bare letters and negated letters first, then "and"s (both parts T), then "or"s and arrows, whose options shrink once some values are known.

Q8(b) · A counterexample

Sheet 1B §3

Question. Give a counterexample to show that this argument is invalid: premises $p \vee (q \wedge r)$ and $\sim(p \wedge q)$; conclusion r.

Answer

Counterexample: p = **true**, $q = r =$ **false**. Both premises $p \vee (q \wedge r)$ = T ∨ F = T and $\sim(p \wedge q)$ = ~F = T are true, but the conclusion r is false.

Break it down

1. A counterexample to an argument is one row where every premise is T and the conclusion is F: a critical row with a false conclusion. One is enough to make the argument invalid.
2. Work backwards from the conclusion. Want r = F.
3. Then premise 1 is $p \vee (q \wedge F) = p \vee F = p$, so p must be T.
4. Then premise 2 is $\sim(T \wedge q) = \sim q$, so q must be F. Every value was forced, so this is the only counterexample.

Trick

To show invalid: set the conclusion false first, then try to make every premise true. Success gives the counterexample; a forced clash means valid.

Q8(c) · Going to the beach

Sheet 1B §3

Question. Valid or invalid? If I go to the beach, I will take my shades or my sunscreen. I am taking my shades but not my sunscreen. ∴ I will go to the beach.

Answer

Let p: I go to the beach; q: I take my shades; r: I take my sunscreen.

```
p → (q ∨ r)
q ∧ ~r
∴ p
```

Critical row: p = **false**, q = **true**, r = **false**. Both premises are true there and the conclusion is false. The argument is invalid.

Break it down

1. "Shades or sunscreen" is $q \vee r$. Bracket it: $p \to (q \vee r)$. (The official key writes $p \to q \vee r$, which means the same because $\to$ is done last, but brackets never hurt.) "Shades but not sunscreen" is $q \wedge \sim r$.
2. Premise 2 forces q = T, r = F. Then $q \vee r$ = T, and $p \to$ T is T whether p is T or F. So there are two critical rows: p = T and p = F.

3. In the row with p = F the conclusion p is F. One bad critical row is enough: invalid.
4. In disguise it is the converse error: you know $q \vee r$ (the "then" part) and conclude p (the "if" part). Taking shades to go shopping does not mean you went to the beach.

Trick

Once the "then" part of an if-then is forced true, that premise holds whatever its "if" part is, so the "if" variable is left free. If the conclusion is exactly that free variable, set it false: invalid.

Q8(d) · The goat and the Yugo

Sheet 1B §3

Question. Valid or invalid? I will buy a new goat or a used Yugo. If I buy both a new goat and a used Yugo, I will need a loan. I bought a used Yugo but I don't need a loan. ∴ I didn't buy a new goat.

Answer

Let p: I buy a new goat; q: I buy a used Yugo; r: I need a loan.

```
p ∨ q
(p ∧ q) → r
q ∧ ~r
∴ ~p
```

Critical row: p = **false**, q = **true**, r = **false**. The conclusion $\sim p$ is true there. The argument is valid.

Break it down

1. Premise 3 ("but" means $\wedge$) forces q = T and r = F. Ignore the tense changes ("will buy", "bought"); logic does not track time.
2. Premise 2 becomes $(p \wedge T) \rightarrow F$, which is $p \rightarrow F$. An arrow into F is T only when its hypothesis is F, so p = F.
3. Premise 1: $F \vee T$ = T. So all premises hold in exactly one row, (F, T, F), and the conclusion $\sim p$ is T there. Valid.
4. In words: buying both would need a loan; no loan, so not both (modus tollens); I did buy the Yugo, so not the goat.

Trick

$X \rightarrow \mathit{false} \equiv \sim X$. Whenever the "then" part of a true premise is known F, its "if" part must be F.

Q9(a) · The model answer

Sheet 1B §5

Question. Knights always tell the truth; knaves always lie; every native is a knight or a knave, but not both. A says: "Both of us are knights." B says: "A is a knave." What are A and B? (Solved in the tutorial; study its format.)

Answer

```
Proof (by contradiction).
1. If A is a knight, then:
 1.1 What A says is true.
      (by definition of knight)
 1.2 ∴ B is a knight too.   (that's what A says)
 1.3 ∴ What B says is true.
      (by definition of knight)
 1.4 ∴ A is a knave.        (that's what B says)
 1.5 ∴ A is not a knight.
      (since A is either a knight or a knave,
       but not both)
 1.6 ∴ Contradiction to 1.
2. ∴ A is not a knight.
3. ∴ A is a knave.
      (since A is either a knight or a knave,
       but not both)
4. ∴ What B says is true.
5. ∴ B cannot be a knave.
      (as B has said something true)
6. ∴ B is a knight.
      (one is a knight or a knave)
7. Conclusion: A is a knave and B is a knight.
```

Break it down

1. Translate the rules. Knight: everything he says is true. Knave: everything he says is false. "Knight or knave, but not both" means "not a knight" is the same as "a knave".
2. **Proof by contradiction**: assume something, reach an impossibility, and conclude the assumption was false. Here the assumption is "A is a knight" (line 1).
3. Lines 1.1 to 1.4 just follow the speeches: a knight's words are true, so B is a knight, so B's words are true, so A is a knave.
4. Line 1.5 is the step people skip. A **contradiction** must be a statement together with its own negation, $P \wedge \sim P$. "A is a knave" is not literally the negation of "A is a knight"; the "not both" rule turns it into "A is not a knight". Only then does 1.6 clash with line 1.
5. Line 3 uses "knight or knave" plus "not a knight": that is elimination (D3(b)).
6. Line 4: B said "A is a knave", and line 3 shows that is true. A knave never says anything true, so B is not a knave (5), so B is a knight (6).
7. Always check the answer against both speeches. A (knave) says "both of us are knights": false, as a knave's words must be. B (knight) says "A is a knave": true. Consistent.

Trap

Writing "contradiction" straight after line 1.4 loses the mark. Knave is not the negation of knight until you cite "a knight or a knave, but not both".

Trick

Template: suppose one native is a knight and follow the speeches one line at a time, citing "by definition of knight/knave" or "that's what X says". Turn "X is a knave" into "X is not a knight" before you call a contradiction. Finish by checking the answer against every speech.

Q9(b) · Two natives accuse each other

Sheet 1B §5

Question. C says: "D is a knave." D says: "C is a knave." How many knights and knaves are there? Use the same format as part (a).

Answer

```
Proof (by division into cases).
1. If C is a knight:
 1.1 What C says is true.
      (by definition of knight)
 1.2 ∴ D is a knave.   (that's what C says)
2. If C is not a knight:
 2.1 Then C is a knave.
      (one is either a knight or a knave)
 2.2 ∴ What C says is false.
      (by definition of knave)
 2.3 ∴ D is not a knave.
      (C says D is a knave, but what C says
       is false)
 2.4 ∴ D is a knight.
      (one is either a knight or a knave)
3. In both cases, there is one knight and
   one knave.
```

Break it down

1. **Proof by division into cases**: list possibilities that cover everything, show each one leads to the same conclusion, and conclude it. "C is a knight" and "C is not a knight" cover every possibility.
2. Why split on C? C's words are about D, so knowing C's type tells you D's type.
3. This is not a proof by contradiction: neither case breaks down. Both are possible. Check D's speech in each: in case 1 D is a knave and "C is a knave" is false, fine; in case 2 D is a knight and "C is a knave" is true, fine.
4. The question asks how many, not who. Every case gives one knight and one knave, so that is the answer, even though you cannot tell which person is which.

Trick

Split on one person and push their words through. If a case produces a clash, kill it (contradiction, as in (a)). If every case survives, report only what all cases have in common.

Q10 · Odd × odd is odd

Sheet 3 §4

Question. Using "n is even iff $\exists k \in \mathbb{Z}$ such that $n = 2k$; n is odd iff $\exists k \in \mathbb{Z}$ such that $n = 2k + 1$", prove that the product of any two odd integers is an odd integer.

Answer

```
Proof (direct proof).
1. Take any two odd integers n, m.
2. Then n = 2k + 1 and m = 2p + 1 for some
   k, p ∈ ℤ.   (by definition of odd integers)
3. Hence nm = (2k + 1)(2p + 1)
            = 2k(2p + 1) + (2p + 1)
            = (4kp + 2k) + (2p + 1)
            = 2(2kp + k + p) + 1
   (by basic algebra)
4. Let q = 2kp + k + p, which is an integer
   (by closure of integers under + and ×)
5. Then nm = 2q + 1, which is odd
   (by definition of odd integers)
6. Therefore, the product of any two odd
   integers is an odd integer.
```

Break it down

1. $\mathbb{Z}$ is the set of all integers {..., −2, −1, 0, 1, 2, ...}. "$\exists k \in \mathbb{Z}$" reads "there is some integer k". So "odd" means "can be written as 2 × (some integer) + 1".
2. The claim is a "for all" statement: for all odd n, m, the product nm is odd. You cannot check every pair, so line 1 takes an arbitrary pair and uses nothing about it except that both are odd. Whatever works for this pair then works for every pair. The official key notes that line 1 means "for all odd integers n, m".
3. Line 2 **unwraps** the definition. The letters must be different (k and p): n and m are different numbers, and using k for both would secretly assume $n = m$.
4. Line 3 aims for the target shape 2 × (an integer) + 1: expand, then take a factor 2 out of every term except the final 1.
5. Line 4 is **closure**: adding or multiplying integers always gives an integer. Without this line "$2q + 1$" proves nothing, since q must be an integer (with $q = 1/2$, $2q + 1 = 2$ is even).
6. Line 5 **rewraps**: nm now matches the definition of odd. Line 6 restates the claim.

Trap

Writing $n = 2k + 1$ and $m = 2k + 1$ with the same k proves only the case $n = m$. Fresh letter for each number.

Trick

Every even/odd/divisibility proof has one skeleton: unwrap each definition with a fresh letter, do algebra towards the target shape, name the bracket, justify that it is an integer by closure, rewrap. A reason in brackets on every line.

Q11(a) · Smart's proof

Sheet 0C §5

Question. Claim: for an integer n, n^2 is odd iff n is odd. Smart's "Proof (by contradiction)": 1. Suppose n is even. 2. Then $\exists k \in \mathbb{Z}$ s.t. $n = 2k$. 3. $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$. 4. Since k is an integer, so is $2k^2$. 5. Hence $n^2 = 2p$ with $p = 2k^2 \in \mathbb{Z}$. 6. Therefore n^2 is even. 7. So if n is even then n^2 is even, which is the same as: if n^2 is odd then n is odd. 8. Therefore n^2 is odd iff n is odd. Comment.

Answer

1. He proves only one direction. Lines 1 to 7 prove the contrapositive of "if n^2 is odd, then n is odd". The other direction, "if n is odd, then n^2 is odd", is never proved, yet line 8 claims the "iff".
2. Wrong label: it is proof by **contraposition**, not by contradiction.
3. Missing justifications, for example line 2 (by definition of an even integer), line 4 (by closure of integers under multiplication), line 6 (by definition of an even integer).

Break it down

1. The claim is a biconditional, so it is two arrows: (⇒) n^2 odd → n odd, and (⇐) n odd → n^2 odd. Both must be proved.
2. The contrapositive of "n^2 odd → n odd" is "n not odd → n^2 not odd". Since every integer is even or odd but not both, "not odd" means "even". So it is "n even → n^2 even", which is what lines 1 to 6 prove. Line 7 is correct.

3. **Proof by contraposition**: to prove $P \to Q$, prove $\sim Q \to \sim P$ instead. Allowed because they are equivalent.
4. **Proof by contradiction**: assume the whole claim is false and reach an impossibility ($P \wedge \sim P$). Smart never assumed the claim false and never reached an impossibility, so the label is wrong. Mislabelling the method costs marks.

> **Trick**
> For "spot the flaw" on an iff proof, first ask: were both directions proved? Then: does the named method match what was done? Then: does every line have a reason?

Q11(b) · The correct proof

Sheet 3 §7

Question. Write your own proof that for an integer n, n^2 is odd if and only if n is odd.

Answer

```
1. (⇒) Proving the contrapositive of
   "if n² is odd, then n is odd".
 1.1 Suppose n is even.
 1.2 Then ∃k ∈ ℤ s.t. n = 2k.
     (by definition of even integers)
 1.3 Squaring both sides,
     n² = (2k)² = 4k² = 2(2k²).
     (by basic algebra)
 1.4 Hence n² = 2p, with p = 2k² ∈ ℤ.
     (by closure of integers under ×)
 1.5 Therefore, n² is even.
     (by definition of even integers)
 1.6 This proves that if n² is odd,
     then n is odd.
2. (⇐) If n is odd, then n × n = n² is odd.
     (by question 10)
3. Therefore, n² is odd if and only if n is odd.
     (by lines 1.6 and 2)
```

Break it down

1. Layout for any iff: two labelled blocks, (⇒) and (⇐), then a line joining them.
2. Why contraposition for (⇒)? Starting from "n^2 is odd" gives $n^2 = 2k + 1$, and there is no clean way to get back to n (square roots of integers are not integers in general). Starting from "n is even" gives $n = 2k$, and squaring is easy.
3. Line 1.6: line 1.5 showed n even $\to n^2$ even. Its contrapositive is n^2 not even $\to n$ not even, which is n^2 odd $\to n$ odd (each integer is even or odd, not both).
4. (⇐) reuses Q10: the product of any two odd integers is odd. Take both of them to be n. Citing a result already proved is allowed; name it.
5. Line 3 combines the two arrows into $\leftrightarrow$, exactly as in Q1(c).

> **Trick**
> When the hypothesis is hard to unwrap (a square, an "irrational", a "not divisible"), prove the contrapositive instead, and say so in the first line.

T2 · Tutorial 2 walkthrough

Logic of quantified statements · answers from the official key; D1 to D4 worked out here

Tutorial 2 trains one skill from four angles: turning English into $\forall/\exists$ statements, reading them back, deciding whether they are true, and proving them. Three tricks crack almost every question: $\forall$ goes with $\rightarrow$ and $\exists$ goes with $\wedge$; with mixed quantifiers, whoever is quantified first moves first (the order is a game); and one counterexample kills a $\forall$ statement, while examples alone never prove one.

D1 · Divisibility, written symbolically

Sheet 3 §1

Question. For integers d and n, $d \mid n$ if and only if $n = kd$ for some integer k ("d divides n"). (a) State this definition symbolically. (b) According to this definition, does 2 divide $2\sqrt{2}$?

Answer (worked out here, no official key)

```
(a) ∀d, n ∈ ℤ (d | n ↔ ∃k ∈ ℤ (n = kd))

(b) No. The definition only covers integers d
    and n, and 2√2 is not an integer. Even if you
    ignore that, 2√2 = k·2 forces k = √2, which
    is not an integer, so no valid k exists.
```

Break it down

1. **Find the three pieces.** The sentence says who it talks about ("for integers d and n"), a link ("if and only if"), and a condition ("$n = kd$ for some integer k"). Each piece becomes one symbol.
2. "For integers d and n" becomes $\forall d, n \in \mathbb{Z}$. The symbol $\forall$ is the **universal quantifier**, read "for all". $\mathbb{Z}$ is the set of integers and $\in$ means "is an element of". $\forall d, n \in \mathbb{Z}$ is shorthand for $\forall d \in \mathbb{Z}\ \forall n \in \mathbb{Z}$.
3. "If and only if" becomes $\leftrightarrow$ (the **biconditional**): both sides are true together or false together. That is exactly what a definition does: the word and its condition can replace each other.
4. "For some integer k" becomes $\exists k \in \mathbb{Z}$. The symbol $\exists$ is the **existential quantifier**, read "there exists". "Some", "a", "at least one" all mean $\exists$.
5. For (b), check every condition of the definition. It starts "for integers d and n", and $n = 2\sqrt{2}$ is not an integer, so the definition does not even apply. The second reason: the only k with $2\sqrt{2} = k \cdot 2$ is $k = \sqrt{2}$, and the definition demands $k \in \mathbb{Z}$. Either reason gives "no".

Trap

$d \mid n$ is a **statement** (true or false), not a number. "$3 \mid 12$" is true; it does not equal 4. Do not mix up the bar $\mid$ with division $/$.

Trick

Most definitions in this course have the shape "for all things of a type, word $\leftrightarrow$ there is a witness making an equation true". When asked whether something fits a definition, check the type of every variable first, then look for the witness. The witness must live in the stated set (here $\mathbb{Z}$), not just exist somewhere.

D2 · "Every boy loves a girl"

Sheet 2 §4

Question. Explain why English can be ambiguous using the sentence "Every boy loves a girl." Interpret it in two ways and write the quantified statement for each.

Answer (worked out here, no official key)

Let B be the set of boys, G the set of girls, and $Loves(x, y)$ mean "x loves y".

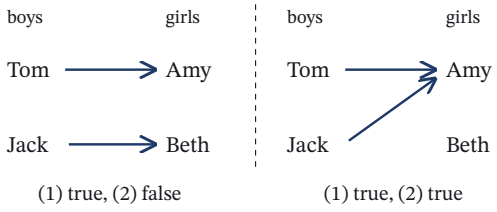
```
(1) Each boy loves some girl (maybe different
    girls for different boys):
    ∀x ∈ B ∃y ∈ G Loves(x, y)

(2) There is one girl whom every boy loves:
    ∃y ∈ G ∀x ∈ B Loves(x, y)
```

The word "a" hides an $\exists$, and English does not say whether that $\exists$ comes before or after the "every". The two orders give two different statements.

Break it down

1. A **predicate** is a sentence with variables, like $Loves(x, y)$, that becomes true or false once you plug in values. It is not yet a statement: "Tom loves Amy" is, "x loves y" is not.
2. "Every boy" is $\forall x \in B$. "A girl" is $\exists y \in G$. Both quantifiers are there; the only question is their **order**.
3. Read $\forall x\ \exists y$ as a game. An opponent names any boy x first, then you find a girl y for that boy. Because you choose after seeing x, your y may change with x.
4. Read $\exists y\ \forall x$ as: you must name one girl first, before seeing any boy, and she must work for every boy.
5. The picture shows the difference. In the left world each boy has his own girl, so (1) is true and (2) is false. In the right world one girl is loved by both, so both are true. So (2) implies (1), but not the other way round.
6. Formal style (Lecture 3) moves the types into the formula: $\forall x\,(Boy(x) \rightarrow \exists y\,(Girl(y) \wedge Loves(x, y)))$ and $\exists y\,(Girl(y) \wedge \forall x\,(Boy(x) \rightarrow Loves(x, y)))$. Notice $\forall$ pairs with $\rightarrow$ and $\exists$ pairs with $\wedge$; D4 explains why.



An arrow means "loves". Left: different girls. Right: one girl for everyone.

Trick

When English mixes "every" with "a/some", write both orders and ask: does the thing I find depend on the thing I was given? If yes, $\forall$ comes first. Swapping $\forall$ with $\exists$ changes the meaning; swapping two $\forall$ (or two $\exists$) never does.

D3 · True/false table for ∀ and ∃

Sheet 2 §3

Question. Given: $\forall x\, P(x)$ is true when $P(x)$ is true for every x, false when some x has $P(x)$ false; $\exists x\, P(x)$ is true when some x has $P(x)$ true, false when $P(x)$ is false for every x. Complete the same table for $\forall x \forall y\, P(x, y)$, $\forall x \exists y\, P(x, y)$, $\exists x \forall y\, P(x, y)$ and $\exists x \exists y\, P(x, y)$.

Answer (worked out here, no official key)

Statement	True when…	False when…
$\forall x\, \forall y\, P(x, y)$	$P(x, y)$ is true for every pair x, y.	There is a pair x, y with $P(x, y)$ false.
$\forall x\, \exists y\, P(x, y)$	For every x there is a y (it may depend on x) with $P(x, y)$ true.	There is an x such that $P(x, y)$ is false for every y.
$\exists x\, \forall y\, P(x, y)$	There is an x such that $P(x, y)$ is true for every y.	For every x there is a y with $P(x, y)$ false.
$\exists x\, \exists y\, P(x, y)$	There is a pair x, y with $P(x, y)$ true.	$P(x, y)$ is false for every pair x, y.

Break it down

1. "True when" is just the statement read aloud, left to right. Each $\forall$ says "for every", each $\exists$ says "there is".
2. A statement is false exactly when its **negation** (its "not" version) is true. So the "False when" column is the negation read aloud.
3. Negation rules: $\sim\forall x\, P(x) \equiv \exists x \sim P(x)$ and $\sim\exists x\, P(x) \equiv \forall x \sim P(x)$. For a chain of quantifiers, walk left to right, flip every $\forall$ to $\exists$ and every $\exists$ to $\forall$, then negate the predicate at the end.

```
~(∀x ∃y P(x, y))
 ≡ ∃x ~(∃y P(x, y))     (negation of ∀)
 ≡ ∃x ∀y ~P(x, y)       (negation of ∃)
```

   Read the last line aloud: "there is an x such that $P(x, y)$ is false for every y". That is row 2's "False when".
4. Two quantifiers of the same kind act like one quantifier over pairs. That is why rows 1 and 4 talk about "pairs".

Trick

Never work out "false when" by intuition. Negate mechanically (flip each quantifier, push the ~ to the end), then read aloud. The result also tells you what a counterexample must look like: to break $\forall x \exists y$, you need ONE x that beats EVERY y.

D4(a) · Why "every" takes → and not ∧

Sheet 2 §2

Question. Domain D = all NUS students; $M(s)$: s is a Math major; $C(s)$: s is a Computer Science major; $E(s)$: s is an Engineering major. "Every Computer Science major is an Engineering major" is $\forall s \in D\, (C(s) \to E(s))$. Why is $\forall s \in D\, (C(s) \wedge E(s))$ wrong? Give an example that shows the difference.

Answer (worked out here, no official key)

The wrong version says every NUS student is both a CS major and an Engineering major. The correct version only constrains CS majors. Example: suppose every CS major also majors in Engineering, and Hana majors in History.

check s = Hana	$C(s) \to E(s)$	$C(s) \wedge E(s)$
values	F → F = true	F ∧ F = false
whole ∀	true	false

In this world the correct answer is true and the wrong answer is false, so they say different things.

Break it down

1. The **domain of discourse** is the set a variable ranges over, here all NUS students. $\forall s \in D$ checks every student, not only the CS ones.
2. "Every CS major is an Engineering major" really means "for every student, IF they are CS, THEN they are Engineering". The "if" is a filter: students who are not CS pass automatically.
3. Why do they pass? $p \to q$ is false in only one row: p true and q false. When p is false, $p \to q$ is true whatever q is. A conditional that is true only because its hypothesis is false is called **vacuously true**.
4. $\wedge$ has no filter. $C(s) \wedge E(s)$ demands both for every student, so it claims the whole university is CS and Engineering. One History student breaks it.
5. The tutorial's Additional Notes state the same idea for the domain itself: $\forall x \in D\, P(x) \equiv \forall x\, ((x \in D) \to P(x))$ and $\exists x \in D\, P(x) \equiv \exists x\, ((x \in D) \wedge P(x))$.

Trap

The mirror mistake with $\exists$: writing "some CS major is an Engineering major" as $\exists s \in D\, (C(s) \to E(s))$. That is true as soon as one student is not CS (F → anything is true), even if no CS major studies Engineering. Use $\exists s \in D\, (C(s) \wedge E(s))$.

Trick

$\forall$ goes with $\to$, $\exists$ goes with $\wedge$. "Every P is Q" is $\forall x\, (P(x) \to Q(x))$. "Some P is Q" is $\exists x\, (P(x) \wedge Q(x))$.

D4(b)–(d) · No, some-not, not both

Sheet 2 §2

Question. Same D, M, C, E. (b) No Computer Science majors are Engineering majors. (c) Some Computer Science majors are not Math majors. (d) If a student is not a Math major, then the student is either a Computer Science major or an Engineering major, but not both.

Answer (worked out here, no official key)

```
(b) ∀s ∈ D (C(s) → ~E(s))
    or equivalently ~∃s ∈ D (C(s) ∧ E(s))

(c) ∃s ∈ D (C(s) ∧ ~M(s))

(d) ∀s ∈ D (~M(s) →
        ((C(s) ∨ E(s)) ∧ ~(C(s) ∧ E(s))))
```

Break it down

1. **(b)** "No P is Q" means "every P is not Q". It is a $\forall$ statement in disguise, so it takes $\rightarrow$, and the "not" lands on the conclusion: $C(s) \rightarrow \sim E(s)$. The second form says "there is no student who is both". They match:

```
~∃s ∈ D (C(s) ∧ E(s))
≡ ∀s ∈ D ~(C(s) ∧ E(s))   (negation of ∃)
≡ ∀s ∈ D (~C(s) ∨ ~E(s))  (De Morgan's law)
≡ ∀s ∈ D (C(s) → ~E(s))   (implication law)
```

   The **implication law** is $p \rightarrow q \equiv \sim p \vee q$; read right to left here with $p = C(s)$, $q = \sim E(s)$.
2. **(c)** "Some" is $\exists$, so $\wedge$. The "not" belongs to Math only: $C(s) \wedge \sim M(s)$.
3. **(d)** Two layers. The outer layer "if not Math, then ..." gives $\sim M(s) \rightarrow (\ldots)$. The inner layer "either CS or Engineering, but not both" is **exclusive or**: at least one, $C(s) \vee E(s)$, and not both, $\sim(C(s) \wedge E(s))$, joined by $\wedge$.
4. Brackets: in this course $\wedge$ and $\vee$ have equal precedence, so $(C(s) \vee E(s)) \wedge \sim(\ldots)$ must be bracketed. $\rightarrow$ is the weakest connective, so the outer layer would parse correctly anyway, but brackets cost nothing.
5. "If a student..." with no "some" means any student: a hidden $\forall$. That is why (d) starts with $\forall s \in D$.

Trap (the tutorial's common mistakes)

1. A comma is not a connective. $\forall x\, P(x), Q(x)$ means nothing; write $\forall x\, (P(x) \wedge Q(x))$ or whatever connective you mean. 2. A predicate returns true or false, so never feed it into another predicate. $Loves(x, Reindeer(y))$ is like $Loves(x, true)$. Write $Reindeer(y) \wedge Loves(x, y)$.

Trick

"No P is Q" $\rightarrow \forall x\, (P(x) \rightarrow \sim Q(x))$. "Not all P are Q" $\rightarrow \exists x\, (P(x) \wedge \sim Q(x))$. "A or B but not both" $\rightarrow (A \vee B) \wedge \sim(A \wedge B)$.

Q1(a) · Variants of 6 | n → 2 | n ∧ 3 | n

Sheet 1 §3

Question. For each statement, write its converse, inverse and contrapositive. Say which of the four are true and which are false, with a counterexample for each false one. (a) $\forall n \in \mathbb{Z}\, (6 \mid n \rightarrow 2 \mid n \wedge 3 \mid n)$.

Term

A **universal conditional** is a statement $\forall x \in D\, (P(x) \rightarrow Q(x))$. Its variants keep $\forall x \in D$ and change only the inside. **Converse**: $\forall x \in D\, (Q(x) \rightarrow P(x))$ (swap). **Inverse**: $\forall x \in D\, (\sim P(x) \rightarrow \sim Q(x))$ (negate both). **Contrapositive**: $\forall x \in D\, (\sim Q(x) \rightarrow \sim P(x))$ (swap and negate). The statement $\equiv$ its contrapositive, and the converse $\equiv$ the inverse. The converse is NOT equivalent to the statement.

Answer

Variant	Formula	T/F
Statement	$\forall n \in \mathbb{Z}\, (6 \mid n \rightarrow (2 \mid n \wedge 3 \mid n))$	True
Converse	$\forall n \in \mathbb{Z}\, ((2 \mid n \wedge 3 \mid n) \rightarrow 6 \mid n)$	True
Inverse	$\forall n \in \mathbb{Z}\, (6 \nmid n \rightarrow (2 \nmid n \vee 3 \nmid n))$	True
Contrapositive	$\forall n \in \mathbb{Z}\, ((2 \nmid n \vee 3 \nmid n) \rightarrow 6 \nmid n)$	True

All four are true, so no counterexamples are needed.

Break it down

1. Split at the arrow. $\rightarrow$ is the weakest connective, so the hypothesis is $P(n)$ = "$6 \mid n$" and the conclusion is the whole $Q(n)$ = "$2 \mid n \wedge 3 \mid n$".
2. Apply the recipe in the Term box. Keep $\forall n \in \mathbb{Z}$ in front of every variant.
3. Negating the conclusion needs **De Morgan's law**: $\sim(2 \mid n \wedge 3 \mid n) \equiv 2 \nmid n \vee 3 \nmid n$. The $\wedge$ flips to $\vee$. The symbol $\nmid$ means "does not divide". This flip is where most people drop the mark.
4. Statement true: if $6 \mid n$ then $n = 6k = 2(3k) = 3(2k)$ for some $k \in \mathbb{Z}$, and $3k, 2k \in \mathbb{Z}$ by closure, so $2 \mid n$ and $3 \mid n$.
5. Converse true (proof not required, but here is a neat one):

```
n = 2a and n = 3b, a, b ∈ ℤ
                          (by definition of divisibility)
n = 3n − 2n               (by basic algebra)
  = 3(2a) − 2(3b)         (substitute n = 2a, n = 3b)
  = 6(a − b)              (by basic algebra)
a − b ∈ ℤ, so 6 | n        (closure; def. of divisibility)
```

6. The inverse is the contrapositive of the converse, so it copies the converse's verdict. The contrapositive copies the statement's verdict. You only ever need to judge two of the four.

Trick

The four variants come in two matched pairs: statement with contrapositive, converse with inverse. Decide the statement and the converse, then copy. Whenever you negate a side that contains $\wedge$ or $\vee$, apply De Morgan's law.

Q1(b) · Variants of x ∈ ℚ → x ∈ ℤ

Sheet 1 §3

Question. Same task for (b) $\forall x\, (x \in \mathbb{Q} \rightarrow x \in \mathbb{Z})$.

Answer

Variant	Formula	T/F
Statement	$\forall x\, (x \in \mathbb{Q} \rightarrow x \in \mathbb{Z})$	False
Converse	$\forall x\, (x \in \mathbb{Z} \rightarrow x \in \mathbb{Q})$	True
Inverse	$\forall x\, (x \notin \mathbb{Q} \rightarrow x \notin \mathbb{Z})$	True
Contrapositive	$\forall x\, (x \notin \mathbb{Z} \rightarrow x \notin \mathbb{Q})$	False

Counterexample for the statement and the contrapositive: $x = 1/2$.

Break it down

1. $\mathbb{Q}$ is the set of **rational numbers**: numbers that can be written a/b with $a, b \in \mathbb{Z}$ and $b \neq 0$. The statement claims "every rational number is an integer".
2. A **counterexample** to $\forall x\, (P(x) \rightarrow Q(x))$ is one x that makes $P(x)$ true and $Q(x)$ false, because that is the only row where $\rightarrow$ is false. For $x = 1/2$: $1/2 \in \mathbb{Q}$ is true, $1/2 \in \mathbb{Z}$ is

false. T → F is false, and one failing x makes the whole $\forall$ false.

3. Contrapositive at $x = 1/2$: $1/2 \notin \mathbb{Z}$ is true, $1/2 \notin \mathbb{Q}$ is false. T → F again. The same x must work, since the two statements are equivalent. ($\notin$ means "is not an element of": $x \notin \mathbb{Q}$ is $\sim(x \in \mathbb{Q})$.)
4. Converse: every integer n is rational because $n = n/1$ with $1 \neq 0$. True. The inverse is the converse's contrapositive, so it is true too.

Trick

To find a counterexample to $\forall x\,(x \in S \to x \in T)$, ask: what is inside S but outside T? Here: a fraction that is not whole.

Q1(c) · Variants of "even + even is even"

Sheet 1 §3

Question. Same task for (c) $\forall p, q \in \mathbb{Z}\,(Even(p) \wedge Even(q) \to Even(p+q))$, where $Even(n)$ means n is an even integer.

Answer

Variant	Formula (inside $\forall p, q \in \mathbb{Z}$)	T/F
Statement	$(Even(p) \wedge Even(q)) \to Even(p+q)$	True
Converse	$Even(p+q) \to (Even(p) \wedge Even(q))$	False
Inverse	$(\sim Even(p) \vee \sim Even(q)) \to \sim Even(p+q)$	False
Contrapositive	$\sim Even(p+q) \to (\sim Even(p) \vee \sim Even(q))$	True

Counterexample for the converse and the inverse: $p = q = 1$.

Break it down

1. $Even(n)$ means $n = 2k$ for some integer k. Statement true: $p = 2k$, $q = 2m$ give $p + q = 2(k+m)$, and $k + m \in \mathbb{Z}$ by closure of integers under $+$.
2. Converse: "if the sum is even, both are even". At $p = q = 1$: $p + q = 2$ is even (T), but $Even(1) \wedge Even(1)$ is false. T → F, so false.
3. Inverse: "if at least one is odd, the sum is odd". At $p = q = 1$: $\sim Even(1) \vee \sim Even(1)$ is true, $\sim Even(2)$ is false. T → F, so false. Same counterexample, because inverse ≡ converse.
4. The inverse's hypothesis is $\sim(Even(p) \wedge Even(q)) \equiv \sim Even(p) \vee \sim Even(q)$ by De Morgan's law. Writing $\sim Even(p) \wedge \sim Even(q)$ is wrong.
5. Two quantified variables means your counterexample must give a value to both p and q. "$p = 1$" on its own is not a counterexample.

Trick

To break a converse about parity, try two odd numbers: odd + odd is even. In general, test the "opposite" case of what the hypothesis looked for.

Q2(a) · "There is no biggest number"

Sheet 2 §4

Question. (a) There is no biggest number: no matter how big a number is, there is always another number that is bigger. Write this symbolically. Which domains make it true?

Answer

```
~∃x ∈ ℕ ∀y ∈ ℕ (y ≤ x)
or, equivalently,
∀x ∈ ℕ ∃y ∈ ℕ (y > x)
```

$\mathbb{N}$ can be replaced by $\mathbb{Z}$, $\mathbb{Q}$ or $\mathbb{R}$, but not by the complex numbers $\mathbb{C}$. Any appropriate domain is accepted.

Break it down

1. "A biggest number" is an x that every y is at most: $\exists x\,\forall y\,(y \leq x)$. "There is no" puts ~ in front.
2. The second form is the "no matter how big, always a bigger one" reading. Push the ~ inward:

```
~∃x ∀y (y ≤ x)
≡ ∀x ~∀y (y ≤ x)      (negation of ∃)
≡ ∀x ∃y ~(y ≤ x)      (negation of ∀)
≡ ∀x ∃y (y > x)       (trichotomy)
```

   Trichotomy: for real numbers exactly one of $y < x$, $y = x$, $y > x$ holds, so "not $y \leq x$" is "$y > x$".
3. Order: you are handed x, then you reply with $y = x + 1$. y depends on x, which $\forall x\,\exists y$ allows. The reversed $\exists y\,\forall x\,(y > x)$ would need one number bigger than every number, including itself: false.
4. Domains: $\mathbb{N} = \{0, 1, 2, \ldots\}$ in this course. In $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, the number $x + 1$ is in the same set and bigger, so the statement holds. $\mathbb{C}$ fails because complex numbers have no "bigger than" at all. A finite set like $\{1, 2, 3\}$ also fails: 3 is biggest.

Trick

"There is no X" is $\sim\exists$. Pushing the ~ inward (flip each quantifier, negate the end) gives the version with no leading ~, which many questions ask for. "No matter what x, there is a y" is always $\forall x\,\exists y$.

Q2(b) · A number between any two

Sheet 2 §4

Question. (b) Given any two distinct numbers, you can always find another number between them. Write this symbolically. Which domains make it true?

Answer

```
∀x ∈ ℚ ∀y ∈ ℚ (x < y →
            ∃z ∈ ℚ ((x < z) ∧ (z < y)))
```

$\mathbb{Q}$ can be replaced by $\mathbb{R}$, but not by $\mathbb{Z}$ or $\mathbb{C}$. ($\forall x \in \mathbb{Q}\,\forall y \in \mathbb{Q}$ can be shortened to $\forall x, y \in \mathbb{Q}$.)

Break it down

1. "Given any two numbers" is $\forall x\,\forall y$. "Distinct" means $x \neq y$, but writing $x < y$ is cleaner: of two distinct numbers one is smaller, and since $\forall$ covers every pair, you can always call the smaller one x.
2. "Find another number between them" is $\exists z$ with $x < z < y$. Logic has no chained inequality, so split it: $(x < z) \wedge (z < y)$.
3. $\exists z$ sits inside, after x and y are chosen, so z may depend on them. The midpoint $z = (x+y)/2$ always works: $x < (x+y)/2$ is the same as $2x < x + y$, i.e. $x < y$.
4. Why $\mathbb{Q}$ works: if $x, y \in \mathbb{Q}$ then $(x+y)/2 \in \mathbb{Q}$ (sum of rationals is rational, half of a rational is rational). $\mathbb{R}$ works

the same way. $\mathbb{Z}$ fails: $x = 1, y = 2$ have no integer strictly between them. $\mathbb{C}$ fails again: no order.

Trap

$\exists z\ \forall x\ \forall y\ (\ldots)$ would claim one single z sits between every pair, which is nonsense. Put each $\exists$ right after the variables its value depends on.

Trick

"$x < z < y$" is two comparisons joined by $\wedge$. "For any ... you can find ..." is $\forall \ldots\ \exists$, with the $\exists$ placed after everything it depends on.

Q3 · Reflexive, symmetric, transitive

Sheet 5 §1

Question. For a binary relation R on a set A: (a) R is reflexive iff "xRx for any x in A". (b) R is symmetric iff "for every x and y in A, if xRy then yRx". (c) R is transitive iff "for all x, y and z in A, if xRy and yRz, then xRz". Rewrite each "..." symbolically with quantifiers and connectives.

Answer

```
(a) ∀x ∈ A (xRx)
(b) ∀x, y ∈ A (xRy → yRx)
(c) ∀x, y, z ∈ A ((xRy ∧ yRz) → xRz)
```

"For any", "for every" and "for all" mean the same thing: $\forall$.

Break it down

1. A **binary relation R on A** decides, for each ordered pair (x, y) of elements of A, whether x is related to y. xRy means "x is related to y". Example: $\leq$ on $\mathbb{Z}$ is a relation, and $3 \leq 5$ is xRy with $x = 3, y = 5$. Lecture 6 defines R as a set of pairs; for now read xRy as a predicate $R(x, y)$.
2. Swap words for symbols: "for any/every/all x in A" $\rightarrow$ $\forall x \in A$; "if ... then" $\rightarrow$ $\rightarrow$; "and" $\rightarrow$ $\wedge$.
3. In (c) the brackets around $xRy \wedge yRz$ are optional ($\rightarrow$ is weakest) but make the hypothesis obvious.
4. Sanity check: $=$ has all three properties. $\leq$ is reflexive and transitive but not symmetric ($1 \leq 2$ but not $2 \leq 1$). $<$ is transitive but not reflexive.

Trick

Each property is a universal statement, so one counterexample disproves it: one x without xRx; one pair with xRy but not yRx; one triple with xRy, yRz but not xRz. You will use exactly this in Lecture 6.

Q4(a) · Is $\mathbb{Z}$ closed under division?

Sheet 0C §1

Question. Recall: r is rational $\Leftrightarrow \exists a, b \in \mathbb{Z}$ s.t. $r = a/b$ and $b \neq 0$. Prove or disprove: (a) Integers are closed under division.

Answer

False. Counterexample: let $a = 3$, $b = 2$. Then $a/b = 1.5$, which is not an integer.

Break it down

1. A set S is **closed under** an operation if applying the operation to any members of S always gives a member of S. "$\mathbb{Z}$ closed under division" means $\forall a, b \in \mathbb{Z}\ (a/b \in \mathbb{Z})$. ("s.t." is short for "such that".)
2. That is a $\forall$ statement, so one counterexample disproves it. Choose integers whose quotient is not whole: $3/2 = 1.5$.
3. Check the counterexample is legal: 3 and 2 are integers, and 1.5 is not, since no integer lies strictly between 1 and 2.
4. $b = 0$ would also work ($3/0$ is undefined, so not in $\mathbb{Z}$), but $3/2$ is the cleaner choice.

Trick

For "prove or disprove" on a $\forall$ claim, spend thirty seconds trying small, awkward values first. If one breaks it, the answer is one line. If nothing breaks, switch to a general proof.

Q4(b) · Rationals are closed under addition

Sheet 3 §4

Question. (b) Prove or disprove: rational numbers are closed under addition.

Answer

True. Proof:

```
1. Let r and s be arbitrarily chosen rational
   numbers.
2. Then ∃a, b, c, d ∈ ℤ s.t. r = a/b, s = c/d
   and b ≠ 0, d ≠ 0.
   (by definition of rational numbers)
3. Hence r + s = a/b + c/d = (ad + bc)/(bd).
   (by basic algebra)
4. ad + bc ∈ ℤ and bd ∈ ℤ.
   (by closure of integers under + and ×)
5. Moreover, bd ≠ 0 since b ≠ 0 and d ≠ 0.
   (by zero product property, Appendix A T11)
6. Hence r + s is rational.
   (by definition of rational numbers)
7. Therefore, rational numbers are closed under
   addition.
```

The official key's line 4 reads "ad + bd", a typo for ad + bc.

Break it down

1. The goal is $\forall r, s \in \mathbb{Q}\ (r + s \in \mathbb{Q})$. To prove a $\forall$, take **arbitrary** elements: ones you know nothing about except that they are rational. Whatever you prove for them holds for all (this is universal generalization, see Q8(c)).
2. Line 2 **unwraps** the definition: "r is rational" becomes "$r = a/b$ for some integers a, b with $b \neq 0$". Use new letters c, d for s. Reusing a, b would secretly force $r = s$.
3. Line 3 is ordinary fraction addition over the common denominator bd.
4. Lines 4 and 5 check every condition of the definition for the new fraction. **Closure**: adding or multiplying integers gives an integer, so top and bottom are integers. **Zero product property**: if $bd = 0$ then $b = 0$ or $d = 0$; since neither is 0, $bd \neq 0$.
5. Line 6 **rewraps**: $r + s$ = (integer)/(nonzero integer), which is the definition of rational.
6. The key flags line 5 as the one students miss. Without it, the "$b \neq 0$" clause of the definition is unchecked and line 6 does not follow.

Trick

Definition proofs are unwrap, compute, rewrap. When you rewrap, tick off every clause of the definition one by one (integer on top, integer below, below $\neq 0$). A missing clause costs marks even when the algebra is right.

Q4(c) · Is $\mathbb{Q}$ closed under division?

Sheet 0C §1

Question. (c) Prove or disprove: rational numbers are closed under division.

Answer

False. Counterexample: let $r, s \in \mathbb{Q}$ with $s = 0$ (for instance $r = 1$). Then r/s is undefined, so it is not a rational number.

Break it down

1. "Closed under division" means $\forall r, s \in \mathbb{Q}\,(r/s \in \mathbb{Q})$.
2. 0 is rational ($0 = 0/1$). Dividing by it gives no number at all, so $r/0 \notin \mathbb{Q}$. One pair breaks the $\forall$.
3. If you exclude 0 as the divisor, the claim becomes true: with $r = a/b$ and $s = c/d \neq 0$ (so $c \neq 0$), $r/s = ad/(bc)$ with $bc \neq 0$. The question as stated includes 0, so the answer is false.

Trick

Whenever a claim involves division, test 0 as the divisor first. It is the cheapest counterexample in the course.

Q5(a)–(b) · $\forall\forall$ over finite sets

Sheet 0A §4

Question. $A = \{1, 3, 5, 7, 11, 13\}$, $B = \{0, 2, 4, 6\}$. True or false? (a) $\forall x \in B\ \forall y \in B\,(x - y \in B)$. (b) $\forall x \in A\ \forall y \in A\,((x < y) \wedge (y < 10) \to y - x \in B)$.

Answer

(a) False. Counterexample: $x = 2, y = 4$ gives $x - y = -2 \notin B$.

(b) True. The pairs with $x < y < 10$ are:

```
x = 1, y = 3: y − x = 2 ∈ B
x = 1, y = 5: y − x = 4 ∈ B
x = 1, y = 7: y − x = 6 ∈ B
x = 3, y = 5: y − x = 2 ∈ B
x = 3, y = 7: y − x = 4 ∈ B
x = 5, y = 7: y − x = 2 ∈ B
```

Break it down

1. Over a finite set, $\forall$ is one long $\wedge$ (every value must pass) and $\exists$ is one long $\vee$ (one value is enough). $\forall x\ \forall y$ covers every ordered pair, including $x = y$ and both orders.
2. (a) Subtraction goes negative whenever $x < y$, and B has no negatives. One pair suffices: $2 - 4 = -2$.
3. (b) The hypothesis $(x < y) \wedge (y < 10)$ filters the pairs. $y < 10$ leaves $y \in \{1, 3, 5, 7\}$. Then $x < y$ leaves exactly the six pairs above. Every difference is 2, 4 or 6, all in B.
4. What about $x = 1, y = 11$, where $y - x = 10 \notin B$? There $y < 10$ is false, so the hypothesis is false and the implication is vacuously true. Such pairs cannot break the statement.
5. Checking every relevant case of a finite domain is a **proof by exhaustion**. It is valid here because A is finite; it would not be valid over $\mathbb{R}$ (see Q7).

Trick

For $\forall x\,(P(x) \to Q(x))$ over a finite set, list only the values of x that make the hypothesis $P(x)$ true. Only they can fail.

Q5(c)–(g) · Swaps and negation pairs

Sheet 0A §4

Question. $A = \{1, 3, 5, 7, 11, 13\}$, $B = \{0, 2, 4, 6\}$. (c) $\forall x \in A\ \forall y \in B\,(x = y + 1)$ (d) $\exists x \in A\ \exists y \in B\,(x = y + 1)$ (e) $\exists y \in B\ \exists x \in A\,(x = y + 1)$ (f) $\forall x \in A\ \forall y \in B\,(x \neq y + 1)$ (g) $\exists x \in A\ \exists y \in B\,(x \neq y + 1)$

Answer

Part	Verdict	Reason
(c)	False	$x = 1, y = 2$: $1 \neq 3$
(d)	True	$x = 3, y = 2$: $3 = 2 + 1$
(e)	True	equivalent to (d)
(f)	False	$x = 3, y = 2$; (f) is the negation of (d)
(g)	True	$x = 1, y = 2$; (g) is the negation of (c)

Break it down

1. (c) claims every x in A equals every $y + 1$ at the same time. One pair that fails is enough: $x = 1, y = 2$.
2. (d) An $\exists$ statement is proved by one **example** (also called a **witness**): values that make it true. $x = 3, y = 2$.
3. (e) is (d) with the two $\exists$ swapped. Neighbouring quantifiers of the same kind can swap without changing meaning, so (e) $\equiv$ (d).
4. (f) Negate (d): $\sim\exists x\ \exists y\,(x = y + 1) \equiv \forall x\ \forall y\,(x \neq y + 1)$, which is (f). Since (d) is true, (f) is false, and (d)'s witness is (f)'s counterexample.
5. (g) is the negation of (c) by the same rule. (c) is false, so (g) is true, and (c)'s counterexample is (g)'s witness.

Trick

Spot negation pairs before evaluating. Of a statement and its negation, exactly one is true, and the witness for one is the counterexample for the other: one computation, two answers.

Q5(h)–(j) · Mixed quantifiers on finite sets

Sheet 0A §4

Question. $A = \{1, 3, 5, 7, 11, 13\}$, $B = \{0, 2, 4, 6\}$. (h) $\forall x \in A\ \exists y \in B\,(x > y)$ (i) $\forall x \in A\ \exists y \in B\,(x \leq y)$ (j) $\exists y \in B\ \forall x \in A\,(x > y)$

Answer

(h) True. Pick $y = 0$: every element of A is larger than 0.

(i) False. Counterexample: $x = 11$. No $y \in B$ has $11 \leq y$.

(j) True. Example: $y = 0$.

Break it down

1. (h) $\forall x\ \exists y$ is the game where the opponent picks x and you answer with y. Here the same answer $y = 0$ works for every x, because the smallest element of A is 1.
2. (i) To show $\forall x\ \exists y\ P$ is false, show its negation $\exists x\ \forall y\ \sim P$: one x that every y fails for. $x = 11$: $11 \leq 0$, $11 \leq 2$, $11 \leq 4$, $11 \leq 6$ are all false. ($x = 7$ or 13 also work, since the largest y is 6.)
3. (j) $\exists y\ \forall x$: you commit to one y first, and it must work for every x. $y = 0$ does.

4. Compare (h) and (j): same predicate, quantifiers swapped. (j) is the stronger claim: if one y works for all x, then each x certainly has some y. The reverse can fail in general; here both are true because a single universal answer ($y = 0$) exists.

> **Trick**
> $\forall x\ \exists y$: "can I answer every x?" $\exists y\ \forall x$: "is there one answer that works for all?" To kill $\forall x\ \exists y$, find one x that no y can answer.

Q6 · Readers and book genres

Sheet 2 §4

Question. A figure links six readers (Mr Aiken, Ms Betty, Mr Carl, Ms Dueet, Ms Emily, Mr Fandi) to book titles in four genres: Children's Fiction (5 titles), Fantasy (3), Mystery (4) and Classic Literary Fiction (3). A line means "reads". True or false, and why? (a) Some title is read by all the female readers. (b) Every reader reads some title in every genre. (c) Some reader reads all titles of some genre. (d) There is some genre for which some reader does not read any of its titles.

The figure's lines did not survive the text extraction of the tutorial, so the reasons below are the key's; check them against the figure in your copy.

Answer

Part	Verdict	Why (official key)
(a)	False	No title is read by all three female readers.
(b)	False	Ms Dueet does not read any Fantasy title.
(c)	True	Ms Dueet reads all the Mystery titles.
(d)	True	No Fantasy title is read by Ms Dueet (nor by Mr Fandi).

Break it down

1. The quantified forms are not required, but they tell you exactly what to check. Let $Reads(r, t)$ mean "r reads t", and let $t \in g$ mean "title t is in genre g".

```
(a) ∃t ∀f ∈ Female Reads(f, t)
(b) ∀r ∀g ∃t ∈ g Reads(r, t)
(c) ∃r ∃g ∀t ∈ g Reads(r, t)
(d) ∃g ∃r ∀t ∈ g ~Reads(r, t)
```

2. (a) The female readers are Ms Betty, Ms Dueet and Ms Emily. You need one title with lines from all three. To show false, you must go through every title and find, for each, a female reader with no line to it (the negation $\forall t\ \exists f\ \sim Reads(f, t)$). The key reports that no title passes.
3. (b) Every (reader, genre) pair needs at least one line. To show false, one bad pair is enough: (Ms Dueet, Fantasy).
4. (c) You pick one reader and one genre, then check every title of that genre. Ms Dueet with Mystery: lines to all four Mystery titles.
5. (d) is exactly the negation of (b): $\sim\forall r\ \forall g\ \exists t\ Reads \equiv \exists r\ \exists g\ \forall t\ \sim Reads$, and $\exists r\ \exists g$ may be swapped to $\exists g\ \exists r$. So (b) false forces (d) true, with the same witness (Ms Dueet, Fantasy).

> **Trap**
> In (c), "all titles of some genre" means one genre chosen first, then all of ITS titles ($\exists g$ before $\forall t$). It does not mean all titles in the whole figure. Read the scope of every "all" and "some" before looking at the lines.

> **Trick**
> For diagram questions, translate to quantifiers first, then play the game: each $\exists$ is your pick, each $\forall$ is the opponent's. After that, checking the diagram is mechanical. Look for negation pairs, as (b) and (d) are here.

Q7(a)–(b) · Proof by examples

Sheet 0C §4

Question. (a) What is wrong with this "proof" of $\forall x \in \mathbb{R}\ (x^2 \geq 0)$? "There are 3 cases: $x < 0$, $x = 0$ and $x > 0$. If $x < 0$, for example $x = -3$, then $x^2 = 9 \geq 0$; if $x = 0$, then $x^2 = 0$; if $x > 0$, say $x = 4$, then $x^2 = 16 \geq 0$. Therefore, in all cases, $x^2 \geq 0$." (b) Use the same logic to "prove" $\forall x \in \mathbb{R}\ (x^3 = x)$.

Answer

(a) It looks like proof by division into cases, but inside each case it only checks one example, so it is really a "proof by examples", which is invalid for a universal statement. The error: it never considers an arbitrary real number x.

(b) "There are 3 cases: $x < 0$, $x = 0$ and $x > 0$. If $x < 0$, for example $x = -1$, then $x^3 = -1 = x$; if $x = 0$, then $x^3 = 0 = x$; if $x > 0$, say $x = 1$, then $x^3 = 1 = x$. Therefore, in all cases, $x^3 = x$."

Break it down

1. **Division into cases** is a valid method: split the domain into pieces that together cover everything, then prove the claim for an arbitrary element of each piece.
2. The three cases here do cover $\mathbb{R}$, so the split is fine. The failure is inside: "for example, $x = -3$" proves something about -3 only. The case $x < 0$ contains infinitely many numbers.
3. A correct case would read: "Let x be any real number with $x < 0$. Then $-x > 0$, and $x^2 = (-x)(-x) > 0$, since the product of two positive reals is positive." Nothing depends on a specific value.
4. (b) shows why the method is broken: it "proves" something false. At $x = 2$, $x^3 = 8 \neq 2$. The chosen examples -1, 0, 1 happen to be the only three reals with $x^3 = x$. Cherry-picked examples can make anything look true.

> **Trick**
> Examples can disprove a $\forall$ (one counterexample) or prove an $\exists$ (one witness). They can never prove a $\forall$ over an infinite domain. In a proof of a $\forall$, the words "for example" or "say $x = 4$" are a red flag.

Q7(c)–(d) · The wrong negation

Sheet 0C §4

Question. (c) What is wrong with this "proof" of $\forall x \in \mathbb{R}\ (x^2 \geq 0)$? "Prove by contradiction. Suppose $x^2 < 0$ for all real numbers x. Let $x = 3$, then $x^2 = 9 > 0$, which is a contradiction. Therefore, $\forall x \in \mathbb{R}\ (x^2 \geq 0)$." (d) Use the same logic to "prove" $\forall x \in \mathbb{R}\ (x^3 = x)$.

Answer

(c) To prove by contradiction, one must assume $\sim(\forall x \in \mathbb{R}\ (x^2 \geq 0))$, which is $\exists x \in \mathbb{R}\ \sim(x^2 \geq 0)$. What the proof assumes is $\forall x \in \mathbb{R}\ \sim(x^2 \geq 0)$, the wrong negation.

(d) "Prove by contradiction. Suppose $x^3 \neq x$ for all real numbers x. Let $x = 0$, then $x^3 = 0 = x$, which is a contradiction. Therefore, $\forall x \in \mathbb{R}\,(x^3 = x)$."

Break it down

1. **Proof by contradiction**: assume the claim is false (assume its exact negation), derive something impossible, and conclude the claim is true. It only works with the exact negation.
2. Negating $\forall$ gives $\exists$: $\sim\forall x\, P(x) \equiv \exists x \sim P(x)$. "Not every x has $x^2 \geq 0$" means "some x has $x^2 < 0$", not "every x has $x^2 < 0$".

```
Claim:          ∀x ∈ ℝ (x² ≥ 0)
Negation:       ∃x ∈ ℝ (x² < 0)    assume this
"Proof" used:   ∀x ∈ ℝ (x² < 0)    too strong
```

3. "Every x has $x^2 < 0$" is an easy target: one example ($x = 3$) contradicts it. But contradicting it only shows $\exists x\,(x^2 \geq 0)$, that SOME real has a non-negative square. That is far weaker than the claim.
4. A correct start would be "Suppose there is a real x with $x^2 < 0$". Then you must reach a contradiction for that unknown x, which takes real work (the case split of Q7(a), done properly).
5. (d) The same pattern "proves" a false claim; all it really shows is $\exists x\,(x^3 = x)$, witnessed by $x = 0$. The claim itself fails at $x = 2$.

Trick

Before any proof by contradiction, write the negation mechanically: $\sim\forall$ becomes $\exists$, $\sim\exists$ becomes $\forall$. If the claim starts with $\forall$ and your assumption also starts with $\forall$, you negated it wrong.

Q8(a) · Why skip the case $r^2 \leq r$?

Sheet 2 §5

Question. Partial proof of $\forall x \in \mathbb{R}\,((x^2 > x) \to (x < 0) \vee (x > 1))$: 1. Let r be an arbitrarily chosen real number. 2. Suppose $r^2 > r$. 2.1 Then $r^2 - r > 0$, or $r(r-1) > 0$ (by basic algebra). 2.2 So both r and $r-1$ are positive, or both are negative (by Appendix A, T25). 2.3 ... 3. Therefore, $\forall x \in \mathbb{R}\,((x^2 > x) \to (x < 0) \vee (x > 1))$. (a) Step 2 explores the case $r^2 > r$. Do we need to include the case $r^2 \leq r$? Why?

Answer

No. A conditional $p \to q$ is always true when its hypothesis (antecedent) p is false. So when $r^2 \leq r$, the implication is already true for that r, and there is nothing to examine.

Break it down

1. For the chosen r, the thing to prove is an implication: $(r^2 > r) \to ((r < 0) \vee (r > 1))$. The **hypothesis** (or antecedent) is the part before $\to$; the **conclusion** (or consequent) is the part after.
2. $p \to q$ is false only when p is true and q is false. So the implication can only fail for an r with $r^2 > r$. That is the only case worth checking.
3. If $r^2 \leq r$ (for example $r = 0.5$: $0.25 \leq 0.5$), the hypothesis is false and the implication is vacuously true.
4. This is why every direct proof of "if P then Q" starts with "Suppose P" and never looks at "not P".

Trick

To prove $p \to q$: assume p, derive q. The case "not p" is free, so never write it.

Q8(b) · Complete the proof

Sheet 3 §7

Question. (b) Complete the proof started in (a).

Answer

```
1. Let r be an arbitrarily chosen real number.
2. Suppose r² > r.
 2.1. Then r² − r > 0, or r(r − 1) > 0.
      (by basic algebra)
 2.2. So, both r and r − 1 are positive, or
      both are negative. (by Appendix A, T25)
 2.3. Case 1: Both r and r − 1 are positive.
      2.3.1. (r > 0) ∧ (r − 1 > 0) → (r > 1).
 2.4. Case 2: Both r and r − 1 are negative.
      2.4.1. (r < 0) ∧ (r − 1 < 0) → (r < 0).
 2.5. From lines 2.3.1 and 2.4.1, we have
      (r > 1) ∨ (r < 0).
 2.6. Therefore, (r² > r) → (r > 1) ∨ (r < 0).
3. Therefore,
   ∀x ∈ ℝ ((x² > x) → (x < 0) ∨ (x > 1)).
```

Break it down

1. Line 2.1: subtract r from both sides of $r^2 > r$ to get $r^2 - r > 0$, then factor out r.
2. Line 2.2 quotes **Appendix A**, the list of real-number properties from Epp's textbook that you may cite by number. T25: if $ab > 0$, then a and b are both positive or both negative. Here $a = r$ and $b = r - 1$.
3. Line 2.2 is an "or" of two situations, so the proof splits into two cases, one per situation. This is **proof by division into cases**: whichever situation actually holds, you have handled it.
4. Case 1 (line 2.3.1): from $r - 1 > 0$, adding 1 to both sides gives $r > 1$. The line is written as an implication, but it simply records "in this case, $r > 1$".
5. Case 2 (line 2.4.1): $r < 0$ is one of the two facts the case gives you, so it holds immediately.
6. Line 2.5: in case 1, $r > 1$; in case 2, $r < 0$. Either way at least one holds, so their "or" holds. (Formally, each case reaches $(r > 1) \vee (r < 0)$ by the generalization rule $p \vdash p \vee q$.)
7. Line 2.6 closes the "Suppose" of line 2: we assumed $r^2 > r$ and reached the conclusion, so the implication holds for r. The order inside $\vee$ does not matter (commutative law).
8. Line 3 generalizes from r to every real x; (c) explains why that is allowed.

Trick

When a line hands you "A or B", open one numbered case per option and finish each. The indented numbering (2.3, 2.3.1) shows which assumption each line lives under, and graders read it that way.

Q8(c) · Universal generalization

Sheet 2 §7

Question. (c) Step 3 is an application of universal generalization. Explain what it means.

Answer

Let $P(x)$ be "$x^2 > x$" and $Q(x)$ be "$(x < 0) \vee (x > 1)$". Since we have established $P(r) \to Q(r)$ for an arbitrarily chosen real number r, we may generalize it to all real numbers: $\forall x \in \mathbb{R}\ (P(x) \to Q(x))$.

Break it down

1. **Universal generalization**: if you prove $P(a)$ for an element a about which you assumed nothing except that it is in the domain, then $\forall x\, P(x)$ holds.
2. The key word is **arbitrary**. Line 1 chose r with no special property, and no later line used a particular value of r. So the same argument runs for every real number you could plug in, which is exactly what $\forall$ claims.
3. Contrast Q7(a): there $x = -3$ was a specific number and the argument used its value, so generalizing from it is invalid.
4. The mirror rule is **universal instantiation**: from $\forall x \in D\ P(x)$, conclude $P(a)$ for any particular $a \in D$.

Trick

Every proof of a $\forall$ opens with "Let x be an arbitrarily chosen ..." and closes with "Therefore, $\forall x$ ...". The first line is what makes the last line legal.

Q9(b)–(d) · Universal Studios visitors

Sheet 2 §2

Question. V = visitors to Universal Studios Singapore on one day. $T(v)$: v took the Transformers ride; $G(v)$: v took the Battlestar Galactica ride; $E(v)$: v visited the Ancient Egypt; $W(v)$: v watched the Water World show. (a) "Every visitor watched the Water World show" is $\forall v \in V\ (W(v))$. Express: (b) Every visitor who took the Battlestar Galactica ride also took the Transformers ride. (c) There is a visitor who took both the Transformers ride and the Battlestar Galactica ride. (d) No visitor who visited the Ancient Egypt watched the Water World show.

Answer

```
(b) ∀v ∈ V (G(v) → T(v))
(c) ∃v ∈ V (T(v) ∧ G(v))
(d) ∀v ∈ V (E(v) → ~W(v))
    alternatively ∀v ∈ V (~E(v) ∨ ~W(v))
```

Break it down

1. (b) "Every visitor who took G also took T" is "every G-visitor is a T-visitor". The "who ..." clause narrows down which visitors are meant; under $\forall$ that narrowing becomes the left side of $\to$.
2. (c) "There is a visitor" is $\exists$, so $\wedge$. "Both" is itself an $\wedge$: $T(v) \wedge G(v)$.
3. (d) "No E-visitor watched W" is "every E-visitor did not watch W": $E(v) \to \sim W(v)$. The alternative comes from the implication law $p \to q \equiv \sim p \vee q$. A third correct form is $\sim\exists v \in V\ (E(v) \wedge W(v))$.
4. Why (a) has no $\to$: there is no narrowing, it is every visitor at all. The domain V already restricts to visitors.

Trick

A "who/that" clause after "every" becomes the left side of $\to$. After "some/there is", it becomes one more $\wedge$.

Q9(e) · "Some did, but some did not"

Sheet 2 §2

Question. (e) Some visitors who took the Transformers ride also visited the Ancient Egypt but some (who took the Transformers ride) did not (visit the Ancient Egypt).

Answer

```
(∃v ∈ V (T(v) ∧ E(v)))
    ∧ (∃u ∈ V (T(u) ∧ ~E(u)))
```

It is a conjunction of two existential clauses. Each variable's scope is only its own clause, so reusing v in the second clause is also correct; a different letter just avoids confusion. Incorrect:

```
∃v ∈ V ((T(v) ∧ E(v)) ∧ (T(v) ∧ ~E(v)))
```

Break it down

1. "But" in logic is plain $\wedge$. The sentence is two claims glued together: some Transformers riders visited Egypt, and some Transformers riders did not.
2. Each claim is its own $\exists$ statement. They can be about different people, so each needs its own $\exists$.
3. The **scope** of a quantifier is the part of the formula it controls, normally the bracket right after it. A variable inside a quantifier's scope is **bound**; a variable under no quantifier is **free**. The first $\exists v$'s scope ends at its closing bracket, so the second clause may reuse the letter v for a new, unrelated person.
4. The incorrect version puts both halves under one $\exists v$, so a single visitor must satisfy $E(v)$ and $\sim E(v)$. Since $E(v) \wedge \sim E(v)$ is false for everyone (negation law, after regrouping by the commutative and associative laws), that formula is always false.

Term

Without brackets, a quantifier covers only the smallest well-formed formula after it. In $\forall x\,(\exists y\, P(x, y) \vee Q(x, y))$, $\exists y$ reaches only $P(x, y)$, so the y in $Q(x, y)$ is free. To bind both, write $\forall x\, \exists y\, (P(x, y) \vee Q(x, y))$.

Trick

"Some A are B but some A are not B" is always two separate $\exists$ joined by $\wedge$. A single $\exists$ over both halves forces one object to do both, which is a contradiction.

Q10(a) · Reorder the premises

Sheet 2 §7

Question. Premises: 1. If an object is above all the triangles, then it is above all the blue objects. 2. If an object is not above all the gray objects, then it is not a square. 3. Every black object is a square. 4. Every object that is above all the gray objects is above all the triangles. ∴ If an object is black, then it is above all the blue objects. (a) Reorder the premises (replacing some by their contrapositives) to show that the conclusion follows by universal transitivity.

Answer

```
3. If an object is black, then it is a square.
2. (Contrapositive form) If an object is a
   square, then it is above all the gray objects.
4. If an object is above all the gray objects,
   then it is above all the triangles.
1. If an object is above all the triangles,
   then it is above all the blue objects.
∴ If an object is black, then it is above all
   the blue objects.
```

Break it down

1. **Universal transitivity** (Lecture 3): from $\forall x\,(P(x) \to Q(x))$ and $\forall x\,(Q(x) \to R(x))$, conclude $\forall x\,(P(x) \to R(x))$. It is a chain: each link's conclusion must be the next link's hypothesis, word for word.
2. Put every premise in if-then form. 3 becomes "if black, then square". 4 becomes "if above all gray, then above all triangles".
3. Start from the conclusion's hypothesis, "black". Premise 3 starts there and ends at "square".
4. No premise starts with "square". But premise 2 has square in its conclusion, negated and backwards: "not above all gray → not square". Its contrapositive (swap and negate both sides, then cancel ~~ by the double negative law) is "square → above all gray". It is equivalent to 2, so the replacement is allowed.
5. "Above all gray" starts premise 4, which ends at "above all triangles". That starts premise 1, which ends at "above all blue", exactly the conclusion's ending.

```
black
  → square                 (3)
  → above all gray         (2, contrapositive)
  → above all triangles    (4)
  → above all blue         (1)
```

6. Apply universal transitivity three times along the chain.

Trick

Write the conclusion as "if START then END". From START, repeatedly find the premise whose hypothesis matches where you stand. If a needed link only appears negated and pointing backwards, take its contrapositive.

Q10(b) · The same chain in symbols

Sheet 2 §7

Question. (b) Rewrite your answer to (a) using predicates and quantified statements. You may use self-explanatory predicate names such as $Triangle(x)$, $Square(x)$.

Answer

Let O, the domain, be the set of objects.

```
3. ∀x ∈ O (Black(x) → Square(x))

2. (Contrapositive form)
   ∀x ∈ O (Square(x) →
     (∀y ∈ O (Gray(y) → Above(x, y))))

4. ∀x ∈ O (
     (∀y ∈ O (Gray(y) → Above(x, y)))
   → (∀z ∈ O (Triangle(z) → Above(x, z))))

1. ∀x ∈ O (
     (∀z ∈ O (Triangle(z) → Above(x, z)))
   → (∀w ∈ O (Blue(w) → Above(x, w))))

∴ ∀x ∈ O (Black(x) →
     (∀w ∈ O (Blue(w) → Above(x, w))))
```

Break it down

1. "x is above all the gray objects" is itself a $\forall$ statement about x: for every object y, if y is gray then x is above y. So it is $\forall y \in O\,(Gray(y) \to Above(x, y))$, with $\forall$ paired with $\to$ as always.
2. $Above(x, y)$ is a two-place predicate, and order matters: it means "x is above y", not the reverse.
3. Inside that formula y is bound but x is free, so it behaves like a predicate in x. That is why it can play the role of $Q(x)$ in the chain. All the $Above(x, \ldots)$ must sit inside the scope of the outer $\forall x$, because they talk about the same object.
4. Fresh letters y, z, w for the inner quantifiers keep things readable. Reusing a letter would be legal (each has its own scope) but invites mistakes.
5. To see the skeleton, abbreviate: $AG(x)$, $AT(x)$, $AB(x)$ for "above all gray / triangles / blue". The premises are then $\forall x\,(Black(x) \to Square(x))$, $\forall x\,(Square(x) \to AG(x))$, $\forall x\,(AG(x) \to AT(x))$, $\forall x\,(AT(x) \to AB(x))$. Three rounds of universal transitivity give $\forall x\,(Black(x) \to AB(x))$.
6. Premise 2 before rewriting is $\forall x \in O\,(\sim AG(x) \to \sim Square(x))$. Swap and negate: $\forall x \in O\,(\sim\sim Square(x) \to \sim\sim AG(x))$, which is $\forall x \in O\,(Square(x) \to AG(x))$ by the double negative law.

Trick

"Above all the X" = $\forall y\,(X(y) \to Above(x, y))$. While planning, give the long inner formula a short name; expand it only when writing the final answer.

Q11 · $a \le \sqrt{n}$ or $b \le \sqrt{n}$ (past midterm)

Sheet 0C §5

Question. Prove that if n is a product of two positive integers a and b, then $a \le n^{1/2}$ or $b \le n^{1/2}$.

Answer

Write $\sqrt{n}$ for $n^{1/2}$. Either method gets full credit.

```
Proof by contraposition (p → q ≡ ~q → ~p)
1. The contrapositive of the given statement is:
   if a > √n and b > √n, then n is not a
   product of a and b.   (by De Morgan's law)
2. Suppose a > √n and b > √n. Then
   ab > √n · √n = n.   (by Appendix A T27)
3. Since ab ≠ n, the contrapositive statement
   is true.
4. Therefore, the original statement is true.
```

```
Proof by contradiction (~(p → q) ≡ p ∧ ~q)
1. Suppose not. Then n = ab and a > √n and
   b > √n.   (by De Morgan's law)
2. Since a > √n and b > √n,
   ab > √n · √n = n.   (by Appendix A T27)
3. This contradicts n = ab in line 1.
4. Therefore, the original statement is true.
```

Break it down

1. $n^{1/2}$ is the square root $\sqrt{n}$. Symbolically the claim is $\forall n, a, b \in \mathbb{Z}^+ (n = ab \to (a \le \sqrt{n} \vee b \le \sqrt{n}))$, where $\mathbb{Z}^+$ is the set of positive integers.
2. Why go indirect? The conclusion is an "or", which is awkward to prove directly (which of the two?). Its negation is an "and" of two facts, $a > \sqrt{n}$ and $b > \sqrt{n}$, which is easy to use. Both methods start from that "and".
3. De Morgan's law: $\sim(a \le \sqrt{n} \vee b \le \sqrt{n}) \equiv \sim(a \le \sqrt{n}) \wedge \sim(b \le \sqrt{n})$, and $\sim(a \le \sqrt{n})$ is $a > \sqrt{n}$ by trichotomy.
4. Appendix A T27: for real numbers, if $0 < a < c$ and $0 < b < d$, then $0 < ab < cd$. Here use $0 < \sqrt{n} < a$ and $0 < \sqrt{n} < b$ to get $\sqrt{n} \cdot \sqrt{n} < ab$, i.e. $n < ab$. ($\sqrt{n} > 0$ because n is a positive integer.)
5. **Proof by contraposition**: prove $\sim q \to \sim p$ instead of $p \to q$; they are equivalent. So "if both factors exceed $\sqrt{n}$, then $ab \neq n$" is enough.
6. **Proof by contradiction**: assume the whole statement false. The negation of $p \to q$ is $p \wedge \sim q$: $n = ab$ holds AND the conclusion fails. Line 2 then gives $ab > n$, clashing with $n = ab$.
7. Sanity check: $36 = 4 \times 9$ and $\sqrt{36} = 6$, so $4 \le 6$. $36 = 6 \times 6$: both equal 6. Both factors can never exceed $\sqrt{n}$, or the product would overshoot n.

Trap

The negation of an implication is not another implication: $\sim(p \to q) \equiv p \wedge \sim q$. Starting with "Suppose that if $n = ab$ then $a > \sqrt{n}$ and $b > \sqrt{n}$" is wrong and loses the mark.

Trick

When the conclusion is an "or", go indirect. Negating it turns the "or" into an "and", which hands you two facts to use at once.

T3 · Tutorial 3 walkthrough

Sets · Q1 to Q10 from the official key · D1 to D3 worked out here

Tutorial 3 trains two skills: reading set notation one brace at a time, and proving facts about sets in the course's numbered style. Three moves carry almost every question. For ∈, list the right side's top-level elements and look for the left side among them; for ⊆, check every element of the left side. To prove two sets equal, prove ⊆ in both directions by taking an arbitrary element. To prove an identity with laws, first turn every \ into ∩ with a complement (Set Difference Law), then work as in a logic law proof, one law per line.

D1 · ∈ versus ⊆, twelve quick checks

Sheet 4 §5

Question. Which are true, which are false? (a) $\varnothing \in \varnothing$ (b) $\varnothing \subseteq \varnothing$ (c) $\varnothing \in \{\varnothing\}$ (d) $\varnothing \subseteq \{\varnothing\}$ (e) $\{\varnothing, 1\} = \{1\}$ (f) $1 \in \{\{1,2\}, \{2,3\}, 4\}$ (g) $\{1,2\} \subseteq \{3,2,1\}$ (h) $\{3,3,2\} \subsetneq \{3,2,1\}$ (i) $\varnothing \in \{1,2\}$ (j) $\{1,2,3\} = \{3,1,1,2\}$ (k) $\{5\} \in \{2,5\}$ (l) $\{1,2\} \in \{1, \{2,1\}\}$

Answer (worked out here, no official key)

#	Claim	T/F	Reason
a	$\varnothing \in \varnothing$	F	∅ has no elements at all.
b	$\varnothing \subseteq \varnothing$	T	∅ is a subset of every set.
c	$\varnothing \in \{\varnothing\}$	T	{∅} lists one element, and it is ∅.
d	$\varnothing \subseteq \{\varnothing\}$	T	∅ is a subset of every set.
e	$\{\varnothing, 1\} = \{1\}$	F	∅ is in the left set only (sizes 2 and 1).
f	$1 \in \{\{1,2\}, \{2,3\}, 4\}$	F	The elements are {1,2}, {2,3}, 4. The number 1 is not one of them.
g	$\{1,2\} \subseteq \{3,2,1\}$	T	1 and 2 are both in {3,2,1}.
h	$\{3,3,2\} \subsetneq \{3,2,1\}$	T	{3,3,2} = {2,3}: inside {1,2,3}, and misses 1.
i	$\varnothing \in \{1,2\}$	F	The elements are 1 and 2 only.
j	$\{1,2,3\} = \{3,1,1,2\}$	T	Order and repeats do not matter.
k	$\{5\} \in \{2,5\}$	F	The elements are the numbers 2 and 5; {5} is a set, not the number 5.
l	$\{1,2\} \in \{1, \{2,1\}\}$	T	The elements are 1 and {2,1}, and {2,1} = {1,2}.

Break it down

1. **Set**: an unordered collection of objects, written between braces. **Element**: one of those objects. $x \in S$ ("x is an element of S") means x is one of the items written directly inside S's outer braces, separated by the top-level commas. ∈ never looks inside an item.
2. **Subset**: $X \subseteq S$ means every element of X is also an element of S. In symbols, $\forall x\,(x \in X \to x \in S)$. So the left side of ⊆ must be a set, and you test its elements one at a time.
3. **Empty set** $\varnothing = \{\}$: the set with no elements. "$x \in \varnothing$" is false for every x, including x = ∅. That settles (a).
4. $\varnothing \subseteq S$ is true for every set S (Theorem 6.2.4). Reason: it says "for all x, if $x \in \varnothing$ then $x \in S$". The if-part is never true, so no x can break the statement. A "for all, if ... then ..." statement that holds because the if-part never happens is called **vacuously true**. That settles (b) and (d).
5. ∅ versus {∅}: ∅ is an empty box. {∅} is a box holding one empty box, so it has exactly one element, namely ∅. So (c) is true, while (i) is false because {1, 2} lists only 1 and 2.
6. **Roster notation** (listing the elements in braces): order and repeats do not matter, so {3,1,1,2} = {1,2,3} for (j), and {3,3,2} = {2,3}. **Proper subset**: $X \subsetneq S$ means $X \subseteq S$ and $X \neq S$. {2,3} sits inside {1,2,3} and misses 1, so (h) is true. (g) holds for the same reason.
7. **Set equality**: two sets are equal iff they have exactly the same elements. In (e), ∅ is an element of {∅, 1} but not of {1}, so the sets differ.
8. Nested sets, (f), (k), (l): write out the top-level list first. {{1,2},{2,3},4} has elements {1,2}, {2,3} and 4. {2,5} has elements 2 and 5. {1,{2,1}} has elements 1 and {2,1}, and {2,1} is the same set as {1,2}, so (l) is true.

Trap

Membership does not pass through layers. 1 ∈ {1,2} and {1,2} ∈ S do not give 1 ∈ S, which is exactly why (f) is false. And swapping the symbol flips (k): $\{5\} \subseteq \{2,5\}$ is true.

Trick

Rewrite the relevant side as a bare list before answering: for ∈, list the right side's elements; for ⊆, list the left side's elements. Then compare item by item. "∅ ⊆ S" is always true. "∅ ∈ S" is true only if you can see ∅ written as one of S's items.

D2 · Sizes of an intersection and a union

Question. $A = \{0,1,4,5,6,9\}$ and $B = \{0,2,4,6,8\}$. Find $|A \cap B|$ and $|A \cup B|$.

Answer (worked out here, no official key)

```
A ∩ B = {0, 4, 6}                |A ∩ B| = 3
A ∪ B = {0,1,2,4,5,6,8,9}        |A ∪ B| = 8
check:  |A| + |B| − |A ∩ B| = 6 + 5 − 3 = 8
```

Break it down

1. **Cardinality** $|S|$: the number of elements of S, each counted once.
2. **Intersection** $A \cap B$: the elements in both A and B. Walk through A and keep each element that also appears in B: 0 yes, 1 no, 4 yes, 5 no, 6 yes, 9 no. So {0, 4, 6}.
3. **Union** $A \cup B$: the elements in at least one of A and B. Take all of A, then add the elements of B not already there (2 and 8). The shared elements 0, 4, 6 are written once.
4. Check: adding $|A| + |B|$ counts each shared element twice, once from each set, so subtract the shared ones once: $|A \cup B| = |A| + |B| - |A \cap B|$. This counting idea drives Q3(a) and (b).

Trick

Find ∩ first. Then $|A \cup B| = |A| + |B| - |A \cap B|$ gives the union's size without listing it. Sizes add exactly when the sets share nothing.

D3 · Hidden duplicates, then A × A

Sheet 4B §1

Question. Let $A = \{1, \{1,2\}, 2, \{2,1,1\}\}$. Find $|A|$ and write out $A \times A$ in its simplest form (no duplicate elements).

Answer (worked out here, no official key)

```
{2,1,1} = {1,2}, so A = {1, 2, {1,2}}
|A| = 3

A × A = { (1,1), (1,2), (1,{1,2}),
          (2,1), (2,2), (2,{1,2}),
          ({1,2},1), ({1,2},2), ({1,2},{1,2}) }
|A × A| = 3 × 3 = 9
```

Break it down

1. Clean the inner sets first. {2,1,1} is roster notation for the set whose elements are 1 and 2, so {2,1,1} = {1,2}. The second and fourth items of A are therefore the same set, and it counts once. A has three distinct elements: 1, 2 and {1,2}. So $|A| = 3$, not 4.
2. **Ordered pair** (x, y): a pair in which order matters. $(a, b) = (c, d)$ iff $a = c$ and $b = d$. So (1,2) ≠ (2,1), even though {1,2} = {2,1}.
3. **Cartesian product**: $X \times Y = \{(x, y) : x \in X \wedge y \in Y\}$, every pair whose first entry comes from X and second entry from Y. In $A \times A$ both entries come from A.
4. List systematically: fix the first entry (1, then 2, then {1,2}) and run through all three second entries. That gives 3 rows of 3. The count rule $|X \times Y| = |X| \cdot |Y|$ tells you to expect 9 before you start.
5. {1,2} is one element of A, so it sits in a pair as one entry, as in ({1,2}, 1). Do not split it into 1 and 2.

Trap

Skipping step 1 gives 16 "pairs", such as (1,{1,2}) and (1, {2,1,1}), which are the same pair written twice.

Trick

Clean the set (collapse repeats inside elements, then equal elements), count it, then build. The size of a product is the product of the sizes, which gives a free check on your list.

Q1(a) · Power set of {a, b, c}

Sheet 0B §3

Question. $\mathcal{P}(A)$ denotes the power set of A. Find $\mathcal{P}(\{a, b, c\})$.

Answer

$\mathcal{P}(\{a, b, c\}) = \{\varnothing, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$

The key prints the last element as {a. b, c}; that is a typo for {a, b, c}.

Break it down

1. **Power set** $\mathcal{P}(A)$: the set of all subsets of A. So every element of $\mathcal{P}(A)$ is itself a set.
2. List the subsets by size so none go missing. Size 0: only ∅ (∅ is a subset of every set, D1). Size 1: {a}, {b}, {c}. Size 2: {a,b}, {a,c}, {b,c}. Size 3: {a,b,c} (every set is a subset of itself).
3. Count check: $|\mathcal{P}(A)| = 2^{|A|}$ (cardinality of power set). Why: to build a subset you decide, for each element, "in" or "out". Three elements with two choices each give 2 × 2 × 2 = 8 subsets, and 1 + 3 + 3 + 1 = 8.

Trick

Sort subsets by size and match the count against 2^n. For n = 3 the size rows hold 1, 3, 3, 1 subsets, so a missing one shows at once. The two most often forgotten are ∅ and the whole set.

Q1(b) · $\mathcal{P}(\mathcal{P}(\mathcal{P}(\varnothing)))$, innermost first

Sheet 4B §1

Question. Find $\mathcal{P}(\mathcal{P}(\mathcal{P}(\varnothing)))$.

Answer

$\mathcal{P}(\mathcal{P}(\mathcal{P}(\varnothing))) = \{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}, \{\varnothing, \{\varnothing\}\}\}$

Break it down

Layer	Result	Size
$\mathcal{P}(\varnothing)$	$\{\varnothing\}$	$2^0 = 1$
$\mathcal{P}(\{\varnothing\})$	$\{\varnothing, \{\varnothing\}\}$	$2^1 = 2$
$\mathcal{P}(\{p, q\})$	$\{\varnothing, \{p\}, \{q\}, \{p, q\}\}$; with p = ∅ and q = {∅} this is $\{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}, \{\varnothing, \{\varnothing\}\}\}$	$2^2 = 4$

1. Work from the inside out and write each layer down before applying the next $\mathcal{P}$.
2. $\mathcal{P}(\varnothing)$: the only subset of ∅ is ∅ itself, so $\mathcal{P}(\varnothing) = \{\varnothing\}$. This set has one element. It is not empty.
3. $\mathcal{P}(\{\varnothing\})$: {∅} has one element, namely ∅. Its subsets: take nothing, giving ∅; take that one element, giving {∅}. So $\mathcal{P}(\{\varnothing\}) = \{\varnothing, \{\varnothing\}\}$.
4. $\mathcal{P}(\{\varnothing, \{\varnothing\}\})$: this set has two elements. Calling them p = ∅ and q = {∅} stops the braces from blurring together. The subsets of {p, q} are ∅, {p}, {q} and {p, q}. Substituting back: {p} = {∅}, {q} = {{∅}}, {p, q} = {∅, {∅}}.
5. The four answers really are different sets: ∅ has 0 elements, {∅} has the element ∅, {{∅}} has the element {∅}, and {∅, {∅}} has 2 elements.

Trap

$\mathcal{P}(\varnothing) = \{\varnothing\}$, not ∅. The empty set still has one subset, itself.

Trick

Rename the elements of a nested set with letters, take the power set with letters, then substitute back. Check the size 2^n at every layer.

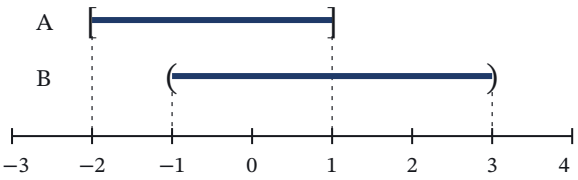
Q2(a)(b) · Union and intersection of intervals

Sheet 4 §1

Question. Universal set $\mathbb{R}$. $A = \{x \in \mathbb{R} : -2 \le x \le 1\} = [-2, 1]$ and $B = \{x \in \mathbb{R} : -1 < x < 3\} = (-1, 3)$. Write each answer in set-builder notation and, if appropriate, in interval notation: (a) $A \cup B$; (b) $A \cap B$; (c) $\bar{A}$; (d) $\bar{A} \cap \bar{B}$; (e) $A \setminus B$.

Answer

	Set-builder	Interval
a	$A \cup B = \{x \in \mathbb{R} : -2 \le x < 3\}$	$[-2, 3)$
b	$A \cap B = \{x \in \mathbb{R} : -1 < x \le 1\}$	$(-1, 1]$



A = [−2, 1] and B = (−1, 3). Square bracket: endpoint in. Round bracket: endpoint out.

Break it down

1. **Universal set** U: the set that every object in the discussion comes from; here $\mathbb{R}$, the real numbers. **Set-builder notation** $\{x \in U : P(x)\}$: all x in U for which the condition P(x) is true. **Interval notation**: [a, b] includes both ends, (a, b) excludes both, [a, b) includes a only, (a, b] includes b only. Square bracket means the endpoint is in; round bracket means it is out.
2. (a) Union: $x \in A \cup B$ iff $x \in A \vee x \in B$. On the number line, A covers −2 to 1 and B covers just after −1 to just before 3. They overlap between −1 and 1, so together they make one unbroken stretch from −2 to 3.
3. Endpoints of (a): −2 is in A (A uses $\leq$), so it is in the union: "[". 3 is in neither set (B uses strict <, and A stops at 1), so it is out: ")". Result [−2, 3).
4. (b) Intersection: x must satisfy both $-2 \leq x \leq 1$ and $-1 < x < 3$. Of the two lower bounds, the tighter one wins: $x > -1$ already forces $x \geq -2$. Of the two upper bounds, $x \leq 1$ already forces $x < 3$. So $-1 < x \leq 1$.
5. Endpoints of (b): −1 is in A but not in B, so it is not in both: "(". 1 is in A and in B: "]". Result (−1, 1].

Trick

Intersection of two intervals: take the larger left end and the smaller right end, each keeping its own bracket. Union of two overlapping intervals: the smaller left end and the larger right end.

Q2(c)(d)(e) · Complements and a difference

Sheet 4 §1

Question. Same A = [−2, 1] and B = (−1, 3) inside $\mathbb{R}$. Find (c) $\bar{A}$, (d) $\bar{A} \cap \bar{B}$, (e) $A \setminus B$.

Answer

	Set-builder	Interval
c	$\bar{A} = \{x \in \mathbb{R} : (x < -2) \vee (x > 1)\}$	not one interval
d	$\bar{A} \cap \bar{B} = \{x \in \mathbb{R} : (x < -2) \vee (x \geq 3)\}$	not one interval
e	$A \setminus B = \{x \in \mathbb{R} : -2 \leq x \leq -1\}$	$[-2, -1]$

Break it down

1. **Complement** $\bar{A}$: every element of the universal set that is not in A. Here $\bar{A} = \{x \in \mathbb{R} : x \notin A\}$. A complement only makes sense once U is fixed, which is why the question says "the universal set is $\mathbb{R}$".
2. (c) $x \notin A$ means $\sim(-2 \leq x \leq 1)$. The chain $-2 \leq x \leq 1$ is shorthand for $(-2 \leq x) \wedge (x \leq 1)$. De Morgan's law turns $\sim(p \wedge q)$ into $\sim p \vee \sim q$. Then $\sim(-2 \leq x)$ is $x < -2$ and $\sim(x \leq 1)$ is $x > 1$.
3. Why no interval for (c): $\bar{A}$ is two separate pieces, (−∞, −2) and (1, ∞). An interval is one unbroken stretch, so the best you can write is the union $(-\infty, -2) \cup (1, \infty)$. That is what "if appropriate" in the question is about.
4. (d), fast route: De Morgan's law for sets (Theorem 6.2.2) says $\bar{A} \cap \bar{B}$ is the complement of $A \cup B$. From (a), $A \cup B = [-2, 3)$. Its complement is $x < -2$ (−2 was in, so now it is out) or $x \geq 3$ (3 was out, so now it is in).
5. (d), slow route, as a check: $\bar{B}$ is $(x \leq -1) \vee (x \geq 3)$. A point in both $\bar{A}$ and $\bar{B}$ must fit one piece of each. $x < -2$ with $x \leq -1$ gives $x < -2$. $x > 1$ with $x \geq 3$ gives $x \geq 3$. The mixed pairs ($x < -2$ with $x \geq 3$, and $x > 1$ with $x \leq -1$) are impossible. Same answer.
6. (e) **Set difference** $A \setminus B$: the elements in A and not in B. So $-2 \leq x \leq 1$ and $(x \leq -1) \vee (x \geq 3)$. The option $x \geq 3$ cannot happen when $x \leq 1$, which leaves $-2 \leq x \leq -1$.
7. Endpoint −1 in (e): −1 is in A, and it is not in B because B needs $x > -1$ strictly. So −1 belongs to $A \setminus B$ and gets a square bracket.

Trap

Taking a complement flips every endpoint: $\leq$ becomes >, and < becomes $\geq$. In (e), −1 is included even though B "starts" there, because B's bracket at −1 is round.

Trick

Draw both intervals on one number line with their brackets, as in the figure above. Every part then becomes "which stretches are covered by which bars", and you settle each endpoint by reading one bracket at a time.

Q3(a) · When the sizes add up

Sheet 3 §7

Question. (AY22/23 Sem 2 midterm.) Prove true or false: there exist non-empty finite sets A and B such that $|A \cup B| = |A| + |B|$.

Answer

True. One example: $A = \{a\}$, $B = \{b\}$. Then $|A \cup B| = |\{a, b\}| = 2 = 1 + 1 = |A| + |B|$. Written out in numbered lines:

```
1. Let A = {a} and B = {b}, where a ≠ b.
2. A and B are non-empty and finite
   (each has exactly one element).
3. A ∪ B = {a, b}, so |A ∪ B| = 2
   (a ≠ b, so the two elements are distinct).
4. |A| + |B| = 1 + 1 = 2 = |A ∪ B|.
5. So such sets exist: the statement is true.
```

Break it down

1. Read the shape first. "There exist A and B such that ..." is an **existential statement**: it is true as soon as one pair of sets works.
2. **Constructive proof**: to prove an existential statement, produce a specific example (a **witness**) and check it meets every condition. So line 1 is "Let A = ..., B = ...". You are choosing, not assuming. There is no "Suppose", because nothing needs supposing.
3. How to pick: from D2, $|A \cup B| = |A| + |B| - |A \cap B|$. The sizes add exactly when $A \cap B = \varnothing$, so choose two **disjoint** sets (sets with no common element). Keep them tiny: one element each.
4. Check every condition the claim names. Non-empty: each set has one element. Finite: yes. The equation: lines 3 and

4.

5. "a ≠ b" matters: if a = b then $A \cup B = \{a\}$, and the sizes are 1 versus 2.

Q3(b) · When the sizes do not add up

Sheet 3 §7

Question. Prove true or false: there exist non-empty finite sets A and B such that $|A \cup B| \neq |A| + |B|$.

Answer

True. One example: $A = \{a, c\}$, $B = \{b, c\}$. Then $|A \cup B| = |\{a, b, c\}| = 3 \neq 4 = 2 + 2 = |A| + |B|$.

```
1. Let A = {a, c} and B = {b, c},
   where a, b, c are distinct.
2. A and B are non-empty and finite.
3. A ∪ B = {a, b, c}, so |A ∪ B| = 3.
4. |A| + |B| = 2 + 2 = 4 ≠ 3 = |A ∪ B|.
5. So such sets exist: the statement is true.
```

Break it down

1. Same shape as (a): an existential statement, so one example settles it, and the proof opens by choosing the sets.
2. To break the equation, make the sets overlap. The shared element c is counted twice in $|A| + |B|$ (once in each set) but only once in $|A \cup B|$, so the two sides differ by exactly $|A \cap B| = 1$.
3. An even smaller witness also works: A = B = {a} gives $|A \cup B| = 1 \neq 2$. The claim never asks for A ≠ B.

Trick

"There exist" questions: pick the smallest sets that work, then tick off, in writing, every condition in the claim (non-empty, finite, the equation). Disjoint sets make sizes add; one shared element breaks it.

Q3(c) · Aiken's proof: right verdict, broken proof

Sheet 0C §4

Question. Prove true or false: there exist finite sets A and B such that $A \times \mathcal{P}(B) = \mathcal{P}(A \times B)$. Aiken says false, with this proof.

1. Let $|A| = n$, $|B| = k$.

2. Then $|\mathcal{P}(B)| = 2^k$ (cardinality of power set).

3. $|A \times B| = nk$ (cardinality of Cartesian product), hence $|\mathcal{P}(A \times B)| = 2^{nk}$ (cardinality of power set).

4. Then $|A \times \mathcal{P}(B)| = n2^k$ (cardinality of Cartesian product).

5. Since $|A \times \mathcal{P}(B)| = n2^k \neq 2^{nk} = |\mathcal{P}(A \times B)|$, the statement is false.

Is his conclusion correct? Is his proof correct? If not, write a correct proof.

Answer

Aiken's conclusion is correct, but his proof is not. If (i) n = 1, or (ii) n = 2 and k = 1, then $n2^k = 2^{nk}$, so line 5 is incorrect.

```
1. Members of A × 𝒫(B) are ordered pairs.
2. Members of 𝒫(A × B) are sets. Note that
   𝒫(A × B) cannot be an empty set.
3. Ordered pairs and sets are different
   mathematical objects, and since 𝒫(A × B)
   cannot be an empty set, hence
   A × 𝒫(B) ≠ 𝒫(A × B).
```

The same proof with the key element named (this also answers the key's two "to think about" questions):

```
1. Let A and B be any finite sets.
2. ∅ ⊆ A × B
   (the empty set is a subset of every set).
3. So ∅ ∈ 𝒫(A × B)
   (by the definition of power set).
4. Every element of A × 𝒫(B) is an ordered
   pair (by the definition of Cartesian product).
5. ∅ is a set, not an ordered pair,
   so ∅ ∉ A × 𝒫(B).
6. ∅ is in 𝒫(A × B) but not in A × 𝒫(B), so
   A × 𝒫(B) ≠ 𝒫(A × B)
   (by the definition of set equality).
7. A and B were arbitrary, so no finite A, B
   make the sets equal: the statement is false.
```

Break it down

1. What must be shown. The claim is "∃A ∃B ($A \times \mathcal{P}(B) = \mathcal{P}(A \times B)$)". To show it is false, prove its negation. Negating ∃ gives ∀ with the inside negated: "for all finite A, B, $A \times \mathcal{P}(B) \neq \mathcal{P}(A \times B)$". So the argument must work for every pair of finite sets. One example proves nothing here.
2. Why line 1 of the second proof is "Let A and B be any finite sets": to prove a "for all" statement, take arbitrary objects, assume nothing about them except the hypothesis (here: finite), and prove the property for them. This is the **generic particular**. Whatever you show about it holds for every choice.
3. Checking Aiken, lines 1 to 4: all correct facts. $|\mathcal{P}(X)| = 2^{|X|}$ and $|X \times Y| = |X| \cdot |Y|$, applied to B, then to A × B, then to A and $\mathcal{P}$(B).
4. Line 5 claims $n2^k \neq 2^{nk}$ for all n, k. Test the smallest values. n = 1 gives $1 \cdot 2^k = 2^k = 2^{1 \cdot k}$, equal. n = 2, k = 1 gives 2 · 2 = 4 = 2^2, equal. For those sizes line 5 is false, so the proof says nothing about those sets. A proof of a "for all" statement must cover every case; one uncovered case breaks it, even when the conclusion happens to be true.
5. In those cases the sets are still different, but size cannot show it. Take A = {1}, B = ∅. Then $A \times \mathcal{P}(B) = \{1\} \times \{\emptyset\} = \{(1, \emptyset)\}$ and $\mathcal{P}(A \times B) = \mathcal{P}(\emptyset) = \{\emptyset\}$. Both have size 1, yet the pair (1, ∅) is not the set ∅. Equal sizes never prove equal sets; only different sizes prove different sets. So no arithmetic patch of line 5 can work for n = 1.
6. The official proof instead asks what kind of object each element is. Line 1: by the definition of Cartesian product, every element of $X \times Y$ is an ordered pair. Line 2: by the definition of power set, every element of $\mathcal{P}(S)$ is a subset of S, hence a set. Line 3: the course treats ordered pairs and sets as different kinds of object, so no object can be an element of both.
7. To think about, first question: why can $\mathcal{P}(A \times B)$ never be empty? Because ∅ is a subset of every set, so $\emptyset \subseteq A \times B$, so $\emptyset \in \mathcal{P}(A \times B)$. Every power set has at least one element.
8. To think about, second question: why does the proof need that? Two sets whose elements would be of different kinds are still equal if both have no elements, since ∅ = ∅. One side really can be empty: if A = ∅ then $A \times \mathcal{P}(B) = \emptyset$. The argument works only because $\mathcal{P}(A \times B)$ is guaranteed an

element (∅) that $A \times \mathcal{P}(B)$ cannot contain. The second proof names that element in lines 2 to 5.

Trap

In "spot the flaw" questions, plug the smallest values (0, 1, 2) into every claimed ≠ or <. Here n = 1 breaks line 5 at once.

Trick

To show "there exist ... such that ..." is false, prove "every choice fails" and open with "Let ... be any ...". For "is set P equal to set Q?", first ask what kind of element each holds (pairs, sets, numbers). Different kinds plus one guaranteed element, such as $\varnothing \in \mathcal{P}(anything)$, settles it.

Q4 · Two descriptions of the odd numbers

Sheet 4B §5

Question. Let $A = \{2n + 1 : n \in \mathbb{Z}\}$ and $B = \{2n - 5 : n \in \mathbb{Z}\}$. Is A = B? Prove your answer. What does this tell us about how odd numbers may be defined?

Answer

Yes, A = B. (Recall: $A = B \Leftrightarrow A \subseteq B \wedge B \subseteq A$.)

```
1. (⊆)
1.1 Let a ∈ A.
1.2 Use the definition of A to find an
    integer n such that a = 2n + 1.
1.3 Then a = 2n + 1 = 2(n + 3) − 5.
1.4 n + 3 ∈ ℤ
    (by closure of integers under +).
1.5 Therefore, a ∈ B
    (by the definition of B).
2. (⊇)
2.1 Let b ∈ B.
2.2 Use the definition of B to find an
    integer n such that b = 2n − 5.
2.3 Then b = 2n − 5 = 2(n − 3) + 1.
2.4 n − 3 ∈ ℤ
    (by closure of integers under -).
2.5 Therefore, b ∈ A
    (by the definition of A).
3. Therefore, A = B
   (by the definition of set equality).
```

Odd numbers: the course defines "n is odd iff n = 2k + 1 for some integer k". The proof shows other forms describe exactly the same set, so the odd numbers may be defined in many other ways, such as n = 2k − 5 for some integer k.

Break it down

1. **Replacement notation** $\{t(n) : n \in \mathbb{Z}\}$: the set of all values t(n) as n runs through the integers. So "$x \in A$" means "x = 2n + 1 for some integer n", and "$x \in B$" means "x = 2n − 5 for some integer n". This unpacking is what "by the definition of A" refers to.
2. Sanity check before proving. n = −1, 0, 1, 2 give A's elements −1, 1, 3, 5; n = 0, 1, 2, 3 give B's elements −5, −3, −1, 1. Both look like all odd integers, so guess "yes".
3. Plan: **set equality** means $A \subseteq B$ and $B \subseteq A$. So the proof has two halves, labelled (⊆) and (⊇), and a final line joining them.
4. Why line 1.1 is "Let a ∈ A": $A \subseteq B$ means $\forall x\,(x \in A \rightarrow x \in B)$. To prove a "for all x, if ... then ...", take an arbitrary a that satisfies the if-part and show it satisfies the then-part. You assume only $a \in A$; the goal is $a \in B$.
5. Line 1.2 unpacks the assumption: $a \in A$, so by the definition of A there is an integer n with a = 2n + 1.
6. Line 1.3 reshapes toward the goal. To be in B, a must look like 2(an integer) − 5. Scratch work: solve 2m − 5 = 2n + 1 to get m = n + 3. Check the algebra: 2(n + 3) − 5 = 2n + 6 − 5 = 2n + 1.
7. Line 1.4: B's definition needs the bracketed quantity to be an integer. n and 3 are integers, and **closure of integers under +** (the sum of two integers is an integer) gives $n + 3 \in \mathbb{Z}$.
8. Line 1.5: a = 2(n + 3) − 5 with $n + 3 \in \mathbb{Z}$ is exactly B's membership condition. Since a was arbitrary, this proves $A \subseteq B$.
9. Part 2 mirrors part 1: from $b \in B$ get b = 2n − 5, rewrite as 2(n − 3) + 1 (check: 2n − 6 + 1 = 2n − 5), and $n - 3 \in \mathbb{Z}$ by closure under −, so $b \in A$.
10. Line 3 joins the halves by the definition of set equality.
11. The same argument works for 2k − 1, 2k + 3, or any 2k + c with c odd: each describes exactly the odd integers, so each is a valid definition of "odd".

Trap: the short proof

The key also shows a four-line version: $A = \{2n + 1 : n \in \mathbb{Z}\} = \{2(n + 3) - 5 : n \in \mathbb{Z}\} = \{2k - 5 : k \in \mathbb{Z}\} = B$. The step from the second to the third form renames n + 3 as k. That needs two facts: every n + 3 is an integer (stated), and every integer k equals n + 3 for some integer n (true with n = k − 3, but never stated). The second fact is the (⊇) half in disguise, so the key prefers the two-direction proof.

Trick

Set equality by elements, per direction: let a ∈ first set; unpack with its definition; reshape into the second set's form (solve for the new variable in scratch work); justify that the new variable is an integer (closure); repack with the second set's definition. Then repeat in the other direction.

Q5 · Element method: $A \cap (B \setminus C) = (A \cap B) \setminus C$

Sheet 4B §5

Question. Using the definitions of set operations (the element method), prove that for all sets A, B, C, $A \cap (B \setminus C) = (A \cap B) \setminus C$.

Answer

```
1. A ∩ (B \ C) = {x : x ∈ A ∧ x ∈ (B \ C)}
   by the definition of ∩
2.    = {x : x ∈ A ∧ (x ∈ B ∧ x ∉ C)}
   by the definition of \
3.    = {x : (x ∈ A ∧ x ∈ B) ∧ x ∉ C}
   by the associativity of ∧
4.    = {x : (x ∈ A ∩ B) ∧ x ∉ C}
   by the definition of ∩
5.    = (A ∩ B) \ C
   by the definition of \
```

Break it down

1. **Element method**: prove a set fact by translating "x is in this set" into a logical statement about x (using the definitions of ∩, ∪, \ and complement), doing the work with logic laws, and translating back.

2. The definitions used: $x \in X \cap Y \Leftrightarrow x \in X \wedge x \in Y$ and $x \in X \setminus Y \Leftrightarrow x \in X \wedge x \notin Y$ (the procedural versions from Lecture 5).
3. Why the proof opens with the left side: the goal is "left = right", and every line is an equality, so start from one side and rewrite until you reach the other. No "Let" line is needed because nothing is assumed; every line holds for any sets A, B, C.
4. Line 1: definition of ∩ with X = A and Y = B \ C. Treat B \ C as a single set for now.
5. Line 2: definition of \ replaces "$x \in B \setminus C$" by "$(x \in B \wedge x \notin C)$". The brackets keep it as one unit.
6. Line 3, the only real step: the associative law for ∧, $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$, with p = "$x \in A$", q = "$x \in B$", r = "$x \notin C$". Two sets described by logically equivalent conditions have the same elements, so they are equal.
7. Line 4: definition of ∩ read backwards: $x \in A \wedge x \in B$ is $x \in A \cap B$.
8. Line 5: definition of \ read backwards, with X = A ∩ B and Y = C.
9. Why no separate (⊆) and (⊇) halves: every step is an "if and only if" (definitions and logical equivalences work both ways), so both inclusions come out at once. If any step were one-way, such as generalization, you would need the two-part proof of Q4.

Trap

Dropping the brackets (writing $x \in A \wedge x \in B \wedge x \notin C$) hides the very step the proof is about. The course brackets every ∧ and ∨ group, so keep them and name the law.

Trick

Definitions out, one logic law in the middle, definitions back in. Look at the target to see which law you need: the right side needs $x \in A \wedge x \in B$ grouped together, so the middle step must regroup (associativity).

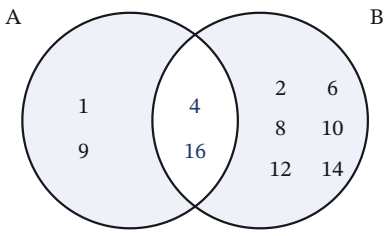
Q6(a) · Compute $A \oplus B$

Sheet 4 §2

Question. For sets A and B, define $A \oplus B = (A \setminus B) \cup (B \setminus A)$. Let $A = \{1,4,9,16\}$ and $B = \{2,4,6,8,10,12,14,16\}$. Find $A \oplus B$.

Answer

$A \setminus B = \{1, 9\}$; $B \setminus A = \{2, 6, 8, 10, 12, 14\}$; $A \oplus B = \{1, 2, 6, 8, 9, 10, 12, 14\}$.



Shaded: $A \oplus B$, the elements in exactly one of A and B. The overlap (4, 16) is left out.

Break it down

1. **Symmetric difference** $A \oplus B$: the elements in exactly one of A and B. The definition builds it from "in A but not B" together with "in B but not A".
2. $A \setminus B$: go through A. 1 is not in B (keep), 4 is in B (drop), 9 is not in B (keep), 16 is in B (drop). So {1, 9}.
3. $B \setminus A$: go through B and drop 4 and 16, the only ones also in A. So {2, 6, 8, 10, 12, 14}.
4. Take the union of the two pieces and write it in increasing order, which makes gaps and repeats easy to spot.
5. Check with part (b): $A \cup B$ minus $A \cap B = \{4, 16\}$ gives the same set.

Trick

$A \oplus B$ is the union with the overlap removed. Find the common elements first (here 4 and 16), then list everything else from both sets.

Q6(b) · $A \oplus B = (A \cup B) \setminus (A \cap B)$ by set identities

Sheet 4B §6

Question. Using set identities (Theorem 6.2.2), prove that for all sets A and B, $A \oplus B = (A \cup B) \setminus (A \cap B)$.

Answer

```
1.  A ⊕ B
2.  = (A \ B) ∪ (B \ A)
      by the definition of ⊕
3.  = (A ∩ B̄) ∪ (B ∩ Ā)
      by the Set Difference Law
4.  = ((A ∩ B̄) ∪ B) ∩ ((A ∩ B̄) ∪ Ā)
      by the Distributive Law
5.  = (B ∪ (A ∩ B̄)) ∩ (Ā ∪ (A ∩ B̄))
      by the Commutative Law
6.  = ((B ∪ A) ∩ (B ∪ B̄)) ∩ ((Ā ∪ A) ∩ (Ā ∪ B̄))
      by the Distributive Law
7.  = ((B ∪ A) ∩ (B ∪ B̄)) ∩ ((A ∪ Ā) ∩ (Ā ∪ B̄))
      by the Commutative Law
8.  = ((B ∪ A) ∩ U) ∩ (U ∩ (Ā ∪ B̄))
      by the Complement Law
9.  = ((A ∪ B) ∩ U) ∩ ((Ā ∪ B̄) ∩ U)
      by the Commutative Law
10. = (A ∪ B) ∩ (Ā ∪ B̄)
      by the Identity Law
11. = (A ∪ B) ∩ (overline(Ā ∩ B))
      by De Morgan's Law
12. = (A ∪ B) \ (A ∩ B)
      by the Set Difference Law
```

Break it down

1. **Set identities**: the named laws of Theorem 6.2.2 (commutative, associative, distributive, identity, complement, double complement, idempotent, universal bound, De Morgan's, absorption, complements of U and ∅, set difference). "Using set identities" means every line is one of these laws or a definition, named, with no element chasing.
2. Plan backwards from the target. The only law that mentions \ is the Set Difference Law, $X \setminus Y = X \cap \bar{Y}$. So the target is $(A \cup B) \cap$ (complement of $A \cap B$), which by De Morgan's law is $(A \cup B) \cap (\bar{A} \cup \bar{B})$. The job is to turn $(A \cap \bar{B}) \cup (B \cap \bar{A})$ into $(A \cup B) \cap (\bar{A} \cup \bar{B})$. Lines 10 to 12 are this plan run forwards.
3. Line 2: the definition of ⊕. Line 3: the Set Difference Law on both pieces, $A \setminus B = A \cap \bar{B}$ and $B \setminus A = B \cap \bar{A}$. Using the same law in two places on one line is fine; the key does it too.
4. Line 4: the Distributive Law in the form $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$, with X = $(A \cap \bar{B})$, Y = B, Z = $\bar{A}$. The whole first

bracket plays the role of X. This is the "OR of ANDs into AND of ORs" move the plan needs.

5. Line 5: the Commutative Law moves $(A \cap \bar{B})$ to the right of each ∪. The distributive law is stated with the single set on the left, $X \cup (Y \cap Z)$, and the course wants each law applied in exactly its stated form, so you commute first.
6. Line 6: the Distributive Law inside each big bracket: $B \cup (A \cap \bar{B}) = (B \cup A) \cap (B \cup \bar{B})$ and $\bar{A} \cup (A \cap \bar{B}) = (\bar{A} \cup A) \cap (\bar{A} \cup \bar{B})$.
7. Line 7: the Commutative Law turns $\bar{A} \cup A$ into $A \cup \bar{A}$, the exact form of the Complement Law.
8. Line 8: the Complement Law twice, $B \cup \bar{B} = U$ and $A \cup \bar{A} = U$, where U is the universal set.
9. Line 9: the Commutative Law twice: $B \cup A$ becomes $A \cup B$ (to match the target) and $U \cap (\bar{A} \cup \bar{B})$ becomes $(\bar{A} \cup \bar{B}) \cap U$ (to match the Identity Law $X \cap U = X$).
10. Line 10: the Identity Law twice drops both "∩ U".
11. Line 11: De Morgan's Law read right to left: $\bar{A} \cup \bar{B}$ is the complement of $A \cap B$.
12. Line 12: the Set Difference Law read right to left, $X \cap \bar{Y} = X \setminus Y$, with X = $A \cup B$ and Y = $A \cap B$.

Logic mirror

Put p = "$x \in A$", q = "$x \in B$". The identity says $(p \wedge \sim q) \vee (q \wedge \sim p) \equiv (p \vee q) \wedge \sim(p \wedge q)$: both describe "exactly one of p, q", exclusive or. $\oplus$ is the set version of exclusive or, and every line above matches a line of the logic proof with ∪ as ∨, ∩ as ∧, complement as ~, and U as true.

Trick

For identity proofs: (1) remove every \ with the Set Difference Law; (2) rewrite the target in ∪, ∩, complement form and aim at it; (3) commute before each law so it matches the textbook form exactly; (4) one kind of law per line, even when it is used in two places.

Q7 · $A \subseteq B$ iff $A \cup B = B$

Sheet 3 §7

Question. Let A and B be sets. Show that $A \subseteq B$ if and only if $A \cup B = B$.

Answer

```
1. (⇒)
1.1 Suppose A ⊆ B.
 1.2 (To show A ∪ B ⊆ B)
  1.2.1 Let z ∈ A ∪ B.
  1.2.2 Then z ∈ A or z ∈ B
        (by the definition of ∪).
  1.2.3 Case 1: Suppose z ∈ A, then z ∈ B
        as A ⊆ B (from line 1.1).
  1.2.4 Case 2: Suppose z ∈ B, then z ∈ B.
  1.2.5 In either case, we have z ∈ B.
 1.3 (To show A ∪ B ⊇ B)
  1.3.1 Let z ∈ B.
  1.3.2 Then z ∈ A or z ∈ B
        (by generalization).
  1.3.3 So z ∈ A ∪ B
        (by the definition of ∪).
 1.4 Therefore, A ∪ B = B
     (by the definition of set equality).
2. (⇐)
2.1 Suppose A ∪ B = B.
2.2 Let z ∈ A.
  2.2.1 Then z ∈ A or z ∈ B
        (by generalization).
  2.2.2 So z ∈ A ∪ B
        (by the definition of ∪).
  2.2.3 So z ∈ B since A ∪ B = B
        (from line 2.1).
2.3 Therefore, A ⊆ B.
3. Therefore, A ⊆ B if and only if A ∪ B = B
   (from 1 and 2).
```

For 1.3 you may instead quote Theorem 6.2.1, Inclusion in Union: $B \subseteq A \cup B$.

Break it down

1. Shape: "P if and only if Q" means (P → Q) ∧ (Q → P). So there are two parts: (⇒) assume $A \subseteq B$ and prove $A \cup B = B$; (⇐) assume $A \cup B = B$ and prove $A \subseteq B$. Line 3 joins them.
2. Why 1.1 is "Suppose A ⊆ B": to prove P → Q directly, assume P and derive Q. Here P is $A \subseteq B$. You now hold that fact and should expect to use it; it gets used in 1.2.3.
3. The goal $A \cup B = B$ is a set equality, so by the definition of set equality it splits into $A \cup B \subseteq B$ (block 1.2) and $B \subseteq A \cup B$ (block 1.3). Each block gets its goal written in brackets.
4. 1.2.1 "Let z ∈ A ∪ B": to prove a subset statement, take an arbitrary element of the left set (the same move as Q4).
5. 1.2.2: the definition of ∪ unpacks $z \in A \cup B$ into "$z \in A$ or $z \in B$".
6. 1.2.3 to 1.2.5: an "or" fact is used by **proof by division into cases**: show the goal in each case separately. Case 1: $z \in A$, and $A \subseteq B$ (line 1.1) says every element of A is in B, so $z \in B$. Case 2: $z \in B$ already. Both cases end at $z \in B$, so 1.2.5 concludes $z \in B$, and $A \cup B \subseteq B$ is proved.
7. 1.3: let $z \in B$. **Generalization** is the rule of inference "from q, conclude p ∨ q": if $z \in B$ is true, then "$z \in A$ or $z \in B$" is true. The definition of ∪ then packs it into $z \in A \cup B$.
8. 1.4: both inclusions hold, so $A \cup B = B$ by the definition of set equality.
9. Part 2 starts fresh. 2.1 assumes the other side, $A \cup B = B$. The goal $A \subseteq B$ is a subset statement, so 2.2 takes an arbitrary $z \in A$.
10. 2.2.1: generalization gives "$z \in A$ or $z \in B$". 2.2.2: the definition of ∪ gives $z \in A \cup B$. 2.2.3: $A \cup B$ and B are the

same set (line 2.1), so $z \in B$. 2.3: z was arbitrary, so $A \subseteq B$.

11. The key warns against jumping from $z \in A$ straight to $z \in A \cup B$. That jump uses two rules (generalization, then the definition of ∪), and the course wants each on its own line.

Trap

In (⇐) you may not use $A \subseteq B$: it is the goal. Each direction assumes only its own starting side. Using the goal as an assumption makes the proof circular.

Trick

The numbering mirrors the goals: 1 and 2 are the two directions, 1.2 and 1.3 are the two inclusions inside direction 1, and 1.2.x are the steps of one inclusion. When lost, write each block's goal in brackets first, as the key does. The same template proves $A \subseteq B$ iff $A \cap B = A$.

Q8 · Hogwarts houses as a partition

Sheet 4B §4

Question. Hogwarts is divided into 4 houses: Gryffindor, Hufflepuff, Ravenclaw and Slytherin. HSWW is the set of Hogwarts students, and G, H, R, S are the sets of students in the four houses. What are the necessary conditions for {G, H, R, S} to be a partition of HSWW? Explain in English, then write logical statements.

Answer

Every student is in exactly one of the four houses, and every house has at least one student.

```
G ∩ H = G ∩ R = G ∩ S
      = H ∩ R = H ∩ S = R ∩ S = ∅
  (the houses are mutually disjoint sets)
G ∪ H ∪ R ∪ S = HSWW
  (every Hogwarts student is in one of the houses)
G ≠ ∅ ∧ H ≠ ∅ ∧ R ≠ ∅ ∧ S ≠ ∅
  (every house has at least one student)
```

Break it down

1. **Partition** of a set X: a collection of non-empty subsets of X that are mutually disjoint and whose union is X. Picture cutting a cake: no crumb left out, no crumb in two slices, no empty slice.
2. **Disjoint**: two sets are disjoint iff their intersection is ∅. **Mutually disjoint** (or pairwise disjoint): every two different sets in the collection are disjoint. Four houses give 6 pairs (GH, GR, GS, HR, HS, RS), and all 6 must be written.
3. The union line has two halves. "⊇": every student is in some house. "⊆": the houses hold only Hogwarts students (true here by how G, H, R, S are defined).
4. "Exactly one house" splits into "at least one" (the union line) and "at most one" (the disjointness line).
5. The definition of partition requires each piece to be non-empty, hence the third line: no house is empty.
6. If asked for quantified statements instead: $\forall x \in HSWW\,(x \in G \vee x \in H \vee x \in R \vee x \in S)$ for coverage; $\sim\exists x\,(x \in G \wedge x \in H)$ and likewise for the other five pairs; $\exists x\,(x \in G)$ and likewise for H, R, S.

Trap

$G \cap H \cap R \cap S = \varnothing$ is not enough. It only says no student is in all four houses at once; a student in both G and H would slip through. Disjointness is checked pair by pair.

Trick

A partition is three checks: covers everything (union), no overlaps (every pair), no empty piece. Write one line for each.

Q9 · Indexed union and intersection of integer runs

Question. For sets $A_m, A_{m+1}, \ldots, A_n$: $\bigcup_{i=m}^{n} A_i = A_m \cup A_{m+1} \cup \ldots \cup A_n$ and $\bigcap_{i=m}^{n} A_i = A_m \cap A_{m+1} \cap \ldots \cap A_n$. Let $A_i = \{x \in \mathbb{Z} : i \leq x \leq 2i\}$ for all integers i. Write A_{-2}, $\bigcup_{i=3}^{5} A_i$ and $\bigcap_{i=3}^{5} A_i$ in set-roster, set-builder and interval notation.

Answer

```
A₋₂ = ∅

A₃ ∪ A₄ ∪ A₅ = {3,4,5,6,7,8,9,10}
  = {x ∈ ℤ : 3 ≤ x ≤ 10} = [3,10]

A₃ ∩ A₄ ∩ A₅ = {5,6}
  = {x ∈ ℤ : 5 ≤ x ≤ 6} = [5,6]
```

Break it down

1. **Indexed union** $\bigcup_{i=3}^{5} A_i$: "the union of A_i for i from 3 to 5", which unrolls to $A_3 \cup A_4 \cup A_5$. The letter i is a counter; the numbers below and above $\bigcup$ say where it starts and stops. The indexed intersection $\bigcap$ works the same way with ∩.
2. A_i is a **family of sets**: one set for each integer i, obtained by substituting i into the rule. Here A_i is the integers from i to 2i.
3. A_{-2}: substitute i = −2 to get $\{x \in \mathbb{Z} : -2 \leq x \leq -4\}$. x would have to be at least −2 and at most −4, but −4 < −2, so no integer qualifies. $A_{-2} = \varnothing$, in roster form {} or ∅.
4. Unroll the three sets:

i	A_i
3	{3, 4, 5, 6}
4	{4, 5, 6, 7, 8}
5	{5, 6, 7, 8, 9, 10}

5. Union: everything that appears in at least one row. The smallest is 3 (from A_3), the largest is 10 (from A_5), and there are no gaps because each row is a run of consecutive integers that overlaps the next.
6. Intersection: what appears in every row. It must be at least the largest start (5) and at most the smallest end (6), giving {5, 6}.
7. Interval notation: the key writes [3,10] and [5,6], read inside $\mathbb{Z}$, meaning the integers from 3 to 10 and from 5 to 6. The lecture defines [a, b] as a set of reals, so write the set-builder form $\{x \in \mathbb{Z} : \ldots\}$ next to it to make the integer meaning clear.

Trap

A negative index flips which bound is bigger: for i < 0, 2i < i, so every such A_i is empty. Always substitute first, then read the inequality, then list.

Trick

For a family of runs from start to end: the union of overlapping runs goes from the smallest start to the largest end; the intersection goes from the largest start to the smallest end, and is empty if that start exceeds that end.

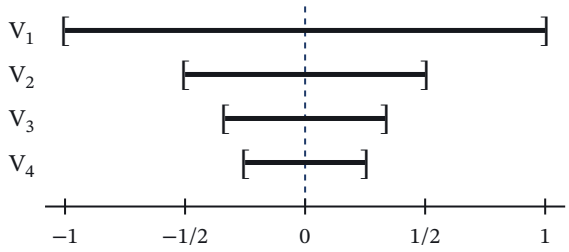
Q10 · Shrinking intervals V_i

Question. $V_i = \{x \in \mathbb{R} : -1/i \le x \le 1/i\} = [-1/i, 1/i]$ for all positive integers i. (a) $\bigcup_{i=1}^{4} V_i$? (b) $\bigcap_{i=1}^{4} V_i$? (c) $\bigcup_{i=1}^{n} V_i$, n a positive integer? (d) $\bigcap_{i=1}^{n} V_i$? (e) Are $V_1, V_2, V_3, \ldots$ mutually disjoint?

Answer

a	$\bigcup_{i=1}^{4} V_i = [-1, 1]$	c	$\bigcup_{i=1}^{n} V_i = [-1, 1]$
b	$\bigcap_{i=1}^{4} V_i = [-1/4, 1/4]$	d	$\bigcap_{i=1}^{n} V_i = [-1/n, 1/n]$

(e) No. $V_1, V_2, V_3, \ldots$ are not mutually disjoint: they have the common element 0. $V_1 = [-1, 1]$, $V_2 = [-1/2, 1/2]$, $V_3 = [-1/3, 1/3]$, $V_4 = [-1/4, 1/4]$, ..., $V_n = [-1/n, 1/n]$.



$V_1 \supseteq V_2 \supseteq V_3 \supseteq V_4$. The dashed line at 0 passes through every one of them.

Break it down

1. Write out the first few: $V_1 = [-1, 1]$, $V_2 = [-1/2, 1/2]$, $V_3 = [-1/3, 1/3]$, $V_4 = [-1/4, 1/4]$. As i grows, 1/i shrinks, so each interval is narrower than the one before.
2. **Nested** sets: each one contains the next. If i < j then 1/j < 1/i, so −1/i < −1/j, and $[-1/j, 1/j]$ lies inside $[-1/i, 1/i]$. So $V_1 \supseteq V_2 \supseteq V_3 \supseteq \ldots$
3. For nested sets the union is the biggest member and the intersection is the smallest. This is Q7 in action: $X \subseteq Y$ gives $X \cup Y = Y$, and in the same way $X \cap Y = X$.
4. (a) The union of V_1 to V_4 is $V_1 = [-1, 1]$. (b) The intersection is $V_4 = [-1/4, 1/4]$.
5. (c) and (d) are the same idea for any n: the union is always $V_1 = [-1, 1]$ (even when n = 1), and the intersection is $V_n = [-1/n, 1/n]$.
6. (e) Mutually disjoint would need $V_i \cap V_j = \varnothing$ for every pair i ≠ j. But $-1/i \le 0 \le 1/i$ for every i, so 0 is in every V_i; for example $V_1 \cap V_2$ contains 0. One overlapping pair is enough to answer no.

Trick

Check how the endpoints move as i grows. If the sets are nested, the union is the largest member and the intersection is the smallest. "Mutually disjoint?" is answered no by one element shared by two of the sets. (A common follow-up: the intersection over all positive integers i is {0}, since 0 is in every V_i and any x ≠ 0 falls outside V_i once 1/i < |x|.)

T4 · Tutorial 4 walkthrough

Relations & equivalence relations · official key (D1, D2 worked out here)

This tutorial trains one habit: treat a relation as nothing more than a set of ordered pairs, then flip it (inverse), chain it (composition) and test it (reflexive, symmetric, transitive). Three moves crack almost every question. Read $S \circ R$ as "walk one R arrow, then one S arrow". Kill a property with one concrete counterexample, but prove it with "Let x …". Prove two relations equal by showing each is a subset of the other.

D1 · Cartesian products

Sheet 0B §3

Question. Let $A = \{0, 1\}$, $B = \{a, b, c\}$ and $C = \{01, 10\}$.
Determine (a) $B \times C$, (b) $A \times B \times C$, (c) $\varnothing \times A$, (d) $\mathcal{P}(\{\varnothing\}) \times A$.

Answer (worked out here, no official key)

```
(a) B × C = {(a,01), (a,10), (b,01),
             (b,10), (c,01), (c,10)}
    3 × 2 = 6 pairs

(b) A × B × C =
      {(0,a,01), (0,a,10), (0,b,01),
       (0,b,10), (0,c,01), (0,c,10),
       (1,a,01), (1,a,10), (1,b,01),
       (1,b,10), (1,c,01), (1,c,10)}
    2 × 3 × 2 = 12 triples

(c) ∅ × A = ∅

(d) 𝒫({∅}) = {∅, {∅}}, so
    𝒫({∅}) × A =
      {(∅,0), (∅,1), ({∅},0), ({∅},1)}
```

Break it down

1. An **ordered pair** (x, y) is two objects written in a fixed order. Order matters: $(0, a) \neq (a, 0)$. Two pairs are equal only when the first entries match and the second entries match.
2. The **Cartesian product** $X \times Y$ is the set of all ordered pairs whose first entry comes from X and whose second entry comes from Y: $X \times Y = \{(x, y) : x \in X \wedge y \in Y\}$. It has $|X| \cdot |Y|$ elements, where $|X|$ (the **cardinality**) is the number of elements of X.
3. (a) To list $B \times C$ without missing anything, fix the first entry (a), run through every second entry (01, 10), then move to b, then c. Six pairs, matching $3 \cdot 2 = 6$.
4. 01 and 10 are just two labels, like bit strings. Copy them exactly as written; do not turn 01 into 1.
5. (b) $A \times B \times C$ is the set of **ordered triples** (x, y, z) with $x \in A$, $y \in B$, $z \in C$. Count first: $2 \cdot 3 \cdot 2 = 12$. List like a car odometer: the last entry changes fastest, the first entry slowest.
6. (c) Every pair in $\varnothing \times A$ would need a first entry taken from $\varnothing$. The empty set has no elements, so no pair can be built, and the product is empty. The count agrees: $0 \cdot 2 = 0$.
7. (d) Work from the inside out. The **power set** $\mathcal{P}(X)$ is the set of all subsets of X. The set $\{\varnothing\}$ has exactly one element (the empty set), so it has $2^1 = 2$ subsets: $\varnothing$ (the empty set is a subset of every set) and $\{\varnothing\}$ itself. Now pair each of those two with 0 and with 1: $2 \cdot 2 = 4$ pairs.

Trap

$(\varnothing, 0)$ is a perfectly good pair whose first entry happens to be the empty set. It is not "empty". Only a product with an empty *factor* is empty. Also, $A \times B \times C$ is a set of triples $(0, a, 01)$, not of nested pairs $((0, a), 01)$: Lecture 5 treats $(A \times B) \times C$ as a different set.

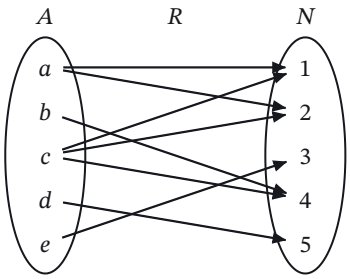
Trick

Before listing any product, write down its size $|X| \cdot |Y| \cdot \ldots$. Then list in odometer order and check that you hit that number exactly.

D2(a) · Inverse from an arrow diagram

Sheet 5 §1

Question. Let $A = \{a, b, c, d, e\}$ and $N = \{1, 2, 3, 4, 5\}$. A relation R from A to N is given by the arrow diagram below. Determine R^{-1}.



The given relation R: eight arrows.

Answer (worked out here, no official key)

```
R    = {(a,1), (a,2), (b,4), (c,1), (c,2),
        (c,4), (d,5), (e,3)}
R⁻¹ = {(1,a), (2,a), (4,b), (1,c), (2,c),
        (4,c), (5,d), (3,e)}
    = {(1,a), (1,c), (2,a), (2,c), (3,e),
       (4,b), (4,c), (5,d)}      (same set, sorted)
```

Break it down

1. A **relation from A to N** is any subset of $A \times N$: a chosen collection of pairs (x, n) with $x \in A$ and $n \in N$. Writing $x \, R \, n$ is shorthand for $(x, n) \in R$, read "x is R-related to n".
2. An **arrow diagram** draws each pair $(x, n) \in R$ as an arrow from x to n. So reading the diagram means writing one pair per arrow. Count the arrows (8) and check you wrote 8 pairs. Note that c sends out three arrows (to 1, 2 and 4), so c appears in three pairs.
3. The **inverse** R^{-1} is R with every pair flipped: $R^{-1} = \{(n, x) : (x, n) \in R\}$. Equivalently, $n \, R^{-1} \, x \Leftrightarrow x \, R \, n$. It is a relation from N to A (the two sets swap roles). In the picture it is the same set of lines with every arrowhead reversed.
4. Flip all 8 pairs. The order in which you list elements of a set does not matter; sorting just makes checking easier.

Trick

An inverse needs no thinking: swap the two coordinates pair by pair. The number of pairs never changes.

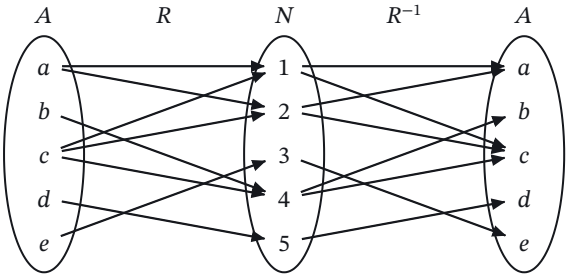
D2(b) · Composing R with its inverse

Sheet 5 §3

Question. Same R as in D2(a). Determine $R^{-1} \circ R$.

Answer (worked out here, no official key)

```
R⁻¹∘R = {(a,a), (a,c), (b,b), (b,c), (c,a),
         (c,b), (c,c), (d,d), (e,e)}
```



$R^{-1} \circ R$: walk one R arrow into N, then one R^{-1} arrow back into A.

Break it down

1. **Composition.** If $R \subseteq X \times Y$ and $S \subseteq Y \times Z$, then $S \circ R = \{(x, z) \in X \times Z : \exists y \in Y\,(x\,R\,y \wedge y\,S\,z)\}$. In words: $(x, z) \in S \circ R$ when you can get from x to z by one R arrow followed by one S arrow. The middle stop y is called a **witness**. The course reads $S \circ R$ right to left: R first, then S, just like $f(g(x))$ applies g first.
2. Here the first relation is R (from A to N) and the second is R^{-1} (from N back to A). So $R^{-1} \circ R$ goes $A \to N \to A$: it is a relation on A.
3. Unwrap the definition: $x\,(R^{-1} \circ R)\,z \Leftrightarrow \exists n\,(x\,R\,n \wedge n\,R^{-1}\,z) \Leftrightarrow \exists n\,(x\,R\,n \wedge z\,R\,n)$. So x and z are related exactly when their arrows hit a **common number**.
4. Walk out from each letter and come back along every arrow that ends at the same number:

```
a → 1,2   → a, c      (a,a), (a,c)
b → 4     → b, c      (b,b), (b,c)
c → 1,2,4 → a, b, c   (c,a), (c,b), (c,c)
d → 5     → d         (d,d)
e → 3     → e         (e,e)
```

5. Sanity check: each letter has at least one arrow, so walking out and straight back along the same arrow always works. That is why all five loops (x, x) appear.

Trap

$R^{-1} \circ R$ is **not** the identity relation in general. With a function, undoing brings you back where you started. With a relation, you can walk back along a *different* arrow and land somewhere new ($a \to 1 \to c$). This result is not even transitive: (a, c) and (c, b) are in it, but (a, b) is not.

Trick

Read $X^{-1} \circ X$ as "share a target" and $X \circ X^{-1}$ as "share a source". Then list by grouping, not by grinding through every pair of pairs.

Q1 · Building R and R^{-1} from a rule

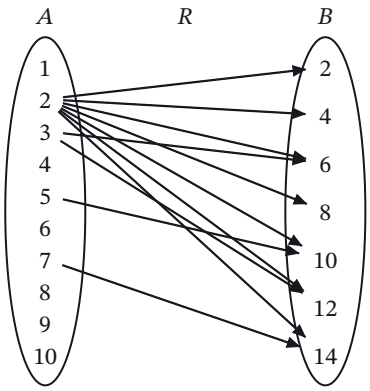
Sheet 5 §1

Question. $A = \{1, 2, \ldots, 10\}$, $B = \{2, 4, 6, 8, 10, 12, 14\}$. For $x \in A$ and $y \in B$: $x\,R\,y \Leftrightarrow x$ is prime and $x \mid y$. Write R and R^{-1} in roster notation, without ellipses.

Answer

```
R = {(2,2), (2,4), (2,6), (2,8), (2,10),
     (2,12), (2,14), (3,6), (3,12), (5,10),
     (7,14)}

R⁻¹ = {(2,2), (4,2), (6,2), (8,2), (10,2),
       (12,2), (14,2), (6,3), (12,3), (10,5),
       (14,7)}
```



Only the primes 2, 3, 5, 7 send arrows. R^{-1} is the same picture with arrowheads reversed.

Break it down

1. **Roster notation** means listing every element between braces. "No ellipses" means you may not write "…": every pair must appear.
2. A **prime** is an integer $n > 1$ whose only positive divisors are 1 and n. In A the primes are $2, 3, 5, 7$. 1 is not prime (primes must be greater than 1), and $4, 6, 8, 9, 10$ each have an extra divisor. So only four values of x can produce pairs.
3. $x \mid y$ ("x divides y") means $y = xk$ for some integer k. Go prime by prime and pick its multiples in B:

```
x = 2: every element of B is even
       2, 4, 6, 8, 10, 12, 14      7 pairs
x = 3: multiples of 3 in B: 6, 12  2 pairs
x = 5: 10                          1 pair
x = 7: 14                          1 pair
total                              11 pairs
```

4. $(2, 2)$ belongs: $2 = 2 \cdot 1$, so $2 \mid 2$.
5. R^{-1}: flip each of the 11 pairs. R^{-1} is a relation from B to A, and $y\,R^{-1}\,x$ reads "x is a prime divisor of y".

Trap

Common slips: including 1 (not prime), forgetting $(2, 2)$, or adding $(3, 3)$ or $(3, 9)$ (3 and 9 are not in B). Always check the second entry lies in the target set.

Trick

Build a relation from a rule row by row: fix x, find every y that works, write those pairs, move to the next x. Count the pairs, then flip them all for the inverse.

Q2 · Symmetric $\Leftrightarrow R = R^{-1}$

Sheet 5 §1

Question. Let R be a relation on a set A. Show the following are logically equivalent by proving (i) $\Rightarrow$ (ii), (ii) $\Rightarrow$ (iii) and (iii) $\Rightarrow$ (i). (i) R is symmetric, i.e. $\forall x, y \in A\ (x\,R\,y \Rightarrow y\,R\,x)$. (ii) $\forall x, y \in A\ (x\,R\,y \Leftrightarrow y\,R\,x)$. (iii) $R = R^{-1}$.

Answer

```
1. ((i) ⇒ (ii))
1.1 Suppose R is symmetric.
1.2 Let x, y ∈ A.
1.3 (⇒) If x R y, then y R x
        (by the symmetry of R).
1.4 (⇐) If y R x, then x R y
        (by the symmetry of R).
1.5 From 1.3 and 1.4, x R y ⇔ y R x.
2. ((ii) ⇒ (iii))
2.1 Suppose ∀x, y ∈ A (x R y ⇔ y R x).
2.2 Then for all x, y ∈ A,
    (x,y) ∈ R ⇔ x R y
                (by definition of x R y)
              ⇔ y R x     (by 2.1)
              ⇔ x R⁻¹ y
                (by definition of R⁻¹)
              ⇔ (x,y) ∈ R⁻¹
                (by definition of x R⁻¹ y)
2.3 Hence R = R⁻¹.
3. ((iii) ⇒ (i))
3.1 Suppose R = R⁻¹.
3.1.1 Let x, y ∈ A such that x R y.
3.1.2 Then x R⁻¹ y, as R = R⁻¹.
3.1.3 ∴ y R x  (by definition of R⁻¹)
3.2 Hence R is symmetric.
4. Therefore (i), (ii) and (iii) are
   logically equivalent.
```

Break it down

1. A **relation on A** is a relation from A to itself, i.e. a subset of $A \times A$.
2. **Symmetric**: whenever $x\,R\,y$, also $y\,R\,x$. In a picture, every arrow has a return arrow.
3. Statements are **logically equivalent** when each one implies every other, so they are true in exactly the same situations. Proving all six implications would be wasteful. A **cycle** (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (i) is enough, because implications chain: from (ii) $\Rightarrow$ (iii) and (iii) $\Rightarrow$ (i) you get (ii) $\Rightarrow$ (i) for free, and likewise for every other pair.
4. Each part is its own small proof: assume only the left-hand statement of that part, and aim for the right-hand one.
5. **(i) $\Rightarrow$ (ii).** A "$\Leftrightarrow$" is two implications, so prove both directions. ($\Rightarrow$) is literally the symmetry assumption. ($\Leftarrow$) is the one clever moment: symmetry says "for *all* x, y", so it also holds with the letters swapped. Plug y in for x and x in for y and you get $y\,R\,x \Rightarrow x\,R\,y$.
6. **(ii) $\Rightarrow$ (iii).** $R = R^{-1}$ is an equality of *sets*. Two sets are equal when they have exactly the same elements. Both R and R^{-1} live inside $A \times A$, so their elements are pairs (x, y). A chain of "$\Leftrightarrow$" from "$(x, y) \in R$" to "$(x, y) \in R^{-1}$" proves both subset directions at once. Each link is either notation, the assumption 2.1, or the definition of inverse:

   ```
   x R⁻¹ y   ⇔   y R x        definition of inverse
   (R⁻¹ holds (x,y) exactly when R holds (y,x))
   ```

7. **(iii) $\Rightarrow$ (i).** "Symmetric" is a "for all … if … then" statement, so open with the standard line: let x, y be arbitrary with $x\,R\,y$, and aim for $y\,R\,x$. Since R and R^{-1} are the same set, $(x, y) \in R$ gives $(x, y) \in R^{-1}$, and unwrapping the definition of inverse turns that into $(y, x) \in R$.

Trap

Get the direction of the inverse right: $x\,R^{-1}\,y$ means $y\,R\,x$, not $x\,R\,y$. And in (ii) $\Rightarrow$ (iii), "the arrows go both ways so the sets match" earns nothing. The grader wants the element chain with one reason per line.

Trick

"Show these are equivalent" = prove a cycle of implications. "Show two sets are equal" = a chain of $\Leftrightarrow$ on an arbitrary element, or $\subseteq$ in both directions. Memorise the result: **R is symmetric $\Leftrightarrow R = R^{-1}$**. Q5 and Q9 lean on it.

Q3 · Property check on five relations

Sheet 5 §4

Question. Decide whether each relation is (i) reflexive, (ii) symmetric, (iii) transitive, (iv) an equivalence relation, giving a counterexample for each property that fails. (a) $Q = \{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\}$ on $A = \{1, 2, 3\}$. (b) E on $\mathbb{Q}$: $x\,E\,y \Leftrightarrow x = y$. (c) R on $\mathbb{Q}$: $x\,R\,y \Leftrightarrow xy \geq 0$. (d) S on $\mathbb{Q}$: $x\,S\,y \Leftrightarrow xy > 0$. (e) T on $\mathbb{Z}$: $x\,T\,y \Leftrightarrow -2 \leq x - y \leq 2$.

Answer

	reflexive	symmetric	transitive
Q	yes	no	yes
E	yes	yes	yes
R	yes	yes	no
S	no	yes	yes
T	yes	yes	no

```
Q not symmetric:  1 Q 2, but (2,1) ∉ Q
R not transitive: 1 R 0 and 0 R -1,
                  but (1,-1) ∉ R
S not reflexive:  (0,0) ∉ S
T not transitive: -2 T 0 and 0 T 2,
                  but (-2,2) ∉ T
Equivalence relation: only E.
```

Break it down

1. The three properties of a relation R on a set A, each a "for all" statement:

   ```
   reflexive    ∀x ∈ A (x R x)
                loop at every vertex
   symmetric    ∀x, y ∈ A (x R y → y R x)
                every arrow has a return arrow
   transitive   ∀x, y, z ∈ A ((x R y ∧ y R z) → x R z)
                every 2-step path has a shortcut
   ```

2. An **equivalence relation** is one that is reflexive, symmetric and transitive, all three. One "no" is enough to say it is not.
3. A **counterexample** is one specific instance where a "for all" statement fails: values that satisfy the "if" part but not the "then" part. Because a "for all" is false as soon as one instance fails, one counterexample settles a "no". A "yes" needs a general argument starting "Let $x \in A$ …".
4. The next five articles do each relation in full.

Trick

For a rule on numbers, test in this order. Reflexive: put $y = x$ in the rule. Symmetric: swap x and y in the rule and see if it reads the same. Transitive: chain three numbers, trying 0, negatives, and points right at the edge of the rule. For products the culprit is usually 0; for distances, it is two steps to the edge.

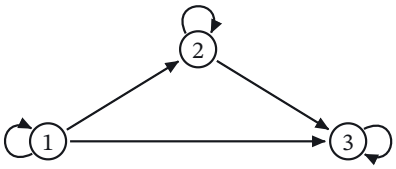
Q3(a) · Q on {1, 2, 3}

Sheet 0B §4

Question. $Q = \{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\}$ on $A = \{1, 2, 3\}$.

Answer

Reflexive: yes. Symmetric: no, $1\ Q\ 2$ but $(2, 1) \notin Q$. Transitive: yes. Equivalence relation: no.



Q: a loop at every vertex, and arrows only from smaller to larger.

Break it down

1. **Reflexive** on $A = \{1, 2, 3\}$ needs $(1,1)$, $(2,2)$, $(3,3)$. All three are listed.
2. **Symmetric** needs the reverse of every pair. $(1, 2)$ is in Q but $(2, 1)$ is not. One failure is enough.
3. **Transitive**: every two-step path $x \to y \to z$ must have the arrow $x \to z$. Save work: a path that uses a loop never causes trouble ($x \to x \to z$ needs $x \to z$, which is the second arrow itself). So only check paths made of two non-loop arrows. The non-loop arrows are $1 \to 2$, $1 \to 3$, $2 \to 3$. The only such path is $1 \to 2 \to 3$, and $1 \to 3$ is present. Transitive.
4. Faster still: $Q = \{(x, y) : x \le y\}$ on $\{1, 2, 3\}$. You know $\le$ is transitive ($x \le y$ and $y \le z$ give $x \le z$) and not symmetric.
5. Not symmetric, so not an equivalence relation. (It is reflexive, antisymmetric and transitive, which makes it a partial order: Tutorial 5.)

Trick

Loops never break transitivity, so skip them. But a two-way pair $x \rightleftarrows y$ forces both loops (x, x) and (y, y): that is the first place to look for a failure.

Q3(b) · E: equality on $\mathbb{Q}$

Question. $x\ E\ y \Leftrightarrow x = y$, for all $x, y \in \mathbb{Q}$.

Answer

Reflexive, symmetric and transitive: yes to all. E is an equivalence relation.

Break it down

1. $\mathbb{Q}$ is the set of rational numbers (fractions p/q of integers with $q \neq 0$). It does not matter much here; the rule is plain equality.
2. Reflexive: let $x \in \mathbb{Q}$. Then $x = x$, so $x\ E\ x$.
3. Symmetric: if $x = y$ then $y = x$.
4. Transitive: if $x = y$ and $y = z$, substitute to get $x = z$.
5. All three hold, so E is an equivalence relation. Each element is related only to itself, so every equivalence class is a single element: $[x] = \{x\}$.

Term

An equivalence relation is a generalised "=": it says two things are "the same in some respect" (same parity, same sign, same product). Equality itself is the model case.

Trick

Any rule of the form $x \sim y \Leftrightarrow f(x) = f(y)$ is automatically an equivalence relation, because it inherits the three properties from "=". E is the case $f(x) = x$; Q8 is the case $f(a, b) = ab$.

Q3(c) · R: $xy \ge 0$ on $\mathbb{Q}$

Question. $x\ R\ y \Leftrightarrow xy \ge 0$, for all $x, y \in \mathbb{Q}$.

Answer

Reflexive: yes. Symmetric: yes. Transitive: no, $1\ R\ 0$ and $0\ R\ -1$ but $(1, -1) \notin R$. Equivalence relation: no.

Break it down

1. Reflexive: $x\ R\ x$ means $x \cdot x \ge 0$, i.e. $x^2 \ge 0$. A square is never negative (positive × positive > 0, negative × negative > 0, $0 \cdot 0 = 0$), so this holds for every x.
2. Symmetric: $x\ R\ y$ means $xy \ge 0$. Multiplication is commutative, $yx = xy$, so $yx \ge 0$, which is $y\ R\ x$.
3. Transitive: first understand the rule. $xy \ge 0$ means x and y have the same sign, *or* at least one of them is 0. So 0 is related to every number ($0 \cdot y = 0 \ge 0$). That makes 0 a bridge between a positive and a negative, which are not related to each other:

```
1 R 0      since 1 · 0 = 0 ≥ 0
0 R -1     since 0 · (-1) = 0 ≥ 0
1 R -1 ?   1 · (-1) = -1 < 0, so no
```

4. Not transitive, so not an equivalence relation.

Trick

If one element is related to *everything*, and some two elements are not related to each other, transitivity fails through that element. For product rules that element is 0: always try it first.

Q3(d) · S: $xy > 0$ on $\mathbb{Q}$

Question. $x\ S\ y \Leftrightarrow xy > 0$, for all $x, y \in \mathbb{Q}$.

Answer

Reflexive: no, $0 \cdot 0 = 0$, so $(0, 0) \notin S$. Symmetric: yes. Transitive: yes. Equivalence relation: no.

Break it down

1. Reflexive needs $x \cdot x > 0$ for *every* rational x. It fails at exactly one point, $x = 0$, and one point is enough. It does not matter that $x\ S\ x$ holds for every other x.

2. Symmetric: $xy = yx$, so $xy > 0$ gives $yx > 0$.
3. Transitive: $xy > 0$ means x and y are both positive or both negative (same sign, and neither is 0). If x has the same sign as y, and y the same sign as z, then x and z share a sign, so $xz > 0$. Written as a proof:

```
1. Let x, y, z ∈ ℚ with x S y and y S z.
2. xy > 0 and yz > 0     (by definition of S)
3. (xy)(yz) > 0          (product of positives)
4. (xz)y² > 0            (by basic algebra)
5. y ≠ 0 (else xy = 0), so y² > 0
6. Hence xz > 0          (if xz ≤ 0, then
   (xz)y² ≤ 0, contradicting 4)
7. So x S z              (by definition of S)
```

4. Not reflexive, so not an equivalence relation.

Trap

"Same sign is obviously an equivalence" is wrong here, because 0 has no sign and $0 \in \mathbb{Q}$. Always check the awkward element. Q10 uses the same rule on $\mathbb{Z} \setminus \{0\}$, where 0 has been removed, and there it *is* an equivalence relation.

Q3(e) · T: within distance 2 on $\mathbb{Z}$

Question. $x\,T\,y \Leftrightarrow -2 \le x - y \le 2$, for all $x, y \in \mathbb{Z}$.

Answer

Reflexive: yes. Symmetric: yes. Transitive: no, $-2\,T\,0$ and $0\,T\,2$ but $(-2, 2) \notin T$. Equivalence relation: no.

Break it down

1. Read the rule: $-2 \le x - y \le 2$ says $|x - y| \le 2$, i.e. x and y are at most 2 apart on the number line.
2. Reflexive: $x - x = 0$, and $-2 \le 0 \le 2$.
3. Symmetric: $y - x = -(x - y)$. Multiplying $-2 \le x - y \le 2$ by -1 flips both inequalities: $2 \ge y - x \ge -2$. That is exactly the condition for $y\,T\,x$.
4. Transitive fails: two steps of size 2 add up to a distance of 4. Choose points right at the edge:

```
-2 T 0    since -2 - 0 = -2   (allowed)
0 T 2     since 0 - 2 = -2    (allowed)
-2 T 2 ?  -2 - 2 = -4 < -2, so no
```

   Any three points spaced 2 apart work, e.g. $0, 2, 4$.
5. Not transitive, so not an equivalence relation.

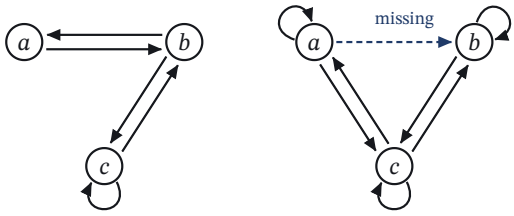
Trick

"Close to" relations (at most d apart) are always reflexive and symmetric but never transitive on $\mathbb{Z}$: step to the edge twice in the same direction.

Q4(a) · $R \circ R$ from a directed graph

Sheet 5 §3

Question. The directed graph of a relation R on $A = \{a, b, c\}$ has arrows $a \to b$, $b \to a$, $b \to c$, $c \to b$ and a loop at c (left graph below). Draw the directed graph of (a) $R \circ R$, (b) $R \circ R \circ R$, (c) $(R \circ R) \cup R$, and decide whether each is transitive; if not, explain.



R $\qquad$ $R \circ R$

Left: the given R. Right: $R \circ R$; the dashed arrow is the shortcut that would be needed but is missing.

Answer

```
R∘R = {(a,a), (a,c), (b,b), (b,c),
       (c,a), (c,b), (c,c)}
```

Not transitive: $a\,(R \circ R)\,c$ and $c\,(R \circ R)\,b$, but $(a, b) \notin R \circ R$.

Break it down

1. A **directed graph** of a relation on A has one dot per element and one arrow $x \to y$ for each pair (x, y) in the relation. A pair (x, x) is drawn as a loop.
2. Read R off the graph and write where each vertex points (its **out-list**):

```
R = {(a,b), (b,a), (b,c), (c,b), (c,c)}
a → {b}     b → {a, c}     c → {b, c}
```

3. $R \circ R$ is "R first, then R": $x\,(R \circ R)\,z \Leftrightarrow \exists y\,(x\,R\,y \wedge y\,R\,z)$. So $(x, z) \in R \circ R$ exactly when you can walk from x to z in **exactly two** arrows. Using the same arrow twice (such as the loop) is allowed.
4. Walk two steps from each vertex. The first step uses the out-list of x; the second uses the out-list of wherever you landed:

```
a → b → {a, c}              a: {a, c}
b → a → {b}
b → c → {b, c}              b: {b, c}
c → b → {a, c}
c → c → {b, c}              c: {a, b, c}
```

5. Transitive? Look for a two-step path inside $R \circ R$ with no shortcut. $a \to c$ and $c \to b$ are both in $R \circ R$, but $a \to b$ is not. So $R \circ R$ is not transitive.
6. Why a cannot reach b in two steps: a's only arrow goes to b, and from b you must move to a or c. There is no loop at b to "stay put".

Trap

"Exactly two arrows", not "at most two". $a \to b$ is a one-step path, so it is in R but not in $R \circ R$.

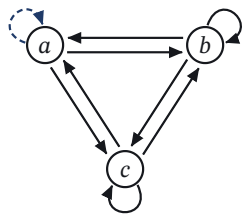
Trick

Write the out-lists first. Then the two-step list of x is the union of the out-lists of x's targets. That turns composition into a lookup.

Q4(b) · $R \circ R \circ R$

Question. Same R. Draw $R \circ R \circ R$ and decide whether it is transitive.

missing (a, a)



$R \circ R \circ R$

$R \circ R \circ R$: every arrow except the loop at a.

Answer

```
R∘R∘R = {(a,b), (a,c), (b,a), (b,b), (b,c),
         (c,a), (c,b), (c,c)}
       = every pair except (a,a)
```

Not transitive: $a\ (R \circ R \circ R)\ c$ and $c\ (R \circ R \circ R)\ a$, but $(a, a) \notin R \circ R \circ R$.

Break it down

1. $R \circ R \circ R$ means walks of **exactly three** arrows. Composition is associative (proved in Q7): $R \circ (R \circ R) = (R \circ R) \circ R$, so no brackets are needed and you can reuse (a). Take the two-step lists and walk one more R arrow:

```
2-step: a:{a,c}  b:{b,c}  c:{a,b,c}
a: a→{b}, c→{b,c}          ⇒ {b, c}
b: b→{a,c}, c→{b,c}        ⇒ {a, b, c}
c: a→{b}, b→{a,c}, c→{b,c} ⇒ {a, b, c}
```

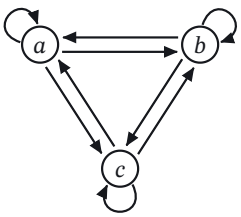
2. Transitive? $a \to c \to a$ is a two-step path inside $R \circ R \circ R$. The shortcut it needs is the loop (a, a), which is missing.
3. Why no three-step walk returns to a: a walk $a \to x \to y \to a$ needs $x = b$ (a's only arrow) and $y = b$ (the only arrow into a comes from b). The middle step would then be $b \to b$, but b has no loop in R.

Trick

In a transitive relation, a two-way pair $x \rightleftarrows y$ forces the loop (x, x). Missing loops are the fastest place to spot a transitivity failure.

Q4(c) · $(R \circ R) \cup R$

Question. Same R. Draw $(R \circ R) \cup R$ and decide whether it is transitive.



$(R \circ R) \cup R$

$(R \circ R) \cup R = A \times A$: every possible arrow.

Answer

```
(R∘R) ∪ R = {(a,a), (a,b), (a,c), (b,a), (b,b),
             (b,c), (c,a), (c,b), (c,c)}
           = A × A
```

Transitive.

Break it down

1. A **union** $X \cup Y$ contains a pair when the pair is in X or in Y (or both). So merge the two lists: $R \circ R$ has 7 pairs, and R adds only (a, b) and (b, a) (its other pairs (b, c), (c, b), (c, c) are already there).
2. The result is all $3 \times 3 = 9$ pairs of $A \times A$: every possible arrow, including every loop.
3. $A \times A$ is transitive: whatever pair (x, z) transitivity asks for, it is already in $A \times A$. The "then" part of the condition is always true, so the condition always holds.

Trick

Recognise the extremes on sight. The full relation $A \times A$ is reflexive, symmetric and transitive. The empty relation is symmetric and transitive (there is no pair to break either rule), but not reflexive on a non-empty set.

Q5(a) · $R^{-1} \circ R = R \circ R^{-1}$

Sheet 5B §1

Question. (AY23/24 midterm) Which of the following are true for *all* equivalence relations R? (a) $R^{-1} \circ R = R \circ R^{-1}$ (b) $R \subseteq R \circ R$ (c) $R \circ R \subseteq R$ (d) $R \circ R^{-1} = R$.

Answer

All four are true. For (a): equivalence relations are symmetric, and R is symmetric iff $R = R^{-1}$ (Q2). So $R^{-1} \circ R = R \circ R = R \circ R^{-1}$.

Break it down

1. "True for all equivalence relations R" means: whichever equivalence relation you pick, on whatever set, the statement holds. To answer "true" you need an argument that uses only reflexive, symmetric and transitive. To answer "false" you need one specific equivalence relation where it fails.
2. Composition is not commutative in general: $R^{-1} \circ R$ ("share a target", see D2(b)) and $R \circ R^{-1}$ ("share a source") are usually different. So something special about R must make the two sides match.
3. That something is symmetry. By Q2, $R = R^{-1}$, so replace every R^{-1} by R. Both sides become $R \circ R$.

Trick

Ask of each option: which of the three properties does it use? R^{-1} uses symmetry, $R \subseteq R \circ R$ uses reflexivity, $R \circ R \subseteq R$ uses transitivity. An equivalence relation has all three, so each passes.

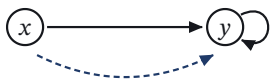
Q5(b) · $R \subseteq R \circ R$

Question. For every equivalence relation R on a set A: is $R \subseteq R \circ R$?

Answer

True, because equivalence relations are reflexive.

```
1. Let (x,y) ∈ R, where x, y ∈ A.
2. Since (x,x), (y,y) ∈ R (by reflexivity
   of R), composing (x,x) with (x,y), or
   (x,y) with (y,y), gives (x,y) ∈ R∘R
   (by definition of composition of
   relations).
```



Pad the arrow $x \to y$ with the loop at y: a two-step path from x to y.

Break it down

1. To prove $X \subseteq Y$: take an arbitrary element of X and show it lies in Y (the definition of subset is $\forall p\,(p \in X \to p \in Y)$). Elements of R are pairs, so open with "Let $(x, y) \in R$".
2. The goal is $(x, y) \in R \circ R$, which means: find a middle element w with $x\,R\,w$ and $w\,R\,y$. You choose this witness yourself, because the $\exists$ is in what you are proving.
3. Use a loop. Reflexivity gives $y\,R\,y$, so $w = y$ works: $x \to y$ (the arrow you have) then $y \to y$ (the loop). Equally, $w = x$ works: $x \to x$ then $x \to y$. You need only one.

> **Trick**
> To land in $R \circ R$, pad your arrow with a loop. Only reflexivity is needed, so $R \subseteq R \circ R$ holds for every reflexive relation.

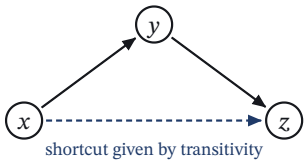
Q5(c) · $R \circ R \subseteq R$

Question. For every equivalence relation R on a set A: is $R \circ R \subseteq R$?

Answer

True, because equivalence relations are transitive.

```
1. Let (x,z) ∈ R∘R, where x, z ∈ A.
2. There exists some y ∈ A such that
   (x,y) ∈ R and (y,z) ∈ R
   (by definition of composition).
3. Hence (x,z) ∈ R (by transitivity of R).
```



Every two-step path $x \to y \to z$ has the shortcut $x \to z$.

Break it down

1. Start with an arbitrary element of the left set, $R \circ R$. Unwrapping composition hands you a middle element y. This time you do not choose y: the $\exists$ is in your assumption, so you simply name the y it promises and use it.
2. $x\,R\,y$ and $y\,R\,z$ is exactly the "if" part of transitivity, which gives $x\,R\,z$.
3. The reverse holds too: $R \circ R \subseteq R$ is true exactly when R is transitive. It is transitivity restated as "every two-step path has a shortcut".

> **Trick**
> $R \circ R \subseteq R$ is transitivity in disguise. $R \subseteq R \circ R$ comes from reflexivity. Together they give $R \circ R = R$.

Q5(d) · $R \circ R^{-1} = R$

Question. For every equivalence relation R: is $R \circ R^{-1} = R$?

Answer

True.

```
1. As R is symmetric, R⁻¹ = R (by Q2).
2. By (b) and (c), R∘R = R.
3. Therefore R∘R⁻¹ = R∘R = R
   (by lines 1 and 2).
```

Break it down

1. Line 2: (b) says $R \subseteq R \circ R$ and (c) says $R \circ R \subseteq R$. Two sets that are each a subset of the other are equal.
2. Line 3: replace R^{-1} by R (line 1), then use line 2.
3. This part uses all three properties: symmetry (line 1), reflexivity (through (b)) and transitivity (through (c)). That is why it needs an equivalence relation and not just any relation.

> **Trick**
> For an equivalence relation, $R^{-1} = R$ and $R \circ R = R$. So every expression built from R, R^{-1} and $\circ$ collapses to R. MCQs like this one become a ten-second question.

Q6 · $R \circ R \circ R \circ R \circ R \circ R$ in one line

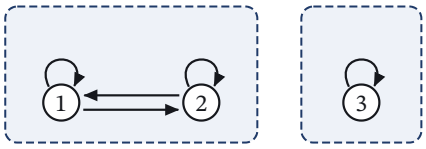
Sheet 5B §1

Question. (AY23/24 exam) $R = \{(1,1), (1,2), (2,1), (2,2), (3,3)\}$ on $A = \{1, 2, 3\}$. Find $R \circ R \circ R \circ R \circ R \circ R$. How can an earlier question give the answer quickly?

Answer

$\{(1,1), (1,2), (2,1), (2,2), (3,3)\}$, which is R itself. R is an equivalence relation, so $R \circ R = R$ by Q5(d) (via (b) and (c)). By associativity of composition,

```
R∘R∘R∘R∘R∘R
  = ((R∘R)∘(R∘R))∘((R∘R)∘R)
  = (R∘R)∘(R∘R)
  = R∘R
  = R
```



R splits A into the classes {1, 2} and {3}.

Break it down

1. First check that R really is an equivalence relation (never assume it):

```
reflexive:  (1,1), (2,2), (3,3) all present
symmetric:  only non-loop pairs are (1,2)
            and (2,1), a matched pair
transitive: 1→2→1 needs (1,1): present
            2→1→2 needs (2,2): present
```

2. Q5(b) and (c) give $R \subseteq R \circ R$ and $R \circ R \subseteq R$, so $R \circ R = R$.
3. **Associative** means brackets can go anywhere: $T \circ (S \circ R) = (T \circ S) \circ R$ (proved in Q7). So bracket any two neighbouring R's, replace $R \circ R$ by R, and repeat. Seven R's become six, then five, and so on down to one.
4. Picture check: a walk can never leave its class ({1, 2} or {3}), and the loops let you wait in place as long as you like. So a walk of exactly 7 arrows connects exactly the pairs in the same class, which is R again.

Trick

"Find $R \circ R \circ \ldots \circ R$" with many R's: check whether R is reflexive and transitive (an equivalence relation qualifies). If so, the answer is R. When a question says "use a question above", it is pointing at a general result you proved.

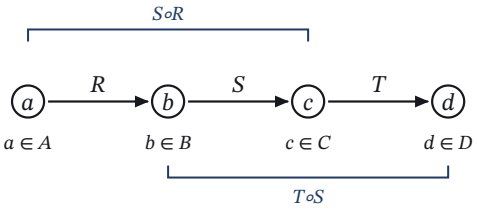
Q7 · Composition is associative

Sheet 5 §1

Question. Let A, B, C, D be sets, $R \subseteq A \times B$, $S \subseteq B \times C$ and $T \subseteq C \times D$. Prove that $T \circ (S \circ R) = (T \circ S) \circ R$.

Answer

```
1. Note S∘R ⊆ A × C and T∘S ⊆ B × D.
2. (⊆) Suppose (a,d) ∈ T∘(S∘R).
2.1 Then there is c ∈ C such that
    (a,c) ∈ S∘R and (c,d) ∈ T
    (by def. of composition of relations)
2.2 From (a,c) ∈ S∘R, there is b ∈ B such
    that (a,b) ∈ R and (b,c) ∈ S.
2.3 From (b,c) ∈ S in 2.2 and (c,d) ∈ T in
    2.1, (b,d) ∈ T∘S.
2.4 From (a,b) ∈ R in 2.2 and (b,d) ∈ T∘S
    in 2.3, (a,d) ∈ (T∘S)∘R.
2.5 Therefore T∘(S∘R) ⊆ (T∘S)∘R.
3. (⊇) Suppose (a,d) ∈ (T∘S)∘R.
3.1 Then there is b ∈ B such that
    (a,b) ∈ R and (b,d) ∈ T∘S
    (by def. of composition of relations)
3.2 From (b,d) ∈ T∘S, there is c ∈ C such
    that (b,c) ∈ S and (c,d) ∈ T.
3.3 From (a,b) ∈ R in 3.1 and (b,c) ∈ S in
    3.2, (a,c) ∈ S∘R.
3.4 From (a,c) ∈ S∘R in 3.3 and (c,d) ∈ T
    in 3.2, (a,d) ∈ T∘(S∘R).
3.5 Therefore (T∘S)∘R ⊆ T∘(S∘R).
4. Therefore T∘(S∘R) = (T∘S)∘R.
```



One chain of three arrows, bracketed two ways.

Break it down

1. Line 1 checks that the pieces fit. $S \circ R$ goes from A to C, and T goes from C to D, so $T \circ (S \circ R)$ makes sense and goes from A to D. Likewise R goes A to B and $T \circ S$ goes B to D. Both sides are subsets of $A \times D$, so comparing them makes sense.
2. Equality of sets is proved as two subset proofs, ($\subseteq$) and ($\supseteq$). In each, take an arbitrary element of the left side and show it lies in the right side. The elements are pairs, so name one (a, d) with $a \in A$, $d \in D$.
3. **Unwrap.** Membership in a composition gives you an $\exists$: "there is a middle element". Name it and keep it. Unwrapping twice (outer $\circ$, then inner $\circ$) produces the whole chain $a \to b \to c \to d$ with $(a, b) \in R$, $(b, c) \in S$, $(c, d) \in T$.
4. **Rewrap.** Build the other bracketing from the same chain. First glue the two arrows the new inner bracket needs ($b \to c \to d$ gives $(b, d) \in T \circ S$), then glue the remaining arrow onto it ($a \to b$ then $b \to d$ gives $(a, d) \in (T \circ S) \circ R$). Each gluing is the definition of composition read backwards, using the middle element you already have as the witness.
5. ($\supseteq$) is the mirror image: unwrap the other bracketing, rewrap into the first one.

Trap

Watch the order. $(x, y) \in S \circ R$ means the R step comes first: $(x, m) \in R$ and then $(m, y) \in S$. Writing $(x, m) \in S$ is the most common slip in composition proofs.

Trick

Every composition proof is "unwrap each $\exists$ into a chain of arrows, then regroup the chain". Tag each letter with its set ($a \in A$, $b \in B$, ...) and cite the line numbers you glue.

Q8 · Classes of "same product"

Sheet 5B §3

Question. (AY20/21 exam) The equivalence relation $\sim$ on $\mathbb{Z}^+ \times \mathbb{Z}^+$ is defined by $(a, b) \sim (c, d) \Leftrightarrow ab = cd$, for all $a, b, c, d \in \mathbb{Z}^+$. Write the equivalence classes $[(1,1)]$ and $[(4,3)]$ in roster notation.

Answer

```
[(1,1)] = {(x,y) ∈ ℤ⁺ × ℤ⁺ : (1,1) ~ (x,y)}
          (by def. of equivalence class)
        = {(x,y) ∈ ℤ⁺ × ℤ⁺ : 1 × 1 (= 1) = xy}
          (by definition of ~)
        = {(1,1)}

[(4,3)] = {(x,y) ∈ ℤ⁺ × ℤ⁺ : (4,3) ~ (x,y)}
          (by def. of equivalence class)
        = {(x,y) ∈ ℤ⁺ × ℤ⁺ : 4 × 3 (= 12) = xy}
          (by definition of ~)
        = {(1,12), (2,6), (3,4), (4,3),
           (6,2), (12,1)}
```

The official key writes "$= ab$" in the middle lines; it means xy.

Break it down

1. $\mathbb{Z}^+$ is the set of positive integers $\{1, 2, 3, \ldots\}$. The relation lives on $\mathbb{Z}^+ \times \mathbb{Z}^+$, so the things being related are *pairs*: $\sim$ relates one pair to another. In this question $\sim$ is just the name of the relation, not negation.
2. The **equivalence class** of an element p is the set of everything related to it: $[p] = \{q : p \sim q\}$. Roster notation means listing every member.
3. Unwrap $\sim$: $(4, 3) \sim (x, y)$ means $4 \cdot 3 = xy$, i.e. $xy = 12$. So $[(4, 3)]$ is every ordered pair of positive integers whose product is 12.
4. List systematically: run $x = 1, 2, 3, \ldots, 12$, keep x when $x \mid 12$, and set $y = 12 / x$. The divisors are $1, 2, 3, 4, 6, 12$: six pairs. They are ordered pairs, so $(3, 4)$ and $(4, 3)$ are different elements and both belong.
5. $[(1,1)]$: $xy = 1$ with $x, y \geq 1$. If either factor were 2 or more the product would be at least 2, so $x = y = 1$: a single element.
6. Why $\sim$ is an equivalence relation (given, but worth seeing): it is "$f(a, b) = f(c, d)$" with $f(a, b) = ab$, so it inherits all three properties from "=".

Trap

No negatives such as $(-3, -4)$: the set is $\mathbb{Z}^+ \times \mathbb{Z}^+$. And $[(4, 3)]$ is a set of *pairs*, not the number 12.

Trick

For a rule "$p \sim q \Leftrightarrow f(p) = f(q)$", the class of p is "everything with the same f-value as p". Compute $f(p)$, then solve $f(q) = f(p)$ inside the given set.

Q9(a) · S^{-1} for $m^3 + n^3$ even

Sheet 5 §1

Question. $S = \{(m, n) \in \mathbb{Z}^2 : m^3 + n^3$ is even$\}$, where $\mathbb{Z}^2 = \mathbb{Z} \times \mathbb{Z}$. Determine (a) S^{-1}, (b) $S \circ S$, (c) $S \circ S^{-1}$. Facts about sums of even and odd integers may be used without quoting.

Answer

```
S⁻¹ = {(x,y) ∈ ℤ² : (y,x) ∈ S}
      (by definition of inverse relation)
    = {(x,y) ∈ ℤ² : y³ + x³ is even}
      (by definition of S)
    = {(x,y) ∈ ℤ² : x³ + y³ is even}
      (by the commutative law of addition)
    = S     (by definition of S)
```

Break it down

1. "Determine" means: find the set and justify it. Here it turns out to be a set you already have, S itself.
2. Line 1 is the definition of inverse in set-builder form: $(x, y) \in S^{-1}$ exactly when the flipped pair $(y, x) \in S$.
3. Line 2 plugs (y, x) into S's rule. The rule is about (m, n), so $m = y$ and $n = x$: "$m^3 + n^3$ is even" becomes "$y^3 + x^3$ is even". Careful substitution is the whole step.
4. Line 3: addition is commutative, $y^3 + x^3 = x^3 + y^3$.
5. Line 4: that is exactly S's condition for the pair (x, y), so the set is S.
6. Link to Q2: $S^{-1} = S$ says S is symmetric. Any rule that reads the same after swapping its two variables gives a symmetric relation.

Trick

Prove a set-builder equality by rewriting the condition one line at a time, one reason per line, until it matches the target's condition word for word.

Q9(b) · $S \circ S = S$

Sheet 4B §5

Question. Same S. Determine $S \circ S$.

Answer

$S \circ S = S$.

```
1. (⊆) Suppose (x,z) ∈ S∘S.
1.1 Then (x,y) ∈ S and (y,z) ∈ S for some
    y ∈ ℤ  (by def. of composition of relations)
1.2 So x³ + y³ is even and y³ + z³ is even.
1.3 This implies x³ + 2y³ + z³ is even.
1.4 This implies x³ + z³ is even,
    as 2y³ is even.
1.5 Therefore (x,z) ∈ S (by definition of S).
2. (⊇) Suppose (x,z) ∈ S.
2.1 Then x³ + z³ is even (by definition of S).
2.2 Case 1: x³ is odd.
2.2.1 Then z³ is also odd.
2.2.2 So x³ + 1³ is even and 1³ + z³ is even.
2.2.3 Thus (x,1) ∈ S and (1,z) ∈ S
      (by definition of S).
2.2.4 So (x,z) ∈ S∘S
      (by def. of composition of relations).
2.3 Case 2: x³ is even.
2.3.1 Then z³ is also even.
2.3.2 So x³ + 0³ is even and 0³ + z³ is even.
2.3.3 Thus (x,0) ∈ S and (0,z) ∈ S
      (by definition of S).
2.3.4 So (x,z) ∈ S∘S
      (by def. of composition of relations).
2.4 In all cases, (x,z) ∈ S∘S.
3. ∴ S∘S = S.
```

The key also gives a shorter (⊇):

```
2.1 Note (x,x) ∈ S as x³ + x³ is even.
2.2 Since (x,x) ∈ S and (x,z) ∈ S, we have
    (x,z) ∈ S∘S (by definition of
    composition of relations).
2.3 Hence S ⊆ S∘S.
```

Break it down

1. To show two sets are equal, prove $\subseteq$ and $\supseteq$. Elements are pairs of integers, so start each half with "Suppose $(x, z) \in$ …".
2. **($\subseteq$)** Unwrap composition: some integer y sits in the middle. You now hold two even numbers. The goal mentions x^3 and z^3 but not y^3, so **add** the two facts: $(x^3 + y^3) + (y^3 + z^3) = x^3 + 2y^3 + z^3$, even (even + even). The unwanted y now appears as $2y^3$, which is even. Subtract it: $x^3 + z^3 = (x^3 + 2y^3 + z^3) - 2y^3$ is even − even, so even.
3. **($\supseteq$)** Now the $\exists$ is in the goal, so *you* must supply the middle integer y. You need $x^3 + y^3$ even and $y^3 + z^3$ even. Two integers add up to an even number exactly when they have the same **parity** (both even or both odd). So choose y with y^3 of the same parity as x^3.
4. That choice depends on x, hence the two cases. 1 is the easy odd choice ($1^3 = 1$); 0 is the easy even choice ($0^3 = 0$). In Case 1, x^3 odd and $x^3 + z^3$ even force z^3 odd (odd + even would be odd), so both $x^3 + 1$ and $1 + z^3$ are odd + odd = even. Case 2 is the same with even numbers.
5. The shorter ($\supseteq$) takes $y = x$. $(x, x) \in S$ because $x^3 + x^3 = 2x^3$ is even, i.e. S is reflexive. Then $x \to x \to z$ is a two-step path. This is the loop trick of Q5(b).

Big picture

x^3 has the same parity as x (a product of odd numbers is odd; a product with an even factor is even). So S is really "m and n have the same parity", an equivalence relation. Q5 then predicts $S^{-1} = S$ and $S \circ S = S$ before you write a line. Use this to know the answer; the proof above is still what earns the marks.

Trick

($\subseteq$) direction: the middle element is *given*; combine the two facts to cancel it. ($\supseteq$) direction: the middle element is *yours to choose*; try the simplest values (0, 1, or x itself).

Q9(c) · $S \circ S^{-1}$

Question. Same S. Determine $S \circ S^{-1}$.

Answer

It follows from (a) and (b) that $S \circ S^{-1} = S \circ S = S$.

Break it down

1. By (a), S^{-1} and S are the same set, so S^{-1} can be replaced by S: $S \circ S^{-1} = S \circ S$.
2. By (b), $S \circ S = S$.

Trick

Multi-part questions are built as a chain. Before starting a later part, ask which earlier part makes it one line.

Q10(a) · $ab > 0$ on $\mathbb{Z} \setminus \{0\}$ is an equivalence

Sheet 5B §2

Question. Define $\sim$ on $\mathbb{Z} \setminus \{0\}$ by $\forall a, b \in \mathbb{Z} \setminus \{0\}\ (a \sim b \Leftrightarrow ab > 0)$. (a) Prove that $\sim$ is an equivalence relation, using the order axioms and theorems of Appendix A (Properties of the Real Numbers). (b) Determine all the distinct equivalence classes.

Answer

```
1. (Reflexivity)
1.1 Let a ∈ ℤ \ {0}. Since a ≠ 0, a² > 0
    (by T21).
1.2 Thus a ~ a (by definition of ~).
1.3 Hence ~ is reflexive.
2. (Symmetry)
2.1 For any a, b ∈ ℤ \ {0}, if a ~ b, then
    ab > 0 (by definition of ~).
2.2 Then ba > 0 (by the commutative law
    of multiplication).
2.3 So b ~ a (by definition of ~).
2.4 Hence ~ is symmetric.
3. (Transitivity)
3.1 For any a, b, c ∈ ℤ \ {0}, suppose a ~ b
    and b ~ c.
3.2 Then ab > 0 and bc > 0
    (by definition of ~).
3.3 Multiplying ab with bc (both positive)
    gives ab²c > 0 (by Ord1).
3.4 Then (ac)b² > 0 (by the associative
    and commutative laws of multiplication).
3.5 Then (ac) and b² are both positive, or
    both negative (by T25).
3.6 Since b² > 0 (by T21, as b ≠ 0),
    (ac) must also be positive.
3.7 Thus a ~ c (by definition of ~).
3.8 Hence ~ is transitive.
4. Therefore ~ is an equivalence relation.
```

Appendix A facts used

```
T21   If a ≠ 0, then a² > 0.
Ord1  If a and b are positive, so are
      a + b and ab.
T25   If ab > 0, then a and b are both
      positive or both negative.
```

Break it down

1. Here $\sim$ is the name of the relation (not negation), and $\mathbb{Z} \setminus \{0\}$ is the set of all integers except 0.
2. An equivalence-relation proof is three mini-proofs with fixed openers. Reflexive: "Let a be in the set". Symmetric: "Suppose $a \sim b$". Transitive: "Suppose $a \sim b$ and $b \sim c$". Each ends by rewrapping the definition of $\sim$.
3. Appendix A lists the facts about order (positive, negative, $<$) you may cite by label. The question explicitly allows them, so cite T21, Ord1, T25 by name, as the key does.
4. Reflexive: $a \sim a$ means $a \cdot a > 0$. T21 needs $a \neq 0$, which holds because $a \in \mathbb{Z} \setminus \{0\}$. This is exactly where Q3(d) failed: there 0 was allowed.
5. Symmetric: $ba = ab$ (commutative law), so $ab > 0$ gives $ba > 0$.
6. Transitive, the only real work. You know $ab > 0$ and $bc > 0$ and want $ac > 0$. Multiply the two facts (Ord1: a product of positives is positive) so the two b's pair up into b^2. Rearrange to $(ac) \cdot b^2 > 0$. T25 says a positive product has two factors of the same sign. b^2 is positive (T21, since $b \neq 0$), so ac cannot be negative: it is positive.
7. The idea is just "same sign as b, twice". The proof says it in the vocabulary the question allows.

Trick

For product rules ($ab > 0$, $ab \geq 0$, ...), multiply the two hypotheses so the middle variable appears squared, then remove it with a "squares are positive" fact.

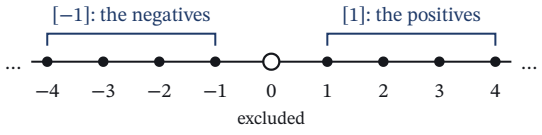
Q10(b) · The distinct classes

Sheet 5B §3

Question. Same $\sim$ on $\mathbb{Z} \setminus \{0\}$. Determine all the distinct equivalence classes.

Answer

By T25, if $ab > 0$ then a and b are both positive or both negative. So all positive integers are related to one another, and all negative integers are related to one another. The two distinct classes are $\{a \in \mathbb{Z} \setminus \{0\} : a > 0\}$ and $\{a \in \mathbb{Z} \setminus \{0\} : a < 0\}$, or with representatives 1 and -1: $[1]$ and $[-1]$.



Two classes; 0 is not in the set at all.

Break it down

1. Compute one class directly from the definition $[p] = \{a : p \sim a\}$:

```
[1]  = {a ∈ ℤ \ {0} : 1 · a > 0}
     = {a ∈ ℤ \ {0} : a > 0}
[−1] = {a ∈ ℤ \ {0} : (−1) · a > 0}
     = {a ∈ ℤ \ {0} : a < 0}
```

2. Every element of $\mathbb{Z} \setminus \{0\}$ is positive or negative (0 is excluded), so these two classes cover the whole set. They are different, because $1 \cdot (-1) = -1 < 0$ means 1 and -1 are not related.
3. "Distinct" matters. $[1] = [2] = [7]$ (all positives are related), so listing $[1], [2], [3], \ldots$ would repeat one set many times. Pick one **representative** per class.
4. The classes form a **partition** of $\mathbb{Z} \setminus \{0\}$: non-empty pieces that do not overlap and together cover the set. Equivalence classes always do.

> **Trick**
> To find all classes: compute $[r]$ for a simple r, see what it misses, pick a representative of the leftovers, and repeat until everything is covered.

T5 · Tutorial 5 walkthrough

Relations & partial orders · no official key, answers worked out here

Tutorial 5 trains two skills: reading relation properties as rules about arrows, and reading partial orders off a Hasse diagram. Three tricks carry almost every question. An "if … then" property whose "if" part never happens is vacuously true. One concrete pair is enough to kill a "for all" property. And for any question about a partial order, draw the Hasse diagram first, then read the answer off the picture.

D1 · The empty relation

Sheet 5 §1

Question. Let R be a binary relation on a non-empty set A. If $R = \varnothing$, then is R reflexive? Symmetric? Transitive?

Answer (worked out here, no official key)

Not reflexive. Symmetric: yes (vacuously). Transitive: yes (vacuously).

```
Reflexive:  ∀ x ∈ A (x R x)                NO
  Pick any a ∈ A (A is non-empty): (a,a) ∉ ∅
Symmetric:  ∀ x,y ∈ A (x R y → y R x)      YES
  x R y is false for every x, y,
  so each implication is true.
Transitive: ∀ x,y,z ∈ A
            (x R y ∧ y R z → x R z)        YES
  x R y ∧ y R z is always false,
  so each implication is true.
```

Break it down

1. **Binary relation on A**: any set of ordered pairs taken from $A \times A$. We write $x\,R\,y$ as shorthand for $(x,y) \in R$. Picture it as arrows: $x\,R\,y$ is an arrow from x to y. Here $R = \varnothing$, so there are no arrows at all, and $x\,R\,y$ is false for every x and y.
2. **Reflexive** means every element is related to itself: $\forall x \in A\ (x\,R\,x)$. In the picture, a loop on every element. To break a "for all" statement you only need one element that fails. A is non-empty, so it contains some element a, and $(a,a) \notin \varnothing$. That one missing loop means R is not reflexive.
3. **Symmetric** means every arrow has its reverse: $\forall x,y \in A\ (x\,R\,y \to y\,R\,x)$. Look at the "if" part, $x\,R\,y$. It is false for every choice of x, y, because R has no pairs. An implication with a false "if" part is true, whatever the "then" part says ($\mathit{false} \to q$ is true in the truth table). A statement that is true for this reason is called **vacuously true**. Every instance is true, so the whole ∀ statement is true.
4. **Transitive** means every two-step path has a shortcut: $\forall x,y,z \in A\ (x\,R\,y \wedge y\,R\,z \to x\,R\,z)$. The "if" part needs $x\,R\,y$, which is always false. So the $\wedge$ is false, and the implication is vacuously true again.
5. The same reasoning shows $\varnothing$ is also antisymmetric and asymmetric (both are "if … then" rules, see D2 and D3). Only reflexivity demands that arrows actually exist.

Trap

The words "non-empty" matter. If $A = \varnothing$, there is no x to check, so $\forall x \in A\ (x\,R\,x)$ is vacuously true and $\varnothing$ *would* be reflexive.

Trick

Sort properties into two kinds. Reflexive is a "must have" property: it forces certain arrows to exist. Symmetric, transitive, antisymmetric and asymmetric are "if you have this arrow, then …" properties: they only restrict arrows that are present. A relation with no arrows passes every "if … then" property for free.

D2 · All four properties at once

Sheet 5 §1

Question. A binary relation R on a non-empty set A is reflexive, transitive, symmetric and antisymmetric. What can you conclude about R? Explain.

Answer (worked out here, no official key)

R is the identity relation on A (plain equality): $R = \{(x,x) : x \in A\}$, so $x\,R\,y \Leftrightarrow x = y$.

```
1. (every loop is in R)
1.1 Let x ∈ A.
1.2 (x,x) ∈ R.          (by reflexivity of R)
2. (nothing else is in R)
2.1 Let (x,y) ∈ R.
2.2 (y,x) ∈ R.            (by symmetry of R)
2.3 x = y.    (by antisymmetry of R, 2.1, 2.2)
2.4 So (x,y) is the loop (x,x).
3. R = {(x,x) : x ∈ A}.          (by 1 and 2)
```

Transitivity is never used. (The identity relation is transitive anyway.)

Break it down

1. **Antisymmetric** means $\forall x,y \in A\ (x\,R\,y \wedge y\,R\,x \to x = y)$. In words: arrows in both directions between x and y are only allowed when x and y are the same element. In the picture: no two-way arrow between two different elements. Loops are fine.
2. Symmetric says every arrow comes with its reverse, so every arrow is half of a two-way pair. Antisymmetric says a two-way pair must be a loop. Together: every arrow is a loop. That is step 2 of the proof, and it shows $R \subseteq \{(x,x) : x \in A\}$.
3. Reflexive says every loop is present. That is step 1: $\{(x,x) : x \in A\} \subseteq R$. A subset in both directions means the two sets are equal.
4. The **identity relation** $\{(x,x) : x \in A\}$ relates each element only to itself. It is an equivalence relation (reflexive, symmetric, transitive) and a partial order (reflexive, antisymmetric, transitive) at the same time. The AY23/24 midterm Q10(iv) claimed "no relation is both an equivalence relation and a partial order". This is the counterexample, so that claim is false.

Trick

Translate each listed property into an arrow rule and stack the rules. Symmetric plus antisymmetric leaves "loops only". Adding reflexive gives "exactly all the loops".

D3 · Symmetric and asymmetric together?

Sheet 5 §1

Question. R on A is asymmetric iff $\forall x,y \in A\ (x\,R\,y \Rightarrow \sim(y\,R\,x))$ (the sheet says "question 6"; in this sheet it is Q7, and the PDF text drops the "not" slash). Are there binary relations that are both symmetric and asymmetric?

Answer (worked out here, no official key)

Yes, but only one on each set: the empty relation $R = \varnothing$.

```
1. ∅ is symmetric and asymmetric: both are
   "if x R y then ..." rules and x R y never
   holds, so both are vacuously true.
2. Conversely, let R be symmetric and
   asymmetric. Suppose (x,y) ∈ R.
2.1 y R x.                (by symmetry of R)
2.2 ~(y R x).             (by asymmetry of R)
2.3 2.1 and 2.2 contradict each other.
3. So R has no pairs at all: R = ∅
```

Break it down

1. **Asymmetric** in words: if x is related to y, then y is not related to x. In the picture: no arrow is allowed to have its reverse. Take $x = y$: the rule reads $x\,R\,x \to \sim(x\,R\,x)$, which can only hold if $x\,R\,x$ is false. So an asymmetric relation has no loops either.
2. Symmetric demands that every arrow's reverse is present. Asymmetric demands that every arrow's reverse is absent. No arrow can satisfy both, so there can be no arrows.
3. "Are there ...?" is an existence question: one example answers it. $\varnothing$ is that example. Step 2 is a proof by contradiction (assume a pair exists, derive something impossible), and it shows $\varnothing$ is the only one.

Trick

When two properties make opposite demands on the same arrow, the only survivor is the relation with no arrows. On any "does such a relation exist?" question, try $\varnothing$ first.

D4 · Another way to say antisymmetric

Sheet 5 §1

Question. Let R be a relation on A. Prove that $\forall x,y \in A\ ((x \neq y) \Rightarrow ((x,y) \in R \Rightarrow (y,x) \notin R))$ is an alternative definition of antisymmetry. (The PDF text garbles the brackets; this is the reconstructed statement.)

Answer (worked out here, no official key)

```
Fix any x, y ∈ A and name the parts:
  p: (x,y) ∈ R   q: (y,x) ∈ R   e: x = y
Class definition:    (p ∧ q) → e
New definition:      ~e → (p → ~q)

(p ∧ q) → e
≡ ~e → ~(p ∧ q)        (contrapositive)
≡ ~e → (~p ∨ ~q)       (De Morgan's law)
≡ ~e → (p → ~q)        (implication law)
```

$\sim e$ says $x \neq y$ and $\sim q$ says $(y,x) \notin R$, so the last line is the new definition. The two bodies are equivalent for every x, y, so the two $\forall$ statements are equivalent.

Break it down

1. "Alternative definition" means "a logically equivalent statement". Both statements start with the same $\forall x,y \in A$. So it is enough to show that the parts inside the brackets (the **bodies**) are equivalent for any fixed x and y.
2. Give each basic claim a letter. Logic laws only care about the shape ($\wedge$, $\to$, $\sim$), not about what the claims mean. This turns a relations question into a Tutorial 1 style law chain.
3. **Contrapositive**: $p \to q \equiv \sim q \to \sim p$. If you want it from named laws: $p \to q \equiv \sim p \vee q$ (implication law) $\equiv q \vee \sim p$ (commutative law) $\equiv \sim(\sim q) \vee \sim p$ (double negative law) $\equiv \sim q \to \sim p$ (implication law).
4. **De Morgan's law**: $\sim(p \wedge q) \equiv \sim p \vee \sim q$. **Implication law**: $p \to r \equiv \sim p \vee r$; with $r = \sim q$ it gives $p \to \sim q \equiv \sim p \vee \sim q$, which is the last step read backwards.
5. What the new version says in plain words: "for two different elements, if the arrow $x \to y$ is there, then the arrow $y \to x$ is not". Same arrow rule as before: no two-way arrows between different elements.

Trick

To prove two definitions are equivalent: (1) strip the identical quantifiers, (2) replace the basic claims by letters, (3) run a law chain. When one version has an $\wedge$ in the "if" part and the other version starts with "$x \neq y \to$", reach for the contrapositive first.

Q1 · Strings compared by length

Sheet 6 §1

Question. S is the set of all strings over the alphabet $\{s, u\}$, including the empty string ε (e.g. s, u, sus, $ussuu$). Define R on S by $\forall a,b \in S\ (a\,R\,b \Leftrightarrow len(a) \leq len(b))$, where $len(x)$ is the number of characters in x. Is R a partial order? Prove or disprove.

Answer (worked out here, no official key)

No. R is not antisymmetric.

```
1. s, u ∈ S.              (strings of length 1)
2. len(s) = 1 ≤ 1 = len(u), so s R u.
                          (by definition of R)
3. len(u) = 1 ≤ 1 = len(s), so u R s.
                          (by definition of R)
4. s ≠ u.                 (different strings)
5. By 2, 3, 4, R is not antisymmetric,
   so R is not a partial order.
                (by definition of partial order)
```

(For the record, R is reflexive and transitive. Antisymmetry is the only property that fails.)

Break it down

1. **Partial order**: a relation that is reflexive, antisymmetric and transitive. Think of it as "behaves like $\leq$": everything is $\leq$ itself, two things that are $\leq$ each other must be equal, and $\leq$ chains together. A set with a partial order is a **poset** (partially ordered set), written $(A, \preccurlyeq)$.
2. "Prove or disprove": first decide which. Test each property in your head. Reflexive: $len(a) \leq len(a)$ always, fine. Transitive: $len(a) \leq len(b) \leq len(c)$ gives $len(a) \leq len(c)$, fine. Antisymmetric: $a\,R\,b$ and $b\,R\,a$ give $len(a) = len(b)$, but equal length does not make the strings equal. Fails.

3. Antisymmetry is $\forall a,b\,(a\,R\,b \wedge b\,R\,a \to a = b)$. Its negation is $\exists a,b\,(a\,R\,b \wedge b\,R\,a \wedge a \neq b)$. So a disproof is one concrete pair of different strings with the same length.
4. Pick the smallest such pair: s and u. (ε is useless here: it is the only string of length 0, so it has no partner.)
5. For a disproof, one failed property is enough. You do not have to prove the other two.

Trap

"$\leq$ is a partial order, so R is one too" is wrong. R compares the *lengths*, and many different strings share a length.

Trick

Any relation of the form "$x\,R\,y \Leftrightarrow f(x) \leq f(y)$" (compare a measurement) is automatically reflexive and transitive. It is antisymmetric exactly when no two different inputs have the same measurement. So hunt for two different things with the same size. Q3 dodges this trap by defining $x \preccurlyeq z \Leftrightarrow (F_x = F_z) \vee (|F_x| < |F_z|)$: ties are left unrelated instead of related both ways.

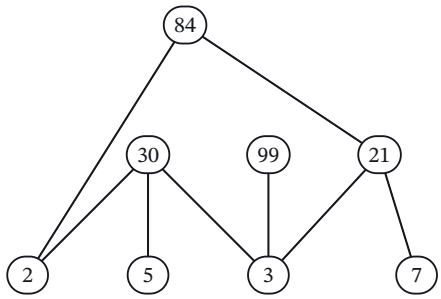
Q2 · Linearizations of "divides"

Sheet 6 §4

Question. [AY22/23 Sem 1 midterm] $A = \{2,3,5,7,21,30,84,99\}$ with $x \preccurlyeq y \Leftrightarrow x \mid y$. Which are true? The partial order has a linearization $\preccurlyeq^*$ such that (i) $21 \preccurlyeq^* 7$; (ii) $3 \preccurlyeq^* 2$; (iii) $21 \preccurlyeq^* 5 \preccurlyeq^* 84$; (iv) $99 \preccurlyeq^* 84 \preccurlyeq^* 30$.

Answer (worked out here, no official key)

(ii), (iii) and (iv) are true. (i) is false.



Hasse diagram of divides on A. Covers: 2–30, 2–84, 3–21, 3–30, 3–99, 5–30, 7–21, 21–84.

```
(i)   7 | 21, so 7 ≼ 21, so 7 ≼* 21 in EVERY
      linearization. Also having 21 ≼* 7
      would force 7 = 21.              FALSE
(ii)  3 2 5 7 21 30 84 99               TRUE
(iii) 2 3 7 21 5 30 84 99               TRUE
(iv)  2 3 5 7 21 99 84 30               TRUE
Check each list against the 8 covers:
the lower number must come first. All ✓.
```

Term

Hasse diagram: a picture of a partial order on a finite set. If $x \preccurlyeq y$, draw x lower than y. Draw a line only for a **cover**: y covers x when $x \preccurlyeq y$, $x \neq y$, and nothing sits strictly between them. Loops and lines implied by transitivity are left out, so "$x \preccurlyeq y$" means "you can walk upward from x to y along lines".
Comparable: $x \preccurlyeq y$ or $y \preccurlyeq x$. **Total order**: a partial order where every two elements are comparable (the Hasse diagram is one vertical line).
Linearization of $\preccurlyeq$: a total order $\preccurlyeq^*$ on the same set that keeps every relationship of $\preccurlyeq$: $x \preccurlyeq y \to x \preccurlyeq^* y$.

Break it down

1. Draw the Hasse diagram first. For every pair where $x \mid y$, ask whether some z in A fits between ($x \mid z$ and $z \mid y$). If so, no line. Example: $3 \mid 84$, but $3 \mid 21 \mid 84$, so there is no line from 3 to 84. Likewise $7 \mid 84$ goes through 21. Factorising helps: $21 = 3{\cdot}7$, $30 = 2{\cdot}3{\cdot}5$, $84 = 2^2{\cdot}3{\cdot}7$, $99 = 3^2{\cdot}11$.
2. A linearization is "line everyone up in one queue so that each element comes after everything below it in the diagram". It copies all of $\preccurlyeq$, and only decides the order of pairs that $\preccurlyeq$ leaves **incomparable** (neither divides the other, like 2 and 3).
3. (i): $7 \mid 21$, so 7 is below 21, and every linearization must keep $7 \preccurlyeq^* 21$. A linearization is itself a partial order, so it is antisymmetric: $7 \preccurlyeq^* 21$ and $21 \preccurlyeq^* 7$ together would give $7 = 21$. Impossible.
4. (ii): 2 and 3 are incomparable, so $\preccurlyeq$ does not fix their order. We just need one valid queue that starts with 3. The list in the answer works.
5. (iii): we need 21 before 5 and 5 before 84. 5 is unrelated to both ($5 \nmid 21$, $5 \nmid 84$), so nothing forbids it. Build the queue with **Kahn's algorithm**: repeatedly write down any element with nothing left below it (a minimal element of what remains), then remove it. Take 2, 3, 7; now 21 has nothing below it, take it; then take 5; then 30, 84, 99.
6. (iv): 99, 84 and 30 have nothing above them, and none divides another. Queue everything else first (2, 3, 5, 7, 21), then put those three last in the order you want.
7. Checking a queue: go through the 8 covers. For (iii), 2 before 30 and 84, 3 before 21, 30 and 99, 5 before 30, 7 before 21, 21 before 84. All hold. You only need to check covers; the rest follows by transitivity.

Trap

21 and 7 look unrelated if you only scan the list. Always factorise: $21 = 3{\cdot}7$, so 7 sits below 21.

Trick

"Can some linearization put a before b?" Yes, unless you can walk *up* from b to a in the Hasse diagram. For a longer requirement like $a \preccurlyeq^* b \preccurlyeq^* c$, just try to build the queue with Kahn's algorithm, holding back the elements that must come later. If you never get stuck, the answer is yes, and your queue is the proof.

Q3 · Ordering by number of divisors

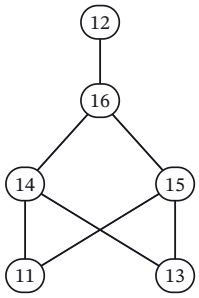
Sheet 6 §2

Question. [AY23/24 Sem 1 midterm] $A = \{11,12,13,14,15,16\}$. For $x \in A$, $F_x = \{k \in \mathbb{Z}^+ : k \mid x\}$. The partial order is $x \preccurlyeq z \Leftrightarrow (F_x = F_z) \vee (|F_x| < |F_z|)$. What are the minimal, smallest, maximal and largest elements of A with respect to $\preccurlyeq$?

Answer (worked out here, no official key)

```
x    F_x                  |F_x|
11   {1, 11}                2
13   {1, 13}                2
14   {1, 2, 7, 14}          4
15   {1, 3, 5, 15}          4
16   {1, 2, 4, 8, 16}       5
12   {1, 2, 3, 4, 6, 12}    6

Minimal: 11, 13     Smallest: None
Maximal: 12         Largest: 12
```



Hasse diagram: levels by number of divisors, 2 < 4 < 5 < 6.

Cross-check: the AY23/24 midterm answer sheet (Q18) gives exactly these four answers.

Term

c is	plain words	formally
maximal	nothing strictly above c	$\forall x\,(c \preccurlyeq x \to x = c)$
minimal	nothing strictly below c	$\forall x\,(x \preccurlyeq c \to x = c)$
largest	c is above everything	$\forall x\,(x \preccurlyeq c)$
smallest	c is below everything	$\forall x\,(c \preccurlyeq x)$

Maximal means "top of its own branch". Largest means "top of the whole thing". Every largest element is maximal, not the other way round.

Break it down

1. Decode the notation. F_x is the set of positive divisors of x ($\mathbb{Z}^+$ = positive integers, $k \mid x$ = "k divides x"). $|F_x|$ is its size, the number of divisors.
2. When is $F_x = F_z$? The largest element of F_x is x itself, so equal sets force $x = z$. So $x \preccurlyeq z$ just means "$x = z$, or x has strictly fewer divisors than z".
3. Consequence: two different numbers with the same number of divisors are **incomparable**. For 11 and 13: $F_{11} \neq F_{13}$, and $2 < 2$ is false, so $11 \not\preccurlyeq 13$. The same argument gives $13 \not\preccurlyeq 11$. Likewise 14 and 15.
4. So the elements sit in levels: $\{11, 13\}$ (2 divisors), $\{14, 15\}$ (4), $\{16\}$ (5), $\{12\}$ (6). Every element of a lower level is below every element of a higher level. The Hasse diagram draws lines only between neighbouring levels.
5. Minimal: nothing in A has fewer than 2 divisors, so nothing is strictly below 11 or 13. Every other element has 11 below it. Minimal: 11 and 13.
6. Smallest: it would have to be below everything, including the other minimal element. But 11 and 13 are incomparable. So there is no smallest element. General rule: two different minimal elements means no smallest.
7. Maximal: 12 has 6 divisors, strictly more than anything else, so nothing is above it. 16 is not maximal because $16 \preccurlyeq 12$.
8. Largest: every other element has fewer divisors than 12, so $x \preccurlyeq 12$ for all x. 12 is largest (and also maximal).

Trap

11 is not "smallest" just because it is the smallest number. The order here is $\preccurlyeq$, not the usual $\leq$ on numbers. Also, never leave a box blank: write "None". A blank is marked as no answer.

Trick

When the order compares a measurement, sort the elements into levels by that measurement, then draw the Hasse diagram. Read off: bottom elements are minimal, top elements are maximal. In a finite poset a smallest (largest) element exists exactly when there is only one minimal (maximal) element.

Q4 · Every linearization of Q3

Sheet 6 §4

Question. For the partial order of Q3, one linearization is $11 \preccurlyeq^* 13 \preccurlyeq^* 14 \preccurlyeq^* 15 \preccurlyeq^* 16 \preccurlyeq^* 12$. Write out all the other possible linearizations.

Answer (worked out here, no official key)

```
11 ≼* 13 ≼* 15 ≼* 14 ≼* 16 ≼* 12
13 ≼* 11 ≼* 14 ≼* 15 ≼* 16 ≼* 12
13 ≼* 11 ≼* 15 ≼* 14 ≼* 16 ≼* 12
```

So there are 4 linearizations in total, counting the given one.

Break it down

1. A linearization must keep every relationship of $\preccurlyeq$. From Q3's levels: both 11 and 13 come before both 14 and 15; 14 and 15 come before 16; 16 comes before 12.
2. The only freedom is inside a level, between incomparable elements: the order of 11 and 13 (2 ways) and the order of 14 and 15 (2 ways). That gives $2 \times 2 = 4$ queues. Remove the given one: 3 left.
3. Kahn's algorithm view: at the start, 11 and 13 have nothing below them, so pick either (2 choices). Next, only the other of the two is free (14 and 15 still have it below them), so that step is forced. Then 14 and 15 are both free (2 choices), then the other one, then 16, then 12.
4. Why can 14 never come before 13? Because $13 \preccurlyeq 14$ (2 divisors against 4), so $13 \preccurlyeq^* 14$ in every linearization.

Trick

When the Hasse diagram is a stack of levels where the elements inside a level are incomparable and every element of one level is below every element of the next (as here), the number of linearizations is the product of (level size)! over the levels: $2! \times 2! \times 1! \times 1! = 4$. For irregular diagrams, count by running Kahn's algorithm and branching on each choice. Exam version: AY23/24 midterm Q15 asked which of four lists are linearizations of this same order. Only the lists that match one of the four above are valid.

Q5(a) · Coordinate-wise order on $\{0,1\}^2$

Sheet 6 §1

Question. $B = \{0,1\}$. Define R on $B \times B$ by $(a,b)\ R\ (c,d) \Leftrightarrow (a \leq c) \wedge (b \leq d)$. (a) Prove that R is a partial order.

Answer (worked out here, no official key)

```
1. (reflexive)
1.1 Let (a,b) ∈ B × B.
1.2 a ≤ a and b ≤ b.          (≤ is reflexive)
1.3 So (a,b) R (a,b).   (by definition of R)
2. (antisymmetric)
2.1 Let (a,b), (c,d) ∈ B × B with
    (a,b) R (c,d) and (c,d) R (a,b).
2.2 a ≤ c, b ≤ d, c ≤ a, d ≤ b.
                            (by definition of R)
2.3 a = c and b = d.    (≤ is antisymmetric)
2.4 So (a,b) = (c,d).
              (by equality of ordered pairs)
3. (transitive)
3.1 Let (a,b), (c,d), (e,f) ∈ B × B with
    (a,b) R (c,d) and (c,d) R (e,f).
3.2 a ≤ c, b ≤ d, c ≤ e, d ≤ f.
                            (by definition of R)
3.3 a ≤ e and b ≤ f.        (≤ is transitive)
3.4 So (a,b) R (e,f).   (by definition of R)
4. By 1, 2 and 3, R is a partial order.
            (by definition of partial order)
```

Break it down

1. $B \times B = \{(0,0), (0,1), (1,0), (1,1)\}$: four ordered pairs. R compares two pairs coordinate by coordinate: (a,b) is below (c,d) when the first coordinates satisfy $\leq$ *and* the second coordinates satisfy $\leq$.
2. "Partial order" means three separate $\forall$ statements to prove. Each one uses the same three moves: unpack R into $\leq$ facts (definition of R), apply the matching property of $\leq$ on numbers, pack back into R.
3. Antisymmetry needs one extra fact: two ordered pairs are equal exactly when both coordinates are equal, $(a,b) = (c,d) \Leftrightarrow a = c \wedge b = d$.
4. Because $B \times B$ is finite, you could also list R (9 pairs: each element with itself, $(0,0)$ with the other three, and $(0,1)$, $(1,0)$ with $(1,1)$) and check each property. The general proof is shorter and is what the tutorial wants.

Trick

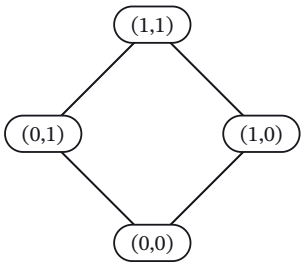
For a relation built from $\leq$ (or $\subseteq$, or $|$), every property "passes through" to the familiar relation: unpack, use the property of $\leq$, repack. Write the three blocks in the same shape every time.

Q5(b)(c) · The diamond and its extremes

Sheet 6 §1

Question. (b) Draw the Hasse diagram for R. (c) Find the maximal, largest, minimal and smallest elements.

Answer (worked out here, no official key)



A diamond. No line from (0,0) to (1,1): it is implied by transitivity.

```
Maximal: (1,1)       Largest:  (1,1)
Minimal: (0,0)       Smallest: (0,0)
```

Break it down

1. $(0,0)\ R$ everything, because $0 \leq$ anything in B. Everything $R\ (1,1)$, because anything in B is ≤ 1.
2. $(0,1)$ and $(1,0)$ are incomparable: $(0,1)\ R\ (1,0)$ needs $1 \leq 0$ in the second coordinate, and $(1,0)\ R\ (0,1)$ needs $1 \leq 0$ in the first. Both false. So they sit side by side on the middle level.
3. Covers: $(0,0)$ to each middle element, each middle element to $(1,1)$. Nothing fits between these. $(0,0)$ to $(1,1)$ is not a cover, because $(0,1)$ sits between them.
4. $(0,0)$ is below everything, so it is smallest, and a smallest element is also the only minimal one. Same for $(1,1)$ at the top: largest and the only maximal element.

Trick

A coordinate-wise order on pairs has "incomparable twins" as soon as each coordinate can take two different values, like $(0,1)$ and $(1,0)$ here: one is bigger in the first coordinate, the other in the second. That pair is the usual witness for "not a total order".

Q5(d) · Is it well-ordered?

Sheet 6 §1

Question. (d) Is $(B \times B, R)$ well-ordered?

Answer (worked out here, no official key)

No. R is not a total order, since $(0,1)$ and $(1,0)$ are incomparable. Concretely, the non-empty subset $\{(0,1), (1,0)\}$ has no smallest element: neither is below the other.

Break it down

1. **Well-ordered**: a total order in which every non-empty subset has a smallest element. Example: $(\mathbb{N}, \leq)$ is well-ordered. $(\mathbb{Z}, \leq)$ is not, because $\mathbb{Z}$ itself has no smallest element.
2. First requirement fails: a total order needs every two elements comparable, and $(0,1)$, $(1,0)$ are not.
3. Even if your notes phrase the definition without "total", the second requirement fails too: pick the subset $\{(0,1), (1,0)\}$. A smallest element of it would have to be below the other one, and neither is.

4. Note that the whole set $B \times B$ does have a smallest element, $(0,0)$. That is not enough: *every* non-empty subset needs one.

> **Trick**
> For a finite poset, well-ordered ⇔ total order ⇔ the Hasse diagram is one vertical line. Any side-by-side pair (like the middle of the diamond) kills it, and that pair is your counterexample subset.

Q6(a)(b) · Adding every loop

Sheet 5 §2

Question. R is a relation on a non-empty set A. Define S on A by $x\,S\,y \Leftrightarrow (x = y) \vee (x\,R\,y)$. Show that (a) S is reflexive; (b) $R \subseteq S$.

Answer (worked out here, no official key)

```
(a)
1. Let x ∈ A.
2. x = x is true, so (x = x) ∨ (x R x)
   is true.     (an "or" is true if one part is)
3. So x S x.              (by definition of S)
4. Hence S is reflexive.
                    (by definition of reflexive)
(b)
1. Let (x,y) ∈ R, that is, x R y.
2. So (x = y) ∨ (x R y) is true.
3. So x S y, that is, (x,y) ∈ S.
                          (by definition of S)
4. Hence R ⊆ S.     (by definition of subset)
```

Break it down

1. In the arrow picture, S is R plus a loop on every element: $S = R \cup \{(x,x) : x \in A\}$. Both parts are then obvious in the picture; the proof just says it formally.
2. Both proofs use one logic fact: $p \vee q$ is true as soon as one of p, q is true. In (a) the true part is $x = x$. In (b) it is $x\,R\,y$.
3. (b) is a subset proof. Subset proofs always have the same skeleton: "let an element of the left set; show it belongs to the right set". For relations, the elements are ordered pairs.

> **Trick**
> For any "show X has property P" or "show $X \subseteq Y$" question, first write the definition of the target with arbitrary elements ("let $x \in A$", "let $(x,y) \in R$"). The rest is usually one line of logic.

Q6(c) · The smallest such relation

Sheet 5 §2

Question. (c) If S' is another reflexive relation on A and $R \subseteq S'$, then $S \subseteq S'$. What is this relation S called? (Hint: transitive closure in Lecture 6.)

Answer (worked out here, no official key)

```
1. Let S′ be a reflexive relation on A
   with R ⊆ S′.
2. Let (x,y) ∈ S.
3. (x = y) ∨ (x R y).     (by definition of S)
4. Case 1: x = y.
4.1 (x,x) ∈ S′, i.e. (x,y) ∈ S′.
                      (by reflexivity of S′)
5. Case 2: x R y.
5.1 (x,y) ∈ R, so (x,y) ∈ S′.
                              (since R ⊆ S′)
6. In both cases (x,y) ∈ S′.  (proof by cases)
7. Hence S ⊆ S′.     (by definition of subset)
```

S is the **reflexive closure** of R: the smallest reflexive relation on A that contains R.

Break it down

1. Step 3 gives an "or". When you know "P or Q" but not which, use **proof by cases**: show the goal assuming P, then show it assuming Q. Either way the goal holds.
2. Case 1 is a loop, and S' has every loop because it is reflexive. Case 2 is an original arrow of R, and S' has every arrow of R because $R \subseteq S'$.
3. A **closure** of R for a property P is the smallest relation that (1) has property P, (2) contains R, and (3) sits inside every other relation that has P and contains R. Parts (a), (b), (c) check exactly these three clauses for "reflexive".
4. The three closures you meet: reflexive closure = add every loop, $R \cup \{(x,x) : x \in A\}$. Symmetric closure = add every reverse arrow, $R \cup R^{-1}$. Transitive closure = keep adding shortcuts $x \to z$ for every path $x \to y \to z$ until nothing changes.

> **Trick**
> Part (3) of any closure proof always goes the same way: take an element of the closure, ask why it got in (it was in R, or it was added to force the property), and show the other relation must contain it for the same reason.

Q7(a)(b)(d) · Three example relations

Sheet 0B §4

Question. Antisymmetric: $\forall x,y \in A\ (x\,R\,y \wedge y\,R\,x \Rightarrow x = y)$. Asymmetric: $\forall x,y \in A\ (x\,R\,y \Rightarrow \sim(y\,R\,x))$. Find a relation on A that is (a) both asymmetric and antisymmetric; (b) not asymmetric but antisymmetric; (d) neither asymmetric nor antisymmetric.

Answer (worked out here, no official key)

Let a, b be two different elements of A (for example $A = \{1,2\}$, $a = 1$, $b = 2$).

part	R	asym?	antisym?
(a)	$\{(a,b)\}$	yes	yes
(b)	$\{(a,a)\}$	no	yes
(d)	$\{(a,b), (b,a)\}$	no	no

Other valid answers: (a) $\varnothing$, or $<$ with $A = \mathbb{Z}$; (b) the identity relation on A, or $\leq$ with $A = \mathbb{Z}$.

Break it down

1. Turn both definitions into arrow rules. Antisymmetric: no two-way arrows between *different* elements; loops allowed. Asymmetric: no arrow may have its reverse. A loop is its own reverse (put $x = y$: $x\,R\,x \to \sim(x\,R\,x)$), so asymmetric also bans loops.
2. So there are three basic shapes to build with:

shape	asym?	antisym?
one-way arrow $a \to b$	ok	ok
loop at a	breaks	ok
$a \rightleftarrows b$ with $a \neq b$	breaks	breaks

3. (a): use only one-way arrows. $\{(a,b)\}$: the only arrow is $a \to b$ and $b \to a$ is absent, so asymmetric; no two-way pair, so antisymmetric.

4. (b): use a loop. $\{(a,a)\}$: $a\ R\ a$ holds, so asymmetry would demand $\sim(a\ R\ a)$, which fails. Antisymmetric: the only case with arrows both ways is $x = y = a$, and then $x = y$ holds.
5. (d): use a two-way arrow between different elements. $a\ R\ b$ and $b\ R\ a$ break asymmetry, and since $a \neq b$ they also break antisymmetry. This needs A to have at least two elements; on a one-element set every relation is antisymmetric.

> **Trick**
> Asymmetric = antisymmetric + no loops. To fail exactly one of the two, add a loop. To fail both, add a two-way arrow between different elements.

Q7(c) · The impossible combination

Sheet 0B §4

Question. (c) Find a binary relation on A that is asymmetric but not antisymmetric.

Answer (worked out here, no official key)

No such relation exists: every asymmetric relation is antisymmetric.

```
1. Suppose R is asymmetric.
2. Let x, y ∈ A with x R y and y R x.
3. From x R y: ~(y R x).
                        (by asymmetry of R)
4. 3 contradicts y R x from 2. So no x, y
   satisfy the "if" part of antisymmetry.
5. So x R y ∧ y R x → x = y is vacuously
   true for all x, y: R is antisymmetric.
6. Hence no relation is asymmetric but
   not antisymmetric.
```

Break it down

1. Antisymmetry is an "if ... then" rule whose "if" part is "$x\,R\,y \wedge y\,R\,x$" (arrows both ways). Asymmetry guarantees that this "if" part never happens. So antisymmetry is vacuously true, exactly as in D1.
2. In arrow terms: asymmetric bans all two-way arrows and all loops; antisymmetric bans only two-way arrows between different elements. The asymmetric rules include the antisymmetric ones, so asymmetric is the stronger property.
3. A "find an example" question can have "impossible" as its answer. Say so, and prove it.

> **Trap**
> This exact fact was examined: AY23/24 midterm Q10(i) "Every asymmetric relation is antisymmetric" is **true**. The same paper's Q10(ii) used $\{(a,b)\}$ on $\{a,b\}$ as a relation that is not reflexive but is asymmetric.

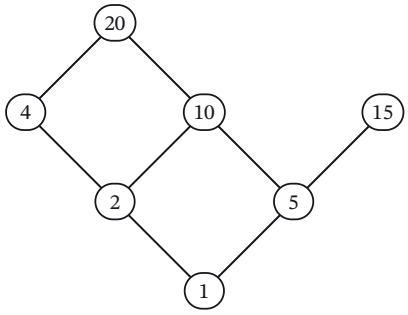
> **Trick**
> When one property's arrow rules contain all of another's, the stronger implies the weaker, and "strong but not weak" is impossible. Check for this before hunting for an example that cannot exist.

Q8 · Comparable versus compatible

Sheet 6 §5

Question. For a partial order $\preccurlyeq$ on A: a, b are **comparable** iff $a \preccurlyeq b$ or $b \preccurlyeq a$; **compatible** iff some $c \in A$ has $a \preccurlyeq c$ and $b \preccurlyeq c$. For "divides" on $A = \{1,2,4,5,10,15,20\}$, list the pairs $\{x,y\}$ of distinct elements that are (a) comparable; (b) compatible. Are two comparable elements always compatible? Are two compatible elements always comparable?

Answer (worked out here, no official key)



Hasse diagram of divides on A. Maximal elements: 15 and 20.

```
(a) comparable: 14 pairs
{1,2} {1,4} {1,5} {1,10} {1,15} {1,20}
{2,4} {2,10} {2,20} {4,20}
{5,10} {5,15} {5,20} {10,20}

(b) compatible: 17 pairs
all 14 above, plus {2,5} {4,5} {4,10}
(not compatible: {2,15} {4,15} {10,15} {15,20})
```

Comparable ⇒ compatible: always. Compatible ⇒ comparable: not always, e.g. $\{2,5\}$: both divide 10, but $2 \nmid 5$ and $5 \nmid 2$.

Break it down

1. Comparable here means "one divides the other". List it systematically: for each element, list its multiples further along in A. 1 divides all six others; 2 divides 4, 10, 20; 4 divides 20; 5 divides 10, 15, 20; 10 divides 20; 15 and 20 divide nothing else.
2. Compatible means the two have a common "upper element" inside A. For divides, that is a common multiple that is in A. In the Hasse diagram: some element you can reach by walking up from both.
3. Shortcut: every element sits below some maximal element (keep walking up until you cannot). So a, b are compatible exactly when both sit below the same maximal element. Below 20: $\{1,2,4,5,10,20\}$. Below 15: $\{1,5,15\}$. Pairs inside the first group: $6{\cdot}5/2 = 15$. Inside the second: 3. $\{1,5\}$ is counted twice, so $15 + 3 - 1 = 17$.
4. Count check: 7 elements give $7{\cdot}6/2 = 21$ pairs. $21 - 14 = 7$ incomparable pairs. Of those, $\{2,5\}$ (via 10), $\{4,5\}$ and $\{4,10\}$ (via 20) are compatible. The other four all pair 15 with an even number; their common multiples (30, 60) are not in A.
5. Comparable ⇒ compatible, proof: suppose $a \preccurlyeq b$ (the case $b \preccurlyeq a$ is the same with roles swapped). Also $b \preccurlyeq b$ by reflexivity. Take $c = b$: then $a \preccurlyeq c$ and $b \preccurlyeq c$. So a, b are compatible.
6. Compatible ⇏ comparable: one counterexample settles a "for all" question. $\{2,5\}$ with $c = 10$.

Trick

Comparable = one path going up from one element to the other. Compatible = two upward paths that meet. A comparable pair is compatible because the meeting point can be the upper element itself. To count compatible pairs, group the elements under each maximal element.

Q9(a) · Maximal chains in a power set

Sheet 6 §3

Question. A **chain** is a subset C of A in which every two elements are comparable. A **maximal chain** is a chain M such that $t \notin M \Rightarrow M \cup \{t\}$ is not a chain. The **length** of a chain is one less than its number of elements. (a) $A = \{a,b,c,d\}$. Write out two maximal chains in $(\mathcal{P}(A), \subseteq)$.

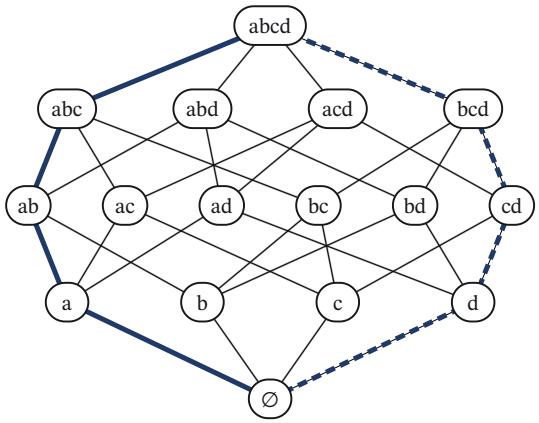
Answer (worked out here, no official key)

$M_1 = \{\varnothing, \{a\}, \{a,b\}, \{a,b,c\}, \{a,b,c,d\}\}$

$M_2 = \{\varnothing, \{d\}, \{c,d\}, \{b,c,d\}, \{a,b,c,d\}\}$

Each has 5 elements, so length 4. (There are $4! = 24$ maximal chains, one for each order of adding a, b, c, d.)

Why M_1 is maximal: let $T \subseteq A$ with $T \notin M_1$. Then T is not $\varnothing$ or A, so $|T|$ is 1, 2 or 3. M_1 has exactly one set X of that size. $X \neq T$ and they have the same size, so neither is a subset of the other. So $M_1 \cup \{T\}$ contains an incomparable pair and is not a chain.



Hasse diagram of $(\mathcal{P}(A), \subseteq)$, with "ab" short for $\{a,b\}$. Solid navy: M_1. Dashed navy: M_2.

Break it down

1. $\mathcal{P}(A)$, the **power set**, is the set of all subsets of A: $2^4 = 16$ of them. The order is $\subseteq$. So the elements of a chain here are sets, and a chain is a set of sets that are all nested inside each other.
2. In the Hasse diagram, a chain is a set of elements lying on one upward path. A maximal chain is one you cannot extend: adding any outside element creates an incomparable pair.
3. Maximal is not the same as longest. It only means "cannot be extended". In $\mathcal{P}(A)$ all maximal chains happen to have the same length, but Q9(b) shows this is not true in general.
4. How to build one. It must contain $\varnothing$: $\varnothing \subseteq$ every set, so if $\varnothing$ were missing you could add it and still have a chain. Same for A at the top. And it must climb one element at a time: if it jumped from $\{a\}$ to $\{a,b,c\}$, you could insert $\{a,b\}$.
5. Write the answer as a set of sets, with braces: $\{\varnothing, \{a\}, \ldots\}$. Writing $\varnothing \subseteq \{a\} \subseteq \ldots$ shows the idea, but a chain is a subset, so give it in set notation.

Trick

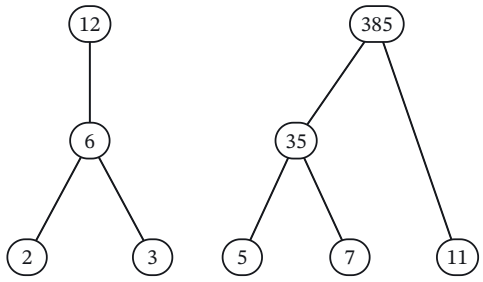
A maximal chain in a finite poset is a path in the Hasse diagram that starts at a minimal element, ends at a maximal element, and uses cover lines only. In $\mathcal{P}(A)$: start at $\varnothing$, add one element at a time, stop at A.

Q9(b) · Chains of different lengths

Sheet 6 §3

Question. [AY22/23 Sem 2 midterm] $B = \{2,3,5,6,7,11,12,35,385\}$ with the divides relation. Draw the Hasse diagram and write out two maximal chains of different lengths in $(B, |)$.

Answer (worked out here, no official key)



Covers: 2–6, 3–6, 6–12, 5–35, 7–35, 35–385, 11–385.

```
{11, 385}     length 1
{2, 6, 12}    length 2
All five maximal chains:
{2,6,12} {3,6,12} {5,35,385} {7,35,385}
(length 2) and {11,385} (length 1).
```

Break it down

1. Factorise first: $6 = 2 \cdot 3$, $12 = 2^2 \cdot 3$, $35 = 5 \cdot 7$, $385 = 5 \cdot 7 \cdot 11$. Then $x \mid y$ exactly when every prime factor of x (with repeats) also appears in y.
2. Covers: for each divisibility pair, check for an element in between. $2 \mid 12$, but $2 \mid 6 \mid 12$, so no line 2–12. $5 \mid 385$ goes through 35, and so does $7 \mid 385$. For $11 \mid 385$, the only candidate in between is 35, and $11 \nmid 35$. So 11–385 is a cover that skips a level.
3. Nothing in $\{2,3,6,12\}$ divides or is divided by anything in $\{5,7,11,35,385\}$, so the diagram falls into two pieces.
4. Maximal chains are the upward paths from a minimal element (2, 3, 5, 7, 11) to a maximal element (12, 385) along covers. That gives the five listed.
5. Why $\{11, 385\}$ is maximal: in B, the only elements comparable with 11 are 11 and 385 (1 is not in B). So any other t is incomparable with 11, and adding t breaks the chain.
6. Why $\{2, 6, 12\}$ is maximal: every other element (3, 5, 7, 11, 35, 385) is incomparable with 2.

Trick

A short maximal chain appears where a cover line skips a level, like 11 straight up to 385. When asked for chains of different lengths, look for the element with the fewest steps to the top.

T6 · Tutorial 6 walkthrough

Functions (Epp ch. 7) · no official key, answers worked out here

Tutorial 6 is about functions: deciding whether a rule even is a function, computing images and preimages, and proving injective, surjective or bijective. Three moves crack almost every question: check that every input gets exactly one output inside the codomain; start an injectivity proof from $f(x_1) = f(x_2)$ and a surjectivity proof from an arbitrary element of the codomain; and remember that f^{-1} of a *set* always exists, while f^{-1} as a *function* needs a bijection.

D1 · Is it even a function?

Sheet 8 §1

Question. Which of these is a function? If not, explain.
(a) $f: \mathbb{Z} \to \mathbb{Z}$, $f(z) = 1$ if $2 \mid z$, and $f(z) = 2$ if $3 \mid z$.
(b) $f: \mathbb{Z} \to \mathbb{Z}$, $f(z) = 1$ if $2 \mid z$, and $f(z) = 2$ if $2 \nmid z$.
(c) $f: \mathbb{R} \to \mathbb{Z}$, $f(x) = 2x$.
(d) $f: \mathbb{Z} \to \mathbb{R}$, $f(x) = 2x$.

Answer (worked out here, no official key)

Part	Function?	Reason
(a)	No	$f(6)$ gets two values: $2 \mid 6$ says 1, $3 \mid 6$ says 2. Also $f(1)$ gets no value at all.
(b)	Yes	Every integer is even or odd, never both, so exactly one case applies, and $1, 2 \in \mathbb{Z}$.
(c)	No	$f(1/4) = 1/2 \notin \mathbb{Z}$: the output escapes the codomain.
(d)	Yes	$2x$ is one definite number, and $2x \in \mathbb{Z} \subseteq \mathbb{R}$.

Term

Function $f: A \to B$: a rule that sends *every* element of A to *exactly one* element of B. Formally it is a relation (a set of pairs) with $\forall x \in A\ \exists! y \in B\ ((x, y) \in f)$, where $\exists!$ reads "there exists exactly one". A is the **domain** (the allowed inputs). B is the **codomain** (the declared target set; every output must land in it). A rule that really passes this test is called **well defined**.

Break it down

1. **Read the notation.** $2 \mid z$ means "2 divides z": $z = 2k$ for some integer k, so z is even. $2 \nmid z$ means 2 does not divide z, so z is odd. A definition with a big brace and "if" conditions is a **piecewise** definition: to evaluate it, you use the line whose condition your input satisfies.
2. **Three checks for any claimed function.** Gap: is there an input that gets no output? Clash: is there an input that gets two different outputs? Escape: is there an output that is not in the codomain? One concrete input that fails any check is a complete "not a function" answer.
3. **(a)** Try small inputs. $z = 1$: 2 does not divide 1 and 3 does not divide 1, so neither line applies. That is a gap. $z = 6$: both conditions hold and the lines disagree (1 versus 2). That is a clash. Either one alone is enough.
4. **(b)** The two conditions are a statement and its negation, so for every z exactly one of them is true: no gap, no clash. Both possible outputs, 1 and 2, are integers: no escape.
5. **(c)** The formula $2x$ always gives one number, so there is no gap or clash. The problem is the codomain. f promises integer outputs, but $x = 1/4$ gives $1/2$, which is not an integer.
6. **(d)** Same formula, but now the inputs are integers, so $2x$ is an integer, and every integer is a real number. A codomain much bigger than the actual outputs is fine: a function does not have to hit every element of its codomain.

Trap

The domain and codomain are part of the function. The same formula $2x$ is a function in (d) and not in (c). Always read the line $f: A \to B$ before the formula.

Trick

"Is this a function?" means hunt for a gap, a clash or an escape. Test the edge inputs: numbers that satisfy no condition, numbers that satisfy two conditions at once (6 for "$2 \mid z$" and "$3 \mid z$"), and non-integers when the codomain is $\mathbb{Z}$. Overlapping cases are only a problem when they give different values.

D2 · Finding preimages

Sheet 8 §1

Question. $f: \mathbb{Z} \times \mathbb{Z} \to \mathbb{Q}$ is defined by $f(x, y) = (x + y)/3$ for all $x, y \in \mathbb{Z}$. Find three distinct preimages of 2.

The PDF text of this formula is garbled; it decodes as $(x + y)/3$. If your printed sheet shows something else, the method below is unchanged.

Answer (worked out here, no official key)

A preimage of 2 is an input pair (x, y) with $f(x, y) = 2$, that is, $x + y = 6$. Three of them: $(0, 6)$, $(1, 5)$ and $(2, 4)$.

```
f(0, 6) = (0 + 6)/3 = 2
f(1, 5) = (1 + 5)/3 = 2
f(2, 4) = (2 + 4)/3 = 2
```

Break it down

1. $\mathbb{Z} \times \mathbb{Z}$ (the **Cartesian product**) is the set of all **ordered pairs** (x, y) with $x, y \in \mathbb{Z}$. So each input of f is a pair of integers, and $f(x, y)$ is short for $f((x, y))$.
2. **Image** and **preimage**: if $f(a) = b$, then b is the image of a, and a is a preimage of b. Every input has exactly one image (that is what "function" means). An output can have many preimages, or none.
3. So "find preimages of 2" means: solve $f(\text{input}) = 2$ for the input. Here $(x + y)/3 = 2 \Leftrightarrow x + y = 6$ (multiply both sides by 3).
4. Any integer pair that adds to 6 works. The answer must be pairs, not single numbers, because the domain is $\mathbb{Z} \times \mathbb{Z}$.
5. "Distinct" means different as pairs. Ordered pairs care about order, so $(1, 5)$ and $(5, 1)$ are two different preimages.
6. Bonus: the full set of preimages is $\{(x, 6 - x) : x \in \mathbb{Z}\}$, infinitely many. So f is not **injective** (injective means different inputs always give different outputs).

Trick

Preimage of b: solve "$f(input) = b$" for the input, then write the answer in the shape of the domain (here, integer pairs). Plug each answer back in to check.

D3 · Floor, ceiling and equal functions

Sheet 8 §1

Question. For real x: the floor $\lfloor x \rfloor$ is the unique integer n with $n \le x < n+1$; the ceiling $\lceil x \rceil$ is the unique integer n with $n-1 < x \le n$. Let $f, g: \mathbb{Q} \to \mathbb{Q}$ with $f(x) = \lfloor x \rfloor + 1$ and $g(x) = \lceil x \rceil$. What is the range of f? Of g? Is $f = g$? Why?

Answer (worked out here, no official key)

Range of f = $\mathbb{Z}$. Range of g = $\mathbb{Z}$. But $f \neq g$: they have the same domain and codomain, yet $f(0) = \lfloor 0 \rfloor + 1 = 1$ while $g(0) = \lceil 0 \rceil = 0$.

```
Range of f is ℤ.
1. (⊆) Let x ∈ ℚ. ⌊x⌋ ∈ ℤ   (def. of floor)
   so f(x) = ⌊x⌋ + 1 ∈ ℤ
   (closure of integers under +)
2. (⊇) Let n ∈ ℤ. Then n − 1 ∈ ℚ, and
   n − 1 ≤ n − 1 < n, so ⌊n − 1⌋ = n − 1
   (def. of floor). So f(n − 1) = n.
Range of g is ℤ.
1. (⊆) g(x) = ⌈x⌉ ∈ ℤ   (def. of ceiling)
2. (⊇) Let n ∈ ℤ. n − 1 < n ≤ n, so
   ⌈n⌉ = n (def. of ceiling): g(n) = n.
```

Term

The **range** of $f: A \to B$ is the set of outputs that actually occur, $\{f(x) : x \in A\}$. It is a subset of the codomain and can be smaller: here the codomain is $\mathbb{Q}$ but the range is only $\mathbb{Z}$. Two functions are **equal** iff they have the same domain, the same codomain, and the same value at every input.

Break it down

1. Floor rounds down to an integer: $\lfloor 2.7 \rfloor = 2$, $\lfloor 5 \rfloor = 5$, and $\lfloor -2.3 \rfloor = -3$ (down means toward $-\infty$, not toward 0). Ceiling rounds up: $\lceil 2.3 \rceil = 3$, $\lceil 5 \rceil = 5$, $\lceil -2.7 \rceil = -2$. The phrase "the unique integer n with $n \le x < n+1$" says: x sits in exactly one block from one integer up to (not including) the next, and the floor is the left end of that block.
2. "Range = $\mathbb{Z}$" is a set equality, so prove two inclusions. ($\subseteq$): every output is an integer. ($\supseteq$): every integer is an output, which means you must produce an input that hits it. f adds 1 at the end, so feed it $n-1$ to get n out. For g, feed n itself.
3. Now compare f and g on test values. A non-integer, say 2.5: $f(2.5) = 2 + 1 = 3$ and $g(2.5) = 3$. Same. An integer, say 2: $f(2) = 2 + 1 = 3$ but $g(2) = 2$. Different.
4. Why: if x is not an integer, it sits strictly between $\lfloor x \rfloor$ and $\lfloor x \rfloor + 1$, so rounding up gives $\lfloor x \rfloor + 1$. If x is an integer, floor and ceiling both leave it alone, so $f(x) = x + 1$ but $g(x) = x$.
5. $f = g$ needs agreement at every input. Its negation needs just one input where they differ ($\exists x \in \mathbb{Q}\, (f(x) \neq g(x))$), and $x = 0$ does it. Having the same range does not make two functions equal.

Trick

"Is $f = g$?": check the domain, check the codomain, then hunt for one input where they differ. With floor and ceiling, the integers are always the first edge case to test.

D4 · What is wrong with Aiken's proof

Sheet 8 §2

Question. Aiken's "proof" that a composition of two surjections is a surjection:

1. Suppose $f: X \to Y$ and $g: Y \to Z$ are surjections.
2. Then $\forall y \in Y\, \exists x \in X$ such that $f(x) = y$ as f is surjective,
3. and $\forall z \in Z\, \exists y \in Y$ such that $g(y) = z$ as g is surjective.
4. So $(g \circ f)(x) = g(f(x)) = g(y) = z$.
5. Hence $g \circ f$ is a surjection.

Explain the mistakes.

Answer (worked out here, no official key)

- The goal is $\forall z \in Z\, \exists x \in X\, ((g \circ f)(x) = z)$. The proof never says "let $z \in Z$" (an arbitrary element of the codomain), so it never proves the "for every z" part.
- Lines 2 and 3 only restate the definitions. Their x, y, z are placeholders that live inside those quantified sentences. Line 4 then uses x, y, z as if they were specific elements, but nothing was ever picked.
- Line 4 assumes the y of line 2 is the same as the y of line 3. Nothing makes them equal: in line 2, y is any element of Y; in line 3, y is one that g sends to z.
- So line 5 does not follow.

A correct proof:

```
1. Suppose f: X → Y and g: Y → Z are
   surjections.
2. Let z ∈ Z.                    (arbitrary)
2.1. There is y ∈ Y with g(y) = z.
     (g is surjective)
2.2. There is x ∈ X with f(x) = y.
     (f is surjective, and y ∈ Y)
2.3. (g∘f)(x) = g(f(x)) = g(y) = z
     (def. of composition, 2.2, 2.1)
3. So ∀z ∈ Z ∃x ∈ X ((g∘f)(x) = z),
   that is, g∘f is surjective.
   (def. of surjective)
```

Term

Surjective (onto) $f: A \to B$: every element of the codomain is hit, $\forall y \in B\, \exists x \in A\, (f(x) = y)$. **Composition**: for $f: X \to Y$ and $g: Y \to Z$, the function $g \circ f: X \to Z$ is $(g \circ f)(x) = g(f(x))$. Read right to left: f first, then g.

Break it down

1. Before writing anything, write the goal in symbols: "$g \circ f$ is surjective" means $\forall z \in Z\, \exists x \in X\, ((g \circ f)(x) = z)$.
2. A $\forall$ goal is proved by "Let $z \in Z$", assuming nothing else about z (that is what **arbitrary** means). An $\exists$ goal is proved by naming a specific x built from what you have. So the skeleton is: let z; build x; check $(g \circ f)(x) = z$.
3. Using a hypothesis is different from quoting it. To use "g is surjective", plug in your specific z and name the element it gives you: "so there is $y \in Y$ with $g(y) = z$". That named element is called a **witness**. Aiken copied the hypotheses and never plugged anything in.

4. In "$\forall y \in Y\ \exists x \in X\ (f(x) = y)$", the letters x and y are **bound variables**: placeholders that only mean something inside that sentence. They mean nothing on the next line. Also, x depends on y: a different y generally needs a different x.
5. Order of construction: $g \circ f$ goes $X \to Y \to Z$, but you build the preimage backwards. From z, get y using g. From that y, get x using f. Step 2.2 works only because the y from 2.1 is in Y, and f's surjectivity covers every element of Y.

Trap

Starting from some x and computing $(g \circ f)(x) = z$ only shows that some output exists. It says nothing about every z being hit. A surjectivity proof must start from an arbitrary element of the codomain and build its preimage.

Trick

Every surjectivity proof opens with "Let y be an arbitrary element of the codomain" and ends with "so $f(x) = y$" for an x you built. Each time you use an $\exists$ fact, give its witness a name and treat it as fixed from then on.

Q1 · Which relations are functions?

Sheet 8 §1

Question. On $\mathbb{N}$ define $x\ R_1\ y \Leftrightarrow x^2 = y^2$; $x\ R_2\ y \Leftrightarrow y \mid x$; $x\ R_3\ y \Leftrightarrow y = x + 1$. Are R_1, R_2 and R_3 functions? Prove or disprove.

Answer (worked out here, no official key)

R_1: yes (it is $x \mapsto x$). R_2: no. R_3: yes (it is $x \mapsto x + 1$).

```
R₁ is a function.
1. Let x ∈ ℕ.
2. (exists) x² = x², so x R₁ x.  (def. of R₁)
3. (unique) Suppose x R₁ y₁ and x R₁ y₂,
   where y₁, y₂ ∈ ℕ.
3.1. y₁² = x² = y₂²              (def. of R₁)
3.2. (y₁ − y₂)(y₁ + y₂) = 0      (basic algebra)
3.3. y₁ − y₂ = 0 or y₁ + y₂ = 0
     (zero product property)
3.4. If y₁ + y₂ = 0, then y₁ = y₂ = 0,
     as y₁, y₂ ≥ 0               (ℕ has no negatives)
3.5. Either way, y₁ = y₂.
4. Each x ∈ ℕ has exactly one y with x R₁ y,
   so R₁ is a function.

R₂ is not a function.
1. 2 = 1·2 and 2 = 2·1, so 1 | 2 and 2 | 2.
   So 2 R₂ 1 and 2 R₂ 2.         (def. of R₂)
2. Input 2 has two outputs, and 1 ≠ 2.

R₃ is a function.
1. Let x ∈ ℕ.
2. (exists) x + 1 ∈ ℕ and x R₃ (x + 1).
   (closure of ℕ under +; def. of R₃)
3. (unique) If x R₃ y₁ and x R₃ y₂, then
   y₁ = x + 1 = y₂.              (def. of R₃)
4. So R₃ is a function.
```

Term

A **relation** on $\mathbb{N}$ is a set of pairs from $\mathbb{N} \times \mathbb{N}$, and $x\ R\ y$ means $(x, y) \in R$. Read the left element as the input and the right one as the output. The relation is a function iff $\forall x \in \mathbb{N}\ \exists! y \in \mathbb{N}\ (x\ R\ y)$. The arrow $x \mapsto x + 1$ reads "x maps to $x + 1$", another way of writing $f(x) = x + 1$.

Break it down

1. Proving "is a function" is two jobs. **Existence**: show one y that works. **Uniqueness**: assume two outputs y_1 and y_2 and prove $y_1 = y_2$. "Assume two, show they are equal" is the standard way to prove "at most one".
2. Disproving takes one input with no output, or one input with two different outputs.
3. R_1: $x^2 = y^2$ allows $y = x$ and $y = -x$. On $\mathbb{Z}$ that breaks it: both $1\ R\ 1$ and $1\ R\ (-1)$ would hold. But $\mathbb{N} = \{0, 1, 2, \ldots\}$ has no negatives, so $y = x$ is the only candidate. Step 3.4 is exactly where "we are in $\mathbb{N}$" gets used. If your proof never uses it, the proof is wrong, because the same claim is false on $\mathbb{Z}$.
4. R_2: watch the order. $x\ R_2\ y \Leftrightarrow y \mid x$ says the output y is a divisor of the input x, where $y \mid x$ means $x = yk$ for some integer k. Most numbers have several divisors (1 and themselves at least), so uniqueness fails. $x = 2$ with divisors 1 and 2 is the smallest clean counterexample.
5. R_3: the rule names exactly one number, $x + 1$, and adding 1 to a natural number gives a natural number, so the output stays in $\mathbb{N}$.

Trick

"Is relation R a function?" means "$\forall x\ \exists! y\ (x\ R\ y)$?". Prove existence by naming y, and uniqueness by "suppose y_1 and y_2 both work". Watch the underlying set: squares and "divides" behave differently on $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{R}$.

Q2 (a)(b) · Length of strings

Sheet 8 §1

Question. Let $A = \{s, u\}$ and define $len: A^* \to \mathbb{Z}_{\geq 0}$ by $len(\sigma)$ = the length of σ. (a) What is $len(suu)$? (b) What is $len(\{\varepsilon, ss, uu, ssss\})$?

The codomain is garbled in the PDF text; it must be $\mathbb{Z}_{\geq 0}$ (the non-negative integers), since the empty string has length 0.

Answer (worked out here, no official key)

(a) $len(suu) = 3$. (b) $len(\{\varepsilon, ss, uu, ssss\}) = \{0, 2, 4\}$.

Term

A **string** over the alphabet A is a finite sequence of symbols from A written side by side, like suu. A^* is the set of all strings over A of every length, including the **empty string** ε (length 0). **Setwise image**: for a set X of inputs, $f(X) = \{f(x) : x \in X\}$, the set of their outputs.

Break it down

1. (a) suu is the sequence s, u, u: three symbols, so its length is 3.
2. (b) Look at what is inside the brackets: a *set* of strings, not a string. len only takes strings, so len applied to a set means the setwise image: apply len to each member and collect the results.
3. $\varepsilon \mapsto 0$, $ss \mapsto 2$, $uu \mapsto 2$, $ssss \mapsto 4$. Collected: $\{0, 2, 2, 4\} = \{0, 2, 4\}$, because a set lists each element once.

4. ε versus $\varnothing$: ε is a string (an element of A^*) with no symbols; $\varnothing$ is a set with no elements. They are different kinds of object.

Trick

A function applied to a set of inputs returns the set of outputs. Your answer must be a set, with duplicates removed.

Q2 (c)(d) · Preimage set versus inverse function

Sheet 8 §1

Question. (c) What is $len^{-1}(\{3\})$? (d) Does len^{-1} exist? Explain.

Answer (worked out here, no official key)

(c) $\{sss, ssu, sus, suu, uss, usu, uus, uuu\}$, all $2^3 = 8$ strings of length 3.

(d) No. An inverse function exists only for a bijection, and *len* is not injective: $len(su) = len(us) = 2$ but $su \neq us$. (*len* is surjective, since n copies of s form a string of length n, but that is not enough.)

Term

Setwise preimage: $f^{-1}(Y) = \{x \in A : f(x) \in Y\}$, all inputs whose output lands in Y. It exists for every function. **Injective** (one-to-one): $\forall x_1, x_2 \in A\ (f(x_1) = f(x_2) \to x_1 = x_2)$, so different inputs give different outputs. **Bijective**: injective and surjective. The **inverse function** $f^{-1}: B \to A$ exists only when f is bijective, and $f^{-1}(y)$ is the unique x with $f(x) = y$.

Break it down

1. (c) Inside $len^{-1}(\ )$ is a set, $\{3\}$. So this is the setwise preimage: every string whose length is in $\{3\}$, that is, length exactly 3.
2. List them without missing any. Each of the 3 positions is s or u, so there are $2 \times 2 \times 2 = 8$ strings. Go in dictionary order: $sss, ssu, sus, suu, uss, usu, uus, uuu$.
3. (d) is a different question: is there a function $len^{-1}: \mathbb{Z}_{\geq 0} \to A^*$ that undoes *len*? It would have to send 3 back to *the* string of length 3. Part (c) shows there are eight. There is no single choice, so there is no inverse function.
4. In the language of definitions: an inverse needs every output to have exactly one preimage. "At least one" is surjective; "at most one" is injective. *len* has at least one but not at most one.
5. The notation f^{-1} is overloaded. $f^{-1}(a\ set)$ is the setwise preimage and always exists. $f^{-1}(an\ element)$ uses the inverse function and needs f bijective. Look at what is inside the brackets.

Trick

"Does f^{-1} exist?" means test injective and surjective separately. A preimage set $f^{-1}(\{y\})$ with two or more elements, like (c), is instant proof that f is not injective.

Q3 · Inverse of a composition

Sheet 8 §1

Question. For bijections $f: A \to B$ and $g: B \to C$, prove that $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Answer (worked out here, no official key)

```
1. f, g injective, so g∘f is injective
   (Theorem 7.3.3); f, g surjective, so
   g∘f is surjective (D4). So g∘f: A → C
   is a bijection and (g∘f)⁻¹: C → A exists.
2. g⁻¹: C → B and f⁻¹: B → A, so
   f⁻¹∘g⁻¹: C → A. Same domain and codomain.
3. Let z ∈ C, and let x = (g∘f)⁻¹(z).
3.1. (g∘f)(x) = z        (def. of inverse)
     so g(f(x)) = z      (def. of composition)
3.2. f(x) ∈ B and g(f(x)) = z, so
     g⁻¹(z) = f(x)       (def. of inverse of g)
3.3. f(x) = g⁻¹(z), so
     f⁻¹(g⁻¹(z)) = x     (def. of inverse of f)
3.4. (f⁻¹∘g⁻¹)(z) = x = (g∘f)⁻¹(z)
     (def. of composition; line 3)
4. Same domain, codomain and values, so
   (g∘f)⁻¹ = f⁻¹∘g⁻¹.
   (def. of equal functions)
```

Term

A **bijection** is injective and surjective, so every element of the codomain has exactly one preimage. The **inverse** of a bijection $F: X \to Y$ is $F^{-1}: Y \to X$ with $F^{-1}(y) = x \Leftrightarrow F(x) = y$.

Break it down

1. This is an equation between functions. The definition of equal functions tells you exactly what to prove: same domain, same codomain, same value at every input.
2. Check the types first (line 2). $g \circ f$ runs $A \to B \to C$, so its inverse runs $C \to A$. On the right, g^{-1} acts first ($C \to B$), then f^{-1} ($B \to A$). Both sides are $C \to A$. If the types did not match, the claim would be false before you start.
3. Line 1 matters: the left side only makes sense if $g \circ f$ is a bijection. Quote the two composition facts (injective from Theorem 7.3.3, surjective from the corrected proof in D4).
4. For the values, pick an arbitrary $z \in C$ and give the left side a name, x. The only thing you know about an inverse is its definition, so use it as a translator: "$x = F^{-1}(z)$" becomes "$F(x) = z$", and back.
5. Then peel off one function at a time. $g(f(x)) = z$ translates to $f(x) = g^{-1}(z)$. Then $f(x) = g^{-1}(z)$ translates to $x = f^{-1}(g^{-1}(z))$.
6. Why the order flips: socks and shoes. You put on socks (f), then shoes (g). To undo, the shoes come off first (g^{-1}), then the socks (f^{-1}).

Trick

In any proof with inverses, translate every "$F^{-1}(z) = x$" into "$F(x) = z$", work with plain equations, then translate back. For an equation between two functions, prove same domain, same codomain, same values.

Q4 (a) · $x \mapsto 12x + 31$ on $\mathbb{Q}$

Sheet 8 §2

Question. Which functions are injective? Which are surjective? Prove it. If a function is bijective, find a formula for its inverse. ($Bool = \{true, false\}$.)
(a) $f: \mathbb{Q} \to \mathbb{Q}, x \mapsto 12x + 31$.

Answer (worked out here, no official key)

Injective and surjective, so bijective. $f^{-1}: \mathbb{Q} \to \mathbb{Q}, y \mapsto (y - 31)/12$.

```
Injective.
1. Let x₁, x₂ ∈ ℚ with f(x₁) = f(x₂).
2. 12x₁ + 31 = 12x₂ + 31        (def. of f)
3. 12x₁ = 12x₂                  (basic algebra)
4. x₁ = x₂                      (divide by 12 ≠ 0)
5. So f is injective.
Surjective.
1. Let y ∈ ℚ.
2. Let x = (y − 31)/12. Then x ∈ ℚ
   (ℚ closed under -, and ÷ by nonzero)
3. f(x) = 12·(y − 31)/12 + 31 = y
   (def. of f; basic algebra)
4. So f is surjective.
```

Break it down

1. **Injective template**: assume two inputs give the same output, $f(x_1) = f(x_2)$, and derive $x_1 = x_2$. This is the contrapositive of "different inputs give different outputs", and it is easier because equations are easier to manipulate than "$\neq$".
2. **Surjective template**: let y be any element of the codomain, then produce an x in the domain with $f(x) = y$.
3. Where does $x = (y - 31)/12$ come from? Scratch work, not part of the proof: solve $y = 12x + 31$ for x. The proof then states that x and checks it.
4. The check has two parts, and people forget the first. (i) x is in the domain. (ii) $f(x) = y$. Part (i) is where the number system matters. On $\mathbb{Q}$, $(y - 31)/12$ is rational. If f were $\mathbb{Z} \to \mathbb{Z}$, then $y = 0$ would need $x = -31/12$, not an integer, so f would not be surjective.
5. The inverse formula is exactly your scratch-work solution. Check both directions: $f(f^{-1}(y)) = 12 \cdot (y - 31)/12 + 31 = y$ and $f^{-1}(f(x)) = (12x + 31 - 31)/12 = x$.

> **Trick**
> Surjective: solve $y = f(x)$ for x on scratch paper, then prove "x is in the domain" and "$f(x) = y$". The solved formula is also the inverse. The domain check is what decides $\mathbb{Z}$-versus-$\mathbb{Q}$ questions.

Q4 (b) · $(p, q) \mapsto p \wedge \sim q$

Sheet 8 §2

Question. (b) $g: Bool^2 \to Bool, (p, q) \mapsto p \wedge \sim q$.

Answer (worked out here, no official key)

Not injective, but surjective. Not bijective, so no inverse.

(p, q)	$g(p, q) = p \wedge \sim q$
(true, true)	false
(true, false)	true
(false, true)	false
(false, false)	false

Not injective: $g(true, true) = false = g(false, false)$, but $(true, true) \neq (false, false)$. Surjective: $true = g(true, false)$ and $false = g(false, false)$, so both elements of $Bool$ are hit.

Break it down

1. $Bool^2 = Bool \times Bool$ is the set of the 4 ordered pairs of truth values. The domain is finite, so you can compute every output, and a complete table is a complete proof.
2. $p \wedge \sim q$ is true only when p is true and q is false (row 2). Every other row gives false.
3. Injective? Look down the output column for a repeat. false appears three times; any two of those rows are a counterexample.
4. Surjective? The codomain is $\{true, false\}$. Check that each appears in the output column. Both do.
5. Size check: 4 inputs going into 2 outputs forces some output to repeat (the pigeonhole principle), so injective was never possible.

> **Trick**
> On a small finite set, write the full input/output table. Injective ⇔ no output repeats. Surjective ⇔ every codomain element appears. If the domain is bigger than the codomain, it cannot be injective; if smaller, it cannot be surjective.

Q4 (c) · $(p, q) \mapsto (p \wedge q, p \vee q)$

Sheet 8 §2

Question. (c) $h: Bool^2 \to Bool^2, (p, q) \mapsto (p \wedge q, p \vee q)$.

Answer (worked out here, no official key)

Neither injective nor surjective. No inverse.

(p, q)	$h(p, q)$
(true, true)	(true, true)
(true, false)	(false, true)
(false, true)	(false, true)
(false, false)	(false, false)

Not injective: $h(true, false) = h(false, true) = (false, true)$. Not surjective: $(true, false)$ is never an output, because if $p \wedge q$ is true then p is true, so $p \vee q$ is true as well.

Break it down

1. The outputs are now pairs. For each of the 4 inputs, compute both coordinates.
2. Injective fails at rows 2 and 3. That is no accident: $\wedge$ and $\vee$ are both unchanged when you swap p and q (commutative laws), so an input with $p \neq q$ always collides with its swap.
3. Surjective fails: list the 4 elements of the codomain and look for one missing from the output column. $(true, false)$ is missing. Explain *why*, not just that it is missing: a true first coordinate forces a true second coordinate.
4. Domain and codomain both have 4 elements. For a function between finite sets of the same size, injective ⇔ surjective. So once one fails, the other must fail too. Use this as a sanity check.

Trick

A formula that does not change when you swap its inputs is never injective on pairs with $p \neq q$. Between finite sets of equal size, injective, surjective and bijective stand or fall together.

Q4 (d) · A piecewise $k: \mathbb{Z} \to \mathbb{Z}$

Sheet 8 §2

Question. (d) $k: \mathbb{Z} \to \mathbb{Z}$, $k(x) = x$ if x is even, and $k(x) = 2x - 1$ if x is odd.

Answer (worked out here, no official key)

Injective, not surjective (3 has no preimage). No inverse.

```
x    : -2 -1  0  1  2  3  4  5
k(x) : -2 -3  0  1  2  5  4  9

Key fact: k(x) has the same parity as x.
  x even: k(x) = x is even.
  x odd:  k(x) = 2x - 1 = 2(x - 1) + 1 is odd
  (x - 1 ∈ ℤ by closure; def. of odd)
Injective.
1. Let x₁, x₂ ∈ ℤ with k(x₁) = k(x₂).
2. x₁ and x₂ both have the parity of
   k(x₁) = k(x₂), so the same parity.
   (key fact)
2.1. Both even: x₁ = k(x₁) = k(x₂) = x₂.
2.2. Both odd: 2x₁ - 1 = 2x₂ - 1, so
     x₁ = x₂.                (basic algebra)
3. So k is injective.
Not surjective.
1. Suppose k(x) = 3 for some x ∈ ℤ.
2. 3 is odd, so x is odd.       (key fact)
3. So 2x - 1 = 3, giving x = 2, which is
   even. Contradiction.
4. So 3 is not in the range: k is not
   surjective.
```

Break it down

1. **Parity** means whether an integer is even or odd.
2. Piecewise functions can fail injectivity in a sneaky way: each piece may be fine on its own, but an even input and an odd input could land on the same output. The real work is ruling out these cross-collisions.
3. The key fact does exactly that: the even piece outputs evens, the odd piece outputs odds, so outputs from different pieces are never equal. After that, each piece is injective on its own (x itself, and $2x - 1$).
4. For surjectivity, compute some values and look for a gap. Every even number appears (it maps to itself). The odd outputs are $\ldots, -3, 1, 5, 9, \ldots$, jumping by 4, since odd $x = 2m + 1$ gives $4m + 1$. So odd numbers like 3 and -1 are missed.
5. Proving 3 is missed: try to solve $k(x) = 3$ in each piece. The even piece needs $x = 3$, but 3 is not even. The odd piece needs $x = 2$, but 2 is not odd. A solution only counts if it satisfies its own piece's condition. The numbered proof packs this into one contradiction.

Trick

Piecewise injectivity: (1) each piece is injective, and (2) different pieces produce non-overlapping outputs. Piecewise surjectivity: solve in every piece, and throw out any solution that breaks that piece's condition.

Q5 (a) · Left inverses

Sheet 8 §1

Question. For $f: A \to B$: $L: B \to A$ is a left inverse of f iff $L(f(a)) = a$ for all $a \in A$; $R: B \to A$ is a right inverse of f iff $f(R(b)) = b$ for all $b \in B$. (The sheet calls them g and h; renamed here to avoid a clash with Q4.) No proofs needed. (a) Which of the four functions in Q4 have a left inverse?

Answer (worked out here, no official key)

f (Q4a) and k (Q4d), exactly the injective ones.

- f: its inverse $y \mapsto (y - 31)/12$ works.
- k: $L(y) = y$ if y is even, $L(y) = (y + 1)/2$ if y is odd. Check: x even gives $L(k(x)) = L(x) = x$; x odd gives $L(2x - 1) = (2x - 1 + 1)/2 = x$.
- g and h are not injective, so they have none.

Term

The **identity function** $\mathrm{id}_A: A \to A$ is $\mathrm{id}_A(a) = a$. So L is a left inverse iff $L \circ f = \mathrm{id}_A$ (do f, then L, and you are back where you started), and R is a right inverse iff $f \circ R = \mathrm{id}_B$.

Break it down

1. Why injective is needed: suppose $f(a_1) = f(a_2) = b$ with $a_1 \neq a_2$. A left inverse must send b back to a_1 (because $L(f(a_1)) = a_1$) and also to a_2. A function cannot give b two outputs. So g (where $g(true, true) = g(false, false)$) and h have no left inverse.
2. Why injective is enough: each b in the range has exactly one preimage, so let L send b back to it. For b outside the range, the definition never looks at $L(b)$ (it only checks $L(f(a))$), so send it anywhere in A. This needs $A \neq \varnothing$, which holds here.
3. Building L for k: undo each piece. Even outputs came from the even piece, which changed nothing, so $L(y) = y$. Odd outputs came from $2x - 1$, so solve $y = 2x - 1$ to get $x = (y + 1)/2$. For odd y outside the range, like 3, the value $L(3) = 2$ is a harmless "don't care", and it is still an integer because $y + 1$ is even.

Trick

Left inverse ⇔ injective (non-empty domain). Right inverse ⇔ surjective. Both ⇔ bijective, and then both are f^{-1}. Memory hook: a left inverse undoes f afterwards, so f must not merge inputs; a right inverse picks a preimage for every target, so every target must have one.

Q5 (b) · Right inverses

Sheet 8 §1

Question. (b) Which of the four functions in Q4 have a right inverse?

Answer (worked out here, no official key)

f (Q4a) and g (Q4b), exactly the surjective ones.

- f: its inverse again.
- g: $R(true) = (true, false)$ and $R(false) = (false, false)$. Check: $g(R(true)) = true \wedge \sim false = true$ and $g(R(false)) = false \wedge \sim false = false$.
- h and k are not surjective, so they have none.

Break it down

1. $f(R(b)) = b$ for all b says every b is an output of f (the output of the input $R(b)$). So a right inverse forces surjectivity. h misses $(true, false)$ and k misses 3, so neither has one.
2. Conversely, if f is surjective, build R by choosing one preimage for each b. When b has several preimages (false has three under g), any choice works, so right inverses need not be unique.
3. This R does not undo g in the other order: $R(g(true, true)) = R(false) = (false, false)$. That is fine; a right inverse only has to work as $f \circ R$.

> **Trick**
> To build a right inverse, go through the codomain one element at a time and write down one preimage for each. If some element has no preimage, there is no right inverse.

Q5 (c) · Does one inverse give the other?

Sheet 8 §1

Question. (c) Which are true? (i) If a function has a left inverse, then it has a right inverse. (ii) If a function has a right inverse, then it has a left inverse.

Answer (worked out here, no official key)

Neither. (i) is false: k has a left inverse but no right inverse. (ii) is false: g has a right inverse but no left inverse.

Break it down

1. Translate using (a) and (b): (i) says "injective → surjective", and (ii) says "surjective → injective".
2. One counterexample disproves a general "if ... then" claim. Q4 already hands you both: k is injective but not surjective; g is surjective but not injective.
3. Why they feel true: for a function between two finite sets of the same size, injective ⇔ surjective (Q4(c) shows this). The claims break once the sizes differ (g has 4 inputs and 2 outputs) or the sets are infinite (k on $\mathbb{Z}$).

> **Trap**
> The equal-size fact does not carry over to infinite sets. $k: \mathbb{Z} \to \mathbb{Z}$ has the same set on both sides, yet it is injective and not surjective.

> **Trick**
> When a "which statements are true" part comes right after a question full of examples, those examples are usually the counterexamples. Reuse them.

Q6 · $g \circ f$ injective forces f injective

Sheet 8 §1

Question. (Theorem 7.3.3: if $f: X \to Y$ and $g: Y \to Z$ are injective, then $g \circ f$ is injective.) Now let $f: B \to C$, and let g be a function with domain C such that $g \circ f$ is injective. Show that f is injective.

Answer (worked out here, no official key)

```
1. Let x₁, x₂ ∈ B with f(x₁) = f(x₂).
2. g(f(x₁)) = g(f(x₂))
     (g is a function: equal inputs give
   equal outputs)
3. (g∘f)(x₁) = (g∘f)(x₂)
     (def. of composition)
4. x₁ = x₂              (g∘f is injective)
5. So f is injective.   (def. of injective)
```

Break it down

1. Goal: f injective. Template: start with $f(x_1) = f(x_2)$, finish with $x_1 = x_2$.
2. Your only tool is "$g \circ f$ is injective", which turns $(g \circ f)(x_1) = (g \circ f)(x_2)$ into $x_1 = x_2$. So you need to get from $f(x_1) = f(x_2)$ to $g(f(x_1)) = g(f(x_2))$.
3. The bridge is "apply g to both sides". It is allowed because $f(x_1)$ and $f(x_2)$ are the same element of C, and a function gives one output per input. It is the same move as "if $a = b$ then $a^2 = b^2$". Since the domain of g is C, where $f(x_1)$ lives, $g(f(x_1))$ makes sense.
4. What you did not need: g injective. In fact g need not be. Example: $B = \{1\}$, $C = \{1, 2\}$, $f(1) = 1$, $g(1) = g(2) = 1$. Then $g \circ f$ has a one-element domain, so it is injective, but g is not.
5. Direction matters. "Apply g to both sides" is always legal. "Cancel g from both sides" ($g(a) = g(b) \Rightarrow a = b$) needs g injective. Mixing these up is the classic error here.

> **Trick**
> Know something about $g \circ f$, want something about f: apply g to both sides. Remember the pair: $g \circ f$ injective ⇒ the inner function f is injective; $g \circ f$ surjective ⇒ the outer function g is surjective.

Q7 · Orders of bijections on {1, 2, 3}

Question. $A = \{1, 2, 3\}$. The order of a bijection $f: A \to A$ is the smallest $n \in \mathbb{Z}^+$ with $f \circ f \circ \cdots \circ f$ (n copies of f) $= \mathrm{id}_A$. Define $g, h: A \to A$ by $g(2) = 1$, $g(1) = 2$, $g(3) = 3$ and $h(3) = 2$, $h(2) = 3$, $h(1) = 1$. Find the orders of g, h, $g \circ h$ and $h \circ g$.

Answer (worked out here, no official key)

Orders: g is 2, h is 2, $g \circ h$ is 3, $h \circ g$ is 3.

x	$h(x)$	$g(h(x))$	$g(x)$	$h(g(x))$
1	1	2	2	3
2	3	3	1	1
3	2	1	3	2

```
g ≠ id_A, since g(1) = 2.
g∘g: 1 ↦ 2 ↦ 1, 2 ↦ 1 ↦ 2, 3 ↦ 3 ↦ 3.
So g∘g = id_A: order 2. Same for h.

c = g∘h: 1 ↦ 2, 2 ↦ 3, 3 ↦ 1.
c ≠ id_A, since c(1) = 2.
c∘c ≠ id_A, since c(c(1)) = c(2) = 3.
c∘c∘c: 1 ↦ 2 ↦ 3 ↦ 1, 2 ↦ 3 ↦ 1 ↦ 2,
       3 ↦ 1 ↦ 2 ↦ 3. Order 3.
h∘g: 1 ↦ 3, 3 ↦ 2, 2 ↦ 1. Same shape,
     one loop of length 3: order 3.
```

Break it down

1. Turn each piecewise rule into an arrow list: g is $1 \mapsto 2$, $2 \mapsto 1$, $3 \mapsto 3$ (it swaps 1 and 2). h swaps 2 and 3.
2. Composition order: $g \circ h$ means h first, then g, just like $g(h(x))$. Fill the table column by column: first $h(x)$, then apply g to that.
3. The **order** is how many times in a row you must apply f before *every* element is back where it started. One element returning is not enough, and you must also check that every smaller number fails (that is the "smallest" part).
4. Cycle picture: a swap is a loop of length 2, so everyone is home after 2 steps. $g \circ h$ sends $1 \to 2 \to 3 \to 1$, one loop of length 3, so it takes 3 steps.
5. $g \circ h \neq h \circ g$: composition is not commutative. They are the same loop run in opposite directions. In fact $h \circ g = (g \circ h)^{-1}$, because by Q3 $(g \circ h)^{-1} = h^{-1} \circ g^{-1} = h \circ g$ (each swap undoes itself).

Trick

For a bijection of a small set, draw its arrows and find the loops (cycles). The order is the loop length, or the lcm of the loop lengths if there are several (a 2-loop plus a 3-loop needs 6 steps).

Q8 (a)(b) · Iterating $g(x) = 3x \,\%\, 5$

Question. $n \,\%\, d$ is the remainder r in $n = dq + r$, $0 \le r < d$ (quotient-remainder theorem). $f^{(n)}(x)$ is f applied n times to x, e.g. $f^{(3)}(x) = f(f(f(x)))$. The order of an input x with respect to f is the smallest positive integer m with $f^{(m)}(x) = x$. Let $g: A \to A$, $g(x) = 3x \,\%\, 5$. (a) What is $g^{(3)}(21)$? (b) What is the order of 3 with respect to g?

$A = \{0, 1, 2, 3, 4\}$ is only given in part (c), and 21 is not in it. Read (a) as applying the formula $3x \,\%\, 5$ to 21; from the first step on, every value is in A anyway.

Answer (worked out here, no official key)

(a) 2. (b) 4.

```
(a) g(21) = 63 % 5 = 3    (63 = 5·12 + 3)
    g(3)  = 9 % 5  = 4    (9 = 5·1 + 4)
    g(4)  = 12 % 5 = 2    (12 = 5·2 + 2)
    So g^(3)(21) = 2.
(b) g(3) = 4, g(4) = 2, g(2) = 6 % 5 = 1,
    g(1) = 3 % 5 = 3.
    3 ↦ 4 ↦ 2 ↦ 1 ↦ 3: first return to 3
    after 4 steps, so the order is 4.
```

Term

$n \,\%\, d$ is the remainder when n is divided by d, always one of $0, 1, \ldots, d-1$. For $n \,\%\, 5$ with $n \ge 0$, subtract the largest multiple of 5 that is at most n. With negatives the remainder is still non-negative: $-7 \,\%\, 5 = 3$, because $-7 = 5 \cdot (-2) + 3$.

Break it down

1. $g^{(3)}(21) = g(g(g(21)))$. Work from the inside out, one application at a time, and write down every intermediate value.
2. "Order of an input" is new vocabulary but simple: start at x, keep applying g, and count the steps until you first land back on x.
3. Do not confuse it with Q7's order of a *function*, which needs all elements back home at the same time. Here only the one input matters.
4. Check you did not stop early: after 1, 2 and 3 steps you are at 4, 2 and 1, none of which is 3.

Trick

For iterated functions, write the chain $x \mapsto g(x) \mapsto g(g(x)) \mapsto \ldots$ with every link shown. The order of x is the length of the loop x lives on.

Q8 (c) · Classes of "same order"

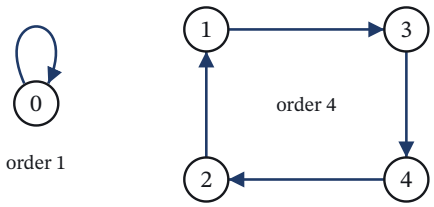
Sheet 5B §3

Question. (c) Let $A = \{0, 1, 2, 3, 4\}$, and let $x \,R\, y$ iff the order of x equals the order of y with respect to g. R is an equivalence relation. Write out all the distinct equivalence classes in set-roster notation (not with [ ]).

Answer (worked out here, no official key)

$\{0\}$ and $\{1, 2, 3, 4\}$.

```
x    : 0  1  2  3  4
3x   : 0  3  6  9  12
g(x) : 0  3  1  4  2
0 ↦ 0: order 1.
1 ↦ 3 ↦ 4 ↦ 2 ↦ 1: one loop of length 4,
so 1, 2, 3, 4 each have order 4.
```



Each arrow is one step $x \to g(x)$. 0 sits on a loop of length 1; 1, 3, 4, 2 share a loop of length 4.

Term

The **equivalence class** of x is $[x] = \{y \in A : x \,R\, y\}$, everything related to x. **Roster notation** lists the elements inside braces. "Distinct" means write each class once, even though $[1] = [2] = [3] = [4]$.

Break it down

1. Compute g on every element of A (5 inputs, quick), then draw an arrow $x \to g(x)$ for each.
2. Every element has exactly one outgoing arrow (because g is a function), and here also exactly one incoming arrow (because g is a bijection on A), so the arrows form loops: 0 loops to itself, and $1 \to 3 \to 4 \to 2 \to 1$ is one loop of length 4.
3. The order of each element is the length of its loop: 0 has order 1; 1, 2, 3, 4 have order 4.
4. $x \,R\, y$ iff same order, so group the elements with equal order: $\{0\}$ and $\{1, 2, 3, 4\}$.
5. Sanity check: the classes must be disjoint and together cover A (they form a partition). They do.

Trick

Any relation of the form "$x\,R\,y$ iff $F(x) = F(y)$" is an equivalence relation, and its classes are just "elements with the same F-value". Compute F for each element and group.

Q9 (a) · Is $X = f^{-1}(f(X))$?

Sheet 8 §1

Question. Let $f\colon A \to B$, $X \subseteq A$, $Y \subseteq B$. Here $f(X)$ is the setwise image and $f^{-1}(Y)$ the setwise preimage. (a) Is it always the case that $X \subseteq f^{-1}(f(X))$? That $f^{-1}(f(X)) \subseteq X$? Justify.

Answer (worked out here, no official key)

$X \subseteq f^{-1}(f(X))$: always. $f^{-1}(f(X)) \subseteq X$: not always.

```
X ⊆ f⁻¹(f(X)):
1. Let x ∈ X.
2. f(x) ∈ f(X)    (def. of image; x ∈ X)
3. x ∈ A and f(x) ∈ f(X), so
   x ∈ f⁻¹(f(X))      (def. of preimage)
4. So X ⊆ f⁻¹(f(X)).

Counterexample to f⁻¹(f(X)) ⊆ X:
A = {1, 2}, B = {0}, f(1) = f(2) = 0,
X = {1}. Then f(X) = {0} and
f⁻¹({0}) = {1, 2} ⊈ {1}.
```

Term

Membership tests, straight from the definitions: $b \in f(X) \Leftrightarrow b = f(x)$ for some $x \in X$; and $a \in f^{-1}(Y) \Leftrightarrow a \in A$ and $f(a) \in Y$.

Break it down

1. Read $f^{-1}(f(X))$ inside out. $f(X)$ is the set of outputs of elements of X. f^{-1} of that is every element of A whose output is one of those outputs.
2. Each $x \in X$ qualifies: its output $f(x)$ is, by definition, one of the outputs of X. That is the whole idea of the first proof; the numbered lines just write it with definitions.
3. The reverse can fail because an element outside X may share an output with something inside X, and then it sneaks into $f^{-1}(f(X))$. That needs two inputs with the same output, so f must fail to be injective.
4. So build the smallest non-injective function (two inputs, one output) and put only one of the inputs in X.
5. When f is injective, nothing can sneak in, and $f^{-1}(f(X)) = X$.

Trick

"Is it always the case that ...?": prove the true one by element chasing (let x be in the left set; use definitions to land in the right set). Kill the false one with a tiny example of one or two elements. Where the proof gets stuck tells you what to build: here, a collision.

Q9 (b) · Is $Y = f(f^{-1}(Y))$?

Sheet 8 §1

Question. (b) Is it always the case that $Y \subseteq f(f^{-1}(Y))$? That $f(f^{-1}(Y)) \subseteq Y$? Justify.

Answer (worked out here, no official key)

$Y \subseteq f(f^{-1}(Y))$: not always. $f(f^{-1}(Y)) \subseteq Y$: always.

```
f(f⁻¹(Y)) ⊆ Y:
1. Let b ∈ f(f⁻¹(Y)).
2. b = f(a) for some a ∈ f⁻¹(Y)
   (def. of image)
3. f(a) ∈ Y          (def. of preimage)
4. So b = f(a) ∈ Y, and f(f⁻¹(Y)) ⊆ Y.

Counterexample to Y ⊆ f(f⁻¹(Y)):
A = {1}, B = {0, 1}, f(1) = 0, Y = {1}.
f⁻¹(Y) = ∅ (nothing maps to 1), and
f(∅) = ∅. But {1} ⊈ ∅.
```

Break it down

1. Inside out: $f^{-1}(Y)$ is every input that lands in Y. Applying f to those gives the elements of Y that actually get hit.
2. Those are all in Y, so the second inclusion always holds. In step 2, "for some a" is the $\exists$ hidden in the definition of image: name the witness a and use it.
3. The first inclusion fails when some element of Y is never hit: it is in Y but cannot appear in $f(f^{-1}(Y))$. That needs f to fail to be surjective.
4. So build the smallest function that misses a codomain element, and put that element in Y.
5. When f is surjective, every element of Y gets hit, and $f(f^{-1}(Y)) = Y$.
6. Handy facts for such examples: $f(\varnothing) = \varnothing$ and $f^{-1}(\varnothing) = \varnothing$.

Trick

Memorise the pair: $X \subseteq f^{-1}(f(X))$ and $f(f^{-1}(Y)) \subseteq Y$ always hold. Equality in the first needs injective; equality in the second needs surjective. When unsure, test a constant function (not injective) and a function that misses a codomain element (not surjective).

Q10 (a) · Is + well defined on $\mathbb{Q}/\sim$?

Sheet 8 §1

Question. The equivalence relation $\sim$ on $\mathbb{Q}$ is $x \sim y \Leftrightarrow x - y \in \mathbb{Z}$. On $\mathbb{Q}/\sim$ define $[x] + [y] = [x + y]$ and $[x] \cdot [y] = [x \cdot y]$. (a) Is + well defined on $\mathbb{Q}/\sim$? (b) Is $\cdot$ well defined? Prove your answers.

Answer (worked out here, no official key)

Yes, + is well defined.

```
1. Let x, x′, y, y′ ∈ ℚ with [x] = [x′] and
   [y] = [y′].
2. x ~ x′ and y ~ y′
   (equal classes ⇔ related elements)
3. x − x′ = m and y − y′ = n for some
   m, n ∈ ℤ              (def. of ~)
4. (x + y) − (x′ + y′)
   = (x − x′) + (y − y′) = m + n
   (basic algebra)
5. m + n ∈ ℤ (closure of integers under +)
6. So x + y ~ x′ + y′ (def. of ~), that is,
   [x + y] = [x′ + y′].
7. So + is well defined.
```

Term

Here ~ is the name of a relation, not "not". The **equivalence class** $[x] = \{y \in \mathbb{Q} : x \sim y\}$ is every rational that differs from x by an integer, e.g. $[1/2] = \{\ldots, -1/2, 1/2, 3/2, \ldots\}$. The **quotient** $\mathbb{Q}/\sim$ is the set of all classes. A **representative** of a class is any one of its members: 1/2 and 3/2 both represent $[1/2]$. An operation defined through representatives is **well defined** if the answer does not depend on which representatives you pick.

Break it down

1. Picture: $x \sim y$ says x and y have the same fractional part. So a class is like a point on a clock face of circumference 1: 1/2, 3/2 and $-1/2$ are all the same point.
2. The formula $[x] + [y] = [x + y]$ reaches inside the classes and grabs representatives. But a class has many names: $[1/2] = [3/2]$. If different names gave different answers, then + would give one input two outputs and would not be a function (the "clash" from D1). Well-definedness rules this out.
3. Test first: $[1/2] + [1/3]$ with names 1/2 and 1/3 gives $[5/6]$; with names 3/2 and 1/3 it gives $[11/6]$. Since $11/6 - 5/6 = 1 \in \mathbb{Z}$, these are the same class. A good sign.
4. Proof pattern: take two names for each input (x, x' and y, y'), turn "same class" into "differ by an integer", and show the two results also differ by an integer.
5. Step 2 uses the lemma from the equivalence-relations chapter: $[a] = [b] \Leftrightarrow a \sim b$.

Trick

Well-defined means "different names, same answer". Take $x \sim x'$ and $y \sim y'$, compute the result both ways, and show the two results are related.

Q10 (b) · Is · well defined on $\mathbb{Q}/\sim$?

Sheet 8 §1

Question. (b) Is $[x] \cdot [y] = [x \cdot y]$ well defined on $\mathbb{Q}/\sim$? Prove your answer.

Answer (worked out here, no official key)

No.

```
1. 0 − 1 = −1 ∈ ℤ, so 0 ~ 1 and [0] = [1].
2. [0] · [1/2] = [0 · 1/2] = [0]
3. [1] · [1/2] = [1 · 1/2] = [1/2]
4. 0 − 1/2 = −1/2 ∉ ℤ, so 0 ~ 1/2 is false,
   and [0] ≠ [1/2].
5. The same pair of classes, ([0], [1/2]) =
   ([1], [1/2]), gives two different results.
   So · is not well defined.
```

Break it down

1. Try the method of (a) and see where it breaks. Write $x' = x + m$ and $y' = y + n$ with $m, n \in \mathbb{Z}$ (that is exactly what $x \sim x'$ and $y \sim y'$ say). Then $x'y' = xy + xn + my + mn$, so $x'y' - xy = xn + my + mn$. The mn part is an integer, but xn and my need not be: an integer times a fraction can be a fraction.
2. That tells you how to build the counterexample: shift one representative by an integer ($m = 1$) and make the other factor a fraction ($y = 1/2$), so $my = 1/2$ escapes $\mathbb{Z}$.
3. Simplest choice: $x = 0$, $x' = 1$, $y = y' = 1/2$. The results 0 and 1/2 differ by 1/2, which is not an integer, so they lie in different classes.
4. To disprove well-definedness you need only one input pair with two different outputs. You must show the outputs are genuinely different classes (line 4), not just different-looking names.
5. Contrast: congruence mod n on $\mathbb{Z}$ does allow both + and $\times$, because there every cross term is an integer multiple of n. Over $\mathbb{Q}$, multiplying by a fraction breaks that.

Trick

For "is this operation on classes well defined?": replace each representative by representative plus an allowed shift, expand, and look at the leftover. If the leftover is always allowed, prove it as in (a). If some term can escape, that term tells you the counterexample.